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# Introduction



Nearly all of us have to work during our lives. We reach adulthood and seek employment, and then we work for the next 30 to 50 years. Work is our primary means of livelihood. It serves an important economic function in the global world of commerce. It creates opportunities for social interactions and friendships. And it provides the products and services that sustain and improve our standard of living.

This text is all about work and the systems by which it is accomplished. The text also examines the principles and programs that allow work to be performed most efficiently and safely, and it discusses the techniques used to measure and manage work. The common denominator in the analysis, design, and measurement of work is time. For many reasons that are enumerated in this chapter time is important in work. In general, it is desirable to accomplish a given task in the shortest possible time.

Work systems constitute the central theme in the discipline of industrial engineering. Each engineering discipline is concerned with its own type of technical system. The correlations are listed in Table 1 for the major engineering disciplines. In most cases, each engineering field has expanded to include several related kinds of systems, and these additional systems are also listed in the table. Their addition has been a natural evolutionary process that has paralleled the development of new technologies. For industrial engineering, the additional subjects include operations research, ergonomics, discrete event simulation, and information systems. We can reasonably argue that all of these fields are related to the study of systems that perform work.

In this chapter, we define (1) work itself, (2) work systems, (3) jobs and occupations, and (4) productivity. In Section 5, we discuss the variety of topics covered in this text and the way in which the topics are organized. All of these topics are concerned with work in one way or another. Some familiar quotations and sayings about work and jobs are presented in Table 2 for the reader's amusement. Work has been the object

**TABLE 1** Engineering Disciplines and the Types of Systems They Design and Analyze

| Engineering Discipline | Type of System | New Systems and Subjects |
|---|---|---|
| Aeronautical engineering | Airplanes | Aerospace systems, missile systems |
| Chemical engineering | Chemical systems, chemical processing systems | Processing of integrated circuits, process control, polymer science, biotechnology |
| Civil engineering | Structural systems (e.g., bridges, buildings, roads) | Environmental systems, transportation systems, fluid systems |
| Electrical engineering | Electrical systems, power generation systems | Computer systems, integrated circuits, control systems |
| Industrial engineering | Work systems | Operations research, ergonomics, simulation, information systems |
| Mechanical engineering | Mechanical systems, thermal systems | Control systems, computer-aided design, micro-electro-mechanical systems |
| Metallurgical engineering[a] | Metals systems | Ceramics, polymers, composites, electron microscopy |

[a]The name *materials science and engineering* has largely replaced *metallurgical engineering*.

**TABLE 2** Notable Observations About Work and Jobs

I do not like work even when someone else does it. (Mark Twain)
All work and no play make Jack a dull boy—to everyone but his employer.
Man may work from sun to sun, but a woman's work is never done.
A woman's work is never done, especially the part she asks her husband to do.
The people who claim that brain work is harder than physical work are generally brain workers.
Many thousands of people are already working a four-day week; the trouble is it takes them five days to do it.
Hard work never hurt anyone who hired someone else to do it.
Hard work pays off in the future; laziness pays off now.
Some people are so eager for success that they are even willing to work for it.
It isn't the hours you put into your work that count, it's the work you put into the hours.
What will happen to work when the trend toward longer education meets the trend toward earlier retirement?
Genius is one percent inspiration and ninety-nine percent perspiration. (Thomas A. Edison)
The difference between a job and a career is the difference between 40 and 60 hours a week. (Robert Frost)
The softer the job, the harder it is to get.
The best man for the job is often a woman.
Two can live as cheaply as one—if they both have good jobs.

*Source:* Compiled from J. Bartlett, *Familiar Quotations*, 14th ed., E. M. Beck, ed. (Boston: Little, Brown, 1968); E. Esar, *20,000 Quips and Quotes*, (New York: Barnes and Noble, 1995); and other sources.

of study for many years, and some of the more significant findings and personalities that have contributed to this field are described in Historical Note 1.

### HISTORICAL NOTE 1 THE STUDY OF WORK

There is evidence that the study of work and some of the basic principles about work originated in ancient times [1]. The Babylonians used the principle of a ***minimum wage*** around 1950 B.C. The Chinese organized work according to the principle of ***labor specialization*** around 1644 B.C. The ancient Romans used a primitive form of ***factory system*** for the production of armaments, textiles, and pottery. They also perfected the military organizational structure, which is the basis for today's ***line and staff organization*** of work.

The ***Industrial Revolution*** started in England around 1770 with the invention of several new machines used in the production of textiles,[1] ***James Watt***'s steam engine, and ***Henry Maudslay***'s screw-cutting lathe. These inventions resulted in fundamental changes in the way work was organized and accomplished: (1) the transfer of skill from workers to machines, (2) the start of the machine tool industry, based on Maudslay's lathe and other new types of metal-cutting machines that allowed parts to be produced more quickly and accurately than the prior handicraft methods, and (3) the introduction of the factory system in textile production and other industries, which used large numbers of unskilled workers (including women and children) who labored long hours for low pay. The factory system employed the specialization of labor principle.

While England was leading the Industrial Revolution, the important concept of ***interchangeable parts manufacture*** was being introduced in the United States. Much credit for this concept is given to ***Eli Whitney*** (1765–1825), although others had recognized its importance [7]. In 1797, Whitney negotiated a contract to produce 10,000 muskets for the U.S. government. The traditional way of making guns at the time was to custom-fabricate each part for a particular gun and then hand-fit the parts together by filing. Each musket was therefore unique. Whitney believed that the components could be made accurately enough to permit parts assembly without fitting. After several years of development in his Connecticut factory, he was able to demonstrate the principle before government officials, including Thomas Jefferson. His achievement was made possible by the special machine tools, fixtures, and gauges that he had developed. Interchangeable parts manufacture required many years of refinement in the early and mid-1800s before becoming a practical reality, but it revolutionized work methods used in manufacturing. It is a prerequisite for mass production of assembled products.

The mid- and late 1800s and early 1900s witnessed the introduction of several consumer products, including the sewing machine, bicycle, and automobile. In order to meet the mass demand for these products, more efficient production methods were required. Some historians identify developments during this period as the Second

[1]The machines included (1) James Hargreaves's Spinning Jenny, patented in 1770; (2) Richard Arkwright's Water Frame, developed in 1771; (3) Samuel Crompton's Mule-Spinner, developed around 1779; and (4) Edmund Cartwright's Power Loom, patented in 1785. Historians sometimes include James Kay's Flying Shuttle introduced in 1733 in the list of great inventions of the Industrial Revolution.

Industrial Revolution, characterized in terms of its effects on work systems by the following: (1) mass production, (2) assembly lines, and (3) scientific management.

***Mass production*** was primarily an American phenomenon. Its motivation was the mass market that existed in the United States, where the population in 1900 was 76 million and growing and by 1920 exceeded 106 million. Such a large population, larger than any western European country, created a demand for large numbers of products. Mass production provided those products. Certainly one of the important technologies of mass production was the moving ***assembly line***, introduced by ***Henry Ford*** (1863–1947) in 1913 at his Highland Park plant. The assembly line was a new form of work system and made possible the mass production of complex consumer products. Use of assembly line methods permitted Ford to sell a Model T automobile for less than $500 in 1916, thus making ownership of cars feasible for a large segment of the American population.

The ***scientific management*** movement started in the late 1800s in the United States in response to the need to plan and control the activities of growing numbers of production workers. The most important members of this movement were Frederick W. Taylor, Frank Gilbreth, and Lillian Gilbreth. The principal approaches of scientific management were the following: (1) motion study, aimed at finding the best method to perform a given task and eliminating delays; (2) time study to establish work standards for a job; (3) extensive use of standards in industry; (4) the piece rate system and similar labor incentive plans; and (5) use of data collection, record keeping, and cost accounting in factory operations. These approaches, while revolutionary at the time they were first implemented, are fundamental and indispensable techniques used today in business and industry for work management.

***Frederick W. Taylor*** (1856–1915) is known as the "father of scientific management" for his application of systematic approaches to the study and improvement of work. His findings and writings have influenced factory management in virtually every industrialized country in the world, especially the United States. It can be argued that much of the mass production power of the United States in the twentieth century is due to the scientific management principles espoused by Taylor.

Born in Philadelphia into an upper middle class family, Taylor attended the local Germantown Academy and then Phillips Exeter Academy in New Hampshire. His father's wish was for young Taylor to pursue a legal career. However, after passing the entrance tests for Harvard University, poor eyesight forced him to abandon those plans.[2] Instead, he became an apprentice patternmaker and machinist in 1874. In 1878, he became an employee at the Midvale Steel Company in Philadelphia and during the next 12 years progressed from machine shop worker to gang boss, foreman, and master mechanic. In 1883, he earned a mechanical engineering degree from Stevens Institute of Technology through part-time study and was promoted to chief engineer at the company.

[2]Taylor's poor eyesight was apparently caused by too much nighttime study at Phillips Exeter. He was at the top of his class at the Academy. There is some speculation that young Taylor did not want to follow his father's profession as a lawyer. He preferred to follow his own career path. His eyesight was largely recovered by 1875.

Taylor introduced time study at Midvale Steel sometime between 1881 and 1883 while he was foreman of the machine shop. By 1883, a given task had been divided into work elements, and the timing of each element and then summing the times had been found to be more useful than timing the whole task. Taylor's belief was that by studying each element, wasted motions could be eliminated and efficiency could thereby be improved in every step of an operation. While at Midvale, Taylor also conducted metal-cutting experiments on the company's products, which included military cannons and locomotive wheels. The experiments resulted in significant improvements in productivity in the plant. However, not all of Taylor's proposals about work were successful. A notable example was his idea to separate the shop foreman's job into eight specialized functions. The resulting structure is called a functional organization.

Taylor left Midvale Steel in 1889. From 1890 to 1893, he served as general manager of a company that processed wood pulp. He then became a management consultant from 1893 to 1901. During this latter period, his most important engagement was with the Bethlehem Iron Works (predecessor to the Bethlehem Steel Company) between 1898 and 1901. Two famous experiments were conducted at Bethlehem: (1) the shoveling experiment and (2) pig iron handling.

In the ***shoveling experiment***, Taylor observed that each yard worker brought his own shovel to work, and the shovels were all different sizes. The workers were required to shovel various materials in the yard, such as ashes, coal, and iron ore. Because the densities of these materials differed, it meant that the weight per shovelful varied significantly. Through experimentation with the workers, he determined that different-sized shovels should be used for the different materials. His conclusion was that the appropriate shovel size was one in which the load was 21 pounds. This load weight maximized the amount of work that could be accomplished each day by a worker and minimized the costs to the company.

In the study of ***pig iron handling***,[3] Taylor believed that the yard workers who loaded pig iron from the storage yard into freight cars were not using the best method. The workers seemed to work too hard and then had to rest for too long to recover from the exertion. Their daily wage was $1.15 (in 1898) and they averaged 12.5 tons per day. Taylor confronted one of the men named Schmidt and offered him the opportunity to earn $1.85 per day if he followed Taylor's instructions on how to perform the work. The instruction consisted of improvements in the way the pig iron was picked up, carried, and dropped off, combined with more frequent but shorter rest breaks. Enticed by the opportunity to earn more money, Schmidt agreed.[4] By using the improved method, Schmidt was able to consistently load 47 tons per day. Other workers were eager to sign on for the higher pay.

[3]Pig iron is the iron tapped from a blast furnace. It contains impurities and must be subsequently refined to make cast iron and steel.

[4]In fact, Taylor dreamed up the name Schmidt. The worker's real name was Henry Knolle. Opponents of Taylor and scientific management circulated reports in 1910 that Schmidt had died from overworking to achieve the 47 tons per day. The truth is that Henry Knolle lived on until 1925, dying at the age of 54.

In 1901, Taylor retired to his estate in Philadelphia but continued to promote the emerging field of scientific management through lectures and publications such as ***Shop Management*** (1903), ***On the Art of Cutting Metals*** (1906), and ***Principles of Scientific Management*** (1911). Perhaps Frederick W. Taylor's most important contribution in the improvement of work was his successful promotion and promulgation of the field of scientific management during the period of his retirement.

***Frank Gilbreth*** (1868–1924) is noted for his pioneering efforts in analyzing and simplifying manual work. He was associated with the scientific management movement in the late 1800s and early 1900s, in particular for his achievements in motion study. He is sometimes referred to as the "father of motion study." Two of his important theories about work were (1) that all work was composed of 17 basic motion elements that he called "therbligs" and (2) the principle that there is "one best method" to perform a given task.

As a young man, Gilbreth had planned to attend college but was forced to get a job due to his father's untimely death. He started as a bricklayer's apprentice in 1885 at age 17. On his first day at work, Gilbreth noticed that the bricklayer assigned to teach him laid bricks in three different ways, one in normal working, a second when working fast, and a third when instructing Gilbreth. He also noticed that other bricklayers used other methods, all of which seemed to be different. It occurred to Gilbreth that there should be one best way to accomplish bricklaying, and all bricklayers should use that one best method. Gilbreth analyzed the work elements that were required in bricklaying, attempting to simplify the task and eliminate wasted motions. He was able to develop a method that reduced the number of steps required to lay one brick by about 70 percent. He participated in the development of a bricklayer's scaffold that could be adjusted in height so the worker would have bricks and mortar at the same elevation level as he was working rather than be required to stoop to ground level to retrieve these materials. The movable scaffold is still used today.

By age 26, Gilbreth decided to go into business for himself and became a widely recognized building contractor in New York City. One of the reasons for his success was the capability of his crews to complete a project quickly, a result of his interest in time and motion study. He was able to apply the principles of time and motion study to the labor of construction workers and other industrial employees so as to increase their efficiency and output. Ultimately, he would become one of the leading public speakers for Taylor's scientific management movement.

In 1904, Gilbreth married Lillian Moller, a teacher and psychologist who collaborated in his research on motion study, contributing an emphasis on the human and social attributes of work. In 1911, Gilbreth published ***Motion Study***, a book that documented many of their research findings. Around 1915, they founded the management consulting firm of Frank B. Gilbreth, Inc. When Frank died prematurely in 1924, Lillian assumed the presidency of the firm and carried on his work as a researcher and advocate of motion and time study. She became a noted educator, author, engineer, and consultant in her own right.

***Lillian Gilbreth*** (1878–1972) was the oldest of nine children. She earned bachelor's and master's degrees at the University of California, Berkeley. During her marriage to Frank, she became the mother of 12 children and earned a doctorate at Brown

University in 1915, a rare achievement for a woman at the time. Her other accomplishments were equally noteworthy. With her husband, she co-authored four books: ***A Primer of Scientific Management*** (1914), ***Fatigue Study*** (1916), ***Applied Motion Study*** (1917), and ***Motion Study for the Handicapped*** (1917). On her own, she wrote ***The Psychology of Management*** (1914) and several other books after Frank's death. She also held faculty positions at Purdue University (1935–1948) and several other universities. Among the many honors and awards she received during her lifetime, one of the most significant was her election to the National Academy of Engineering. She was the first woman ever to be elected to the NAE. With good reason, she has been called the "First Lady of Engineering."

The story of how Frank and Lillian Gilbreth practiced efficiency and motion study in their own home was humorously documented by two of their 12 children in 1949 with the publication of ***Cheaper by the Dozen***. It was made into a motion picture in 1950.

## 1 THE NATURE OF WORK

For our purposes, ***work*** is defined as an activity in which a person exerts physical and mental effort to accomplish a given task or perform a duty. The task or duty has some useful objective. It may involve one or more steps in making a product or delivering a service. The worker performing the task must apply certain skills and knowledge to complete the task or duty successfully. There is usually a commercial value in the work activity, and the worker is compensated for performing it. By commercial value, we mean that the task or duty contributes to the buying and selling of something (e.g., a product or service), which ultimately provides the means of paying for the work. Work is also performed in government, but its value is surely measured on a scale other than commercial.

In physics, ***work*** is defined as the displacement (distance) that an object moves in a certain direction multiplied by the force acting on the object in the same direction. Thus, physical work is measured in units of newton-meters (N-m) in the International System of Units (metric system) or foot-pounds (ft-lb) in U.S. customary units. This definition can be reconciled with our labor-oriented work definition by imagining a material handling worker pushing a cart across a warehouse floor by exerting a force against the cart to move it a certain distance. However, human work includes many activities other than the muscular application of forces to move objects. Nearly all human work activities include both physical and mental exertions by the worker. In some cases, the physical component dominates the activity, while in others the mental component is more important. In virtually all work situations, the activity cannot be performed unless the worker applies some combination of physical and cognitive effort.

### 1.1 The Pyramidal Structure of Work

Work consists of tasks. A ***task*** is an amount of work that is assigned to a worker or for which a worker is responsible. The task can be repetitive (as in a repetitive operation in mass production) or nonrepetitive (performed periodically, infrequently, or only once). A task can be divided into its constituent activities, which form the pyramidal structure

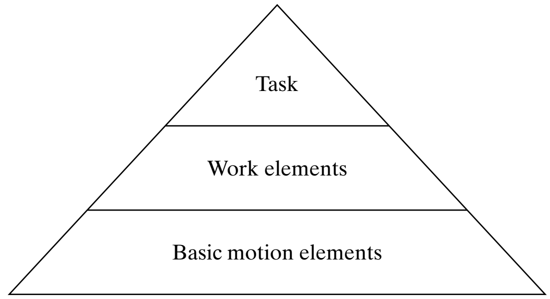



**Figure 1** The pyramidal structure of a task. Each task consists of multiple work elements, which in turn consist of multiple basic motion elements.

shown in Figure 1. Each task consists of several work elements, and each work element consists of basic motion elements. ***Basic motion elements*** are actuations of the limbs and other body parts while engaged in performing the task. These basic motion elements include reaching for an object, grasping an object, or moving an object. Other basic motions include walking and eye movement (e.g., eye focusing, reading). Multiple basic motion elements are generally required to perform a ***work element***, which is defined as a series of work activities that are logically grouped together because they have a unified function within the task. For example, a typical assembly work element consists of reaching for a part, grasping it, and attaching it to a base part, perhaps using one or more fasteners (e.g., screws, bolts, and nuts). Many such work elements make up the total work content of assembling all of the components to the base part. Work elements usually take six seconds or longer, while a basic motion element may take less than a second. The entire task may take 30 seconds to several minutes if it is a repetitive task, while nonrepetitive tasks may require a much longer time to complete.

Just as a task can be divided into its component activities (work elements each consisting of multiple basic motion elements), the typical job of a worker is likely to consist of more than one task. Thus the job adds a next higher level to the existing pyramid. Furthermore, a worker's career is likely to consist of more than one job, as he or she changes employers and/or advances through several jobs with a single employer during a lifetime of working. Accordingly, one might envision the ultimate work pyramid as consisting of the five levels shown in Figure 2.

## 1.2 Importance of Time

In nearly all human endeavors, "time is of the essence." In sports, time is often the major factor in deciding the outcome of a contest. The fastest time wins the race. In games that use time periods, the victor must score the winning points within the limits of those periods. In other aspects of life, time is also important. Students must be in class on time. They must complete an hour quiz in 50 minutes (or other designated time limit). Medical patients must schedule appointments with their doctors at specific times. When we drive to a given

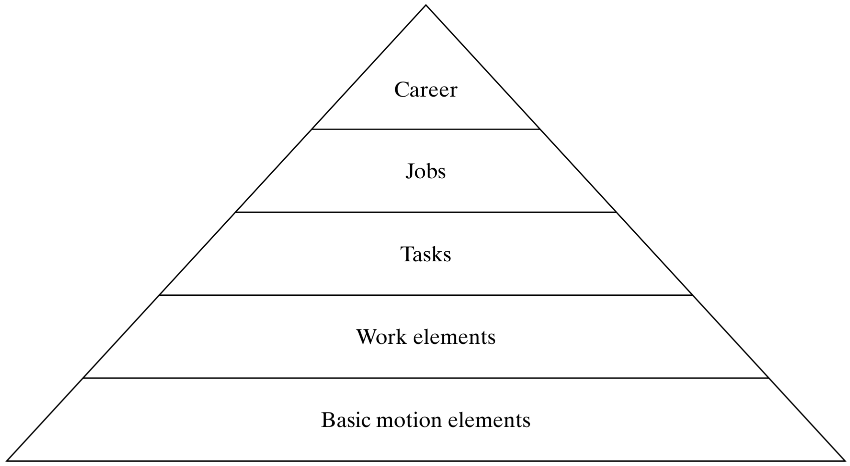



**Figure 2** The pyramidal structure of work.

destination, whether it is a long vacation trip or a nearby shopping expedition, we select the route that will get us there in the shortest possible time. Why waste time and gasoline?

Time is important in business and industry, as the following examples demonstrate:

- *New product introduction.* The manufacturer introducing a new product to the market in the shortest time is usually the one rewarded with the most profits.
- *Product cost.* In many cases, the number of labor hours required to produce a product represents a significant portion of total manufacturing cost, which determines the price of the product. Companies that can reduce the time to make a product can sell it at a more competitive price.
- *Delivery time.* Along with cost and quality, delivery time is a key criterion in vendor selection by many companies. The supplier that can deliver its products in the shortest time is often the one selected by the customer.
- *Overnight delivery.* The success of overnight delivery offered by parcel transport companies (e.g., UPS, Fed Ex) illustrates the growing commercial importance of time.
- *Competitive bidding.* In many competitive bidding situations, proposals must be submitted by a specified date and time. Late proposals will be disregarded.
- *Production scheduling.* The production schedule in a manufacturing plant is based on dates and times.

Time is important in work. The many reasons why this is so include the following:

- The most frequently used measure of work is time. How many minutes or hours are required to perform a given task?
- Most workers are paid according to the amount of time they work. They earn an hourly wage rate or a salary that is paid on a weekly, biweekly, or monthly basis.
- Workers must arrive at work on time. If a worker is a member of a work team, his or her absence or tardiness may handicap the rest of the team.

- When production workers are paid on an incentive plan, they earn their bonuses based on how much time they can save relative to the standard time for a given task.
- Labor and staffing requirements are computed using workloads measured in units of time. For example, how many workers will be required during a 40-hour week to accomplish a workload of 600 hours?

## 2 DEFINING WORK SYSTEMS

A work system can be defined as (1) a physical entity and (2) a field of professional practice. Both definitions are useful in studying the way work is accomplished.

### 2.1 Physical Work Systems

As a physical entity, a ***work system*** is a system consisting of humans, information, and equipment that is designed to perform useful work, as illustrated in Figure 3. The result of the useful work is a contribution to the production of a product or the delivery of a service. Examples of work systems include the following:

- A worker operating a production machine in a factory
- An assembly line consisting of a dozen workers at separate workstations along a moving conveyor
- A robotic spot welding line in an automobile final assembly plant performing spot welding operations on sheet metal car bodies
- A freight train transporting 60 intermodal container cars from Los Angeles to Chicago
- A parcel service agent driving a delivery truck to make customer deliveries in a local area
- A receptionist in an office directing visitors to personnel in the office and answering incoming telephone calls

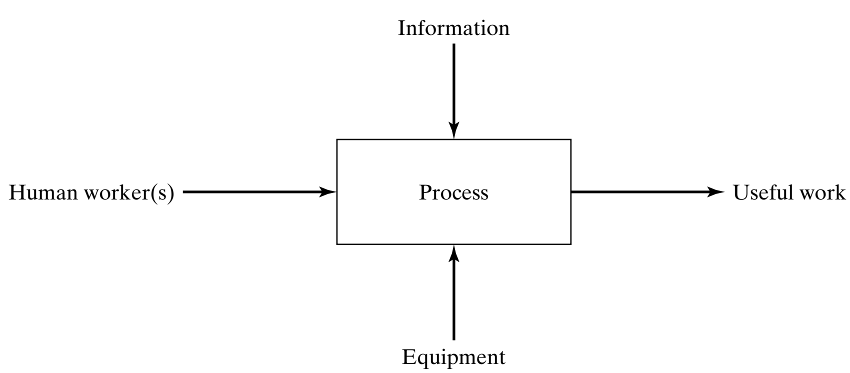



**Figure 3** A work system consists of human workers, information, and equipment designed to accomplish useful work by means of a process.

- A designer working at a computer-aided design (CAD) station to design a new product
- A construction project consisting of a work crew building a highway bridge

As our list indicates, a work system can include one or more human workers. It can also include automated systems that operate for extended periods of time without human attention. Sooner or later, automated systems require the attention of human workers for purposes of maintenance or reprogramming or other reasons. The information associated with the work system may consist simply of a worker's knowledge required to perform an operation, or it may involve the use of databases and programs that must be accessed using computer systems. The equipment in a work system may be a single machine tool operated by one worker or a collection of automated machines that operate under computer control in a coordinated fashion.

### 2.2 Work Systems as a Field of Professional Practice

As a field of professional practice, ***work systems*** includes (1) work methods, (2) work measurement, and (3) work management.[5] The term ***work science*** is often used for this professional practice. The field of ***work methods*** consists of the analysis and design of tasks and jobs involving human work activity. Terms related to work methods include ***operations analysis*** and ***methods engineering***. The term ***motion study*** is also used, but its scope is usually limited to the physical motions, tools, and workplace layout used by a worker to perform a task. The other terms are less restrictive and include the analysis and design of complex processes consisting of material and information flows through multiple operations.

***Work measurement*** is the analysis of a task to determine the time that should be allowed for a qualified worker to perform the task. The time thus determined is called the ***standard time***. Among its many applications, the standard time can be used to compute product costs, assess worker performance, and determine worker requirements (e.g., how many workers are needed to accomplish a given workload). Because of its emphasis on time, work measurement is often referred to as time study. However, in the modern context, ***time study*** has a broader meaning that includes work situations in which an operation is performed by automated equipment, and it is desired to determine the cycle time for the operation. Thus, time study covers any and all work situations in which it is necessary to determine how long it takes to accomplish a given unit of work, whether the work unit is concerned with the production of a product or the delivery of a service. Time is important because it equates to money ("time is money," as the saying goes), and money is a limited resource that must be well managed in any organization.

***Work management*** refers to the various organizational and administrative functions that must be accomplished to achieve high productivity of the work system and effective supervision of workers. Work management includes functions such as (1) organizing workers to perform the specialized tasks that constitute the workload in each

[5]The title of this text emphasizes the two definitions: (1) work systems as physical entities and (2) work systems as a field of professional practice.

department of the company or other organization, (2) motivating workers to perform the tasks, (3) evaluating the jobs in the organization so that each worker is paid an appropriate wage or salary commensurate with the type of work performed, (4) appraising the performance of workers to reward better-performing workers appropriately, and (5) compensating workers using a rational payment system for the work they perform.

## 3 TYPES OF OCCUPATIONS

The Bureau of Labor Statistics of the U.S. Department of Labor identifies 821 occupations in its Standard Occupational Classification (SOC). The SOC covers virtually every type of work performed for pay or profit in the United States, and organizes these occupations into 23 major groups, based on type of work and/or industry sector. The categories are listed in Table 3. Representative positions and job titles are also given.

Rather than use the 23 categories in the SOC system, it is convenient for our purposes to group occupations into the following four broad categories that reflect the work content and function of the jobs rather than type of work and industry sector:

- *Production workers*, in which the work involves making products.
- *Logistics workers*, in which the work involves moving materials, products, or people.
- *Service workers*, in which the work involves one or more of the following functions: (1) providing a service, (2) applying existing information and knowledge, and (3) communicating.
- *Knowledge workers*, in which the work involves one or more of the following functions: (1) creating new knowledge, (2) solving problems, and (3) managing.

These categories are correlated with the types of work systems used in business, industry, and government. Production workers have jobs in production systems (manufacturing, construction), logistics workers are employed in logistics systems (transportation, material handling), service workers work in service operations (retail, government, health care), and knowledge workers are involved in knowledge-oriented activities (creative work, problem solving, management).

The characteristics and representative job titles of the four categories are summarized in Table 4. We see that production and logistics workers engage in mostly physical labor, while service and knowledge workers perform duties that require mostly cognitive activities. Overlaps exist. Production and logistics workers must utilize their brains in their jobs, while the work of service and knowledge workers usually includes physical activities.

A key feature among the four categories is worker discretion [17], which refers to the necessity to make responsible decisions and exercise judgment in carrying out the duties of the position. Jobs that are highly standardized and routine require minimum worker discretion, while jobs in which workers must adapt their behavior in response to variations in the work situation require high discretion. As indicated in Table 4, production and logistics workers are required to use limited or moderate discretion in performing their duties, while service and knowledge workers must exercise much greater discretion in their jobs. In production and logistics operations, the objective is to complete

**TABLE 3** Major Occupation Categories of the U.S. Department of Labor Standard Occupational Classification

| SOC # | Occupation Category | Representative Positions and Job Titles |
|---|---|---|
| 11 | Management Occupations | Chief executive officer, construction manager, funeral director, hotel manager, postmaster, plant manager, sales manager, school principal |
| 13 | Business and Financial Operations Occupations | Accountant, auditor, claims adjuster, cost estimator, financial analyst, loan officer, purchasing agent, tax collector |
| 15 | Computer and Mathematical Occupations | Actuary, computer programmer, systems analyst, database administrator, mathematical analyst, software engineer, statistician |
| 17 | Architecture and Engineering Occupations | Architect, drafter, engineering technician, engineers (e.g., civil, electrical, industrial, mechanical), landscape architect, surveyor |
| 19 | Life, Physical, and Social Science Occupations | Agricultural scientist, astronomer, chemist, economist, geoscientist, materials scientist, physicist, psychologist, sociologist |
| 21 | Community and Social Services Occupations | Clergy, family therapist, marriage counselor, high school counselor, mental health worker, probation officer, social worker |
| 23 | Legal occupations | Court reporter, judge, law clerk, lawyer, legal assistant, magistrate, paralegal, title examiner |
| 25 | Education, Training, and Library Occupations | College professor, elementary school teacher, high school teacher, kindergarten teacher, librarian, special educator, teacher assistant |
| 27 | Arts, Design, Entertainment, Sports and Media Occupations | Actor, artist, athlete, coach, composer, dancer, fashion designer, movie director, musician, news reporter, referee, singer |
| 29 | Healthcare Practitioners and Technical Occupations | Chiropractor, clinical technician, dental hygienist, dentist, paramedic, pharmacist, physician, psychiatrist, nurse, therapist, veterinarian |
| 31 | Healthcare Support Occupations | Dental assistant, medical assistant, nursing aide, orderly, pharmacy aid, physical therapist, veterinary assistant |
| 33 | Protective Services Occupations | Bailiff, detective, firefighter, fire inspector, game warden, jailer, police officer, private investigator, security guard |
| 35 | Food Preparation and Serving Occupations | Bartender, chef, cook, dishwasher, dining room host/hostess, fast food worker, food preparation worker, waiter, waitress |
| 37 | Building and Grounds Cleaning and Maintenance | Groundskeeper, grounds maintenance worker, housekeeping cleaner, janitor, landscape worker, maid, pest control worker, tree trimmer |
| 39 | Personal Care and Service Related Occupations | Animal trainer, barber, bellhop, concierge, cosmetologist, fitness trainer, flight attendant, funeral attendant, hairdresser, usher |
| 41 | Sales and Related Occupations | Cashier, salesperson, insurance agent, model, real estate broker, sales engineer, sales representative, telemarketer, travel agent |
| 43 | Office and Administrative Support Occupations | Bank teller, bill collector, bookkeeper, hotel desk clerk, office manager, receptionist, police dispatcher, secretary, stock clerk |
| 45 | Farming, Fishing, and Forestry Occupations | Agricultural inspector, animal breeder, farm worker, fishing worker, forest and conservation worker, hunter, logging worker, ranch hand |
| 47 | Construction and Extraction Occupations | Bricklayer, building inspector, carpenter, cement mason, construction laborer, electrician, oil drill operator, painter, plumber, roofer |
| 49 | Installation, Maintenance, and Repair Occupations | Aircraft mechanic, automotive mechanic, home appliance repairer, locksmith, office machine repairer, security system installer |
| 51 | Production Occupations | Assembly worker, bookbinder, food processing worker, foundry mold maker, machinist, sewing machine operator, welder |
| 53 | Transportation and Material Moving Occupations | Aircraft pilot, bus driver, crane operator, garbage collector, material handling worker, sailor, ship captain, truck driver |
| 55 | Military Specific Occupations | Air crew member, infantry soldier, military officer, radar technician, sonar technician, special forces commando |

*Source:* From the Bureau of Labor Statistics, www.bls.gov/soc.

**TABLE 4** Comparison of Work Characteristics of Four Categories of Workers

| Worker Category | Production Workers | Logistics Workers | Service Workers | Knowledge Workers |
|---|---|---|---|---|
| Basic functions | Make products | Move materials, products, or people | Provide service<br>Apply information and knowledge<br>Communicate | Create knowledge<br>Solve problems<br>Manage and coordinate |
| Type of work | Mostly physical | Mostly physical | Mostly cognitive | Mostly cognitive |
| Worker discretion | Limited | Limited to moderate | Moderate to broad | Broad |
| Equipment required for basic work | Machinery systems (production tools and machines) | Machinery systems (transportation, material handling) | Computer systems<br>Communication systems | Computer systems<br>Information resources |
| Industry and professional examples | Manufacturing<br>Construction<br>Agriculture<br>Power generation | Transportation<br>Distribution<br>Material handling<br>Storage | Banking<br>Government service<br>Health care<br>Retail | Management<br>Designing<br>Legal<br>Education<br>Consulting |
| Representative positions and job titles | Laborer<br>Machine operator<br>Assembly worker<br>Machinist<br>Quality inspector<br>Construction worker | Truck driver<br>Airplane pilot<br>Ship captain<br>Material handler<br>Order picker<br>Shipping clerk | Bank teller<br>Police officer<br>Nurse<br>Physical trainer<br>Salesperson<br>Foreman | Manager<br>Physician<br>Designer<br>Researcher<br>Lawyer<br>Teacher |

*Source:* Adapted from a figure in [17].

**TABLE 5** Characteristics of Jobs Requiring Low and High Worker Discretion

| Characteristics of Jobs Requiring Low Discretion | Characteristics of Jobs Requiring High Discretion |
|---|---|
| The work is performed at one location. | Workers determine where they will do their jobs. |
| Almost any able-bodied person could perform the work if provided with basic training. | The work depends heavily on technical knowledge and prior experience. |
| The work is dominated by machinery operation, routine procedures, or predetermined activities. | Workers manage their own schedules and the processes used to perform their jobs. |
| The methods, techniques, and materials are specified. | Workers determine the methods, techniques, and materials they use in their jobs. |
| The work requires interactions with the same people every day. | Workers must deal with different types of people in their daily work activities. |
| Work performance is measured primarily in quantitative terms. | Work performance is measured primarily in qualitative terms. |

*Source:* Adapted from [17].

the work precisely as specified and with minimum variation. In service and knowledge operations, the objective is to deal in a discretionary way with situations and people that are inherently variable. Typical characteristics associated with jobs requiring low discretion and jobs requiring high discretion are listed in Table 5.

The relative proportions of workers in various occupational categories are given in Table 6, based on data in an article by Bailey and Barley [2]. The table covers most of the twentieth century for the United States, and indicates trends in the labor force

**TABLE 6** Relative Percentages of Occupations in the U.S. Workforce: 1900–1998

| Occupational Category | Percentage in 1900 | 1950 | 1998 | Net Percentage Change 1900–1998 |
|---|---|---|---|---|
| Production and logistics workers: | | | | |
| Farmworkers | 38 | 12 | 3 | –35 |
| Craft and similar | 11 | 14 | 11 | 0 |
| Operatives and laborers | 25 | 26 | 13 | –12 |
| Total production and logistics workers | 74[a] | 52[a] | 27 | –47 |
| Service and knowledge workers: | | | | |
| Service | 9 | 11 | 16 | +7 |
| Sales workers | 5 | 7 | 11 | +6 |
| Clerical and similar | 3 | 12 | 17 | +14 |
| Professional and technical | 4 | 8 | 18 | +14 |
| Managerial and administrative | 6 | 9 | 11 | +5 |
| Total service and knowledge workers | 27[a] | 47[a] | 73 | +47 |

*Source:* Adapted from [2].

[a]Totals do not add to 100% due to round-off errors.

during this period in which the total U.S. population grew from 76 million to approximately 280 million. The data published in [2] are organized into eight occupational categories, which we have grouped into two major categories, consistent with the worker classification in this text: (1) production and logistics workers and (2) service and knowledge workers. At the beginning of the twentieth century, nearly three-quarters of U.S. workers were employed in farming, skilled trades, factories, and manual labor, which we have grouped as production and logistics workers. Only one-quarter was employed as service and knowledge workers. One hundred years later, the proportions had reversed. Today, most occupations fall within the category of service and knowledge work.

## 4 PRODUCTIVITY

***Productivity*** is defined as the level of output of a given process relative to the level of input. The term *process* can refer to an individual production or service operation, or it can be used in the context of a national economy. Productivity is an important metric in work systems because improving productivity is the means by which worker compensation can be increased without increasing the costs of the products and services they produce. This leads to more products and services at lower prices for consumers, which improves the standard of living for all. In this section we define labor productivity and examine the theoretical basis for increasing productivity.

### 4.1 Labor Productivity

The most common productivity measure is ***labor productivity***, defined by the following ratio:

$$LPR = \frac{WU}{LH} \tag{1}$$

where $LPR$ = labor productivity ratio, $WU$ = work units of output, and $LH$ = labor hours of input. The definition of output work units depends on the process under consideration. For example, in the steel industry, tons of steel is the common measure. In the automobile industry, the number of cars produced is the appropriate output measure. In both industries, it is important to know how many labor hours are required to produce one unit of output. This measure can be used to compare the labor efficiencies of different companies in a given industry, or to compare the same industries among different nations. Obviously, fewer labor hours are better and mean higher productivity. A company or a country that can produce the same output with fewer input labor hours not only has a higher productivity; it also has a competitive advantage in the global economy.

Although labor productivity is a commonly used measure, labor itself does not contribute much to improving productivity. More important factors in determining and improving productivity are capital and technology. ***Capital*** refers to the substitution of machines for human labor; for example, investing in an automated production machine to replace a manually operated machine. The automated machine can probably operate at a higher production rate, so even if a worker is still needed to monitor the operation, productivity has been increased. If the worker is no longer needed at the machine, then labor productivity has been increased even more. ***Technology*** refers to a fundamental change in the way some activity or function is accomplished. It is more than simply using a machine in place of a human worker. It is using a brand new type of machine to replace the previous type. Some examples of how new technologies have made dramatic improvements in productivity are listed in Table 7.

Admittedly, the distinctions between capital improvements and technology improvements are sometimes subtle, because new technologies almost always require capital investments. Anyway, arguing about these differences and subtleties is not as important as recognizing that the important gains in productivity are generally made by the introduction of capital and technology in a work process, rather than by attempting to extract more work in less time from workers. By investing in capital and technology to increase the rate of output work units and/or reduce input labor hours, the labor productivity ratio is increased.

**TABLE 7** Examples of Technology Changes that Dramatically Improved Productivity

| Old Technology | New Technology | Improvement |
|---|---|---|
| Horse-drawn carts | Railroad trains | Substitution of steam power for horse power, use of multiple carts (passenger or freight cars) |
| Steam locomotive | Diesel locomotive | Substitution of diesel power technology for steam power technology |
| Telephone operator | Dial phone | Dial technology allowed "clicks" of the dial to be used to operate telephone switching systems |
| Dial phone | Touch-tone phone | Substitution of tone frequencies in place of "clicks" to operate telephone switching systems for faster dialing |
| Manually operated milling machine | Numerical control milling machine | Substitution of coded numerical instructions to operate the milling machine rather than a skilled machinist |
| DC3 passenger airplane (1930s) | Boeing 747 passenger airplane (1980s) | Substitution of jet propulsion for piston engine for higher speed, larger aircraft for more passenger miles |

Measuring productivity is not as easy as it seems. Although the labor productivity ratio defined by equation (1) appears simple and straightforward, the following problems are often encountered in its measurement and use:

- *Nonhomogeneous output units.* The output work units are not necessarily homogeneous. For example, using the annual production of automobiles as the output measure does not account for differences in models, vehicle sizes, and prices. An expensive luxury model is likely to require more labor hours of assembly time than an inexpensive compact car.
- *Multiple input factors.* As previously indicated, labor is not the only input factor in determining productivity. In addition to capital and technology, other input factors may include materials and energy. For example, in the production of aluminum, electric power and the raw material bauxite are much more important than labor as inputs to the process.[6]
- *Price and cost changes.* The prices of output work units and the costs of input factors (labor, materials, power) change over time, often unpredictably. A company may improve productivity, but if the prices of its products decrease due to market forces, the company could find itself in severe financial difficulty. The steel industry in the United States during the late 1990s and early 2000s provides a perfect example of this case.
- *Product mix changes.* Product mix refers to the relative proportions of products that a company sells. If the mix of expensive and inexpensive products changes from year to year, an annual comparison based on the labor productivity ratio is less meaningful.

An alternative productivity measure is the labor productivity index that compares the output/input ratio from one year to the next. The productivity index is defined as follows:

$$LPI = \frac{LPR_t}{LPR_b} \tag{2}$$

where $LPI$ = labor productivity index, $LPR_t$ = labor productivity ratio during some time period of interest, and $LPR_b$ = labor productivity ratio during some defined base period.

**Example 1 Productivity Measurement**

During the base year in a small steel mill, 326,000 tons of steel were produced using 203,000 labor hours. In the next year, the output was 341,000 tons using 246,000 labor hours. Determine (a) the labor productivity ratio for the base year, (b) the labor productivity ratio for the second year, and (c) the labor productivity index for the second year.

**Solution** **(a)** In the base year, $LPR = \frac{326{,}000}{203{,}000} = 1.606$ tons per labor hour.

**(b)** In the second year, $LPR = \frac{341{,}000}{246{,}000} = 1.386$ tons per labor hour.

[6]Bauxite is the principal ore used to produce aluminum. It consists largely of hydrated aluminum oxide ($Al_2O_3 H_2O$).

**(c)** The productivity index for the second year is $LPI = \dfrac{1.386}{1.606} = 0.863$.

**Comment:** No matter how it is measured, productivity went down in the second year. ■

### 4.2 Productive Work Content

Work is usually not performed in the most efficient way possible. A given task performed by a worker can be considered to consist of (1) the basic productive work content and (2) excess nonproductive activities. In general, the task is associated with the production of a product or the delivery of a service. The ***basic productive work content*** is the theoretical minimum amount of work required to accomplish the task, where the amount of work is expressed in terms of time. Thus, the time required to perform the basic productive work content is the theoretical minimum. It cannot be further reduced. The basic productive work content time is a theoretical concept implying optimal conditions that rarely, if ever, are achieved in practice. Nevertheless, it is an optimum worth seeking.

The ***excess nonproductive activities*** in the task are the extra physical and mental actions performed by the worker that do not add any value to the task, nor do they facilitate the productive work content that does add value. The excess nonproductive activities take time. They add to the basic productive work content time to make up the total time needed to perform the task. The excess nonproductive activities can be classified into three categories, as illustrated in the time line in Figure 4:

- Excess activities caused by poor design of the product or service
- Excess activities caused by inefficient methods, poor work layout, and interruptions
- Excess activities caused by the human factor

Some examples of the three categories of nonproductive activities are listed in Table 8. As several of these examples indicate, nonproductive activities do not necessarily occur during every cycle of the task. For example, industrial accidents occur infrequently in most work situations. However, over the long run, their effects extract a heavy toll on productive work content.

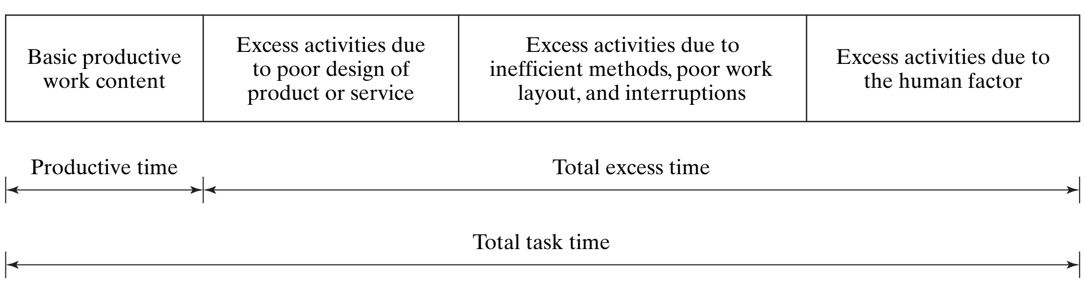



**Figure 4** Total task time consists of basic productive work content and nonproductive activities.

**TABLE 8** Examples of Excess Nonproductive Activities that Add to Total Task Time

| Category of Activity | Examples |
|---|---|
| Poor design of the product or service | Products designed with more parts than necessary, so that excess assembly time is required<br>Product proliferation (e.g., more choices for the customer than necessary)<br>Frequent product design changes, causing changes in tooling, methods, and layout<br>Waste of materials (e.g., machining from bar stock rather than machining from a forging or casting)<br>Quality standards that are more stringent than necessary, requiring excess processing time |
| Inefficient methods, poor work layout, and interruptions | Inefficient plant layout that requires excess movement of materials<br>Inefficient workplace layout that requires excess hand, arm, and body motions<br>Inefficient methods that waste time<br>Wasted space (e.g., using valuable floor space to store materials instead of storing vertically in racks)<br>Inefficient material handling (e.g., using manual methods to move materials rather than mechanized methods)<br>Long setup times between batches of work<br>Frequent breakdowns of equipment<br>Workers waiting for work |
| The human factor | Absenteeism and tardiness<br>Workers spending too much time socializing<br>Workers deliberately working slowly<br>Inadequate training of workers<br>Industrial accidents<br>Hazardous materials that cause occupational illnesses |

*Source:* Adapted from [9].

The concepts of basic productive work content and excess nonproductive activities are applicable not only to individual tasks but also to the entire work sequence required to manufacture a product or provide a service. There are value-adding activities and non-value-adding activities in virtually all work systems. An important objective in the design of a work system is to minimize the non-value-adding activities so that only productive work is accomplished. This objective is achieved through the use of methods engineering, operations analysis, work measurement, and other topics discussed in this text.

## REFERENCES


[1] Amrine, H. T., J. A. Ritchey, and C. L. Moodie. *Manufacturing Organization and Management.* 5th ed. Englewood Cliffs, NJ: Prentice Hall, 1987.

[2] Bailey, D. E., and S. R. Barley. "Return to Work: Toward Post-Industrial Engineering." *IIE Transactions* 37 (2005): 737–52.

[3] Brogan, J. "Distinguishing Functions." *Industrial Engineer* (April 2003): 32–34.

[4] Chase, R. B., and N. J. Aquilano. *Production and Operations Management: A Life Cycle Approach*. 5th ed. Homewood, IL: Irwin, 1989.

[5] Groover, M. P. *Automation, Production Systems, and Computer Integrated Manufacturing*. 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.

[6] Hicks, P. E. *Industrial Engineering and Management*. 2nd ed. New York: McGraw-Hill, 1994.

[7] Hounshell, D. A. *From the American System to Mass Production, 1800–1932*, Baltimore, MD: Johns Hopkins University Press, 1984.

[8] Jay, T. A. *Time Study*. Poole, England: Blandford Press, 1981.

[9] Kanawaty, G., ed. *Introduction to Work Study*. 4th ed. Geneva, Switzerland: Industrial Labour Office, 1992.

[10] Karger, D. W., and F. H. Bayha. *Engineered Work Measurement*. 4th ed. New York: Industrial Press, 1987.

[11] Konz, S., and S. Johnson. *Work Design, Industrial Ergonomics*. 5th ed. Scottsdale, AZ: Holcomb Hathaway Publishers, 2000.

[12] McGinnis, L. "A Brave New Education." *IIE Solutions* (December 2002): 27–32.

[13] Monks, J. G. *Operations Management, Theory and Problems*. New York: McGraw-Hill, 1977.

[14] Moore, F. G. *Manufacturing Management*. 3rd ed. Homewood, IL: Irwin, 1961.

[15] Nadler, G. *Work Design*. Homewood, IL: Irwin, 1963.

[16] Owens, R. N. *Management of Industrial Enterprises*. 4th ed. Homewood, IL: Irwin, 1961.

[17] Pepitone, J. S. "A Case for Humaneering." *IIE Solutions* (May 2002): 39–44.

[18] Russell, R. S., and B. W. Taylor III. *Operations Management*. 4th ed. Upper Saddle River, NJ: Prentice Hall, 2003.

[19] Theriault, P. *Work Simplification*. Norcross, GA: Engineering and Management Press, Institute of Industrial Engineers, 1996.

[20] Wredge, C. D., and R. G. Greenwood. *Frederick W. Taylor: The Father of Scientific Management: Myth and Reality*. Burr Ridge, IL: Irwin Professional Publishing, 1991.

## REVIEW QUESTIONS

1 Define work.
2 What are basic motion elements? Give some examples.
3 What is a work element?
4 Why is time important in work?
5 Define work system as a physical entity.
6 Define work system as a field of professional practice.
7 What are some of the functions included within the scope of work management?
8 Name the four broad categories of worker occupations.
9 Define productivity.
10 Labor is one input factor that determines productivity. What are two other factors that are more important than labor in improving productivity? Define each of these two additional input factors.
11 What is the difference between the labor productivity ratio and the labor productivity index?
12 A given task performed by a worker can be considered to consist of the basic productive work content and excess nonproductive activities. (a) What is meant by the term *basic productive work content*? (b) What is meant by the term *excess nonproductive activities*?
13 What are the three categories of excess nonproductive activities, as they are defined in the text?

## PROBLEMS

1 A work group of 5 workers in a certain month produced 500 units of output working 8 hr/day for 22 days in the month. (a) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? (b) Suppose that in the next month, the same work group produced 600 units but there were only 20 workdays in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.
2 A work group of 10 workers in a certain month produced 7200 units of output working 8 hr/day for 22 days in the month. Determine the labor productivity ratio using (a) units of output per worker-hour and (b) units of output per worker-month. (c) Suppose that in the next month, the same work group produced 6800 units but there were only 20 workdays in the month. For each productivity measure as in (a) and (b), determine the productivity index for the next month using the prior month as a base.
3 A work group of 20 workers in a certain month produced 8600 units of output working 8 hr/day for 21 days. (a) What is the labor productivity ratio for this month? (b) In the next month, the same work group produced 8000 units but there were 22 workdays in the month and the size of the work group was reduced to 14 workers. What is the labor productivity ratio for this second month? (c) What is the productivity index using the first month as a base?
4 There are 20 forging presses in the forge shop of a small company. The shop produces batches of forgings requiring a setup time of 3.0 hours for each production batch. Average standard time for each part in a batch is 45 seconds, and there are an average of 600 parts in a batch. The plant workforce consists of two workers per press, two foremen, plus three clerical support staff. (a) Determine how many forged parts can be produced in 1 month, if there are 8 hours worked per day and an average of 21 days per month at one shift per day. (b) What is the labor productivity ratio of the forge shop, expressed as parts per worker-hour?

**5** A farmer's market is considering the addition of bar code scanners at their check-out counters, which would use the UPC marked on all grocery packaging. Currently, the check-out clerk keypunches the price of each item into the register during check-out. Observations indicate that an average of 50 items are checked out per customer. The clerk currently takes 7 seconds per item to keypunch the register and move the item along the check-out table. On average it takes 25 seconds to total the bill, accept money from the customer, and make change. It then takes 4 seconds per item for the clerk to bag the customer's order. Finally, about 5 seconds are lost to transition to the next customer. Bar code scanners would eliminate the need to keypunch each price, and the time per item would be reduced to 3 seconds with the bar code scanner. (a) What is the hourly throughput rate (number of customers checked out per hour) under the current check-out procedure? (b) What would be the estimated hourly throughput rate if bar code scanners were used? (c) If separate baggers were used instead of requiring the check-out clerk to perform bagging in addition to check-out, what would be hourly throughput rate? Assume that bar code scanners are used by the clerk. (d) Determine the productivity index for each of the two cases in (b) and (c), using (a) as the basis of comparison and hourly customers checked out per labor hour as the measure of productivity.

# *Manual Work and Worker–Machine Systems*



There are various types of work systems used in production, services, offices, projects, and other work situations. All of these work systems utilize the physical and mental capabilities of humans. In terms of the human participation in the tasks performed, work systems can be classified into the three basic categories depicted in Figure 1: (a) manual work systems, (b) worker–machine systems, and (c) automated systems. A ***manual work system*** consists of a worker performing one or more tasks without the aid of powered tools. The tasks commonly require the use of hand tools (e.g., hammers, screwdrivers, shovels). In a ***worker–machine system***, a human worker operates powered equipment (e.g., a machine tool). An ***automated work system*** is one in which a process is performed by a machine without the direct participation of a human worker.

As indicated in Figure 1, the work accomplished by a work system is almost always acted upon some object, called the ***work unit***. The state of the work unit is advanced in some way through the process performed on it. In production, the work may alter the geometry of a work part. In logistics, the work may involve transporting


From Chapter 2 of *Work Systems and the Methods, Measurement, and Management of Work,* First Edition. Mikell P. Groover.

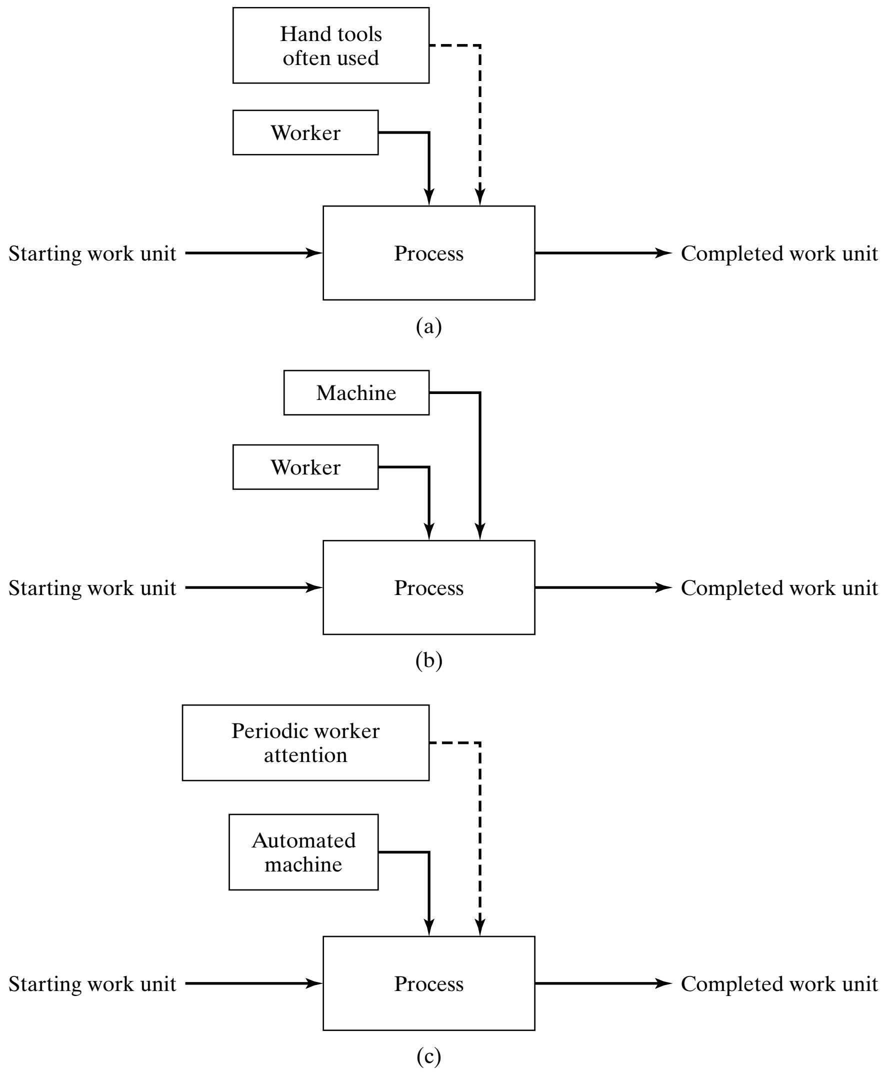



**Figure 1** Three types of work systems: (a) manual work system, (b) worker–machine system, and (c) automated system.

material from a warehouse to a customer. In service work, a sales prospect is transformed into a paying customer by a persuasive salesperson. In knowledge work, a designer takes a product concept and converts it into specifications and engineering drawings.

In this chapter, we discuss the three categories of work systems. These systems are associated mostly with work that is performed by production and logistics workers. The emphasis in our coverage is on unit operations—tasks and processes that are treated as being independent of other work activities in a given facility or work site.

## 1 MANUAL WORK SYSTEMS

Manual work is the most basic form of work, engaging the human body to accomplish some physical task without an external source of power. Hand tools are often used to facilitate the task, but the power to operate them derives from the strength and stamina of a human worker. With or without hand tools, the worker must expend physical energy to accomplish the task. In addition, other human faculties are required, such as hand-eye coordination and mental effort. Our coverage of manual work includes two sections: (1) types of manual work and (2) cycle time analysis of manual work.

### 1.1 Types of Manual Work

Two forms of manual work can be distinguished: (1) pure manual work and (2) manual work using hand tools. ***Pure manual work*** involves only the physical and mental capabilities of the human worker, and no machines, tools, or other implements are employed in performing the task. Examples of pure manual work include the following:

- A material-handling worker moving cartons from the floor onto a conveyor in a warehouse
- Workers loading furniture into a moving van from a house without the use of dollies or other wheeled platforms
- A dealer at a casino table dealing cards
- An office worker filing documents in a file cabinet
- An assembly worker snap-fitting two parts together
- An assembly worker assembling two sheet metal parts with a bolt and nut by hand (tightening is done later using appropriate hand or power tools).

Note that the common characteristic in these examples is that they consist of moving things. Performing manual work without tools almost always involves the movement and handling of objects. Even assembly tasks include moving the parts in order to join them.

Manual tasks are commonly augmented by the use of hand tools. The ability to design and use tools is one of the attributes that distinguish humans from other species on earth. A ***tool*** is a device or implement for making changes to some object (e.g., the work unit), such as cutting, grinding, striking, squeezing, or other process. Instruments used for measurement are also included in the category of tools, even though no physical change in the object results directly from their use. A ***hand tool*** is a small tool that is operated by the strength and skill of the human user. When using hand tools, a ***workholder*** is sometimes employed to grasp the work unit and position it securely during processing. Examples of manual tasks involving the use of hand tools include the following:

- A machinist using a file to round the edges of a rectangular part that has just been milled
- An assembly worker using a screwdriver to tighten a screw
- A painter using a paintbrush to paint the trim around a doorway
- A sculptor using a carving knife to carve a wooden statue
- A grass-cutter using a rake to collect the grass clippings after mowing
- A quality control inspector using a micrometer to measure the diameter of a shaft
- A material-handling worker using a dolly to move cartons in a warehouse
- An office worker using a pen to handwrite entries into a ledger.

## 1.2 Cycle Time Analysis of Manual Work

Manual tasks usually consist of a work cycle that is repeated with some degree of similarity, and each cycle usually corresponds to the processing of one work unit. When a painter is hired to paint the wooden trim around the doorways in a new house, the painting cycle is repeated for each doorway. If the doorways are all the same size, and the wood trim is identical for all doors, then the painting cycle should exhibit a high degree of similarity. If there are differences in the doorways, then the painting cycles will be less similar.

If the work cycle is relatively short, and there is a high degree of similarity from one cycle to the next, we refer to the work as ***repetitive***. If the work cycle takes a long time and the cycles are not similar, the work is ***nonrepetitive***.[1] In either case, the task can be divided into work elements that consist of logical groupings of motions performed by the worker. The cycle time $T_c$ is therefore the sum of the work element times

$$T_c = \sum_{k=1}^{n_e} T_{ek} \tag{1}$$

where $T_{ek}$ = time of work element $k$, where $k$ is used to identify the work elements, min; and $n_e$ = number of work elements into which the cycle is divided. Our focus in this section is on repetitive work. Consider the following example of a pure manual task.

**Example 1 A Repetitive Manual Task**

An assembly worker performs a repetitive manual task consisting of inserting 8 plastic pegs into 8 holes in a flat wooden board. A slight interference fit is involved in each insertion. The worker holds the board in one hand and picks up the pegs from a tray with the other hand and inserts them into the holes, one peg at a time. The workplace layout is shown in Figure 2 (a), and the sequence of work elements is given in the table below. Can the work method be improved in order to reduce the cycle time?

[1]There is no clear boundary between repetitive and nonrepetitive work. When referring to repetitive work cycles, we usually mean a cycle time of a few minutes or less, and the motion patterns are intended to be identical for every cycle, so that cycle-to-cycle variations in time and work content tend to be random.

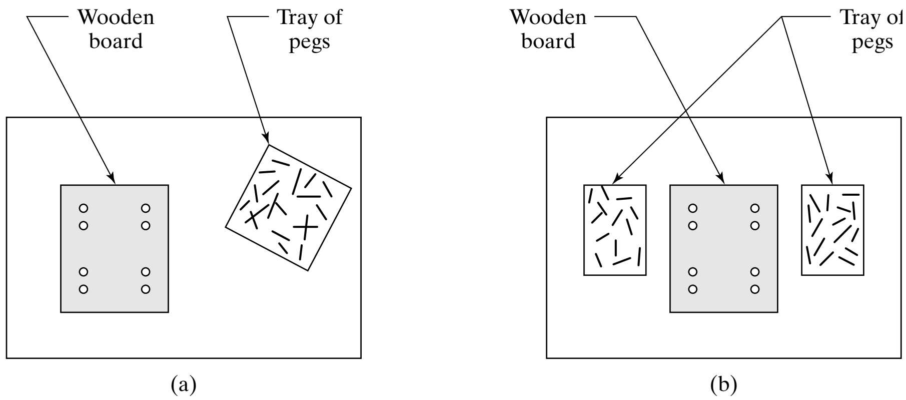



**Figure 2** Workplace layout for assembly task in Example 1: (a) before methods improvement and (b) after methods improvement.

| Sequence | Work Element Description | Work Element Time, $T_{ek}$ (min) |
|---|---|---|
| 1 | Worker picks up board with one hand and holds it. | 0.08 |
| 2 | Worker picks peg from tray and inserts it into hole in board. | 0.06 |
| 3 | Worker picks second peg and inserts it into hole in board. | 0.06 |
| 4 | Worker picks third peg and inserts it into hole in board. | 0.06 |
| 5 | Worker picks fourth peg and inserts it into hole in board. | 0.06 |
| 6 | Worker picks fifth peg and inserts it into hole in board. | 0.06 |
| 7 | Worker picks sixth peg and inserts it into hole in board. | 0.06 |
| 8 | Worker picks seventh peg and inserts it into hole in board. | 0.06 |
| 9 | Worker picks eighth peg and inserts it into hole in board. | 0.06 |
| 10 | Worker lays assembled board into tote pan. | 0.06 |
| Total work cycle time | | 0.62 |

**Solution:** An opportunity for improvement lies in using a work-holding device to hold and position the board while the worker uses both hands simultaneously to insert the pegs. Two trays filled with pegs will be used, one for each hand. Instead of picking the pegs out of the tray one peg at a time, each hand will grab four pegs in order to minimize the number of times the worker's hands must reach to the trays. The revised workplace layout is shown in Figure 2 (b), and the revised sequence of work elements is presented in the table below. The cycle time is reduced from 0.62 min to 0.37 min, a reduction of 40%, which corresponds to an increase in production rate of almost 68%.

| Sequence | Work Element Description | Work Element Time, $T_{ek}$ (min) |
|---|---|---|
| 1 | Worker picks up board and positions it in workholder. | 0.12 |
| 2 | Worker picks 4 pegs each with both hands from 2 trays and inserts them into 8 holes in board. | 0.15 |
| 3 | Worker removes board from workholder and places in tote pan. | 0.10 |
| Total work cycle time | | 0.37 |

■

As the example illustrates, it is important to design the work cycle so as to minimize the time required to perform it. There are many alternative ways to perform a given task, where the differences are in work elements, hand and body motions, tools, workholders, and so forth. Some of the alternative methods are less time-consuming than others. A useful concept in work design is the ***one best method*** principle. According to this principle, of all the possible methods that can be used to perform a task, there is one optimal method that minimizes the time and effort required to accomplish it.[2] One of the primary objectives in work design is to determine the one best method for the task, and then to standardize its use in the workplace.

**Normal Time and Standard Time.** Once the work cycle and associated method are defined, the actual time taken for a given manual cycle is a variable. Variability is inherent in any repetitive human activity, and this variability is manifested in the time to perform the activity. Reasons for variations in work cycle times include the following:

- Differences in worker performance from one cycle to the next
- Variations in hand and body motions
- Worker blunders and bungles
- Variations in the starting work units
- Inclusion of extra elements that are not performed every cycle
- Differences in the physical and cognitive attributes among workers performing the task
- Variations in the methods used by different workers to perform the task
- The learning curve phenomenon.

Topping the list is worker ***performance***, which can be defined simply as the pace or relative speed with which the worker does the task. As worker performance increases, the time to accomplish the work cycle decreases. From the employer's viewpoint, it is desirable for the worker to work at a high level of performance. The question is: what is a reasonable performance or pace to expect from a worker in accomplishing a given task? To answer the question, let us introduce the concept of normal performance. ***Normal performance*** (or ***normal pace***) means a pace of working that can be maintained by a properly trained average worker throughout an entire work shift without deleterious short-term or long-term effects on the worker's health or physical well-being. The work shift is usually assumed to be eight hours, during which periodic rest breaks are allowed. Normal performance refers to the pace of the worker while actually working. When a worker works at a normal performance level, we say he or she is working at 100% performance. A faster pace than normal is greater than 100% and a slower pace is less than 100%. A common benchmark of normal performance is walking on level ground at three miles per hour.

The term ***standard performance*** is often used in place of normal performance. They both refer to the same pace while working, but standard performance acknowledges that periodic rest breaks are included in the work shift. For example, a healthy human

[2]The one best method should also satisfy other criteria in addition to minimizing the work cycle time, such as ensuring a safe and convenient workplace for the worker, and producing a work unit of high quality.

can walk at a pace of three miles per hour for an hour or two. However, walking for a solid eight hours without ever stopping for a rest break would be physically wearing. Accordingly, normal performance and standard performance both mean walking three miles per hour during an eight-hour period, but the walker could stop for a rest a few times and take a lunch break during the eight hours.

When a work cycle is performed at 100% performance, the time taken is called the ***normal time*** for the cycle. If worker performance is greater than 100%, then the time required to complete the cycle will be less than the normal time; and when worker performance is less than 100 percent, the time taken will be greater than the normal time. The actual time to perform the work cycle is a function of the worker's performance as indicated in the equation

$$T_c = \frac{T_n}{P_w} \tag{2}$$

where $T_c$ = actual cycle time, min; $T_n$ = normal time for the work cycle, min; and $P_w$ = pace or performance of the worker, expressed as a decimal fraction (e.g., 100% = 1.0).

**Example 2 Normal Performance**

A man walks in the early morning for health and fitness. His usual route is 1.85 miles long. The route has minimal elevation changes. A typical time to walk the 1.85 miles is 30 min. Using the benchmark of 3 miles/hr as normal performance, determine (a) how long the route would take at normal performance and (b) the man's performance when he completes the route in 30 min.

**Solution:** **(a)** At 3 miles/hr, 1.85 miles can be covered in 1.85/3.0 = 0.6167 hr or 37 min.

**(b)** If the man takes 30 min to complete the walk, then his performance can be determined by dividing the normal time by the actual time, by rearranging equation (2). Thus,

$$P_w = 37/30 = 1.233 \text{ or } 123.3\%$$

**Alternative Solution:** For part (b), we could determine the man's velocity and compare it to the benchmark of 3.0 miles/hr in order to determine his performance. If the man completes 1.85 miles in 30 min, then his walking velocity is 1.85 miles divided by 0.5 hr, which equals 3.7 miles/hr.

$$P_w = 3.7/3.0 = 1.233 \text{ or } 123.3\%$$

■

Workers are allowed periodic rest breaks (e.g., coffee breaks) during their work shift. A typical work shift is eight hours (e.g., 8:00 A.M. to 5:00 P.M. with an hour from noon to 1:00 P.M. for lunch). The shift usually includes a rest break in the morning and another in the afternoon. Unlike the lunch period, these rest breaks are normally included within the eight-hour time of the shift. The employer allows these breaks because it has been found that the overall productivity of the worker during the shift is greater if rest breaks are provided. More work is accomplished by the end of the day and fewer mistakes are made if the worker can take time out periodically from the normal work routine. In addition to the rest breaks, the worker is likely to have other interruptions during the shift, such as equipment breakdowns (if the manual task is somehow dependent on equipment), receiving instructions from the foreman, personal telephone calls, and so on. As a result of all of these factors, the total time actually worked during the shift will be less than the full eight hours, in all likelihood.

To account for these delays and rest breaks, an ***allowance*** is added to the normal time in order to determine an "allowed time" for the worker to perform the task throughout the shift. More commonly known as the ***standard time***, it is defined as follows:

$$T_{std} = T_n(1 + A_{pfd}) \tag{3}$$

where $T_{std}$ = standard time, min; $T_n$ = normal time, min; and $A_{pfd}$ = allowance factor, usually expressed as a percentage but used in equation (3) as a decimal fraction. The allowance is commonly called the personal time, fatigue, and delay allowance (abbreviated ***PFD allowance***), and it is figured in such a way that, if the worker works at 100 percent performance during the portion of the shift that he or she is working, the amount of work accomplished will be eight hours' worth.

Manual work cycles often include ***irregular work elements***, which are elements performed with a frequency of less than once per cycle. Examples of irregular work elements include periodic changing of tools (e.g., changing a knife blade) and replacing tote pans of parts when the containers become full. In determining a standard time for the cycle, the irregular element times are prorated in the regular cycle time. The following examples illustrate these concepts and definitions.

**Example 3 Determining Standard Time and Standard Output**

The normal time to perform the regular work cycle for a certain manual operation is 3.23 min. In addition, an irregular work element whose normal time is 1.25 min must be performed every 5 cycles. The PFD allowance factor is 15%. Determine (a) the standard time and (b) how many work units are produced if the worker's performance in an 8-hour shift is standard.

**Solution:** **(a)** The normal time for the work cycle includes the irregular element prorated according to its frequency:

$$T_n = 3.23 + \frac{1.25}{5} = 3.23 + 0.25 = 3.48 \text{ min}$$

The standard time is $T_{std} = T_n(1 + A_{pfd}) = 3.48(1 + 0.15) = 4.00$ min.

**(b)** The number of work units produced at standard performance in an 8-hour shift is the clock time of the shift divided by the standard time:

$$Q_{std} = \frac{8.0(60)}{4.00} = 120 \text{ work units}$$

■

**Example 4 Determining Lost Time Due to the Allowance Factor**

Determine the anticipated amount of time lost per 8-hour shift when an allowance factor of 15% is used, as in the previous example.

**Solution:** Given that $A_{pfd} = 0.15$, the anticipated amount of time lost per 8-hour shift is determined as follows:

$$8.0 \text{ hr} = (\text{actual time worked})(1 + 0.15)$$

$$\text{Actual time worked} = \frac{8.0}{1.15} = 6.956 \text{ hr}$$

$$\text{Time lost} = 8.0 - 6.956 = 1.044 \text{ hr}$$

This is the anticipated daily amount of time lost due to personal time, fatigue, and delays corresponding to a 15% PFD allowance factor. ■

**Example 5 Production Rate When Worker Performance Exceeds 100%**

Now that the standard is set ($T_{std} = 4.00$ min), and given the data from the previous examples, how many work units would be produced if the worker's average performance during an 8-hour shift were 125% and the hours actually worked were exactly 6.956 hr, which corresponds to the 15% allowance factor.

**Solution:** Based on the normal time $T_n = 3.48$ min, the actual cycle time with a worker performance of 125% is

$$T_c = \frac{3.48}{1.25} = 2.78 \text{ min}$$

Assuming one work unit is produced each cycle, the corresponding daily production rate (symbolized by $R_p$) is

$$R_p = \frac{6.956\,(60)}{2.78} = 150 \text{ work units}$$

Note that 150 units = 125% of 120 units at 100% performance. ■

**Standard Hours and Worker Efficiency.** Two common measures used in industry to assess a worker's productivity are standard hours and worker efficiency. The ***standard hours*** represents the amount of work actually accomplished by the worker during a given period (e.g., shift, week), expressed in terms of time. In its simplest form, it is the quantity of work units produced during the period multiplied by the standard time per work unit; that is,

$$H_{std} = QT_{std} \tag{4}$$

where $H_{std}$ = standard hours accomplished, hr; $Q$ = quantity of work units completed during the period, pc; and $T_{std}$ = standard time per work unit, hr/pc. If the time standard

$T_{std}$ is expressed in min/pc, then conversion of units is required to obtain standard hours $H_{std}$. When a worker works at a performance level greater than 100% and his or her actual time worked during the shift is consistent with or greater than what is provided by the allowance factor, then the number of standard hours accomplished will be greater than the number of hours in the shift.

***Worker efficiency*** is the amount of work accomplished during the shift expressed as a proportion of the shift hours. In equation form,

$$E_w = \frac{H_{std}}{H_{sh}} \tag{5}$$

where $E_w$ = worker efficiency, normally expressed as a percentage; $H_{std}$ = number of standard hours of work accomplished during the shift, hr; and $H_{sh}$ = number of shift hours (e.g., 8 hr).

**Example 6 Standard Hours and Worker Efficiency**

For the worker performance of 125% in the previous example ($T_{std}$ = 4.00 min), determine (a) number of standard hours produced and (b) worker efficiency.

**Solution:** **(a)** $H_{std} = 150(4.0 \text{ min}) = 600 \text{ min} = 10.0 \text{ hr}$

**(b)** $E_w = (10.0 \text{ hr})/(8.0 \text{ hr}) = 1.25 = 125\%$ ■

In Example 5 and 6, the worker's efficiency and performance level are equal for two reasons: (1) the number of hours actually worked is exactly consistent with the 15% allowance factor, and (2) the entire work cycle consists exclusively of manual labor and is therefore entirely operator-controlled. In the absence of either or both of these conditions, worker efficiency will not equal worker performance level (except by coincidence, when certain combinations of values offset each other). In reality, the number of hours actually worked by a worker in an 8-hour shift varies each day, depending on the amount of time lost due to personal reasons, rest breaks, and delays. Worker performance and worker efficiency are different if the time lost is different from what is accounted for by the PFD allowance factor, as the following example illustrates.

**Example 7 Standard Hours and Worker Efficiency as Affected by Hours Actually Worked**

Suppose the worker's pace in the task is 125%, but the actual hours worked is 7.42 hr. Determine (a) the number of pieces produced, (b) the number of standard hours accomplished, and (c) the worker's efficiency.

**Solution:** **(a)** The actual cycle time at 125% performance is 2.78 min, as calculated in Example 5. The number of work units produced in 7.42 hr is

$$Q = \frac{7.42(60)}{2.78} = 160 \text{ units}$$

**(b)** $H_{std} = 160(4.0 \text{ min}) = 640 \text{ min} = 10.67 \text{ hr}$

**(c)** $E_w = 10.67/8.0 = 1.333 = 133.3\%$ ■

Worker efficiency is commonly used to evaluate workers in industry. In many incentive wage payment plans, the worker's earnings are based on his or her efficiency or the number of standard hours accomplished. Worker efficiency and standard hours are easily computed, because the number of hours in the shift and the standard time are known, and the number of work units produced can be readily counted. The two measures are basically equivalent, because either one can be derived from the other. One might think that worker performance would also be a useful measure; however, it is more difficult to assess because it requires data on the amount of time actually worked by the worker during the shift. This varies from day to day because the interruptions and delays vary from day to day. Some method of continuously observing the worker would be required. Aside from the cost, the worker would likely find such observation objectionable. In addition, the effect of worker performance is reduced when machine time is included in the work cycle, as we see in Section 2.

## 2 WORKER–MACHINE SYSTEMS

When a worker operates powered equipment, we refer to the arrangement as a ***worker–machine system***. It is one of the most widely used work systems. Worker–machine systems include combinations of one or more workers and one or more pieces of equipment. The workers and machines are combined to accomplish a desired output. Examples of worker–machine systems include the following:

- A skilled machinist operating an engine lathe in a tool room to fabricate a component (the work unit) for a custom-designed product. The machinist must exercise considerable skill in controlling the feed, speed, and tool position while operating the lathe.
- A construction worker operating a backhoe at a construction site. The worker must continuously operate the machine, using the various levers that control the different hydraulically operated mechanisms.
- A truck driver driving an 18-wheel tractor-trailer on an interstate highway. The driver must constantly be alert while operating the vehicle.
- A factory worker loading and unloading parts at a machine tool. The machine tool operates on semiautomatic cycle to process the parts. At the end of each work cycle, the worker unloads the completed part. Machine processing time is about 3 min. While the machine performs its process, the worker is idle. Loading and unloading the machine takes about 30 sec.
- A crew of workers operating a rolling mill that converts hot steel slabs into flat plates. Each worker has an assigned function. The most important job is the rolling mill operator who must coordinate the gap size (i.e., distance between opposing rolls) and the passing of the slab back and forth between the rolls. Each pass reduces the thickness of the starting slab until the specified thickness has been achieved.
- A secretary using a personal computer with word processor in an office typing pool.
- A clerical worker in a billing center entering data based on checks received by mail from customers into account records on a networked personal computer.
- An industrial engineer creating the design of a plant layout on a computer-aided design (CAD) workstation.

**TABLE 1** Relative Strengths and Attributes of Humans and Machines

| Relative Strengths of Humans | Relative Strengths of Machines |
|---|---|
| Sense unexpected stimuli | Perform repetitive tasks consistently |
| Develop new solutions to problems | Store large amounts of data |
| Cope with abstract problems | Retrieve data from memory reliably |
| Adapt to change | Perform multiple tasks at the same time |
| Generalize from observations | Apply high forces and power |
| Learn from experience | Perform simple computations quickly |
| Make difficult decisions based on incomplete data | Make routine decisions quickly |

*Source:* [2].

Although the last three examples relate to service and knowledge work rather than production and logistics work, they also illustrate the widespread use of worker–machine systems. In these latter examples, the machine is a computer.

In a worker–machine system, the worker and the machine both contribute their own strengths and capabilities to the combination, and the result is synergistic. The relative strengths and attributes of humans and machines are presented in Table 1, and the worker–machine system should be designed to exploit these relative strengths.

### 2.1 Types of Worker–Machine Systems

It is instructive to distinguish the various categories and arrangements of worker–machine systems, some of which are suggested by our list of examples. In this section, we discuss the following classifications: (1) types of powered machinery used in the system, (2) numbers of workers and machines in the system, and (3) level of operator attention required to run the machinery.

**Types of Powered Machinery.** Powered machinery is distinguished from hand tools by the fact that a source of power other than human (or animal) strength is used to operate it. Common power sources are electric, pneumatic, hydraulic, and fossil fuel motors (e.g., gasoline, propane). In most cases, the power source is converted to mechanical energy to process the work unit. Powered machinery can be classified into three categories, summarized in Figure 3: (1) portable power tools, (2) mobile powered equipment, and (3) stationary powered machines.

Portable power tools are light enough in weight that they can be carried by the worker from one location to another and manipulated by hand. Examples include portable power drills, rotary saws, chain saws, and electric hedge trimmers. Common power sources are electric, pneumatic, and gasoline.

Mobile powered equipment can be divided into three categories: (1) transportation equipment, (2) transportable and mobile during operation, and (3) transportable and stationary during operation. They are generally heavy pieces of equipment and cannot be classified as power hand tools. Transportation equipment is a large category that includes cars, taxicabs, buses, trucks, trains, airplanes, boats, and ships. This powered machinery is designed to carry materials and/or people. Equipment in category (2), transportable

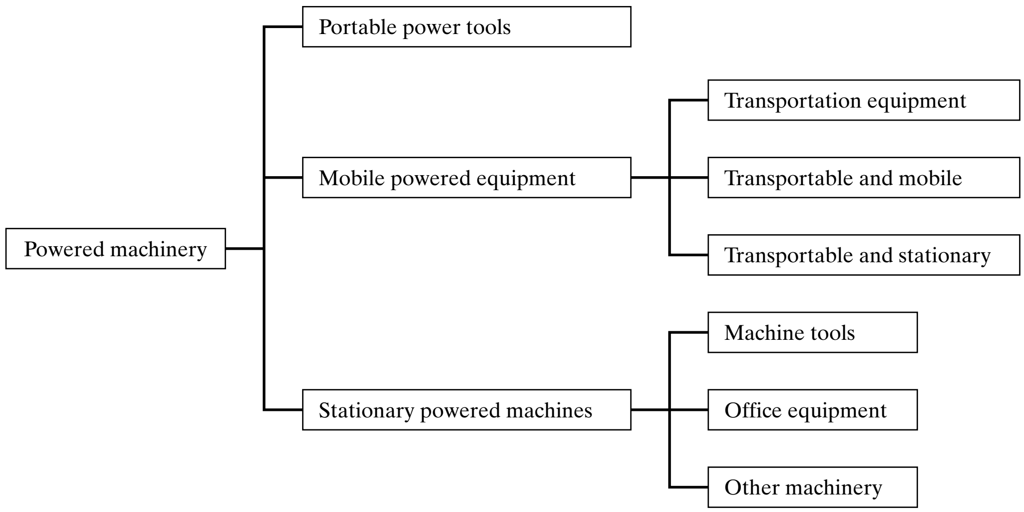



**Figure 3** Classification of powered machinery in worker–machine systems.

and mobile during operation, consists of equipment that can move under its own power but can also be moved by transportation equipment (e.g., tractor and flatbed). Examples include construction equipment (e.g., bulldozers, backhoes), agricultural and lawn-keeping (e.g., small tractors, lawn mowers), and material-handling equipment (e.g., forklift trucks). The third category, transportable and stationary during operation, is equipment that can be transported by highway truck but it performs its function in a stationary location once it is moved (e.g., electric power generator, large power saws used at construction sites). The typical power sources are fossil fuels.

Stationary powered machines stand on the floor or ground and cannot be moved while they are operating, and they are not normally moved between operations. Electricity is the usual power source. We can classify stationary power tools into the following categories: (1) machine tools, (2) office equipment, and (3) other. A ***machine tool*** is a stationary power-driven machine that shapes or forms parts. Machine tools are normally associated with factory production operations such as machining (e.g., turning, drilling, milling), shearing (e.g., blanking, hole-punching), and squeezing (e.g., forging, extrusion). Office equipment (second category) includes personal computers, photocopiers, telephones, fax machines, design workstations, and other equipment and systems normally found in an office facility. The "other" types of equipment (third category) are a miscellaneous group that includes machinery not fitting into the other two categories. Examples include furnaces, ovens, cash registers, and sewing machines.

**Numbers of Workers and Machines.** Another means of classifying worker–machine systems is according to whether there are one or more workers and one or more machines. This provides four categories, as indicated in Table 2.

**TABLE 2** Classification of Worker–Machine Systems According to Number of Machines and Workers

| | One Machine | Multiple Machines |
|---|---|---|
| One worker | One worker–one machine. **Examples**: (1) A worker loading and unloading a machine tool, (2) a truck driver driving a tractor trailer. | One worker–multiple machines **Example**: A worker tending several production machines. |
| Multiple workers | Multiple workers–one machine. **Examples**: (1) Several workers operating a rolling mill, (2) a crew on a ship or airplane. | Multiple worker–multiple machines. **Example**: An emergency repair crew responding to machine breakdowns in a factory. |

For the case of one worker and one machine, good work design attempts to achieve the following objectives:

- Design the controls of the machine to be logical and easy to operate for the worker.
- Design the work sequence so that as much of the worker's task as possible can be accomplished while the machine is operating, thereby minimizing worker idle time.
- Minimize the idle times of both the worker and the machine.
- Design the task and the machine to be safe for the worker. If the task is inherently hazardous to the worker, then an automated work system should be considered.

The same design objectives are applicable when the work system consists of multiple workers and/or multiple machines. An additional objective is to optimize the number of workers or machines in the system according to some appropriate economic objective. For example, if one worker is assigned to attend to multiple machines, how many machines should be assigned to that worker so as to avoid machine idle time?

**Level of Operator Attention Required.** Another way to distinguish among worker–machine systems is by the level of attention required by the worker(s). In this classification, we have the four categories described in Table 3. Full-time attention means that the worker must devote virtually 100% of his or her time to the operation of the equipment during the performance of the task.

### 2.2 Cycle Time Analysis in Worker–Machine Systems

In terms of cycle time analysis, worker–machine systems fall into two categories: (1) systems in which the machine time depends on operator control, and (2) systems in which the machine time is constant and independent of operator control, and the work cycle is repetitive. In the first category, in which machine time depends on operator control, the task can be either (1) repetitive or (2) nonrepetitive. The following examples illustrate repetitive tasks with cycle times that depend on the pace and skill with which the operator applies the powered equipment:

- A typist typing a list of names and telephone numbers on a conventional electric typewriter

**TABLE 3** Levels of Operator Attention in Worker–Machine Systems

| Category | Description | Examples |
|---|---|---|
| Full-time attention | Worker is engaged 100% of the time in operating the equipment. The task can be (1) repetitive and cyclical or (2) nonrepetitive. | Worker operating a drop forge hammer (repetitive and cyclical).<br>Worker on an assembly line whose task time equal to available service time (repetitive and cyclical).<br>Truck driver driving an 18-wheeler (nonrepetitive).<br>Worker mowing the lawn (nonrepetitive). |
| Part-time attention during each work cycle | Worker is engaged less than 100% of the time in operating the equipment. Task is repetitive and cyclical. | Worker loading and/or unloading production machine each cycle. Machine processes work units on mechanized or semiautomatic cycle. Worker is idle during machine cycle. |
| Periodic attention with regular servicing | Worker must service machine at regular intervals that are greater than one work cycle | Crane operator in steel mill moving molten steel ingots after each heat cycle. The operator must spend most of his time waiting for the next heat.<br>Worker loading and/or unloading an automated production machine every 20 cycles. The machine operates on automatic cycle for the 20 cycles, but its storage capacity is limited to 20 work units. |
| Periodic attention with random servicing | Worker must service machine at random intervals that average more than one work cycle | Maintenance worker repairing production equipment when it malfunctions at random times.<br>Firefighters responding to alarms that occur at random times. |

- A metal trades worker operating a power buffer to buff the surface of a metal part
- A carpenter using a power saw to cut standard lengths of lumber
- A forklift driver moving pallet loads from the truck dock to the storage racks in a warehouse.

In these cases when the work cycle is repetitive but the cycle time is not constant, the analysis methods in Section 1.2 can be used.

Examples of worker–machine systems that operate on a nonrepetitive work cycle include the following:

- A trucker driving a tractor-trailer on an interstate highway
- A construction worker operating a backhoe
- A farmer operating a threshing machine to separate seeds from crop
- A carpenter using portable power tools to build a deck on a newly constructed house.

These nonrepetitive work situations do not consist of a regular work cycle that is repeated over and over. The time to accomplish the work depends on the skill and work ethic of the persons performing the tasks. Estimates or historical records based on previous similar jobs are often used to determine how long the work should take to complete.

In this section, we focus attention on the second category of worker–machine system, in which machine time is constant and does not depend on operator control, and the work cycle is repetitive. Two cases are discussed: (1) cycle times with no overlap between worker and machine and (2) worker–machine systems with internal work elements.

**Cycle Times with No Overlap Between Worker and Machine.** In a worker–machine system, the work elements include one or more actions and/or operations performed by the machine. If there is no overlap in work elements between the worker and the machine, then the normal time for the cycle is simply the sum of their respective normal times:

$$T_n = T_{nw} + T_m \tag{6}$$

where $T_{nw}$ = normal time for the worker-controlled portion of the cycle, min; and $T_m$ = machine cycle time (assumed constant).

To determine the standard time for the cycle, a machine allowance is sometimes added to the machine time. If we include such an allowance factor in the standard time calculation, we have

$$T_{std} = T_{nw}(1 + A_{pfd}) + T_m(1 + A_m) \tag{7}$$

where $T_{nw}$ = normal time of the worker, min; $T_m$ = constant time for the machine cycle, min; $A_m$ = machine allowance factor, used in the equation as a decimal fraction; and the other terms have the same meaning as before.

A typical value used by companies for the machine allowance factor is $A_m = 30\%$. This tends to help workers achieve higher worker efficiencies, which is especially important if the worker is paid on an incentive basis. Workers might prefer to work on an entirely manual cycle if the machine allowance were not provided. On the other hand, some companies do not see the need to use a machine allowance, in which case $A_m = 0$. An argument for $A_m = 0$ is that the worker is idle during the machine cycle, and so does not have to expend any effort during this portion of the work cycle. Other companies simply set the $A_m$ value to be the same as $A_{pfd}$. The following examples show how the value of $A_m$ affects the standard time and worker efficiency.

**Example 8 Effect of Machine Allowance on Standard Time**

In the operation of a worker–machine system, the work cycle consists of several manual work elements (operator-controlled) and one machine element performed under semiautomatic control. One workpiece is produced each cycle. The manual work elements total a normal time of 1.0 min and the semiautomatic machine cycle is a constant 2.0 min. The PFD allowance factor $A_{pfd}$ is 15%. Determine the standard time using (a) $A_m = 0$ and (b) $A_m = 30\%$.

**Solution:** The normal time for the work cycle is the normal time for the worker-controlled elements plus the machine cycle time:

$$T_n = 1.0 + 2.0 = 3.0 \text{ min}$$

**(a)** With a machine allowance of 0, the standard time is calculated as

$$T_{std} = 1.0(1 + 0.15) + 2.0 = 3.15 \text{ min}$$

**(b)** With a machine allowance of 30%, the standard time is

$$T_{std} = 1.0(1 + 0.15) + 2.0(1 + 0.30) = 1.15 + 2.60 = 3.75 \text{ min}$$ ■

**Example 9 Effect of Machine Allowance on Worker Efficiency**

Based on the standard times computed in (a) and (b) of the previous example, determine the worker efficiencies for two cases if 150 units are produced in one 8-hour shift.

**Solution:** **(a)** If $T_{std} = 3.15$ min, the number of standard hours accomplished is

$$H_{std} = 150(3.15) = 472.5 \text{ min} = 7.875 \text{ hr}$$

$$\text{Worker efficiency } E_w = 7.875/8.0 = 0.984 = 98.4\%$$

**(b)** If $T_{std} = 3.75$ min, the number of standard hours accomplished is

$$H_{std} = 150(3.75) = 562.5 \text{ min} = 9.375 \text{ hr}$$

$$\text{Worker efficiency } E_w = 9.375/8.0 = 1.172 = 117.2\%$$ ■

When the work cycle includes a machine cycle and the machine time is a constant, then operator performance has no effect on this machine element. The only way the worker's pace can affect the work cycle time is during those elements that are operator-controlled. The operator may be idle during the machine element, as in Examples 8 and 9, unless the work sequence can be designed to include operator work elements that are performed while the machine is running.

**Worker–Machine Systems with Internal Work Elements.** In the operation of a worker–machine system, it is important to distinguish between the operator's work elements that are performed in sequence with the machine's work elements and those that are performed simultaneously with the machine elements. Operator elements that are performed sequentially are called ***external work elements*** while those that are performed simultaneously with the machine cycle are called ***internal work elements***. The distinction is important because it is desirable to construct the work cycle sequence so that as many of the operator elements as possible are performed as internal elements. This tends to minimize the cycle time, as illustrated by the following example.

**Example 10 Internal Versus External Work Elements in Cycle Time Analysis**

The work cycle in a worker–machine system consists of the elements and associated times given in the table below. All of the operator's work elements are external to the machine time. Can some of the worker's elements be made internal to the machine cycle, and if so, what is the expected cycle time for the operation?

| Sequence | Work Element Description | Worker Time (min) | Machine Time (min) |
|---|---|---|---|
| 1 | Worker walks to tote pan containing raw stock | 0.13 | (idle) |
| 2 | Worker picks up raw workpart and transports to machine | 0.23 | (idle) |
| 3 | Worker loads part into machine and engages machine semiautomatic cycle | 0.12 | (idle) |
| 4 | Machine semiautomatic cycle | (idle) | 0.75 |
| 5 | Worker unloads finished part from machine | 0.10 | (idle) |
| 6 | Worker transports finished part and deposits into tote pan | 0.15 | (idle) |
| Totals | | 0.73 | 0.75 |

**Solution:** Since all of the work elements in the cycle are sequential, the total cycle time is the sum of the worker elements and the machine element:

$$T_c = 0.73 + 0.75 = 1.48 \text{ min.}$$

The worker is idle during the entire machine semiautomatic cycle. It should be possible to imbed elements 1, 2, and 6 as internal elements that are performed while the machine is running on a semiautomatic cycle. Using the times from the preceding table, the resulting work cycle can be organized as follows:

| Sequence | Work Element Description | Worker Time (min) | Machine Time (min) |
|---|---|---|---|
| 1 | Worker unloads finished part from machine | 0.10 | (idle) |
| 2 | Worker loads part into machine and engages semiautomatic machine cycle | 0.12 | (idle) |
| 3 | Machine semiautomatic cycle | | 0.75 |
| 4 | Worker transports finished part and deposits it into tote pan, walks to tote pan containing raw stock, and picks up raw workpart and transports it to machine. (This element is internal to the machine semiautomatic cycle.) | 0.15 + 0.13 + 0.23 = 0.51 | |
| Totals | | 0.73 | 0.75 |

Although the total times for the worker and the machine are the same as before, element 4 in the revised cycle (which consists of elements 1, 2, and 6 from the original work cycle) is performed simultaneously with the machine time, resulting in the following new cycle time:

$$T_c = 0.10 + 0.12 + 0.75 = 0.97 \text{ min}$$

This represents a 34% reduction in cycle time, which translates into a 53% increase in production rate.

■

When internal elements are present in the work cycle, it must then be determined whether the machine cycle time or the sum of the worker's internal elements take longer. To determine the normal time for the cycle,

$$T_n = T_{nw} + \text{Max}\,\{T_{nwi}, T_m\} \tag{8}$$

where $T_{nw}$ = normal time for the worker's external elements, min; $T_{nwi}$ = normal time for the worker's internal elements, min; and $T_m$ = machine cycle time. The standard time for the cycle is given by

$$T_{std} = T_{nw}(1 + A_{pfd}) + \text{Max}\,\{T_{nwi}(1 + A_{pfd}),\, T_m(1 + A_m)\} \tag{9}$$

where $A_{pfd}$ and $A_m$ are the worker's allowance factor and the machine allowance factor, respectively. Finally, the actual cycle time depends on the worker's performance level, applied to the normalized times as

$$T_c = \frac{T_{nw}}{P_w} + \text{Max}\left(\frac{T_{nwi}}{P_w},\, T_m\right) \tag{10}$$

where $P_w$ is the worker performance level during the cycle, expressed as a decimal fraction; and the other symbols mean the same as before. We assume in equation (10) that the worker's performance level is the same on the external and internal elements. If these $P_w$ values are different, then the computations must reflect these differences.

## 3 AUTOMATED WORK SYSTEMS

***Automation*** is the technology by which a process or procedure is accomplished without human assistance. It is implemented using a program of instructions combined with a control system that executes the instructions. Power is required to drive the process and to operate the program and control system.

There is not always a clear distinction between worker–machine systems and automated systems, because many worker–machine systems operate with some degree of automation. Let us distinguish between two forms of automation: semiautomated and fully automated. A ***semiautomated machine*** performs a portion of the work cycle under some form of program control, and a human worker tends to the machine for the remainder of the cycle, by loading and unloading it, or performing some other task during each cycle. An example of this category is an automated lathe controlled for most of the work cycle by the part program but requiring a worker to unload the finished part and load the next workpiece at the end of each machine cycle. In these cases, the worker must attend to the machine during every cycle. This type of operation has the same characteristics as a worker–machine system that requires the part-time attention of the worker during each work cycle (Table 3). Its cycle time analysis is discussed in Section 2.2.

The continuous presence of the operator during the cycle may not always be required. If the automatic machine cycle takes, say, 10 min while the part unloading and loading portion of the work cycle takes only 30 sec, then there may be an opportunity for one worker to tend more than one machine. We analyze this possibility in Section 5.

A ***fully automated machine*** is distinguished from its semiautomated cousin by the capacity to operate for extended periods of time with no human attention. By extended periods of time, we mean longer than one work cycle. A worker is not required to be present during each cycle. Instead, the worker may need to tend the machine every tenth cycle, or every hundredth cycle. An example of this type of operation is found in many injection molding plants, where the molding machines run on automatic cycle, but periodically the molded parts at the machine must be collected by a worker. This case is identified in Table 3 as periodic attention with regular servicing.

Certain fully automated processes require one or more workers to be present to continuously monitor the operation and make sure that it performs according to the intended specifications. Examples of these kinds of automated processes are found at chemical-processing facilities, oil refineries, and nuclear power plants. The workers do not actively participate in the process except to make occasional adjustments in the equipment settings, to perform periodic maintenance, and to spring into action if something goes wrong.

## 4 DETERMINING WORKER AND MACHINE REQUIREMENTS

One of the problems faced by any organization is determining the appropriate staffing levels. How many workers are required to achieve the organization's work objectives? If too few workers are assigned to perform a given amount of work, then the work cannot be completed on time, and customer service suffers. If too many workers are assigned, then payroll costs are higher than needed, and productivity suffers. Determining the number of workers or worker–machine systems that will be required to accomplish a specified amount of work is the problem we address in this section. The basic approach consists of two steps:

1. Determine the total workload that must be accomplished in a certain period (hour, week, month, year), where ***workload*** is defined as the total hours required to complete a given amount of work or to produce a given number of work units scheduled during the period.
2. Divide the workload by the available time per worker, where ***available time*** is defined as the number of hours in the same period available from one worker or worker–machine system.

Let us consider two general cases: (1) when setup time is not a factor and (2) when setup time must be included in the determination.

### 4.1 When Setup Is Not a Factor

Workload is figured as the quantity of work units to be produced during the period of interest multiplied by the time (hours) required for each work unit. The time required for each work unit is the work cycle time in most cases, so that workload is given by

$$WL = QT_c \tag{11}$$

U2

# Introduction to Methods Engineering and Operations Analysis



The techniques discussed in this chapter have the following basic objectives:

- To analyze existing work systems
- To make improvements in existing work systems
- To design new work systems
- To plan the facilities in which the work systems operate

Most of the techniques that we discuss are traditional industrial engineering "tools of the trade." Methods engineering is often associated with work measurement. The two areas are often referred to collectively as "motion and time study." Many of the motion and time study techniques can be traced to the origins of industrial engineering, when it was referred to as the "scientific management movement." The techniques have been used to analyze, design, and measure work for many decades.

Methods engineering is a broader term than motion study. The scope of this field has expanded well beyond its original focus on human body motions to perform physical labor. We define ***methods engineering*** as the analysis and design of work methods and systems, including the tooling, equipment, technologies, workplace layout, plant layout, and environment used in these methods and systems.[1] Other names have sometimes been used to indicate the same basic approaches of methods engineering. These other names include ***work study***, ***work simplification***, ***methods study***, ***process re-engineering***, and ***business process re-engineering***.

The traditional objectives in methods engineering, whatever the name for this improvement activity, are the following:

- To increase productivity and efficiency
- To reduce cycle time
- To reduce product cost
- To reduce labor content

In addition, other objectives are frequently defined for process improvement efforts. Some of these additional objectives have a more contemporary appeal in today's society. They include the following:

- To improve customer satisfaction
- To improve product and/or service quality
- To reduce lead times and improve work flow
- To increase work system flexibility
- To improve worker safety
- To apply more ergonomic work methods
- To enhance the environment (both inside and outside the facility)

A term closely related to methods engineering is ***operations analysis***, defined as the study of an operation or group of related operations for the purpose of analyzing their efficiency and effectiveness so that improvements can be developed relative to specified objectives.[2] The specified objectives are basically the same as in methods engineering: to increase productivity, reduce time and cost, and improve safety and quality. Thus, methods engineering and operations analysis are very similar terms, except that methods engineering places more emphasis on design. Both terms are widely used in industrial engineering and this is why both names are included in our chapter title.

[1]The definition of methods engineering developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "That aspect of industrial engineering concerned with the analysis and design of work methods and systems, including the technological selection of operations or processes, specification of equipment type and location, design of manual and worker-machine tasks. May include the design of controls to insure proper level of output, inventory, quality, and cost."

[2]The definition of operations analysis developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "A study of an operation or scenes of operations involving people, equipment, and processes for the purpose of investigating the effectiveness of specific operations or groups so that improvements can be developed which will raise productivity, reduce costs, improve quality, reduce accident hazards, and attain other desired objectives."

## 1 EVOLUTION AND SCOPE OF METHODS ENGINEERING

The initial research in the area of methods engineering by Frank Gilbreth in 1885 dealt with the motions performed by workers in bricklaying. Motion study was truly an appropriate term for Gilbreth's research. The study of manual physical labor in manufacturing and construction was the primary concern of the scientific management movement in the late nineteenth and early twentieth centuries, and motion study was one of the two principal techniques used by the practitioners in that movement (the other, of course, was time study). Today, methods engineering is being applied in many other areas of work, including indirect labor, logistics, service operations, office work, and plant layout design. As these other areas have grown in importance in the economies of industrialized nations, methods engineering has been applied to analyze, improve, and design the work methods.

In terms of the problems that are addressed, methods engineering can be divided into two areas: (1) methods analysis and (2) methods design. ***Methods analysis*** is concerned with the study of an existing method or process, usually by breaking it down into the work elements or basic operations that comprise it.[3] By examining the details of the elements or operations, a systematic search can be carried out to find ways to improve the method or process. The systematic search often consists of checklists of questions and suggestions that offer opportunities for improvement. There are several objectives of using checklists in methods analysis:

- To *eliminate* unnecessary and non-value-adding elements or operations from the larger method or process
- To *combine* multiple elements or operations by performing them at one location and/or simultaneously
- To *rearrange* the elements or operations into a more logical sequence and work flow
- To *simplify* the remaining elements or operations so they can be accomplished more quickly and with minimum effort.

In addition to studying an existing method or process, methods analysis can also be used to analyze a proposed new method for possible improvements. In this regard the two areas of methods analysis and methods design overlap each other.

***Methods design*** is concerned with either of the following situations: (1) the design of a new method or process or (2) the redesign of an existing method or process based on a preceding methods analysis. The design of a new method or process occurs when a new product or service is introduced and/or a new facility or equipment is installed, and there is no existing precedent for the operation. In this case, the method or process must be designed from scratch. This can often be accomplished by referring to best current practice for similar operations and attempting to improve on the current work design. In other cases, an original work design must be developed, including the basic operations,

[3]The definition of methods analysis developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "That part of methods engineering normally involving an examination and analysis of an operation or a work cycle broken down into its constituent parts for the purpose of improvement, elimination of unnecessary steps, and/or establishing and recording in detail a proposed method of performance."

methods, equipment, special tooling, workplace layout, and plant layout. As in other areas of design, guidelines are available, such as the principles of motion economy and the plant layout planning approach, and these can be applied in solving the design problem.

Methods engineering studies have traditionally been the province and responsibility of industrial engineers. Today, methods analysis and design are no longer accomplished exclusively by industrial engineers. They are accomplished, with greater or lesser effectiveness, by a variety of individuals, departments, and teams. The workers themselves often participate in the development and improvement of their own work methods, under the assumption that they know their jobs better than anyone else. Industrial engineers often participate in these improvement activities, serving an important role as technical consultant and mentor.

## 2 HOW TO APPLY METHODS ENGINEERING

In this section we present a systematic approach that can be used to accomplish a methods engineering study. We then survey the tools and techniques used in such a study. Finally, we present a procedure for selecting among alternative possible solutions to a methods engineering problem.

### 2.1 Systematic Approach in Methods Engineering

An underlying assumption in methods engineering is that a systematic approach is more likely to yield operational improvements than an undisciplined approach. Our systematic approach to problem solving in methods engineering has its basis in the ***scientific method*** used in science, research and development, engineering design, and other problem areas. The systematic approach in methods engineering consists of the steps described below.

**Step 1: Define the Problem and Objectives.** The problem is the reason for needing a systematic approach to determine its solution. The problem in a methods engineering study may be low productivity, high cost, inefficient methods, or the need for a new method or a new operation. The objective is the desired improvement or new methods design that would result from the methods engineering project. Possible objectives are to increase productivity, reduce labor content and cost, improve safety, or develop a new method or new operation. These are the typical objectives discussed in our chapter introduction. The problem definition and objectives must be specific to the problem under investigation, although there may be similarities with other problems.

**Step 2: Analyze the Problem.** This step consists of data collection and analysis activities that are most appropriate for the type of problem being studied. The kinds of activities often used in this step include the following:

- Identify the basic function of the operation.
- Gather background information.
- Observe the existing process or observe similar processes if the problem involves a new work design.

- Collect data on the existing operation and document the details in a format that lends itself to examination.
- Conduct experiments on the process.
- Develop a mathematical model of the process or utilize an existing mathematical model such as those on work systems and how they work.
- Perform a computer simulation of the process.
- Use charting techniques.

A survey of the analysis techniques used in step 2 is provided in Section 2.2.

**Step 3: Formulate Alternatives.** There are always multiple ways to perform a task or accomplish a process, some of which are more efficient and effective than others. Only by enumerating the alternative ways and comparing them can the most efficient and effective method or process be determined.[4] However, the purpose of this step in the problem-solving approach is not to identify the best alternative but to formulate all of the alternatives that are feasible.[5]

**Step 4: Evaluate Alternatives and Select the Best.** This step consists of a methodical assessment of the alternatives and the selection of the best solution among them, based on the original definition of the problem and objectives. The selection procedure described in Section 2.3 is useful in this step of the methods engineering approach.

**Step 5: Implement the Best Method.** Implementation means installing the selected solution: introducing the changes proposed in the existing method or operation, or instituting the new method or process. This may involve pilot studies or trials of the new or revised method preliminary to online implementation and application of the method. Implementation also includes complete documentation of the new or revised method and replacement of the previous documentation in the case of a revised method. Unless the old documentation is replaced, the old method may remain as the official method until a new methods engineering study is performed sometime in the future (thus reinventing the wheel).

**Step 6: Audit the Study.** It is desirable to perform an audit or follow-up on the methods engineering project. How successful was the project in terms of the original problem definition and objectives? What were the implementation issues? What should be done differently in the next methods engineering study? For an organization committed to continuous improvement, answers to these kinds of questions help to fine-tune its problem-solving and decision-making skills.

[4]In the scientific method as applied to research, the alternatives are typically the different proposed theories to explain the observed phenomenon. Ultimately, one theory is selected as the most reasonable in terms of explaining the phenomenon.

[5]Even the infeasible solutions should probably be included in the list. Sometimes the alternatives that initially seem infeasible turn out to be quite doable after all. What made them seem unacceptable on first consideration were artificial or false constraints that proved not to be constraints upon closer scrutiny.

## 2.2 The Techniques of Methods Engineering

A variety of techniques are available for operations analysis. The techniques are most closely associated with the analysis step in methods engineering, although they may also be applicable in some of the other steps as well. As a starting point, we have the basic data collection and analysis techniques discussed in Section 3. These are graphical and statistical methods for gathering, plotting, and displaying data. They include histograms and *x-y* plots as well as other charts and diagrams.

Beyond the basic data collection and analysis techniques, which are applicable in many disciplines other than methods engineering, there are the specialized analysis techniques more closely associated with operations analysis and industrial engineering. They are described briefly in the following sections.

**Charting and Diagramming Techniques.** There are many charting techniques available for collecting, displaying, and analyzing data on a given work system or operation sequence of interest. They can be classified into the following categories:

- *Network diagrams.* These are used for analyzing work flow, assembly line balancing, and project scheduling. Special algorithms are often available to analyze these network diagrams.
- *Traditional industrial engineering charting techniques.* These are used to symbolize and summarize the details of an existing operation or sequence of operations. The traditional charting techniques can be used to analyze the activities of one human worker, groups of workers, worker-machine systems, materials, parts, and products.
- *Block diagrams and process maps.* These diagrams represent alternative ways of depicting processes. They are sometimes used in place of the traditional IE charting techniques.

**Motion Study and Work Design.** This area of methods engineering is concerned with the study of the basic motions of a human worker while performing a given task. The basic motions include ***reach*** (using the hand to reach for an object), ***grasp*** (grasping the object), ***move*** (moving the object), and ***release*** (releasing it). All manual tasks performed at a single workplace are composed of these basic motions. There are 17 basic motion elements, most of which involve movements of the arm and hand. By studying the basic motions in the work method used by a worker, unnecessary motions can be eliminated, or some of the motion elements can be combined (for example, using both hands to simultaneously perform motions rather than one arm doing everything), or the method can be otherwise simplified.

Over many years, the study of basic motions in manual operations has resulted in the development of certain principles on how to perform work. Commonly called the ***principles of motion economy***, they provide guidelines for work design in three categories: (1) use of the human body in developing the standard method, (2) workplace layout, and (3) design of the tooling and equipment used in the task. Many of the principles are simple and obvious; for example, "design the work so that both hands are fully utilized." Yet simple work design principles such as these are often neglected in many manual operations performed throughout the world. By having these guidelines available, work methods can be designed to be safer, faster, more efficient, and less fatiguing.

**Facility Layout Planning.** A facility (e.g., factory, office building, warehouse, hospital) is a fixed asset of the organization that owns (or rents) it. ***Facility layout*** refers to the size and shape of a facility, the arrangement of the different functions and/or departments in it, and the way the equipment is positioned. The layout plays a significant role in determining the overall efficiency of the operations accomplished in the facility. Facility layout planning represents an important problem area in industrial engineering, and we discuss the techniques for solving it within the scope of methods engineering. The problem area includes designing a new facility, installing new equipment, retiring old equipment, and expanding (or contracting) an existing facility.

Plant layout planning and design are best accomplished using a systematic approach, similar to the systematic approach in methods engineering. As in other problem areas in methods engineering, there are tools available to use in a plant layout design project. The approach is called "systematic layout planning," developed by Richard Muther [13].

**Work Measurement Techniques.** Several of the work measurement techniques can also be used in methods engineering. We mention two that seem most relevant for studying and analyzing operations and methods: predetermined motion time systems and work sampling.

A ***predetermined motion time system*** (PMTS) is a database of basic motion elements and their associated normal time values, and it includes procedures for applying the database to analyze manual tasks and establish standard times for the tasks. The principal application of a PMTS is to determine standard times. However, some systems also include tools for analyzing methods and motions in the task, which is a methods engineering function.

***Work sampling*** is a statistical technique for determining the proportions of time spent by workers or machines in various categories of activity. It can be applied to determine machine utilization, worker utilization, and the average time spent performing various types of activities. As such it can be a useful tool in methods engineering for identifying areas that need attention. For example, if a work sampling study finds that workers in a facility spend large amounts of their time waiting for work, then this is a management problem that should be addressed.

**New Approaches in Methods Engineering**. Nearly all of the preceding techniques have been used for many decades, some even longer.[6] More recently, alternative approaches for improving production and service operations have evolved from these traditional techniques. These alternative approaches include lean production (based on

[6]An example: Frank Gilbreth's interest in motion study dates from 1885; his book, *Motion Study*, was published in 1911.

the famed Toyota Production System), total quality management, and Six Sigma.

It is the view of this author that most of the tools used in these more recent approaches are basically adaptations and modifications of good industrial engineering techniques and principles. Not to demean their importance, they are often used with good effect to identify and solve problems and to make improvements. And they sometimes successfully serve the function of providing a rallying point for motivating workers who might not be motivated by something called motion and time study. Their objectives are generally the same as the objectives of methods engineering.

## 2.3 Selecting Among Alternative Improvement Proposals

Evaluating the alternative improvement proposals and selecting the best among them (step 4 in the systematic methods engineering approach) can be a difficult process in methods engineering. As always, it is helpful to use a systematic procedure to decide which improvement proposal(s) should be selected. The procedure recommended here is most applicable when the selection is to be made among specific proposals—for example, alternative process designs or different equipment proposals from vendors. The alternatives have been developed to address a specified problem or a need of the customer organization. Each proposed solution has its relative strengths and weaknesses, and the selection procedure must weigh the pros and cons in a fair and logical way. The procedure is explained as follows.[7]

Prior to the development of proposals, a list of technical features and functional specifications for the given application must be prepared by the customer organization. This is an essential part of the problem definition step (step 1) in the methods engineering approach. The operations analyst(s) or equipment vendors will use this list as the basis for formulating alternative solutions and proposals (step 3 in the systematic approach). The same list will be used by the customer organization to evaluate the proposals. The features and specifications should be divided into two categories:

1. *Must features.* These are the features and specifications that the proposal must satisfy. If these are not satisfied, then the proposal is not suitable for the application.
2. *Desirable features.* These are features and specifications that are not necessarily required to satisfy the application, but they are desirable.

After all of the proposals have been submitted, the evaluation process begins. The tabular format shown in Table 1, sometimes referred to as a ***criteria matrix***, is used. First, the proposals are evaluated against the must features. A candidate proposal must satisfy all of the must features or it is dropped from further consideration.

[7]Although we are presenting this selection procedure as a decision-making tool in methods engineering, its applications are much broader and more diverse. For example, it can be used for selecting which new car to buy, deciding which job offer to accept (assuming there is more than one offer), or even choosing a prospective spouse.

**TABLE 1** Evaluation of Alternative Industrial Robots for a Welding Application in Example 1

| | Industrial Robot Candidates | | | |
|---|---|---|---|---|
| | Model A | Model B | Model C | Model D |
| Must features: | | | | |
| Continuous path control | OK | OK | OK | OK |
| Six-axis robot arm | OK | OK | Not OK | OK |
| Walkthrough programming | OK | OK | OK | OK |
| Desirable features: | | | | |
| Ease of programming (0–9) | 6 | 4 | | 6 |
| Capability to edit program (0–5) | 4 | 2 | | 5 |
| Multipass features (0–4) | 2 | 2 | | 2 |
| Work volume (0–9) | 5 | 8 | | 6 |
| Repeatability (0–5) | 5 | 2 | | 4 |
| Lowest price (0–5) | 4 | 5 | | 3 |
| Delivery (0–3) | 1 | 1 | | 3 |
| Evaluation of vendor (0–9) | 6 | 5 | | 8 |
| Totals: | 33 | 29 | | 37 |

*Source:* [6].

Next, the proposals are compared against each of the desirable features. For each feature, a rating score is given to each candidate to indicate how well it satisfies that feature. No doubt there will be differences in the relative importance of the different features, and this is taken into account by assigning a maximum possible point score to each desirable feature. For example, if a particular feature is judged to be twice as important as another, then the more important feature is assigned a maximum score of, say, 10 points, while the less important feature is given a maximum of 5 points. These maximum point scores are judgment calls made by the methods engineer or by the collective wisdom of the project team doing the selection. Scoring each proposal on each feature is also a judgment call based on the relative merits of each candidate.

Finally, for the proposals that still remain after elimination of those that failed one or more of the must features, the scores of each proposal are tallied and the proposal with the highest total score is selected as the winner.

Of course, economics must play a role in the selection process. One might be inclined to simply choose the low cost bidder. However, that is often a mistake. Cost is rarely the only factor in implementing a new process, purchasing a piece of equipment, or making other investment decisions. The cost factor can be included as one of the desirable features in the list and given an appropriate weighting score to reflect its importance. The following example illustrates the use of this procedure to select a robot for a welding application.

**Example 1 Selecting a Welding Robot[8]**

Four industrial robots were being considered to satisfy an arc-welding application at the company. They are identified in Table 1 as Models A, B, C, and D. As suggested in the selection

[8]This example is based on an industrial case study in which the author participated. The case study was first reported in [7]. The company remains anonymous.

procedure, the features and specifications are divided into two categories: "must" and "desirable." The must features were considered essential for the application. The desirable features were assigned maximum point scores as shown in the table. The entries in the table for each robot indicate how that candidate was scored in each of the features. Note that one of the features was price, but it was not considered to be the most important feature.

**Conclusion.** First, model C was eliminated from consideration because it did not satisfy one of the "must" features. For the three remaining models, model D was selected because it had the highest point score among the desirable features. ■

## 3 BASIC DATA COLLECTION AND ANALYSIS TECHNIQUES

The data collection and analysis techniques discussed in this section are used by scientists, engineers, mathematicians, and statisticians. Most of them are statistical charting tools used to record and/or exhibit data so that it can be interpreted more readily. The techniques also have value for analysis. When used to measure and analyze production and service operations, these statistical tools are often associated with a field known as ***statistical process control*** (SPC).[9] Statistical process control and other quality-oriented programs are discussed in Chapter 21. In these programs, worker teams are organized to study operational problems. To enable the teams to be more effective, they are trained in the use of these basic tools. Thus, industrial engineers are not the exclusive users of these techniques in operations analysis. Instead, IEs must often serve as team leaders, training instructors, and expert consultants for the worker teams who carry out the studies.

**Histograms.** A ***histogram*** is a statistical graph consisting of bars representing different values or ranges of values, in which the length of each bar is proportional to the frequency or relative frequency of the value or range, as shown in Figure 1. Also known as a bar chart, it is a graphical display of the ***frequency distribution*** of the numerical data. What makes the histogram such a useful statistical tool is that it enables the analyst to quickly visualize the features of a complete set of data. These features include (1) the shape of the distribution, (2) any central tendency exhibited by the distribution, (3) approximations of the mean and mode of the distribution, and (4) the amount of scatter or spread in the data.

**Example 2 Frequency Distribution and Histogram**

Part dimension data from a manufacturing process are displayed in the frequency distribution of Table 2. The data are the dimensional values of individual parts taken from the process, while the process is running normally. Plot the data as a histogram and draw inferences from the graph.

**Solution:** The frequency distribution in Table 2 is displayed graphically in the histogram of Figure 1. We can see that the distribution is normal (in all likelihood), and that the mean is

[9]Although we are giving due recognition to the field of statistical process control, it should also be noted that many of these basic tools have been taught for many years in industrial engineering courses and used by practicing industrial engineers to study problems in methods engineering and operations analysis.

**TABLE 2** Frequency Distribution of Part Dimension Data

| Range of Dimension | Frequency | Relative Frequency | Cumulative Relative Frequency |
|---|---|---|---|
| $1.975 \leq x < 1.980$ | 1 | 0.01 | 0.01 |
| $1.980 \leq x < 1.985$ | 3 | 0.03 | 0.04 |
| $1.985 \leq x < 1.990$ | 5 | 0.05 | 0.09 |
| $1.990 \leq x < 1.995$ | 13 | 0.13 | 0.22 |
| $1.995 \leq x < 2.000$ | 29 | 0.29 | 0.51 |
| $2.000 \leq x < 2.005$ | 27 | 0.27 | 0.78 |
| $2.005 \leq x < 2.010$ | 15 | 0.15 | 0.93 |
| $2.010 \leq x < 2.015$ | 4 | 0.04 | 0.97 |
| $2.015 \leq x < 2.020$ | 2 | 0.02 | 0.99 |
| $2.020 \leq x < 2.025$ | 1 | 0.01 | 1.00 |

around 2.00. We can approximate the standard deviation to be the range of the values (2.025–1.975) divided by 6, based on the fact that nearly the entire distribution (99.73%) is contained within $\pm 3\sigma$ of the mean value. This gives a $\sigma$ value of around 0.008. ■

**Pareto Charts.** A ***Pareto chart*** is a special form of histogram, illustrated in Figure 2, in which attribute data are arranged according to some criteria such as cost or value. When appropriately used, it provides a graphical display of the tendency for a small proportion of a given population to be more valuable than the much larger majority. This tendency is sometimes referred to as ***Pareto's law***, which can be succinctly stated:

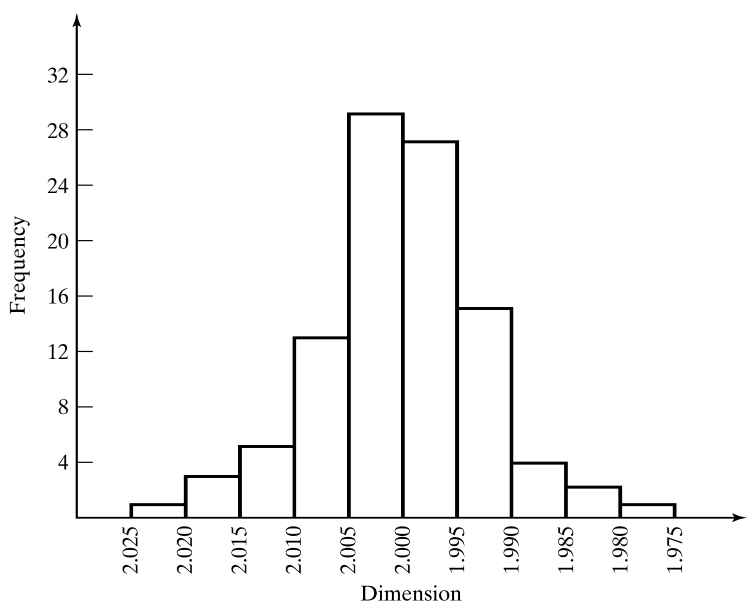



**Figure 1** Histogram of the data in Table 2.

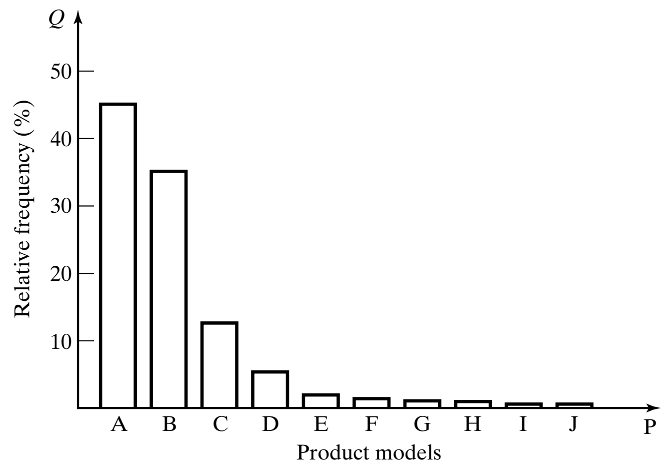



**Figure 2** Typical (hypothetical) Pareto distribution of a factory's production output. Although there are ten models produced, two of the models account for 80% of the total units. This chart is sometimes referred to as a *P*-*Q* chart, where $P$ = products and $Q$ = quantity of production.

"the vital few and the trivial many."[10] The "law" was identified by Vilfredo Pareto (1848–1923), an Italian economist and sociologist who studied the distribution of wealth in Italy and found that most of it was held by a small percentage of the population.

Pareto's law applies not only to the distribution of wealth, but to many other distributions as well. The law is often identified as the 80%–20% rule (although exact percentages may differ from 80 and 20): 80% of the wealth of a nation is in the hands of 20% of its people; 80% of inventory value is accounted for by 20% of the items in inventory; 80% of sales revenues are generated by 20% of the customers; and 80% of a factory's production output is concentrated in only 20% of its product models (as in Figure 2). A Pareto chart identifies the proportion of the population that is the most important, and the focus in any improvement study or project should be on that proportion.

A Pareto distribution can also be plotted as a cumulative frequency distribution, as shown in Figure 3 for the same data shown in Figure 2. The Pareto cumulative distribution can be modeled by the following equation:[11]

$$y = \frac{(1+A)\,x}{A+x} \qquad \text{for} \qquad 0 \le y \le 1 \qquad \text{and} \qquad 0 \le x \le 1 \tag{1}$$

[10]The statement is attributed to J. M Juran [8].

[11]Based on Bender, P., "Mathematical Modeling of the 20/80 Rule: Theory and Practice," *Journal of Business Logistics*, Vol. 2, No. 2, 1981, pp 139–157.

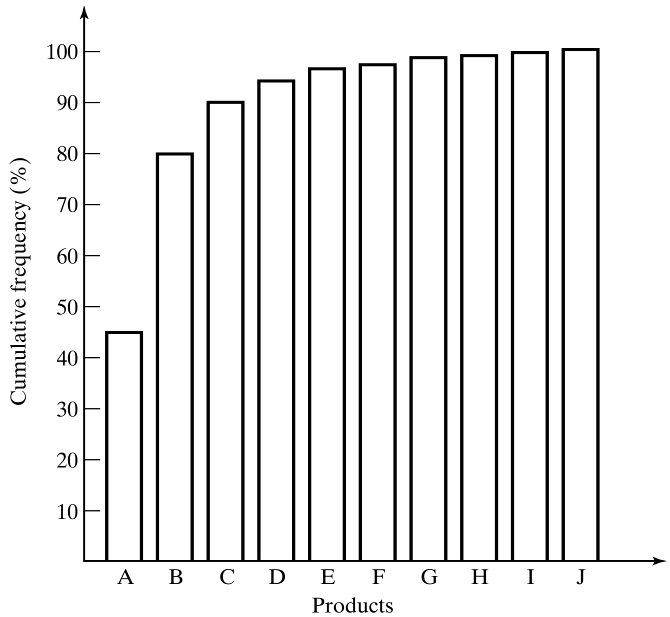



**Figure 3** Pareto chart plotted as a cumulative frequency distribution for the same data shown in Figure 2.

where $y$ = cumulative fraction of the value variable (e.g., wealth, inventory value, sales revenue), $x$ = cumulative fraction of the item variable (e.g., population, inventory items, customers), and $A$ is a constant that determines the shape of the distribution. Values of $A$ between zero and infinity provide shapes that possess the Pareto characteristic, as shown in Figure 4. When $A = 0$, the equation reduces to $y = 1$ for all $x$, and when $A = \infty$, the equation becomes $y = x$.

To determine the appropriate value of $A$ for a given situation or set of data, equation (1) can be rearranged to solve for $A$ as a function of $x$ and $y$, as follows:

$$A = \frac{x(1-y)}{y-x} \tag{2}$$

where $x$ and $y$ are the cumulative frequencies of the two variables at a given point in the distribution. The following example illustrates the approach.

**Example 3 Pareto Cumulative Distribution**

It is known that 20% of the total inventory items in a company's warehouse accounts for 80% of the value of the inventory. (a) Determine the parameter $A$ in the Pareto cumulative distribution equation. (b) Given that the relationship is valid for the remaining inventory, how much of the inventory value is accounted for by 50% of the items?

**Solution** **(a)** To find $A$, we use equation (2) given that $x = 0.20$ and $y = 0.80$ (20% of the items, 80% of the value).

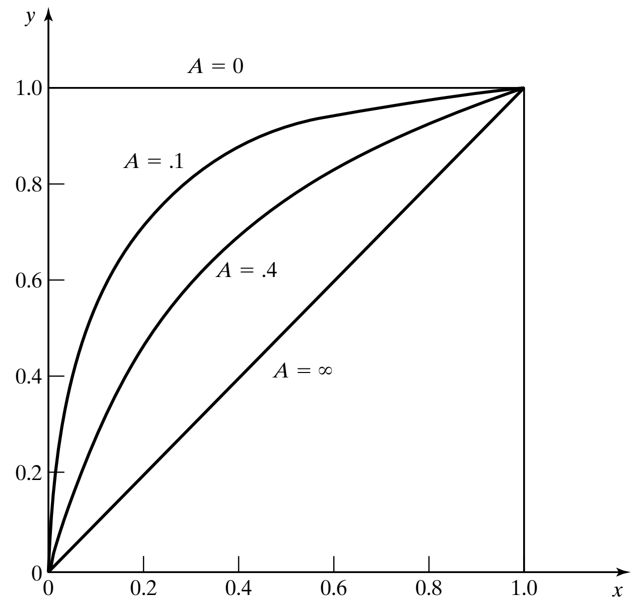



**Figure 4** Shape of plots of equation (1) for several values of $A$.

$$A = \frac{0.20(1 - 0.80)}{0.80 - 0.20} = 0.06667$$

**(b)** Now that we know the value of $A$, the following Pareto cumulative distribution equation can be used:

$$y = \frac{(1.06667)x}{0.06667 + x}$$

For $x = 0.50$, the equation can be used to calculate $y$:

$$y = \frac{(1.06667)(0.50)}{0.06667 + 0.50} = 0.941$$

We expect that 50% of the items in inventory account for 94.1% of the value of the inventory. ■

**Pie Charts.** A ***pie chart*** is a circular (pie-shaped) display that is sliced by radii into segments whose relative areas are proportional to the magnitudes or frequencies of the data categories comprising the total circle. The visual effect is similar to a Pareto chart, in the sense that the important categories can be immediately recognized because of their relative sizes. Multiple pie charts can be displayed side by side to indicate not only the relative category sizes within a circle but also the relative sizes of the circles. Figure 5 shows two consecutive years of company sales, indicating how sales increased the second year and how the increase was distributed among types of customers.

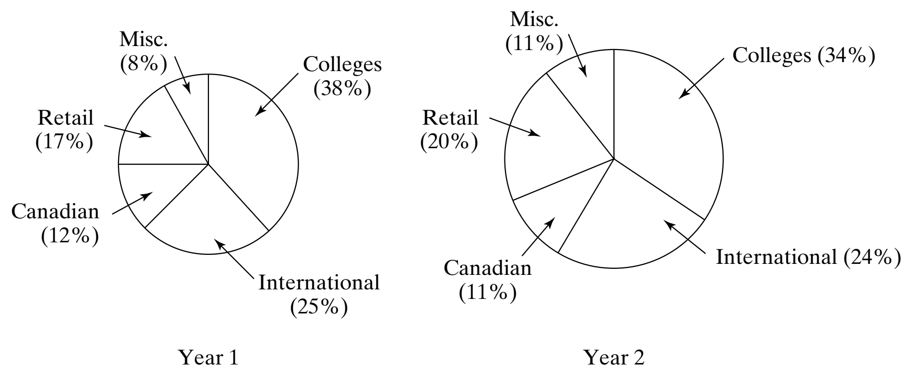



**Figure 5** Pie charts showing two annual sales revenues and the customer distributions for the two years. Sales in year 2 increased 43% over year 1.

**Check Sheets.** The ***check sheet*** (not to be confused with "check list") is a data gathering tool generally used in the preliminary stages of the study of a problem. The operator running a process (for example, the machine operator) is often given the responsibility for recording the data on the check sheet, and the data are often recorded in the form of simple check marks.

**Example 4 Check Sheet Application**

For the dimensional data in the frequency distribution of Table 2, suppose we wanted to see if there were any differences between the three shifts that are responsible for making the parts. A check sheet has been designed for this purpose, and data have been collected. Analyze the data.

**Solution:** The check sheet is illustrated in Table 3. The data include the shift on which each dimensional value was produced (shifts are identified simply as 1, 2, and 3). The data in a check sheet are usually recorded as a function of time periods (days, weeks, months), as in our table.

It is clear from the data that the third shift is responsible for much of the variability in the data. Further analysis, shown in Table 4, substantiates this finding. This should lead to an investigation to determine the causes of the greater variability on the third shift, with appropriate corrective action to address the problem.

We also note from Table 4 that the average daily production rate for the third shift is somewhat below the daily rate for the other two shifts. The third shift seems to be a problem that demands management attention. ■

Check sheets can take many different forms, depending on the problem situation and the ingenuity of the analyst. The form should be designed to allow some interpretation of results directly from the raw data, although subsequent data analysis may be necessary to recognize trends, diagnose the problem, or identify areas of further study.

**Defect Concentration Diagrams.** The defect concentration diagram is a graphical method that has been found to be useful in analyzing the causes of product or part defects. It is a drawing of the product (or other item of interest), with all relevant views displayed,

**TABLE 3** Check Sheet Using Data from Table 2 Recorded According to Shift on Which Parts Were Made

| Range of Dimension | May 5 | May 6 | May 7 | May 8 | May 9 | Weekly Totals |
|---|---|---|---|---|---|---|
| 1.975 ≤ x < 1.980 | | | 3 | | | 1 |
| 1.980 ≤ x < 1.985 | | 2 | | 3 | 3 | 3 |
| 1.985 ≤ x < 1.990 | 1 | 3 | 3 | 1 | 3 | 5 |
| 1.990 ≤ x < 1.995 | 1 2 | 11 2 3 | 1 2 | 1 2 | 1 22 | 13 |
| 1.995 ≤ x < 2.000 | 11 222 3 | 11 22 3 | 111 222 3 | 11 222 3 | 11 22 3 | 29 |
| 2.000 ≤ x < 2.005 | 11 22 3 | 11 22 3 | 111 22 3 | 11 222 3 | 111 22 | 27 |
| 2.005 ≤ x < 2.010 | 1 2 3 | 1 2 3 | 22 3 | 1 33 | 1 2 3 | 15 |
| 2.010 ≤ x < 2.015 | 3 | 3 | 3 | | 3 | 4 |
| 2.015 ≤ x < 2.020 | 3 | | | 3 | | 2 |
| 2.020 ≤ x < 2.025 | 3 | | | | | 1 |
| Total parts/day | 20 | 20 | 21 | 20 | 19 | 100 |

**TABLE 4** Summary of Data from Check Sheet of Table 3 Showing Frequency of Each Shift in Each Dimension Ranges

| Range of Dimension | Shift 1 | Shift 2 | Shift 3 | Totals |
|---|---|---|---|---|
| 1.975 ≤ x < 1.980 | | | 1 | 1 |
| 1.980 ≤ x < 1.985 | | 1 | 2 | 3 |
| 1.985 ≤ x < 1.990 | 2 | | 3 | 5 |
| 1.990 ≤ x < 1.995 | 6 | 6 | 1 | 13 |
| 1.995 ≤ x < 2.000 | 11 | 13 | 5 | 29 |
| 2.000 ≤ x < 2.005 | 12 | 11 | 4 | 27 |
| 2.005 ≤ x < 2.010 | 4 | 5 | 6 | 15 |
| 2.010 ≤ x < 2.015 | | | 4 | 4 |
| 2.015 ≤ x < 2.020 | | | 2 | 2 |
| 2.020 ≤ x < 2.025 | | | 1 | 1 |
| Weekly total parts/shift | 35 | 36 | 29 | 100 |
| Average daily parts/shift | 7.0 | 7.2 | 5.8 | |

onto which the various types of defects or other problems of interest have been sketched at the locations where they each occurred. By analyzing the defect types and corresponding locations, it may be possible to identify the underlying causes of the defects.

Montgomery [11] describes a case study involving the final assembly of refrigerators that were plagued by surface defects. A defect concentration diagram (Figure 6) was utilized to analyze the problem. The defects were clearly shown to be concentrated around the middle section of the refrigerator. Upon investigation, it was learned that a belt was wrapped around each unit for material handling purposes. It became evident that the defects were caused by the belt, and corrective action was taken to improve the handling method.

**Scatter Diagrams.** In many industrial problems involving manufacturing operations, it is useful to identify a possible relationship that exists between two process variables. The scatter diagram is helpful in this regard. A ***scatter diagram*** is an *x*-*y* plot of the data taken of the two variables of interest, as illustrated in Figure 7. The data

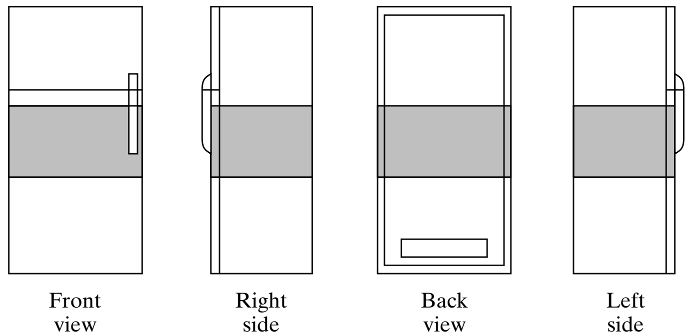



**Figure 6** Defect concentration diagram showing four views of refrigerator with locations of surface defects indicated in shaded areas.

are plotted as pairs; for each $x_i$ value, there is a corresponding $y_i$ value. The shape of the data points considered in aggregate often reveals a pattern or relationship between the two variables. For example, the scatter diagram in Figure 7 indicates that a negative correlation exists between cobalt content and wear resistance of a cemented carbide cutting tool. As cobalt content increases, wear resistance decreases.

One must be circumspect in using scatter diagrams and in extrapolating the trends that might be indicated by the data. For instance, it might be inferred from our diagram that a cemented carbide tool with zero cobalt content would possess the highest wear resistance of all. However, cobalt serves as an essential binder in the pressing and sintering process used to fabricate cemented carbide tools, and a minimum level of cobalt is necessary to hold the tungsten carbide particles together in the final product. There are other reasons why caution is recommended in the use of the scatter diagram, since

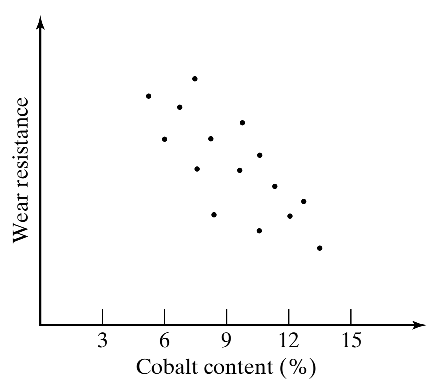



**Figure 7** Scatter diagram showing the effect of cobalt binder content on wear resistance of a cemented carbide cutting tool insert.

only two variables are plotted. There may be other variables in the process whose importance in determining the output is far greater than the two variables displayed.

**Cause and Effect Diagrams.** The ***cause and effect diagram*** is a graphical-tabular chart used to list and analyze the potential causes of a given problem. It is not really a statistical tool, in the sense of the preceding data collection and analysis techniques. As shown in Figure 8, the diagram consists of a central stem leading to the effect (the problem), with multiple branches coming off the stem listing the various groups of possible causes of the problem. Owing to its characteristic appearance, the cause and effect diagram is also known as a ***fishbone diagram***. In application, cause and effect diagrams are often developed by worker teams who study operational problems. The diagram provides a graphical means for discussing and analyzing a problem and listing its possible causes in an organized and understandable way. Members of the team collectively identify the branches of the diagram (causes of the problem) and then attempt to determine which causes are most consequential and how to take corrective action against them.

As a starting point in identifying the causes of the problem (the main branches in the fishbone diagram), six general categories of causes are often used because they are the factors that affect performance of most production and service processes. Called the 5Ms and 1P [4], they are as follows:

1. *Machines.* This refers to the equipment and tooling used in the process.
2. *Materials.* These are the starting materials in the process.

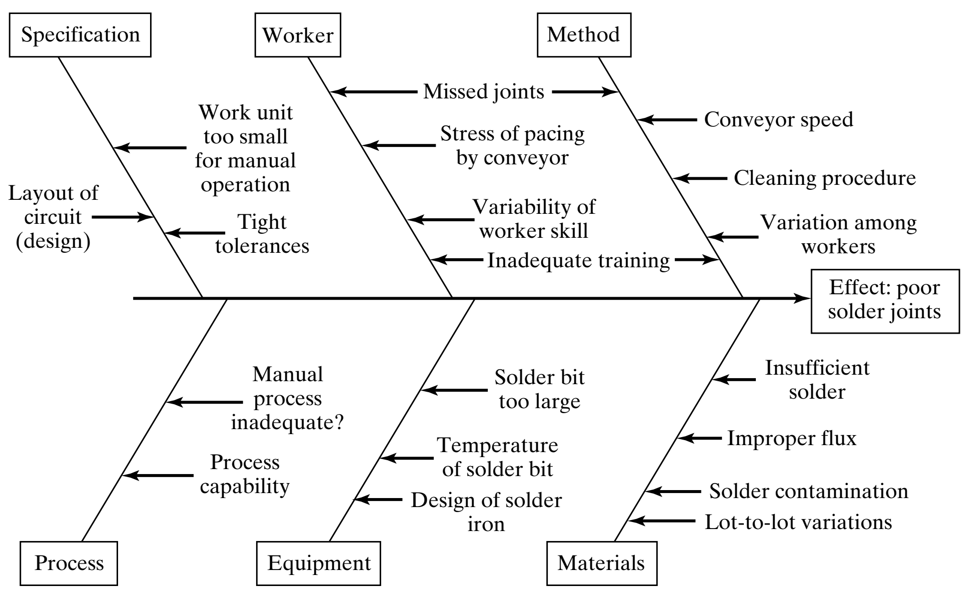



**Figure 8** Cause and effect diagram for a manual soldering operation. The diagram indicates the effect (the problem is poor solder joints) at the end of the arrow and the possible causes are listed on the branches leading toward the effect.

3. *Methods.* This refers to the procedures, sequence of activities, motions, and other aspects of the method used in the process.
4. *Mother Nature.* This is a pseudonym for environmental factors such as air temperature and humidity that might affect the process.
5. *Measurement.* This relates to the validity and accuracy of the data collection procedures.
6. *People.* This is the human factor. Does the worker bring the necessary skills to the process?

During construction of the fishbone diagram, more specific causes and issues are listed on the smaller branches within each of these six categories as the analysis team pursues a solution to the problem.

## 4 METHODS ENGINEERING AND AUTOMATION

The issue of automation arises frequently in methods engineering. The analysis of an operation may lead to the conclusion that an automated or semiautomated work system is preferable to performing a task manually. However, a certain caution and respect must be observed in applying automation technologies. In this section, we offer three approaches for dealing with automation projects in methods engineering: (1) the USA Principle, (2) ten strategies for automation, and (3) an automation migration strategy.

### 4.1 USA Principle

The USA principle is a commonsense approach to automation projects. Similar procedures have been suggested in the manufacturing and automation trade literature, but none have a more captivating title than this one. USA stands for three steps in the analysis and design procedure:

1. *Understand* the existing process.
2. *Simplify* the process.
3. *Automate* the process.

Described in [10], the approach is so general that it is applicable to nearly any automation project. One might argue that the USA principle is basically an abbreviated version of the methods engineering approach (Section 2.1).

The purpose of the first step in the USA principle is to understand the current process in all of its details. What are the inputs? What are the outputs? What exactly happens to the work unit between input and output? What is the function of the process? How does it add value to the product? What are the upstream and downstream operations in the production sequence and can they be combined with the process under consideration?

Some of the basic charting tools used in methods engineering are useful in this step. Applying these kinds of tools to the existing process provides a model of the process that can be analyzed and searched for weaknesses and opportunities. The number of steps in the process, the number and placement of inspections, the number of moves and delays experienced by the work unit, and the time spent in storage can be determined from these charting techniques.

Mathematical models of the process may also be useful to indicate relationships between input parameters and output variables. What are the important output variables? How are these output variables affected by inputs to the process, such as raw material properties, process settings, operating parameters, and environmental conditions? This information may be valuable in identifying what output variables need to be measured for feedback purposes and in formulating algorithms for automatic process control.

Once the existing process is understood, then the search begins for ways to simplify the process (step 2). This often involves a checklist of questions about the existing process. What is the purpose of this operation or this transport? Is that operation necessary? Can this step be eliminated? Is the most appropriate technology being used in this process? How can this step be simplified? Are there unnecessary steps in the process that might be eliminated without detracting from the function? These are basic questions in a methods engineering study.

Some of the ten strategies for automation (Section 4.2) may be used to simplify the process. Can steps be combined? Can steps be performed simultaneously? Can steps be integrated into a manually operated production line? Simplifying the process may lead to the conclusion that automation is not necessary, thus saving the significant investment cost that would be entailed.

When the process has been reduced to its simplest form, then automation can be considered (step 3). The possible forms of automation include those listed in the ten strategies discussed in the following section. An automation migration strategy (Section 4.3) might be implemented for a new product that has yet to prove itself.

### 4.2 Ten Strategies for Automation

The USA Principle is a good first step in any automation evaluation project. As suggested previously, it may turn out that automation is unnecessary or cannot be cost justified after the process has been simplified. If automation seems a feasible solution to improving productivity, quality, or another measure of performance, then the following ten strategies provide a road map to search for these improvements.[13] Although we refer to them as strategies for automation, some of them are applicable whether the process is a candidate for automation or just simplification.

1. *Specialization of operations.* Analogous to the concept of labor specialization for improving labor productivity, this strategy involves the use of special-purpose equipment designed to perform one operation with the greatest possible efficiency.
2. *Combined operations.* Production almost always occurs as a sequence of operations. Complex parts may require dozens, or even hundreds, of processing steps.

[13]These ten strategies were first published in my book *Automation, Production Systems, and Computer-Aided Manufacturing* (Prentice Hall, 1980). They seem as relevant and appropriate today as they did in 1980.

The strategy of combined operations involves reducing the number of distinct workstations through which the work units must be routed. This is accomplished by performing more than one operation at a given workstation, thereby reducing the number of separate workstations needed.

3. *Simultaneous operations.* A logical extension of the combined operations strategy is to simultaneously perform the operations that are combined at one workstation. In effect, two or more operations are performed at the same time on the same work unit, thus reducing total processing time.
4. *Integration of operations.* This strategy involves linking several workstations together into a single integrated mechanism using automated work handling devices to achieve continuous work flow. With an integrated sequence of workstations, several work units can be processed simultaneously (one at each station), thereby increasing the overall output of the system.
5. *Increased flexibility.* This strategy attempts to achieve maximum utilization of human and equipment resources for low and medium volume situations by using the same resources for a variety of work units. It involves the use of the flexible automation concepts that are implemented using computer systems.
6. *Improved material handling and storage.* The use of automated material handling and storage systems is a great opportunity to reduce nonproductive time. Typical benefits include reduced work-in-process and shorter lead times. In information service operations, the counterpart is a the use of advanced database and data processing technologies.
7. *On-line inspection.* Inspection for quality is traditionally performed after the process. This means that any poor quality product or service has already been completed by the time it is inspected. Incorporating inspection into the process permits corrections during the process. This brings the overall quality level closer to the nominal specifications intended by the designer.
8. *Process control and optimization.* This includes a wide range of control schemes intended to operate individual processes and associated equipment more efficiently. By using this strategy, the individual process times can be reduced and quality improved.
9. *Plant operations control.* Whereas the previous strategy is concerned with control of the individual process, this strategy is concerned with control at the plant level. It attempts to manage and coordinate the aggregate operations in the plant more efficiently. Its implementation usually involves a high level of computer networking within the facility.
10. *Computer integrated manufacturing* (*CIM*). Taking the previous strategy one level higher, we have the integration of factory operations with engineering design and the business functions of the firm. CIM involves extensive use of computer applications, computer databases, and computer networking throughout the enterprise.

The ten strategies constitute a checklist of the possibilities for improving the work system, through automation or simplification. They should not be considered as mutually exclusive. For many situations, multiple strategies can be implemented in one improvement project.

### 4.3 Automation Migration Strategy

Because of competitive pressures in the marketplace, a company often needs to introduce a new product in the shortest possible time. The easiest and least expensive way to accomplish this objective is to design a manual production method, using a sequence of workstations operating independently. The tooling for a manual method can be fabricated quickly and at low cost. If more than a single set of workstations is required to make the product in sufficient quantities, as is often the case, then the manual cell is replicated as many times as needed to meet the demand. If the product turns out to be successful and high demand is anticipated in the future, it makes sense for the company to automate production. The improvements are often carried out in phases. Many companies have an ***automation migration strategy***—a formalized plan for evolving the manufacturing systems used to produce new products as demand grows. The following phases are included in the typical automation migration strategy:

Phase 1: ***Manual production*** using single station manned cells operating independently. This is used for introduction of the new product for reasons mentioned above: quick and low-cost tooling to get started.

Phase 2: ***Automated production*** using single station automated cells operating independently. As demand for the product grows and it becomes clear that automation can be justified, the single stations are automated to reduce labor and increase production rate. Work units are still moved between workstations manually.

Phase 3: ***Automated integrated production*** using a multistation automated system with serial operations and automated transfer of work units between stations. When the company is certain that the product will be produced in mass quantities and for several years, then integration of the single station automated cells is warranted to further reduce labor and increase production rate.

This strategy is illustrated in Figure 9. Details of the automation migration strategy vary from company to company, depending on the types of products they make and the processes they perform. But well-managed companies have policies like this one. There are several advantages to such a strategy:

- It allows introduction of the new product in the shortest possible time, since production cells based on manual workstations are the easiest to design and implement.
- It allows automation to be introduced gradually (in planned phases), as demand for the product grows, engineering changes in the product are made, and time is allowed to do a thorough design job on the automated manufacturing system.
- It avoids the commitment to a high level of automation from the start, since there is always a risk that demand for the product will not justify it.

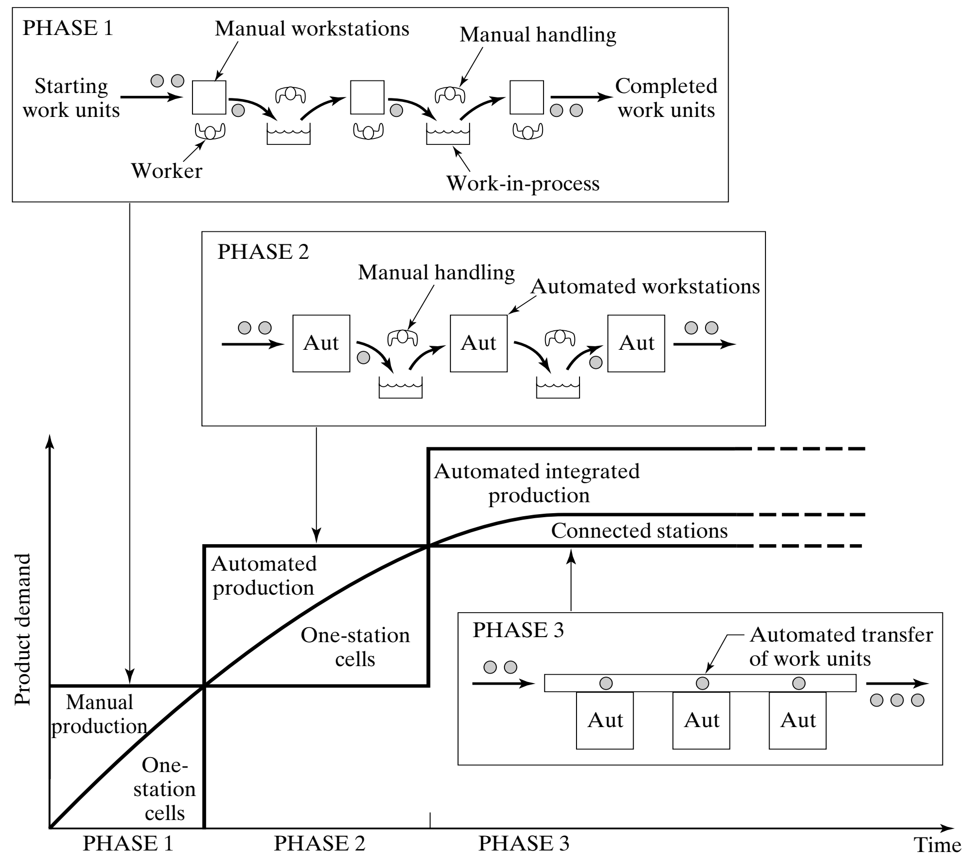



**Figure 9** The three phases of a typical automation migration strategy: (1) manual production with single independent workstations, (2) automated production stations with manual handling between stations, and (3) automated integrated production with automated handling between stations.

## REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] Akiyama, M., and H. Kamata. "Method Engineering and Workplace Design." Pp.4. 3–4.20 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[3] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley, 1980.

[4] Eckes, G. *Six Sigma for Everyone.* Hoboken, NJ: Wiley, 2003.

[5] Geitgey, D. C. "Operation Analysis." Pp. 3.23–3.40 in *Maynard's Industrial Engineering Handbook*, 4th ed., edited by W. K. Hodson. New York: McGraw-Hill, 1992.

[6] Groover, M., M. Weiss, R. Nagel, and N. Odrey. *Industrial Robotics: Technology, Programming,and Applications*. New York: McGraw-Hill, 1986.

[7] Groover, M. P. *Automation, Production Systems, and Computer Integrated Manufacturing*. 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.
[8] Juran, J. M., and F. M. Gryna. *Quality Planning and Analysis*. 3rd ed. New York: McGraw-Hill, 1993.
[9] Kadota, T., and S. Sakamoto. "Methods Analysis and Design." Pp. 1415–45 in *Handbook of Industrial Engineering*, 2nd ed., edited by G. Salvendy. New York: Wiley and Institute of Industrial Engineers, Norcross, GA, 1992.
[10] Kapp, K. M. "The USA Principle." *APICS—The Performance Advantage* (June 1997): 62–66.
[11] Montgomery, D. *Introduction to Statistical Quality Control*. 3rd ed. New York: Wiley, 1996.
[12] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.
[13] Muther, R. *Systematic Layout Planning*. 2nd ed. Boston, MA: Cahners Books, 1973.
[14] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 11th ed. New York: McGraw-Hill, 2003.

## REVIEW QUESTIONS

**1** What is methods engineering?
**2** What are the principal objectives of methods engineering?
**3** What is operations analysis?
**4** What was the operation studied by Frank Gilbreth in his initial research on motion study?
**5** What is methods analysis?
**6** What is methods design?
**7** What are the six steps of the systematic approach in methods engineering?
**8** The procedure offered in the text for selecting among alternatives divides the technical features of proposed equipment alternatives into two categories. What are the two categories?
**9** What is a histogram?
**10** What is a Pareto chart?
**11** What is a check sheet?
**12** What is a defect concentration diagram?
**13** What is a scatter diagram?
**14** What is a cause and effect diagram?
**15** What does "USA" stand for in the USA principle?
**16** What are the three phases in the automation migration strategy?
**17** Why would a company want to use manual production methods instead of automated methods at the beginning of production of a new product?

## PROBLEMS

**1** A factory has 10 departments, all of which have quality problems leading to delays in shipping products to customers. A breakdown of the number of quality problems for each department (listed alphabetically) is as follows: (1) assembly, 16; (2) final packaging, 9; (3) finishing, 37; (4) forging, 73; (5) foundry, 362; (6) machine shop, 294; (7) plastic molding, 120; (8) receiving inspection, 124; (9) sheet metalworking, 86; and (10) tool-making, 42.

(a) Construct a Pareto chart for this data. (b) Assuming that all quality problems are of equal value, in which department would you start to take corrective action to reduce the quality problems? (c) Determine the percentage of total quality problems that are attributable to the two departments (20% of the departments) with the most quality problems.

**2** Using your answer to part (c) of the preceding problem, (a) determine the parameter $A$ in equation (1) representing the Pareto cumulative distribution. Use 20% of the departments as the $x$ value in your computations. (b) Construct the idealized Pareto chart based on your answer to part (a) and discuss the comparison between this idealized chart and the actual data in the previous problem. Use a spreadsheet program to calculate the data for part (b).

**3** Assume that 75% of the sales in a retail company are accounted for by 25% of the customers. (a) Determine the parameter $A$ in the Pareto cumulative distribution equation. (b) Given that the relationship is valid for the remaining sales, how much of the sales value is accounted for by 50% of the customers?

**4** The inventory policy of a retail company is to hold only the highest sales volume items in its distribution center and to ship the remaining lower sales volume items direct from the respective manufacturers to its stores. This policy is intended to reduce transportation costs. Total annual sales of the company are $1 billion. It is known that half of this amount is accounted for by only 15% of the items. In addition, it is assumed that equation (1) in the text can be used to model the Pareto cumulative distribution. (a) If the company wants to stock the top selling 35% of the items in the distribution center, what is the expected value of these items in terms of annual sales? (b) On the other hand, if the company wants to stock only those items accounting for the top 75% of annual sales, what proportion of the items corresponds to this sales volume?

**5** The marketing research department for the Stitch Clothing Company has determined that 22% of the items stocked account for 70% of the dollar sales. A typical outlet store carries 1000 items. The items accounting for the top 60% of sales are replenished from the company's distribution center. The rest are shipped directly from the supplier (manufacturer) to the stores. How many items are represented by the top 60%?

**6** Consider some process or procedure with which you are familiar that manifests some chronic problem. Develop a cause and effect diagram that identifies the possible causes of the problem. This is a project that lends itself to a team activity.

# Charting and Diagramming Techniques for Operations Analysis



Charting and diagramming techniques are useful for analyzing a work process because they graphically illustrate and summarize the activities in that process. Several charting and diagramming techniques include:

- Network diagrams for depicting work flow in sequential operations
- From-To charts for indicating material flows and/or distances among workstations or departments
- Precedence diagrams for assembly line balancing
- Gantt charts and network diagrams for scheduling projects

This chapter discusses the important charting and diagramming techniques used in methods engineering and operations analysis.

## 1 OVERVIEW OF CHARTING AND DIAGRAMMING TECHNIQUES

Charting and diagramming techniques are intended to graphically display relationships among the various entities included in the graphic. Objectives in using charts and diagrams to study work include the following:

1. To permit work processes to be communicated and comprehended more readily
2. To allow the use of algorithms specifically designed for the particular diagramming technique
3. To divide a given work process into its constituent elements for analysis purposes
4. To provide a structure in the search for improvements
5. To represent a proposed new work process or method

In this chapter, the charting and diagramming techniques for operations analysis are classified into three categories: (1) network diagrams, (2) traditional industrial engineering charts and diagrams, and (3) block diagrams and process maps.

How does the analyst create the chart or diagram for the process of interest? Any of four methods or combinations thereof can be used to develop a description of the work process that is ultimately used to create the graphic:

- *The analyst is intimately familiar with the process.* In this case, the analyst works in the area and already knows how the process works. The analyst develops a graphic of the process and asks others who are also familiar with it to review it.
- *The analyst observes and records information about the process.* The analyst spends the time necessary to observe and understand the process, develops a chart or diagram of the process, and asks others familiar with it to react to it.
- *One-on-one interviews of those familiar with the process.* The analyst conducts a series of one-on-one interviews of the people who are intimately familiar with the process. A graphical depiction of the process is developed based on the interviews and subjected to review by the interviewees.
- *Group interviews of those familiar with the process.* The analyst asks those who understand the process to participate in a group meeting and describe it. A skilled facilitator may be used to elicit input from the participants. The analyst records the discussion of the meeting and develops a graphical model of the process. The analyst then asks the participants to review it. The advantage of a group meeting is that the participants can challenge each other, thus ultimately developing a more accurate picture of the actual process.

Once the chart or diagram is created, how is it analyzed? Again, there is more than one approach:

- *Algorithmic.* The specific algorithm for the particular diagram is used. Examples include line balancing algorithms for assembly lines and critical path methods for project scheduling.
- *Checklists.* In this case, the analyst reviews a series of general questions to assess whether they can be applied to the particular problem of interest. We offer a number of these checklist questions in Section 3.

- *Brainstorming.* This is a group or team activity in which participants contribute recommendations about potential improvements in the process.
- *Separating value-added and non-value-added operations.* This approach attempts to distinguish between those steps in the process that actually add value to the product or service from the customer's viewpoint and those that do not. ***Value-added steps*** are operations that (1) the customer considers important and (2) physically change the product or service. Potential non-value-added operations include rework, delays, unnecessary inspections, and unnecessary moves.

## 2 NETWORK DIAGRAMS

A ***network diagram*** consists of (1) nodes representing operations, work elements, or other entities and (2) arrows connecting the nodes indicating relationships among the nodes. When used in the context of work systems analysis, the arrows usually indicate either direction of work flow between nodes or precedence order among them, and the nodes represent work activities (e.g., operations, work elements, tasks).

The network diagram is used to model work flow between operations. In this case, the nodes are processing operations and the arrows indicate the sequence in which the operations are performed and the direction of work transport. A network diagram is used to model the precedence order in which work elements must be performed in assembly operations. Called a precedence diagram, nodes represent the assembly work elements, and arrows indicate the sequence in which the elements must be performed. Critical path method (CPM) and program evaluation and review technique (PERT) diagrams are network diagrams used to schedule the work activities in a project. The methods for analyzing the various network diagrams differ for the different applications, but the same basic format is used in constructing the diagrams.

For network diagrams with two-way flows between nodes (e.g., materials moving in both directions between two departments), the maximum number of arrows is given by

$$\text{Maximum number of arrows possible} = n(n - 1) \tag{1}$$

where $n$ = number of nodes in the diagram. For network diagrams containing only one-way arrows (e.g., arrows indicating precedence order of work elements or activities), the maximum possible number of arrows between nodes in the network is given by the following:

$$\text{Maximum number of arrows possible} = \frac{n(n - 1)}{2} \tag{2}$$

Most network diagrams have fewer than the maximum values given by these equations.

## 3 TRADITIONAL INDUSTRIAL ENGINEERING CHARTING AND DIAGRAMMING TECHNIQUES

The charts and diagrams discussed in this section are commonly used to analyze an existing operation, sequence of operations, or other work activity for the purpose of making improvements. The underlying assumption is that by examining the work situation in detail, and subjecting the details to critical scrutiny, improvements can be found more readily than through macroscopic examination. The possible improvements include reducing cycle time and cost, eliminating unnecessary steps, mitigating safety hazards, and improving product quality. In addition to analyzing existing operations, the charts and diagrams can also be used to present proposals for new ways of accomplishing the same operations, or to design new operations that have never been implemented before.

There are many charting and diagramming techniques that have been developed over the years within the discipline of industrial engineering. For purposes of organization, we classify them into the following four major categories: (1) operation charts, (2) process charts, (3) flow diagrams, and (4) activity charts.[1] Each of these techniques provides a graphical and symbolic means of visualizing the work situation for better understanding of its scope and details. The differences relate to the levels of detail and how the work situation is represented graphically. Also, within a given category there may be variations in format and symbols depending on the subject of the analysis, for example, whether the purpose of the analysis is to study a material or to study a human worker.

In addition to these four categories of charting and diagramming techniques, the Gantt chart is a charting tool associated with traditional industrial engineering that is used in production control and project scheduling.

### 3.1 Operation Charts

The ***operation chart*** is a graphical and symbolic representation of the operations used to produce a product. There are two types of operations in an operation chart: (1) processing and assembly operations, and (2) inspection operations. The operation chart can also be used to analyze the steps involved in the delivery of a service, but this application of the chart is far less common. As shown in Figure 1, the operation chart consists of a series of vertical stems, each one depicting the sequence of operations and inspections performed on a given component of the product. The operation chart uses only two symbols (for operation and inspection), which are defined in Table 1. At the top of each stem is the starting material or purchased part, and the steps performed on it are indicated by symbol and brief description. The time to accomplish the operation (e.g., standard time) is also sometimes included. As each component is completed, it is assembled to other components toward the right. The column at the far right usually represents the base part or chassis of the assembly. It is the component to which all other parts in the chart are joined.

Developing the detailed listing of operations for the components and their assembly into the final entity (e.g., product, subassembly) is only the first step in the operation chart analysis. The second step involves examination of the chart to discover possible

[1]There is considerable variation in some of the terminology among different authors, and we have attempted to rationalize and abbreviate the terms. Where appropriate, we identify the alternative names of the charts.

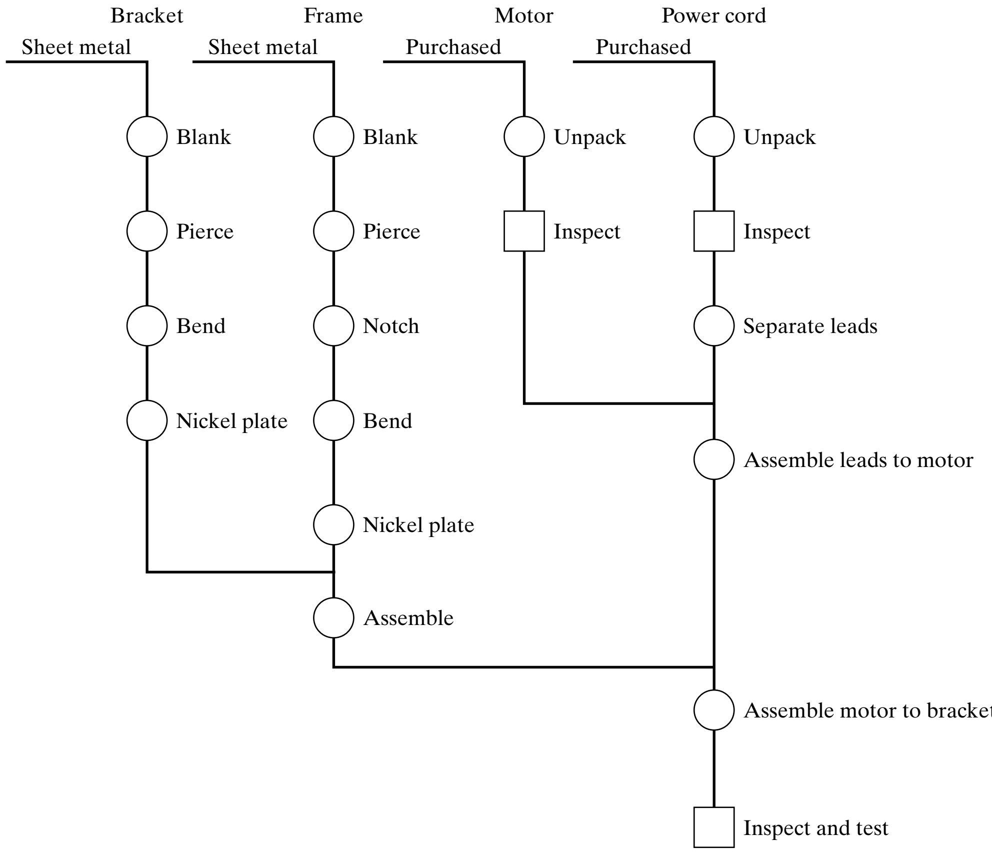



**Figure 1** An operation chart for the subassembly of a product.

**TABLE 1** Symbols Used in Operation Charts

| Symbol | Letter | Description |
|---|---|---|
| ○ | O | ***Processing or assembly operation***. Processing operations consist of changing the shape, properties, or surface of a material or workpart. Assembly operations join two or more parts to form an assembly. |
| □ | I | ***Inspection operation.*** An inspector checks the material, workpart, or assembly for quality or quantity. |

improvements. Because the focus of the operation chart is on the materials of a product and the operations performed on them, the examination step consists of a questioning procedure aimed at the materials and operations. A systematic approach includes questions such as those offered in the checklist of Table 2. The third step in the use of the operation chart for an existing work situation is to develop proposals for improvement based on the results of the questioning procedure.

**TABLE 2** Checklist of Questions Used to Analyze an Operation Chart

| |
|---|
| Questions related to material |
| What alternative starting material could be used (e.g., plastic rather than metal)? |
| Would a design change allow the part to be purchased as a standard commercially available item? |
| Could the functions of several separate components be combined into one component through a design change? |
| Make or buy decision: should this part be produced in our own factory or purchased from an outside vendor? |
| Question related to processing or assembly operations |
| What is the purpose of each processing operation? |
| Is the processing operation necessary? |
| Can operations be eliminated, combined, or simplified? |
| Is the operation time too high? |
| Could the processing operation be automated? |
| Could a different joining method be used for assembly (e.g., snap fit rather than threaded fasteners to save time)? |
| Questions related to inspection operations |
| What is the purpose of the inspection operation? |
| Is the inspection operation necessary? |
| Can the inspection operation be combined with the preceding processing or assembly operation? |
| If the operation is performed 100%, could it be performed on a sampling basis to reduce inspection time? |
| Could the inspection operation be automated? |

### 3.2 Process Charts

A ***process chart*** is a graphical and symbolic representation of the processing activities performed on something or by somebody. The chart consists of a vertical list of the steps performed on or by the work entity using various symbols to represent operations, inspections, moves, delays, and other activities. The principal types of process chart are: (1) the ***flow process chart***, used to analyze a material or workpiece being processed, (2) the ***worker process chart***, used to analyze a worker performing a process, and (3) the ***form process chart***, used to analyze the processing of paperwork forms. These charts are described in the following sections.

**Flow Process Chart.** The flow process chart uses five symbols, as defined in Table 3 to detail the work performed on a material or work part as it is being processed through a sequence of operations and other activities.[2] Either iconic symbols or letter symbols can be used in constructing the chart, depending on the analyst's preference. The characteristic features of the flow process chart are shown in Figure 2. Alternative formats are possible, such as the standardized form shown in Figure 3.

The operation and inspection symbols are sometimes combined if the processing step includes a processing operation combined with an inspection at the same workstation—for example, a worker buffing a part and periodically checking its luster. In this case, the symbol consists of a circle inside a square, with the diameter of the circle equal to the side of the square.[3] Significantly more detail about the steps required to process a material

[2]Other names for the flow process chart include process chart-product analysis [7].
[3]The reverse of this format is also used in the charting literature, in which a square is positioned inside a circle to represent the combination of operation and inspection.

**TABLE 3** Symbols Used in the Flow Process Chart

| Symbol[a] | Letter | Description |
|---|---|---|
| ○ | O | ***Operation***, usually a processing operation performed on the material at one location or workstation in which the physical shape or chemical characteristics of the material are changed. Assembly operations are unusual in a flow process chart. |
| □ | I | ***Inspection***, either to check for quality or quantity, performed at a single location or workstation. |
| → | M | ***Move*** that involves transport of the material from one location to another, but not including moves within an operation at a workstation. |
| D | D | ***Delay*** that occurs when the material does not or cannot proceed to the next activity—for example, a material waiting to be processed at a workstation, but other materials are ahead of it. |
| ▽ | S | ***Storage*** in which the material is kept in a protected location to prevent unauthorized removal. Storage usually involves the use of a requisition to withdraw from storage, whereas a delay does not involve such a transaction. |

[a]Based on symbols developed by the American Society of Mechanical Engineers (ASME). A simple arrow (→) has been substituted for the ASME move symbol (⇒) for ease of drawing.

is provided in the flow process chart than in the operation chart. Consequently, this charting technique is used to study only a single material or work part rather than the multiple components of an assembly. For each symbol, a brief description of the work activity is listed in the flow process chart. In addition, the chart also indicates the distances for move activities and times for the other activities. The time values may be especially relevant for operations, inspections, delays, and storages.

As in the operation chart, the flow process chart is examined for possible improvements and savings. However, the emphasis in the questioning procedure is expanded beyond the coverage of the operation chart because the details in the flow process chart include more types of activities. In addition to the questions shown in Table 2, current questions focus on moves, delays, and storages. Table 4 indicates the types of question that the methods analyst would want to ask. (For completeness, we have included questions from Table 2 in this checklist for the flow process chart.)

**Worker Process Chart.** The ***worker process chart*** is used to analyze the activities of a human worker as he or she performs a task that requires movement around a facility.[4] The symbols are virtually the same as those appearing in Table 3, but they are interpreted in terms of what the human worker does rather than what is done to a material. Table 5 summarizes the interpretations. The storage activity is difficult to interpret in the context of human work activity, so it is omitted. Analysis of the worker process chart in the search for improvements involves the same kinds of questions as in the flow process chart, only in the context of the worker performing the task of interest.

**Form Process Chart.** The ***form process chart*** is used to analyze the flow of paperwork forms and office procedures that normally involve the processing of documents.

[4]Alternative names for the worker process chart include process chart-person analysis [7].

| Flow Process Chart | | | | | |
|---|---|---|---|---|---|
| Part No. 459011 | Material: Steel C1045 forging | Description: Forgings processed in batches of 20 | | | |
| Seq. | Activity Description | Symbol | Time | Distance | Analysis Notes |
| 1 | Forgings transported from forge shop | ⟶ | | 300 m | Forklift truck |
| 2 | Inspection of incoming forgings | □ | 1 hr | | |
| 3 | Forgings moved and placed in storage | ⟶ | | 75 m | Hand truck |
| 4 | Storage | ∇ | 7 days | | Factory warehouse |
| 5 | Forgings retrieved from storage | ⟶ | | 75 m | Hand truck |
| 6 | Transport to machine shop | ⟶ | | 180 m | Forklift truck |
| 7 | Move to milling machine | ⟶ | | 20 m | Hand truck |
| 8 | Delay in queue for milling machine | D | 5 hr | | |
| 9 | Milling operation (roughing and finishing) | ○ | 8 min/pc | | Milling Machine No. 573 |
| 10 | Move to drill press | ⟶ | | 20 m | Hand truck |
| 11 | Delay in queue for drill press | D | 2 hr | | |
| 12 | Drilling and tapping operations (6 holes) | ○ | 3 min/pc | | CNC Drill Press No. 226 |
| 13 | Delay waiting for inspection | D | 4 hr | | |
| 14 | Inspection for machining operations | □ | 0.2 hr | | |
| 15 | Delay waiting for transport to cleaning | D | 3 hr | | |
| 16 | Transport to finishing department | ⟶ | | 75 m | Forklift truck |
| 17 | Move to cleaning operation | ⟶ | | 10 m | Hand truck |
| 18 | Delay in queue for cleaning operation | D | 30 min | | |
| 19 | Cleaning operation (all parts in batch) | ○ | 10 min | | Solvent clean tank |
| 20 | Move to nickel plate operation | ⟶ | | 15 m | Hand truck |
| 21 | Delay in queue for nickel plate operation | D | 45 min | | |
| 22 | Nickel plate operation (all parts in batch) | ○ | 20 min | | Electroplating tank |
| 23 | Delay waiting for transport to storage | D | 30 min | | |
| 24 | Transport to storage | ⟶ | | 200 m | Forklift truck |
| 25 | Storage awaiting assembly | ∇ | | | Factory warehouse |

**Figure 2** A flow process chart used to detail the steps in the processing of a material. In the example shown, the material is a forging, and its processing consists of several machining operations, cleaning, and electroplating, but much of its time is spent in transport, delays, and storage.

Activities that occur in form processing require a change in the interpretation of the symbol. Additional symbols are also sometimes used to cover activities that are not associated with process charts for materials and workers. Symbols that can be used for the form process chart are presented and defined in Table 6.

<table>
<tr><td>Date:</td><td colspan="2">Flow Process Chart</td><td colspan="3">Page ___ of ____</td></tr>
<tr><td>Analyst:</td><td>Approval:</td><td colspan="4">Summary of Activities</td></tr>
<tr><td>Job:</td><td>Part No:</td><td>Activity (symbols)</td><td>Count</td><td>Time</td><td>Distances</td></tr>
<tr><td colspan="2">Material:</td><td>Operations (○, O)</td><td></td><td></td><td></td></tr>
<tr><td colspan="2" rowspan="5">Description:</td><td>Inspections (□ , I)</td><td></td><td></td><td></td></tr>
<tr><td>Moves (→, M)</td><td></td><td></td><td></td></tr>
<tr><td>Delays (D, D)</td><td></td><td></td><td></td></tr>
<tr><td>Storages (∇, S)</td><td></td><td></td><td></td></tr>
</table>

| Seq. | Activity Description | Symbol | Time | Distance | Analysis Notes |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |

**Figure 3** A standardized form for the flow process chart that can be readily created using word processing software.

### 3.3 Flow Diagrams

The ***flow diagram*** is a drawing of the facility layout but with the addition of lines representing movement of materials or workers to specific locations in the facility. Arrows are used on the lines to indicate the direction of movement. The flow diagram is often used in conjunction with a process chart, especially when movement of the material, worker, or form is a major factor in the analysis. An example of a flow diagram for a setup worker is presented in Figure 4. When used in connection with a process chart, the operations, inspections, delays, and storages at specific locations in the layout can be identified by numbers that are referenced to the activity numbers in the process chart.

The flow diagram reveals problems in the work flow that may not readily be identified using the process chart alone. For example, if the work flow involves considerable backtracking, this can be seen in the flow diagram whereas it is indicated only as distances in the process chart. Other work flow problems may include excessive travel, possible traffic congestion, points where delays typically occur, and inefficient layout of workstations.

### 3.4 Activity Charts

An ***activity chart*** is a listing of the work activities of one or more subjects (e.g., workers, machines) plotted against a time scale to indicate graphically how much time is

**TABLE 4** Checklist of Questions Used to Analyze a Flow Process Chart

Questions related to material

What alternative starting material could be used (e.g., plastic rather than metal)?
Would a design change allow the part to be purchased as a standard commercially available item?
Could the functions of several separate components be combined into one component through a design change?
Make or buy decision: should this part be produced in our own factory or purchased from an outside vendor?

Question related to processing operations

What is the purpose of each processing operation?
Is the processing operation necessary?
Can operations be eliminated, combined, or simplified?
Is the operation time too high?
Could the processing operation be automated?
Where else could this operation be performed to reduce move distances?

Questions related to inspection operations

What is the purpose of the inspection operation?
Is the inspection operation necessary?
Can the inspection operation be combined with the preceding processing operation?
If the operation is performed 100%, could it be performed on a sampling basis to reduce inspection time?
Could the inspection operation be automated?

Questions relating to moves

How can moves be shortened or eliminated by combining or eliminating operations?
Can the parts or materials be collected into larger unit loads to increase efficiency in material handling?
What material handling equipment is used to transport materials? Can the level of mechanization be increased (e.g., using forklift trucks rather than manually pushed dollies)?
Can the operation sequence be modified to reduce distances traveled?

Questions relating to delays

What is the reason for the delay? Can the reason be eliminated?
Is the delay avoidable?
Why can't the material be started immediately at the next operation?

Questions relating to storage

Is the storage necessary?
Why can't the material be moved immediately to the next operation?
Can just-in-time delivery be used to eliminate the storage (e.g., can the material received from the supplier be moved immediately to the first operation after delivery rather than into storage)?

spent on each activity. The usual format is to provide brief descriptions of the activities against a vertical time scale, as shown in Figure 5 for a single worker performing a repetitive work cycle. Instead of using symbols for the work activities, as in the other charting and diagramming techniques described previously, the activities are indicated by vertical lines or bars. When bars are used, they are shaded or colored to indicate

**TABLE 5** Symbols Used in the Worker Process Chart

| Symbol | Letter | Description |
|---|---|---|
| ○ | O | ***Operation*** performed by a worker at a single location or workstation. The operation may involve movements of materials within the workstation. |
| □ | I | ***Inspection***, either to check for quality or quantity, performed by a worker at a single location or workstation. |
| → | M | ***Move*** in which the worker moves from one location to another as a regular element required in the task. It does not include moves within a workstation. |
| D | D | ***Delay*** of the worker. Worker is forced by the situation to wait (e.g., waiting for an elevator). The waiting may involve moving, but the move is not a regular element required in the task (e.g., worker goes to the coffee machine while waiting for the elevator). |

**TABLE 6** Symbols Used in the Form Process Chart

| Symbol | Letter | Description |
|---|---|---|
| ◎ | C | ***Creation*** of the form (circle in a circle). This symbol is used for the origination of the form, when the form is first initiated. |
| ○ | O | ***Operation*** performed on the form at a single location or workstation. The operation may involve calculations, data entries, filling out forms, folding, photocopying, stapling, assembling multiple forms into one document, etc. |
| □ | I | ***Inspection*** to read information from the form or check for correctness performed at a single location or workstation. |
| → | M | ***Movement*** of the form from one location to another by mail or human carrier. |
| D | D | ***Delay*** of the form. Form is waiting to be worked on, located in an in-basket or similar location other than a storage file. |
| ▽ | S | ***Storage*** in a file, normally in a file cabinet or other organized filing system. This usually involves storage for a considerable time period, rather than a temporary delay. |
| X | X | ***Disposal*** of the form. The form or a copy is destroyed. |

the kind of activity being performed. Figure 6 indicates some possible shading and color conventions.[5]

Activity charts usually have more than one time scale. In Figure 5 two time scales are used, one for cumulative time during the work cycle and the second to indicate the time taken for each work activity. When activity charts are used to track several participants working together, the general name of the chart is a ***multiple-activity chart***, which consists of multiple columns, one for each participant. In this case, one time scale marks cumulative time during the cycle and a separate time scale indicates activity times for each of the columns.

There are a number of work situations in which the multiple-activity chart is useful for analysis purposes. Because the situations involve multiple entities working together, a common objective of using the chart is to analyze how the workload is coordinated and

[5]There seems to be significant variation in shading and color conventions among different experts and authors, and users are at liberty to develop their own conventions or color schemes to suit the purposes of their respective studies.

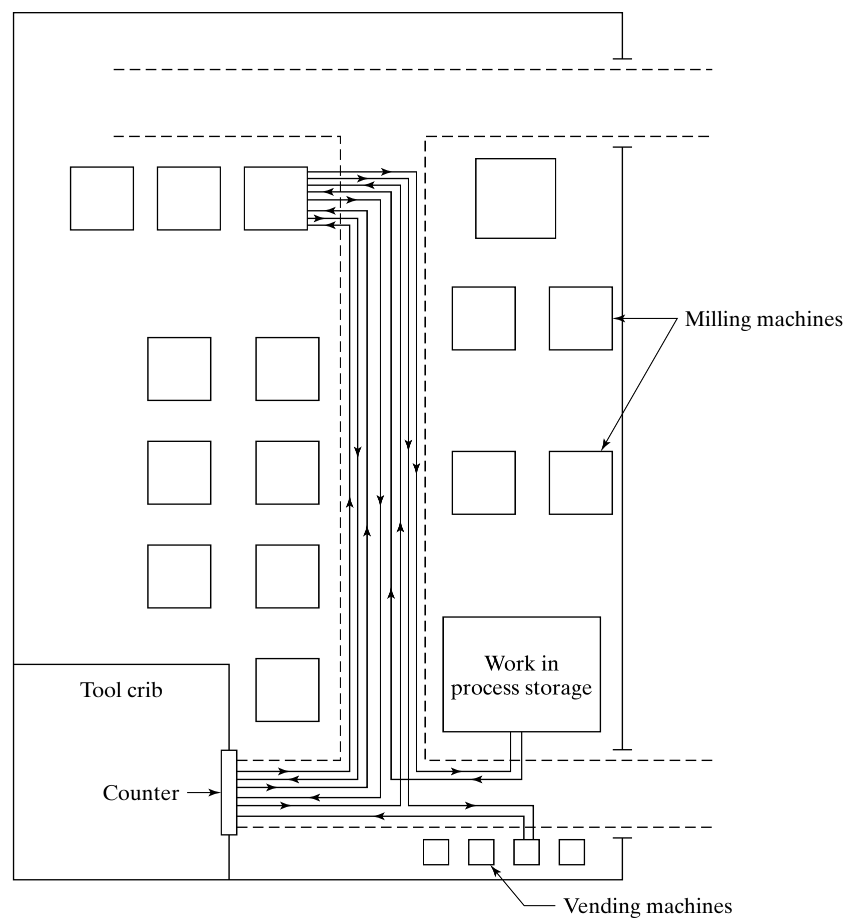



**Figure 4** Flow diagram for a worker setting up a milling machine in the milling department. Note the large number of trips back and forth between the milling machine and the tool crib, which suggests that the setup worker's task might be made more efficient if all of the items needed for the setup were collected in one trip.

shared among them. The following is a listing of the major types of multiple-activity charts:

- *Right-hand/left-hand activity chart.* This chart details the contributions of the right and left hands of one worker performing a task that is highly repetitive. The task is usually performed at a single workplace, and so the chart is sometimes referred to as a ***workplace activity chart***. Figure 7 illustrates a right-hand/left-hand activity chart for a task in which the worker is using his left hand as a workholder while his right hand performs nearly all of the activities. A methods analyst would seek to install a fixture to hold the work unit during the operation

| Activity Description | Chart | Activity Time (min) | Cumulative time (min) |
|---|---|---|---|
| Pick up plate from tote pan. | | 0.05 | 0.05 |
| Carry plate to drill press and load. | | 0.07 | 0.10 |
| Activate press. | | 0.03 | 0.15 |
| Semiautomatic machine cycle. | | 0.20 | 0.20 |
| | | | 0.25 |
| | | | 0.30 |
| | | | 0.35 |
| Remove plate. | | 0.03 | |
| Carry to pallet container. | | 0.05 | 0.40 |
| Place in pallet container. | | 0.02 | 0.45 |
| Walk to tote pan. | | 0.05 | 0.50 |

**Figure 5** Activity chart for one worker performing a repetitive task.

| Shading | Color | Activity |
|---|---|---|
| Black | Blue | ***Operation***: Performing an operation. Worker operating on or handling material at workplace. Machine performing an operation on automatic or mechanized cycle. |
| Gray | Yellow | ***Inspection***: Worker performing an inspection, to check for either quantity or quality. |
| White (blank) | White (blank) | ***Idle time***: Worker or machine is idle, waiting, or stopped. |
| Diagonal lines | Green | ***Moving***: Worker walking outside immediate workplace (e.g., to fetch tools or materials). |
| Horizontal lines | Red | ***Holding***: Worker holding an object in fixed position without performing any work on it. |

**Figure 6** Shading formats for activity charts.

and to achieve a more even balance of the workload between the right and left hands.

- *Worker-machine activity chart.* As the name indicates, this chart shows how the work elements and associated times are allocated between a worker and a machine for the repetitive cycle of a worker-machine system. Like the right-hand/left-hand activity chart, it consists of two main columns, one for the worker and the other for the machine. The worker-machine activity chart can often help to identify opportunities for cycle time improvement, such as the replacement of external work elements by internal elements, where the worker and machine perform parallel rather than sequential activities.

| Left Hand | Time (min) | Right Hand | Cumulative time (min) |
|---|---|---|---|
| Pick up board | 0.08 | | 0.08 |
| Hold board | 0.06 | Pick up peg and insert | 0.14 |
| Hold board | 0.06 | Pick up peg and insert | 0.20 |
| Hold board | 0.06 | Pick up peg and insert | 0.26 |
| Hold board | 0.06 | Pick up peg and insert | 0.32 |
| Hold board | 0.06 | Pick up peg and insert | 0.38 |
| Hold board | 0.06 | Pick up peg and insert | 0.44 |
| Hold board | 0.06 | Pick up peg and insert | 0.50 |
| Hold board | 0.06 | Pick up peg and insert | 0.56 |
| Put assembly in tote pan | 0.06 | | 0.62 |

**Figure 7** Right-hand/left-hand activity chart.

| Worker | Time (min) | Machine 1 | Time | Machine 2 | Time | Cumulative time (min) |
|---|---|---|---|---|---|---|
| Walks to machine 1 | 0.2 | | | | | |
| Services machine 1 | 0.3 | Idle | 0.3 | | | 0.5 |
| Walks to machine 2 | 0.2 | Automatic cycle | | | | |
| Services machine 2 | 0.3 | | | Idle | 0.3 | 1.0 |
| | | | | Automatic cycle | | |
| | 0.5 | | | | | 1.5 |
| Walks to machine 1 | 0.2 | | 1.2 | | | |
| Services machine 1 | 0.3 | Idle | 0.3 | | | 2.0 |
| Walks to machine 2 | 0.2 | Automatic cycle | | | 1.2 | |
| Services machine 2 | 0.3 | | | Idle | 0.3 | 2.5 |
| Idle | | | | Automatic cycle | | |
| | 0.5 | | | | | 3.0 |
| Walks to machine 1 | 0.2 | | 1.2 | | | |
| Services machine 1 | 0.3 | Idle | 0.3 | | | 3.5 |
| Walks to machine 2 | 0.2 | Automatic cycle | | | 1.2 | |
| Services machine 2 | 0.3 | | | Idle | 0.3 | 4.0 |

**Figure 8** Worker-multimachine activity chart.

- *Worker-multimachine activity chart.* This chart is similar to the preceding except that the worker is responsible for more than one machine, and a work cycle must be developed that minimizes or eliminates ***machine interference*** (when one machine must wait for service because the worker is currently servicing another machine). Figure 8 illustrates the worker-multimachine activity chart for two machines. We have previously referred to this worker-multimachine arrangement as a ***machine cluster***.
- *Gang activity chart.* Other names include ***gang chart*** and ***multiworker activity chart***. This chart tracks the activities of two or more workers performing together as a team. Work situations involving the activities of a team or crew are often somewhat complex. The purpose of the activity chart analysis is to better coordinate the activities and balance the workload among the workers.

## 4 BLOCK DIAGRAMS AND PROCESS MAPS

Several additional diagramming techniques are available to show the detailed steps and work flow in a given process. They utilize labeled blocks and other objects, connected by arrows, to indicate processes and their interrelationships. In this section, we describe two types of diagramming techniques: block diagrams and process maps.

### 4.1 Block Diagrams

Block diagrams are commonly applied in linear control theory to portray the relationships among components of a physical system or process. A simple example is illustrated in Figure 9. In control system applications, arrows represent the flow of signals or variables between blocks, and blocks contain transfer functions that define how the input signals are mathematically transformed into output signals. A transfer function is the ratio of the output signal to the input signal.[6] For example, the variable $y$ in Figure 9 equals the variable $x_3$ multiplied by the transfer function $B$.

In addition to blocks and arrows, two other operations are possible in block diagrams. First, two or more separate variables can be summed algebraically using a summing node, indicated by the circle enclosing a summation symbol in Figure 9. Thus, the variable $x_2$ is equal to $x_1$ minus $z$. Second, any given signal can be used as an input to more than one block. In Figure 9, the variable $x_3$ is used as an input to blocks $B$ and $C$. The operation is called a takeoff point, shown in the diagram as a large dot between blocks $A$ and $B$.

In linear control theory and other physical system applications, a form of algebra (called "block diagram algebra") can be used to reduce a complex block diagram to a single block containing the input/output relationship for the entire system. Thus, a block diagram is useful for developing the input/output functions of the individual components of a control system and then analyzing the diagram to determine overall system performance.

Our applications of block diagrams in this book deal with systems that tend to be more subjective and operations oriented. In general, our systems do not lend themselves to the mathematical treatment by which physical systems can be analyzed. Nevertheless, the block diagram format is a powerful means of depicting the signal flows and interrelationships

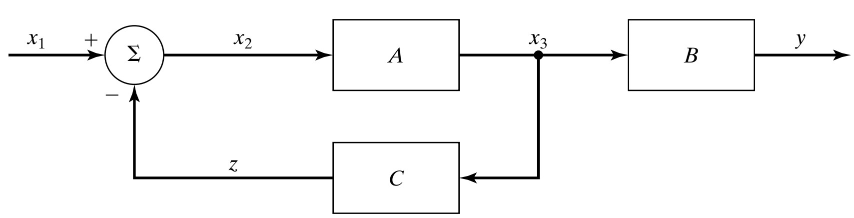



**Figure 9** Block diagram of a feedback control system.

[6]To be more precise, the transfer function is the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable, given that all initial conditions are zero.

among components in complex systems that are related to work. We use them frequently throughout this text.[7] In addition, block diagrams have clearly influenced the development of other diagramming techniques in operations analysis, such as process maps.

### 4.2 Process Maps

Process mapping techniques have been widely applied to business processes. They are also applicable to production, logistics, and service operations. As in other diagramming and charting techniques, process maps provide a detailed picture of the process or system of interest that is helpful for communicating and understanding by those involved in the operations analysis study.

There are several forms of process maps. The basic process map is a block diagram that shows the steps in the process of interest. A ***process*** is defined as a sequence of tasks or activities that add value to one or more inputs to produce outputs. The inputs and outputs may be materials, products, information, services, or other form. The process must consist of more than one step, and the steps are linked together in a logical way. The process has a beginning point and an ending point, and its purpose in the organization is to provide something of value to its customers.

Block symbols used in the basic process map are shown in Figure 10, and an example of their arrangement in a process map is illustrated in Figure 11. The block for

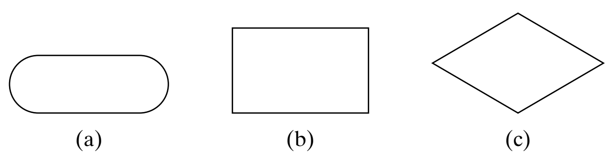



**Figure 10** Symbols in the basic process map: (a) beginning/ending point of the process, (b) task or activity step, and (c) decision point. Brief labels or names are written in the various blocks to identify them.

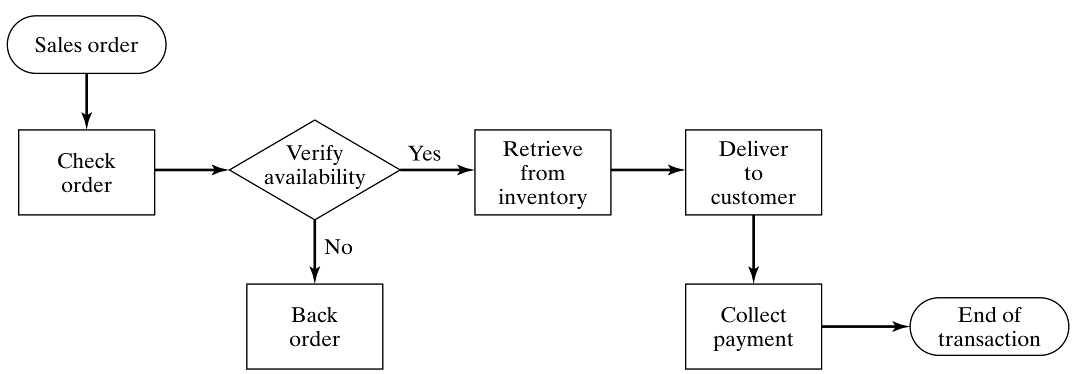



**Figure 11** An example of a basic process map.

[7]The author’s preference for block diagrams may derive from his educational origins in mechanical engineering.

the beginning point of the process is a racetrack oval. It is the boundary between the process under study and the activities that occur outside the process under study. The same symbol and interpretation apply to the end of the process. Rectangular blocks are used to symbolize tasks or activity steps in the process. Decision points are diamond-shaped blocks. Labels in the blocks identify what each step and decision point does, as suggested in Figure 11. Arrows are used to indicate the sequence of the steps and the flow of the work.

Process mapping can be accomplished at various levels of detail. A ***high-level process map*** depicts a macroscopic view of the process of interest and includes only the most important steps in the process. Each step in the high-level map consists of tasks or work elements, which can be examined in greater detail to create a ***low-level process map*** for that step, also called a ***detailed process map***. When this is done, the blocks in the high-level map are shadowed or otherwise highlighted to indicate that the lower-level process map exists. In some cases, more than two levels of process maps may be developed for processes requiring such detail.

Alternative forms of process maps include relationship maps and cross-functional process maps [4]. A ***relationship map*** is a block diagram that shows the input-output connections among the departments or other functional components of an organization. The map consists of blocks that represent the departments and arrows to show the flow of work. A relationship map illustrates the pairs of supplier-customer relationships throughout the organization. Every department is a customer of another department, and it is also a supplier to some other department. Arrows indicate the flow of inputs that are processed and outputs that are produced in these supplier-customer relationships. The relationship map often includes the connections with external suppliers and/or customers of the organization, as well as the associated inputs and outputs. Figure 12 illustrates a relationship map for a custom workshop.

A ***cross-functional process map*** is a block diagram that shows how the steps of a process are accomplished by the various departments or other functional groups that contribute to it. As shown in Figure 13, the departments are listed in rows separated by dashed lines. This format causes the cross-functional process map to also be known as

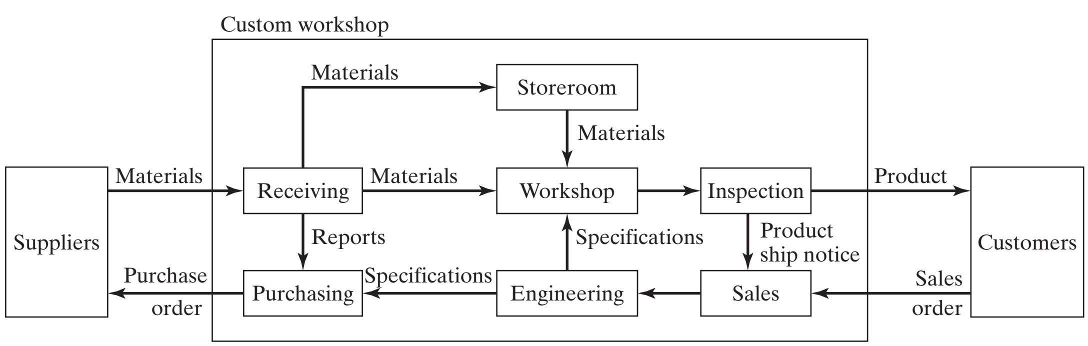



**Figure 12** Relationship map for a custom workshop.

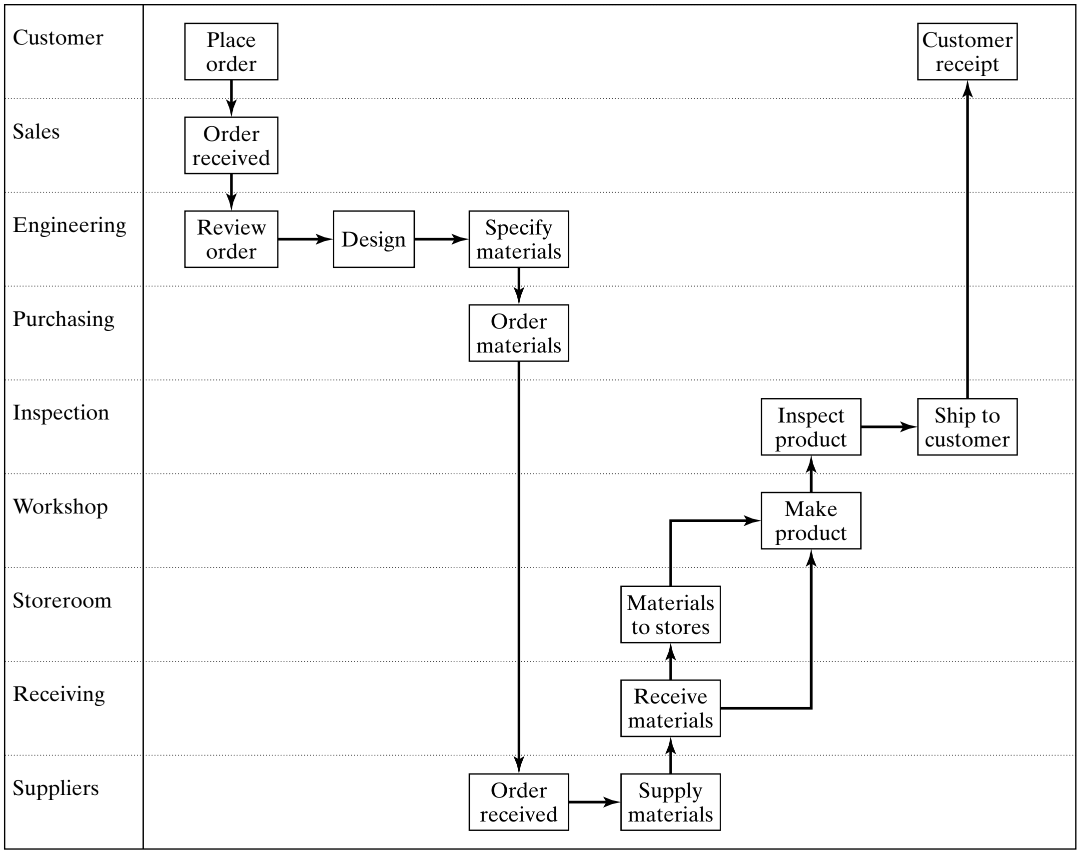



**Figure 13** A cross-functional process map for a custom workshop.

the ***swim-lane chart***. Rectangular blocks represent activity steps in the process, and diamond-shaped blocks represent decision points. Arrows indicate the inputs and outputs for each block, as well as the sequence of steps.

The cross-functional process map differs from the relationship map by showing the blocks as process steps (work activities), whereas the relationship map uses blocks to represent departments. The difference between the cross-functional process map and the basic process map is the use of rows (swim lanes) to show where the process steps are accomplished. In the basic process map, the process steps do not indicate in what department the work is done.

## REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] ANSI Standard Z94.0-1989. *Industrial Engineering Terminology*. Norcross, GA: Industrial Engineering and Management Press, Institute of Industrial Engineers, 1989.

[3] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley 1980.
[4] Damelio, R. *The Basics of Process Mapping*. Portland, OR: Productivity, Inc., 1996.
[5] Harbour, J. L. *Cycle Time Reduction: Designing and Streamlining Work for High Performance*. New York: Quality Resources, 1996.
[6] Meyers, F. E., and J. R. Stewart. *Motion and Time Study for Lean Manufacturing*. 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.
[7] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.
[8] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 11th ed. New York: McGraw-Hill, 2003.
[9] Paradiso, J. "The Essential Process." *Industrial Engineering* (April 2003): 46–48.
[10] Raymond, G. F. "Charting Procedures." Pp. 3.3–3.22 in *Maynard's Industrial Engineering Handbook*. 4th ed., edited by W. K. Hodson. New York: McGraw-Hill, 1992.
[11] Sharp, A., and P. McDermott. *Workflow Modeling: Tools for Process Improvement and Application Development*. Boston: Artech House, 2001.

## REVIEW QUESTIONS

**1** What are the objectives of using charts and diagrams to study work?

**2** What are the four methods indicated in the text by which the analyst develops a description of the work process that is ultimately used to create the graphic?

**3** What are the two characteristics of value-added steps in a given process?

**4** Name some examples of network diagrams.

**5** What are the two types of operations diagrammed in an operation chart?

**6** Identify the five types of symbols used in a flow process chart.

**7** Name the three types of process chart described in the text, and identify the application area for each.

**8** What is a flow diagram?

**9** What are some of the problem areas that can be identified using a flow diagram?

**10** What is an activity chart?

**11** Identify some of the types of multiple activity charts.

**12** What are the three block symbols used in a basic process map?

## PROBLEMS

### Traditional Industrial Engineering Charting and Diagramming Techniques

**1** A motorist experienced a flat tire on the driver side rear wheel of his car and went through the following procedure to replace the flat tire with the spare. The tire change occurred in the middle of the day in his own driveway about 20 ft in front of his garage. He first secured the other three wheels of the car with six bricks from his garage to prevent the vehicle from rolling (two trips back and forth to the garage). He then took out the jack, crank, and lug nut wrench from the trunk of the car, and read over the attached instructions for operating the jack. He removed the spare tire from the trunk and placed it near the rear left wheel.

Next, he proceeded to position the jack under the car at the recommended support beam on the car frame. He began to turn the crank to elevate the car. After the left rear portion of the car was lifted a few inches, but before the flat tire was lifted from the driveway surface, he used the lug nut wrench to loosen the five lug nuts securing the tire to the wheel hub. He then returned to the task of elevating the car, turning the jack crank until the flat tire was completely off the driveway surface. The next step was to remove the loosened lug nuts, placing them in a nearby position within reach. The flat tire was then removed and lifted into the trunk. The motorist moved the spare tire into position and lifted it onto the five studs protruding from the wheel hub. He reached for the five lug nuts, one at a time, placing them onto the studs and rotating them until finger tight. The lug nut wrench was used to tighten the five nuts. With the tire secure, he proceeded to lower the car by cranking the jack down slowly until the spare tire supported the car. For good measure, he again tightened the five lug nuts now that the car was securely on the ground. He collected the hardware, put it back into the trunk, and removed the bricks from the other three wheels and put them back into his garage. Document this tire changing procedure using a worker process chart.

**2** With reference to the tire changing procedure of the previous problem, develop the steps of changing a tire, as they would be accomplished in an automotive tire center, where the worker has access to a hydraulic car lift and pneumatic lug nut wrench instead of the manual tools used by the motorist in his driveway. Document your improved tire changing procedure using a worker process chart.

**3** A foundry uses the following steps in its procedure for high production of investment casting process: (1) Produce wax patterns by injection molding. (2) Transport wax patterns to an assembly work area where they are manually assembled to a wax sprue forming a pattern tree. The entire tree is made of wax. (3) Move pattern tree to a separate room where the tree is coated with a thin layer of refractory material. (4) In the same room, coat tree with successive layers of refractory material to make it a rigid structure that will become the mold for casting. (5) Move tree to a furnace room, where it should be held in an inverted position and heated to melt the wax out of the mold cavities. With the wax removed, the rigid structure is now a multiple-cavity mold with runners leading to each cavity from the sprue cavity. (6) In the same furnace room, heat the mold to a high temperature to ensure that all contaminants are removed from the mold. (7) With the mold still heated at an elevated temperature and in an upright orientation pour the molten metal into the sprue; metal will flow through the runners to each cavity. (8) After the metal cools and solidifies move the assemblage to a finishing room; the mold can be broken away from the cast metal and the parts separated from the runners and sprue. Assignment: (a) Develop the flow process chart for this casting process. (b) Based on your flow process chart, what are some changes in the investment casting procedure that you would recommend?

**4** A supplier of machined components for industrial machinery (e.g., power tools, pumps, motors, compressors) operates a factory that includes a forge shop, machine shop, and finishing department. Many of the parts produced by the company are fabricated through these three departments. Because of this, the factory is laid out as three large square rooms, arranged in-line to form a rectangle with an aspect ratio of three-to-one. Each room is 200 ft by 200 ft. The rectangle runs from north to south, with the forge shop on the south end and the finishing department on the north end. Large doors are located on the south wall for work entering the factory and on the north wall for finished products exiting the factory. For one part of particular interest here, the raw material is a steel billet that is purchased from a steel wholesale supplier. The billets arrive in pallet loads of 100 billets at the shipping and receiving department, which is a building that is 35 ft by 50 ft located

25 ft from the south wall door of the factory. The shipping and receiving department inspects the parts and sends them by forklift truck to be stored in the company's warehouse that is located in another building 500 ft away from the factory in a southerly direction. The warehouse is 200 ft by 200 ft with its entrance door on the north wall. When a production order for the part is received, a factory forklift truck is dispatched to the warehouse to retrieve the billets. The forklift truck must wait while the warehouse crew locates the billets in storage, takes a pallet out of storage using the same type of forklift truck, and delivers the pallet to the dock where it is transferred to the factory forklift. The pallet is then brought back to the factory and delivered to the forge shop. The billets must wait their turn in the production schedule before being pressed into the desired shape by one of the forge presses. From the forge shop, the parts are moved to the machine shop where they are machined on two different machine tools, a milling machine and a drill press. From the machine shop, the parts travel to the finishing department for painting and baking (to cure the paint). From the finishing department, the parts are moved back to the machine shop, where additional milling is accomplished to provide two machined metal surfaces that will mate with other components in the final product. The parts are then moved to the shipping and receiving department for shipment to the customer. Develop the (a) flow process chart and (b) flow diagram for the process, using the centroid of each department to estimate distances between departments. (c) Based on your flow process chart, what are some changes in the production process that you would recommend? (d) Develop a revised procedure for the production process, documenting your revision in the form of a new flow process chart.

**5** The Calm Seas Cruise Ship Line wants to analyze its passenger laundry operation. A typical cruise ship of the line is 850 ft in length and has 600 passenger cabins located on four decks. For every occupied cabin during a cruise, the cabin stewards retrieve the bed linens (sheets and pillowcases) and towels (washcloths, hand towels, and bath towels) when they make up the rooms each day. Each steward is responsible for 15 cabins. The steward separates the bed linens from the towels, putting them into two separate laundry bags. Because of the large size of each bag, the two bags are separately hand-carried to the laundry department, which is located on one of the lower decks. Depending on which deck and deck section a steward serves, the travel distance ranges from several hundred feet to more than a thousand feet, plus several flights of stairs. In the laundry department, the bags are emptied, making sure that the bed linens are not mixed with the towels during laundering. The two categories of laundry are washed separately in large washing machines, using a different wash cycle for each category. After washing, the linens and towels are dried in tumble dryers located on the same deck about 100 ft away, and then they are folded and stacked. They are then moved in stacks to the ironing department, which is located on the deck immediately above the laundry department, where they are separated and individually ironed in large flat ironers. After ironing, they are folded and stacked. From the ironing department they are transported to either of two main storerooms, located fore and aft on one of the main passenger decks for access on the following day by the cabin stewards in making up the rooms. (a) Construct a flow process chart of the laundry operation, using the bed linens and towels as the subject material in the analysis. Estimate distances moved but ignore operation times. (b) Based on your flow process chart, what are some changes in the laundry operation that you would recommend? (c) Develop a revised procedure for the laundry operation, documenting your revision in the form of a new flow process chart.

**6** Given the information and data in previous problem, the cruise ship line wants to analyze the work activities of a cabin steward in its passenger laundry operation. Do not consider all of the activities that are performed in cleaning and making over a passenger cabin (vacuuming, making beds, collecting towels and linens, etc.) as separate operations. Instead, consider making up a room and collecting the laundry in each room as one operation. The focus of the analysis is on the work activities of a cabin steward that relate to the laundry operation. (a) Construct a worker process chart to analyze the duties of the steward that are related to the ship's laundry operation. (b) Based on your worker process chart, what are some changes that you would recommend? (c) Develop a revised procedure for the laundry operation, documenting your revision in the form of a new worker process chart.

**7** The Purchasing Department is required to use the following paperwork procedure for each purchase order (PO) related to a company production order. The purchase order is the legal document used by the company to order raw materials and parts from vendors in specified quantities and to guarantee payment to the vendors upon delivery of the items ordered. The purchase order procedure is triggered by the release of a production order that has been authorized by top management for one of the company's regular products. The production order indicates what product is to be produced, how many units, and when it is to be completed. To produce the product, the raw materials and component parts must be ordered in the correct quantities from suppliers (vendors). To initiate the purchase order procedure, a purchasing agent in the Purchasing Department fills out a blank purchase requisition form for each raw material or component part, obtaining the quantity information from the bill of materials for the product. The purchase requisition is an internal company document used to obtain approvals by several departments that are responsible for the product and/or its production. The purchase requisition is sent first to the Design Engineering Department where it is checked for any engineering changes that may have been made to the item ordered. The requisition is then sent from Design Engineering to the Manufacturing Engineering Department, which checks to make sure that a valid route sheet exists for the item. The route sheet is the process plan that indicates how the item is to be processed in the factory. From Manufacturing Engineering, the requisition is sent to the Production Control Department, which checks the requisition to make sure the quantity information and delivery date agrees with the production order. At each of these departments, the signature of the department manager is required in addition to the regular department employee who performed the check. It takes an average of three days in each department to obtain the necessary checks and approvals. After approval by Production Control, the purchase requisition is returned to the Purchasing Department, where it is used as the authorization to prepare the actual purchase order that will be sent to the vendor. The information on the purchase requisition is transcribed onto a blank purchase order form by the originating purchasing agent, and the PO is then signed by the manager of the Purchasing Department and mailed to the vendor. Each PO is sent in a separate envelope by first class mail, even though there are many vendors receiving more than one purchase order from the company. (a) Construct a form process chart for the current purchase order procedure. (b) Based on your form process chart, what are some changes that you would recommend? (c) Develop a revised purchase order procedure, documenting your revision in a new form process chart.

**8** Consider the allocation of time between the right hand and left hand in the activity chart shown in Figure 7 in the text. (a) If the workplace were redesigned using a workholding fixture, and the worker were trained to use both hands simultaneously to perform the task, construct a right-hand/left-hand activity chart for the revised method, estimating the amounts of time for each step in the method. (b) What is the percent reduction in cycle time?

**9** The repetitive work cycle in a worker-machine system consists of the work elements and associated times given in the table below. As the table shows, all of the operator's elements are external to the machine time. (a) Construct a worker-machine activity chart for this work cycle. (b) Can some of the worker's elements be made internal to the machine cycle? If so, construct a worker-machine activity chart for the revised work cycle. What is the approximate cycle time for the revised cycle?

| Sequence | Work Element Description | Worker Time (min) | Machine Time (min) |
|---|---|---|---|
| 1 | Worker walks to tote pan containing raw stock | 0.13 | (idle) |
| 2 | Worker picks up raw work part and transports it to machine | 0.23 | (idle) |
| 3 | Worker loads part into machine and engages machine semiautomatic cycle | 0.12 | (idle) |
| 4 | Machine semiautomatic cycle | (idle) | 0.75 |
| 5 | Worker unloads finished part from machine | 0.10 | (idle) |
| 6 | Worker transports finished part and deposits into tote pan | 0.15 | (idle) |
| Totals | | 0.73 | 0.75 |

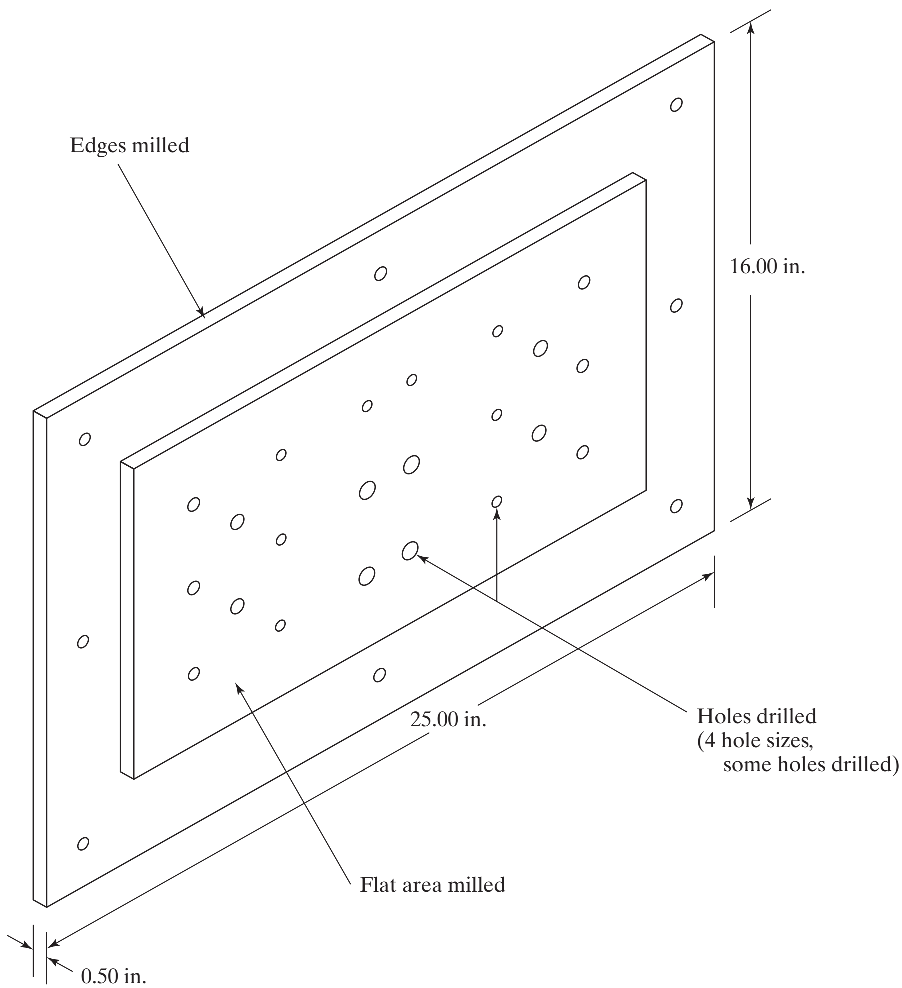
## Case Problem: Flow Process Chart and Flow Diagram

**10** A company produces machinery for industry. Many of the company's products are made in large quantities. This case problem deals with the mechanism plate for one of these machinery products, shown in Figure P10(a). The mechanism plate is a major part in the machine, serving as the backbone of the product. Most of the internal components and subassemblies are fastened to this plate. In the finished product, the plate is completely enclosed within the machine. The mechanism plate is the most expensive part in the product and is the source of numerous quality problems as well. Similar mechanism plates are used on many of the products, and their methods of production are similar, so any improvements made to the subject plate will apply to the other plates as well.

As is typical in batch production, the manufacturing lead time of the mechanism plate (time between release of the order and its completion) is quite long. Since other parts are being processed at the same time in the plant, there are delays in front of each production station as batches of mechanism plates wait their turn in the queues at these stations. Mechanism plates take an average of 18 weeks to complete, once a production order has been released. No information is available about how this time is distributed among the different production steps. In addition, the castings may remain in the warehouse prior to production for as much as 4 months. The following description summarizes the sequence of steps used to fabricate the mechanism plate. Figure P10(b) and (c) show floor layouts for the two stories of the factory building. Key areas and departments relating to the mechanism plate production are indicated in the description and on the floor layouts by numbers in parentheses.

The mechanism plate is a sand casting, supplied by a foundry located 20 miles away. It is made of cast iron and is very heavy, constituting fully one-third the total weight of the product. The designer selected cast iron because of its inherent rigidity and vibration damping characteristics. However, because it is a large flat part, the casting is subject to warping.

The castings are shipped to the company in batches of 50 to 100 units and inspected twice:(1) prior to being stored in the plant warehouse and (2) when an order is placed for




| | | |
|---|---|---|
| Part name: Mechanism Plate | Casting tol: +/−0.075 in. | Drawn by: M.P.G. |
| Material: Cast Iron sand casting | Machining tol: +/−0.010 in. | Date: Jan21/99 |

**Figure p10a** Exhibits for Problem 10: mechanism plate.

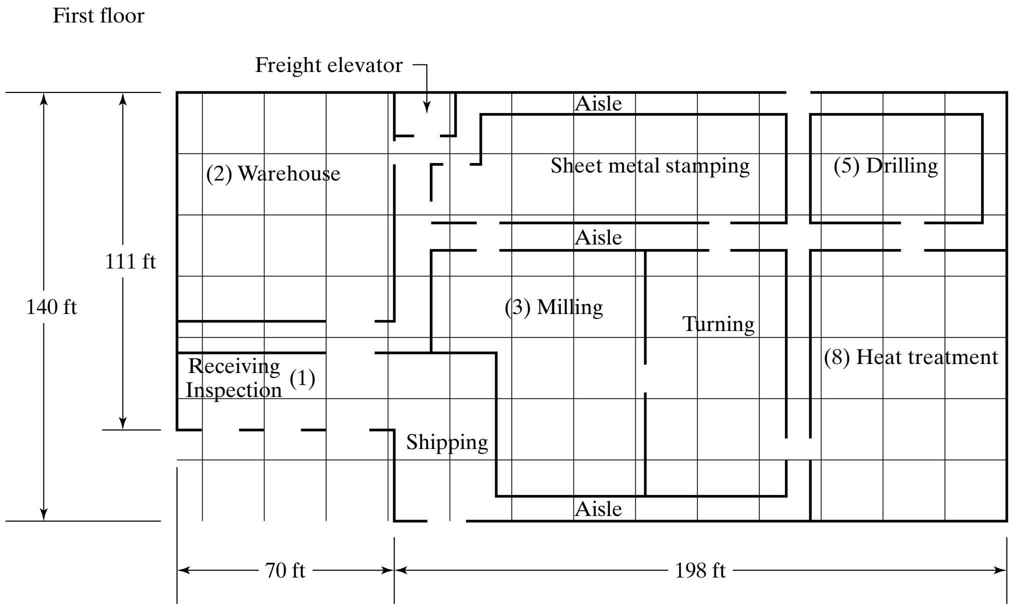



**Figure p10(b)** Exhibits for Problem 10: first floor layout of the factory building

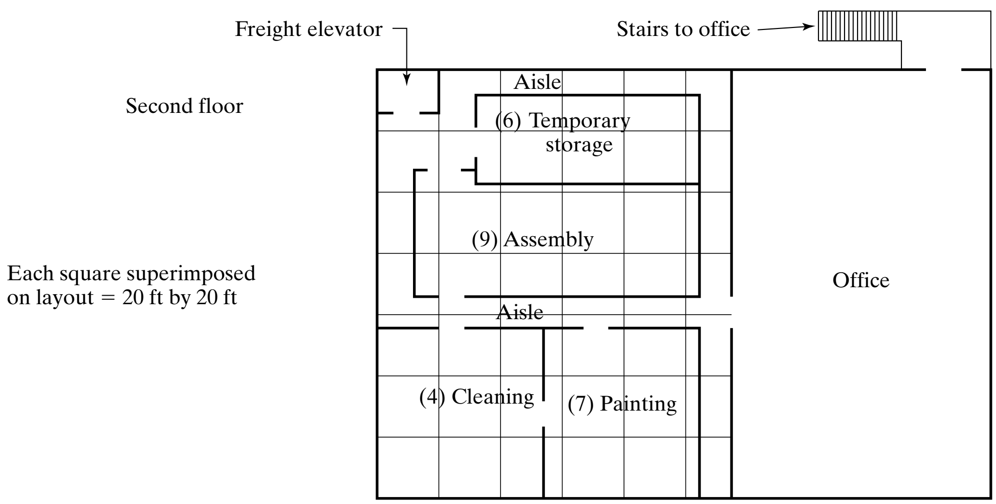



**Figure p10(c)** Exhibits for Problem 10: second floor layout.

their release to production. Production lot sizes are usually 5 to 10 parts. When the castings are picked from the warehouse for an order, the first step is to send them to a local firm (5 miles away) that specializes in sand blasting (which the company is not currently set up to do). This treatment dislodges any sand imbedded in the surface that is left over from the mold, removes rust that has occurred during storage, and smoothes the surface. The plates are then returned to the company warehouse.

The parts are then processed through a series of machining operations, the first and second of which are in the milling department (3). The first milling operation is face milling to make one side of the plate flat and smooth to accept the components that are to be attached. This is accomplished in one setup on a large milling machine in the mill department. The second milling requires a separate setup, and this operation makes the edges of the casting smooth and straight.

As the second milling operation is completed on each part, it is moved individually to the chemical cleaning department (4) to be cleaned and pickled. The cleaning is performed by human workers using cloth wipers soaked with chemical solvents. The purpose of the cleaning is to remove cutting fluids from machining. Pickling is then performed in open tanks filled with relatively dilute acid which etches the cast iron. The pickling brightens the surface texture of the machined surface of the casting. Owing to the size of the casting, each part must be separately dipped into the tank. The parts are then individually returned back to the machining area (drilling department) to reconstitute the batch.

The next step involves a series of hole drilling operations in the drilling department (5). These holes are used to attach the various brackets and components to the plate. This is performed on four different upright drill presses, organized according to drill size. There are four different hole sizes in the plate. After drilling, two of the hole sizes are tapped in the same department (5), using two additional drill presses. After drilling and tapping, the plates are returned to the cleaning department (4) for cleaning and pickling. Again, they must be dipped separately and moved individually to a temporary storage area (6) near the assembly department (9).

The plates are stored in the temporary storage area (6) until the other components that are to be assembled have been collected. When all of the components have been collected, the mechanism plate is sent to the painting department (7). The painting is needed to provide a protective coating and for appearance reasons. The operation is performed by human workers using paintbrushes to work the paint into the rough texture of the sand cast surface. Two coats are applied in this way, with a five hour low temperature baking operation following each coating. The baking ovens are located in the heat treatment department (8) at the rear of the factory. As many as five plates can be baked at one time.

After the final bake, the mechanism plates rejoin the other components in the temporary storage area (6). A "kit" of components is then made up, which consists of one mechanism plate and a set of all the components that are to be assembled to it. The kit is then sent to the assembly department (9), where the subassembly is completed. A total of 75 parts, including fasteners, are attached to the plate. Each mechanism plate subassembly is put together at a single workstation. When finished, it is put back into temporary storage (6) to await final assembly into the machinery product.

*Assignment:* (a) Prepare a flow process chart showing the operations, moves, delays, etc., in the current process. (b) Summarize the flow process chart by determining the number of operations and moves, the total distances moved, and delays, etc. (c) Construct a flow diagram of the path followed by the mechanism plate during its manufacture. (d) Develop a list of possible improvements that might be made in the production of the mechanism plate. (e) Develop a proposed improved method based on your answers to (a), (b), (c), and (d), and document your proposed new method by means of a flow process chart. (f) Summarize the flow process chart for the proposed method, indicating the number of operations and moves, the total distances traveled, and delays, etc. Show how this method is an improvement over the current method in terms of these statistics. (g) Consider the issue of plant layout in more detail. What changes in the overall layout of the plant would you make to improve the efficiency of mechanism plate production?

## Process Mapping

**11** Develop a basic process map for the cruise ship laundry operation described in Problem 5. What recommendations can you make for improving the operation?

**12** Develop a relationship map for the machinery components factory described in Problem 4.

**13** Develop a cross-functional process map for the Purchasing Department and other departments described in Problem 7. What recommendations can you make for improving the paperwork flow?

# Motion Study and Work Design



This chapter focuses on the traditional industrial engineering topics of motion study and work design. ***Motion study*** involves the analysis of the basic hand, arm, and body movements of workers as they perform work.[1] ***Work design*** involves the methods and motions used to perform a task. This design includes the workplace layout and environment as well as the tooling and equipment (e.g., workholders, fixtures, hand tools, portable power tools, and machine tools). In short, work design is the design of the work system. The traditional approaches in motion study and work design are based on the findings of researchers such as Frank and Lillian Gilbreth in the early part of the twentieth century. The approaches also include common sense "principles of motion economy" that are used to simplify and improve the efficiency and effectiveness of manual work. Today, these principles are widely used not only by industrial engineers but also by workers themselves who in recent years are increasingly being given greater responsibility for designing their own work methods.

Work design is commonly associated with manual work, the type of work done in production and logistics. In recent decades, with the growth of service industries,

[1]The definition of motion study developed by the Employee and Industrial Relations Committee [ANSI Standard Z94.13-1989] is the following: "The study of the basic manual movements and procedures associated with the accomplishment of work. Historically associated with the Gilbreths (Frank and Lillian), in contrast to time study, associated with Taylor (Frederick)."

greater attention has been focused on the design of work procedures and environments in those industries. To the extent that service operations usually involve a manual component, the general principles that apply to manual work are applicable to service and office work as well. In addition, work design principles dealing with worker comfort, workplace layout, and equipment operation apply just as well to service and office work as they do to manual tasks.

The traditional approaches in motion study and work design were developed before the advent of ergonomics and human factors, which are also applied in the design of work.

## 1 BASIC MOTION ELEMENTS AND WORK ANALYSIS

As indicated in the pyramidal structure of work, any manual task is composed of work elements, and the work elements can be further subdivided into basic motion elements. In this section, we define the basic motion elements and how they can be used to analyze work.

### 1.1 Therbligs

Frank Gilbreth was the first to catalog the basic motion elements. He called each motion a ***therblig*** (Gilbreth spelled backward except for the "th"). A list of Gilbreth's 17 therbligs is presented in Table 1 along with the letter symbol he used for each as well as a brief description. Therbligs are the basic building blocks of virtually all manual work performed at a single workplace and consisting primarily of hand motions. Therbligs are relatively few in number, but they are performed over and over, often in very similar sequences, during a given task. For example, a common hand motion sequence in repetitive assembly work is to reach for a part in the work area, grasp it, move and position it, and then release it. This sequence may be repeated many times in a work cycle, depending on how many parts are assembled in the cycle. Therbligs include mental elements as well as physical elements. With some modification, these basic motion elements are used today in a number of work measurement systems, such as Methods-Time Measurement (MTM) and the Maynard Operation Sequence Technique (MOST).

The therbligs in a manual task can be classified in several ways that are useful in analyzing the work cycle. One possible classification is shown in Table 2, in which therbligs are classified (1) as to their physical or mental emphasis and (2) whether they are effective or ineffective. Methods analysis at the therblig level seeks to eliminate or reduce ineffective therbligs.

### 1.2 Work Analysis Using Therbligs

Each therblig represents time and energy expended by a worker to perform a task. If the task is repetitive, of relatively short duration, and will be performed many times, it may be appropriate to analyze the therbligs that make up the work cycle as part of the

**TABLE 1** Gilbreth's Seventeen Therbligs

| Therblig | Letter Symbol | Description |
|---|---|---|
| Transport empty | TE | Reaching for an object with empty hand; for example, reach for a part prior to grasping and moving the part. Today, we commonly refer to the transport empty motion element as a "reach." |
| Grasp | G | Grasping an object by contacting and closing the fingers of the active hand about the object until control has been achieved. |
| Transport loaded | TL | Moving an object using a hand motion; for example, moving a part from one location to another at a workstation. Today, we commonly refer to a transport loaded motion element as a "move." |
| Hold | H | Holding an object; for example, holding an object with one hand while the other hand performs some operation on it. |
| Release load | RL | Releasing control of an object, typically by opening the fingers that held it and breaking contact with the object. |
| Preposition | PP | Positioning and/or orienting an object for the next operation and relative to an approximate location; for example, lining up a pin next to a hole for insertion into the hole. Preposition usually follows transport loaded. |
| Position | P | Positioning and/or orienting an object in the defined location that is intended for it. Position is generally performed during transport loaded; for example, moving a pin toward a hole and simultaneously lining it up in preparation for insertion into the hole. |
| Use | U | Manipulating and/or applying a tool in the intended way during the course of working, usually on an object; for example, using a screwdriver to turn a threaded fastener or using a pen to sign one's name. |
| Assemble | A | Joining two parts together to form an assembled entity; for example, using a threaded fastener to assemble two mating parts by hand. |
| Disassemble | DA | Separating multiple components that were previously joined in some way; for example, unfastening two parts held together by a threaded fastener. |
| Search | Sh | Attempting to find an object using the eyes or hand, concluding when the object is found. |
| Select | St | Choosing among several objects in a group, usually involving hand-eye coordination, and concluding when the hand has located the selected object. |
| Plan | Pn | Deciding on a course of action, usually consisting of short pause or hesitation in the motions of the hands and/or body. |
| Inspect | I | Determining the quality or characteristics of an object using the eyes and/or other senses. |
| Unavoidable delay | UD | Waiting due to factors beyond the control of the worker and included in the work cycle; for example, waiting for a machine to complete its feed motion. |
| Avoidable delay | AD | Waiting that is within the worker's control, causing idleness that is not included in the regular work cycle; for example, the worker opening a pack of chewing gum. |
| Rest | R | Resting to overcome fatigue, consisting of a pause in the motions of the hands and/or body during the work cycle or between cycles. |

work design process. The term ***micromotion analysis*** is sometimes used for this type of analysis. For example, the activities of the right and left hands when performing a manual task can be specified in terms of therbligs. Figure 1 illustrates this kind of analysis for

**TABLE 2** Classification of Therbligs

| | Effective Therbligs | Ineffective Therbligs |
|---|---|---|
| Physical basic motion elements | Transport empty (TE)<br>Grasp (G)<br>Transport loaded (TL)<br>Release load (RL)<br>Use (U)<br>Assemble (A)<br>Disassemble (DA) | Hold (H)<br>Preposition (PP) |
| Physical and mental basic motion elements | | Position (P)<br>Search (Sh)<br>Select (St) |
| Mental basic elements | Inspect (I) | Plan (Pn) |
| Delay elements | Rest (R) | Unavoidable delay (UD)<br>Avoidable delay (AD) |

*Source:* Adapted from a classification in [7].

| Sequence | Left hand | Therbligs | | Right hand | Cumulative Time (min) |
|---|---|---|---|---|---|
| 1 | Reach for board | TE | AD | Idle | |
| 2 | Grasp board | G | AD | Idle | |
| 3 | Transport board | TL | AD | Idle | |
| 4 | Position board | PP | AD | Idle | 0.08 |
| 5 | Hold board | H | TE | Reach for peg | |
| 6 | Hold board | H | G | Grasp peg | |
| 7 | Hold board | H | TL | Transport peg to board | |
| 8 | Hold board | H | P | Position peg in hole | |
| 9 | Hold board | H | R | Release peg | 0.14 |
| 10 | Hold board | H | TE | Reach for peg | |
| 11 | Hold board | H | G | Grasp peg | |
| 12 | Hold board | H | TL | Transport peg to board | |
| 13 | Hold board | H | P | Position peg in hole | |
| 14 | Hold board | H | R | Release peg | 0.20 |

**Figure 1** Right-hand/left-hand activity chart where therbligs are used to define the work cycle. Only the first two cycles of the right hand are shown here.

the same task that is described in the right-hand/left-hand activity chart. The following general objectives are involved in micromotion analysis:[2]

- Eliminate therbligs that are ineffective if possible; for example, eliminate the need to search for parts or tools by positioning them in a known and fixed location in the workplace.

[2]The list of objectives is based largely on a list of basic principles in [11, p. 398].

- Avoid the use of a hand for holding parts; use a workholder instead.
- Combine therbligs where possible; for example, perform right-hand and left-hand motions simultaneously.
- Simplify the overall method; for example, resequence therbligs in the cycle.
- Reduce the time required for the motion; for example, shorten distances of therbligs such as transport empty and transport loaded.

As in the analysis associated with the various charting techniques, checklists are useful in micromotion analysis. Table 3 presents a checklist of questions and suggestions for possible improvements for each therblig.

## 2 PRINCIPLES OF MOTION ECONOMY AND WORK DESIGN

The principles of motion economy have been developed over many years of practical experience in work design. They are guidelines that can be used to help determine the work method, workplace layout, tools, and equipment that will maximize the efficiency and minimize the fatigue of the worker. Most of the guidelines are based on common sense. Many are derived from the preceding therblig analysis and will be recognizable to those who have studied Table 3. The reader might question why it is even necessary to state these principles, since some of them seem so obvious. The answers to that question can be found in factories, warehouses, offices, and other workplaces around the world in the many instances where these principles are violated.

The principles of motion economy can be organized into three categories: (1) principles related to the use of the human body, (2) principles related to workplace arrangement, and (3) principles related to the design of tooling and equipment.[3] Our coverage is based on this organization.

### 2.1 Principles Related to the Use of the Human Body

The principles of motion economy that are related to the use of the human body can be used to design the most appropriate methods and motions of the human worker in performing a given task. They are most applicable to manual work, either repetitive or nonrepetitive.

1. ***Both hands should be fully utilized***. The natural tendency of most people is to use their preferred hand (right hand for right-handed people and left hand for left-handed people) to accomplish most of the work. The other hand is relegated to a minor role, such as holding the object, while the preferred hand works on it. This first principle states that both hands should be used as equally as possible.
2. ***The two hands should begin and end their motions at the same time***. This principle follows from the first. To implement, it is sometimes necessary to design the method so that the work is evenly divided between the right-hand side and the left-hand side of the workplace. In this case, the division of work should be organized according to the following principle.

[3]This is a classification suggested by Barnes [4]. Sources for the principles discussed in this section include [1], [4], [9], [11], and [13].

**TABLE 3** Micromotion Analysis Checklist for Possible Improvements

| Therblig | Questions and Suggestions |
|---|---|
| Transport empty (TE) | Minimize number of parts in the product to reduce frequency of TE and TL.<br>Minimize reach distance required.<br>Use parts bins that have easy access.<br>Can abrupt changes in direction of movement of body member be eliminated or minimized?<br>Locate parts and tools used most frequently near their respective points of use.<br>Minimize requirements for hand-eye coordination during reach.<br>Can right and left hands be used simultaneously to accomplish two transport empty motions? |
| Grasp (G) | Use parts bins that have easy access.<br>Use workholders that have fast release mechanism. For example, screw type vises are time consuming to operate, while pneumatic clamps are fast-acting.<br>Locate parts and tools in known locations to save time in searching.<br>Can right and left hands be used simultaneously to accomplish two grasp motions?<br>Avoid transfer of objects from one hand to the other.<br>Design parts that do not tangle. |
| Transport loaded (TL) | Can parts be slid across work surface rather than carried above work surface? This usually saves time.<br>Can abrupt changes in direction of movement of body member be eliminated or minimized?<br>Design parts and tools to be as lightweight as possible to save move time.<br>Minimize number of parts in the product to reduce frequency of TE and TL.<br>Minimize move distance required.<br>Locate parts and tools used most frequently near their respective points of use.<br>Minimize requirements for hand-eye coordination during movement.<br>Can right and left hands be used simultaneously to accomplish two transport loaded motions? |
| Hold (H) | This is considered an ineffective therblig. Can it be eliminated?<br>Can a workholding device (e.g., fixture, jig, vise, clamp) be used instead of holding by hand?<br>Can friction, an adhesive, or a mechanical stop be used instead of holding by hand?<br>If holding by hand cannot be eliminated, can an armrest be provided? |
| Release load (RL) | Is it possible to release the object by dropping it (e.g., into a chute)?<br>Is the delivery point (e.g., bin, workholder) designed for ease of release of the object?<br>Minimize requirements for hand-eye coordination during release. |
| Preposition (PP) | This is considered an ineffective therblig. Can it be eliminated?<br>Can symmetry of prepositioning be increased? For example, it is easier to preposition a round shaft relative to a round hole than a square shaft relative to a square hole because of increased symmetry of the fit.<br>Can a guide be designed to facilitate prepositioning?<br>Can an armrest be used to steady the hand during prepositioning?<br>Design parts and tools to be as lightweight as possible to save prepositioning time.<br>Make sure object is grasped properly to facilitate prepositioning. |
| Position (P) | This is considered an ineffective therblig. Can it be eliminated?<br>Can symmetry of positioning be increased? For example, it is easier to position a round shaft relative to a round hole than a square shaft relative to a square hole because of increased symmetry of the fit.<br>Can a guide be designed to facilitate positioning?<br>Can an armrest be used to steady the hand during positioning?<br>Can tools be suspended from overhead to avoid positioning?<br>Design parts and tools to be as lightweight as possible to save positioning time.<br>Make sure object is grasped properly to facilitate positioning. |
| Use (U) | Can a more efficient hand tool be designed to reduce the time of the use motion?<br>Can a portable power tool be devised to reduce the time of the use motion? |

| Therblig | Questions and Suggestions |
|---|---|
| | The part should be held in a workholder during the use motion.<br>Can a jig be designed to guide the use of the tool? A ***jig*** is a special workholder that has a mechanism for guiding the tool.<br>Can a mechanized or automated operation be used to eliminate the need for the use motion? |
| Assemble (A) | Can a hand tool be designed to reduce the time required for the assembly motion?<br>Can a portable power tool be devised to reduce the time of the assembly motion?<br>The base part or existing subassembly should be positioned in a workholder during the assembly motion.<br>Can the product be designed with fewer components to minimize assembly time?<br>Design the product for automated assembly to eliminate the need for manual assembly. |
| Disassemble (DA) | Can a hand tool be designed to reduce the time required for the disassembly motion?<br>Can a portable power tool be devised to reduce the time required for the disassembly motion?<br>The base part or existing subassembly should be positioned in a workholder during the disassembly motion. |
| Search (Sh) | This is considered an ineffective therblig. Can it be eliminated?<br>Make sure lighting is adequate to facilitate searching.<br>Can parts be fed from magazines or chutes to avoid searching?<br>Locate tools in known positions in the workplace to facilitate searching; for example, suspend tools from overhead.<br>Can different parts be made with different colors to facilitate searching? |
| Select (St) | This is considered an ineffective therblig. Can it be eliminated?<br>Use parts bins that have easy access.<br>Make sure lighting is adequate to facilitate selecting.<br>Can parts be fed from magazines or chutes for one-at-a-time selection?<br>Can different parts be made with different colors to facilitate searching? |
| Plan (Pn) | This is considered an ineffective therblig. Can it be eliminated?<br>Remove the need for the worker to decide on a course of action that causes hesitation in the work cycle. |
| Inspect (I) | Make sure lighting is adequate to facilitate the inspection procedure.<br>Minimize the number of characteristics to inspect. Only the key characteristics of the part should be inspected. Time should not be wasted inspecting unimportant characteristics.<br>Can the object be inspected using gauges instead of actually measuring the characteristics of interest? Gauging takes less time than measuring.<br>Can inspection be combined with another operation so it is not performed separately?<br>Can inspection be automated (e.g., machine vision) to eliminate the need for a worker to accomplish it (e.g., visually)?<br>Can multiple but separate inspection steps be combined into one inspection? |
| Unavoidable delay (UD) | This is considered an ineffective therblig. Can it be eliminated?<br>Eliminate the reason for the delay. For example, can the machine speed be increased to reduce the machine cycle time?<br>Can external work elements be made into internal work elements to fill up the delay time with useful work activities? |
| Avoidable delay (AD) | This is considered an ineffective therblig. Can it be eliminated?<br>Eliminate the reason for the delay.<br>Provide incentives for the worker to minimize delay time. |
| Rest (R) | Reduce metabolic load on worker through the use of machines and tools to minimize need for rest breaks.<br>Improvements in methods and motions through analysis of previous therbligs should reduce need for rest breaks. |

*Source:* Adapted from checklists in [4], [11], and other sources.

3. ***The motions of the hands and arms should be symmetrical and simultaneous.*** This will minimize the amount of hand-eye coordination required by the worker. And since both hands are doing the same movements at the same time, less concentration will be required than if the two hands had to perform different and independent motions. It should be noted that not all tasks can be organized according to this principle. If this is the case, then the principle that follows next is applicable.
4. ***The work should be designed to emphasize the worker's preferred hand.*** The preferred hand is faster, stronger, and more dexterous. If the work to be done cannot be allocated evenly between the two hands, then the method should take advantage of the worker's best hand. For example, work units should enter the workplace on the side of the worker's preferred hand and exit the workplace on the opposite side. The reason is that greater hand-eye coordination is required to initially acquire the work unit, so the worker should use the preferred hand for this element. Releasing the work unit at the end of the cycle requires less coordination.
5. ***The worker's two hands should never be idle at the same time.*** The work method should be designed to avoid periods when neither hand is working. It may not be possible to completely balance the workload between the right and left hands, but it should be possible to avoid having both hands idle at the same time. The exception to this principle is during rest breaks. The work cycle of a worker-machine system may also be an exception, if the worker is responsible for monitoring the machine during its automatic cycle, and monitoring involves using the worker's cognitive senses rather than the hands. If machine monitoring is not required, then internal work elements should be assigned to the worker during the automatic cycle.

The central theme of these first five principles is the use of two hands and the allocation of work between them. Barnes [4] cites the results of a study done at the University of Iowa in which a right-handed worker performed a relatively simple manual task consisting of reaching, selecting, grasping, transporting, and releasing small parts. The task was accomplished using two different containers for the parts: a rectangular bin and a bin with tray (the bin with tray making it easier to perform the select and grasp elements). The worker performed the work cycle with each container using (1) only the right hand, (2) only the left hand, and (3) both hands performing symmetrical and simultaneous motions. The results of the study are presented in Table 4. The normalized times are simply the percentage values of the left hand and both hands using the right- hand time as the base. As expected, the right-handed worker was able to complete the work cycle in the shortest time using the right hand. Also as expected, the work cycle

**TABLE 4** Study Results of a Task Performed One-Handed and Two-Handed

| | Rectangular Bin | | | Bin with Tray | | |
|---|---|---|---|---|---|---|
| | Right Hand | Left Hand | Both Hands | Right Hand | Left Hand | Both Hands |
| Time to perform | 1.04 sec | 1.10 sec | 1.48 sec | 0.81 sec | 0.91 sec | 1.07 sec |
| Normalized time | 100% | 106% | 142% | 100% | 112% | 132% |

*Source:* [4].

using both hands took the longest time. However, two parts were completed using both hands, so the time per part for each tray type is one-half the value shown. Accordingly, using both hands was the most productive method.

The next five principles of motion economy attempt to utilize the laws of physics to assist in the use of the hands and arms while working.

6. ***The method should consist of smooth continuous curved motions rather than straight-line motions with abrupt changes in direction***. It takes less time to move through a sequence of smooth continuous curved paths than through a sequence of straight paths that are opposite in direction, even though the actual total distance of the curved paths may be longer (since the shortest distance between two points is a straight line). This is especially true when an object is being moved and the mass of the object affects the motions. The reason behind this principle is that the straight-line path sequence includes start and stop actions (accelerations and decelerations) that consume the worker's time and energy. Motions consisting of smooth continuous curves minimize the lost time in starts and stops.
7. ***Momentum should be used to facilitate the task wherever possible.*** When carpenters strike a nail with a hammer, they are using ***momentum***, which can be defined as mass times velocity. Imagine trying to apply a static force to press the nail into the wood. Not all work situations provide an opportunity to use momentum as a carpenter uses a hammer, but if the opportunity is present, exploit it. The previous principle dealing with smooth continuous curved motions illustrates a beneficial use of momentum (i.e., conservation of momentum) to make a task easier.
8. ***The method should take advantage of gravity instead of opposing it***. Less time and energy are required to move a heavy object from a higher elevation to a lower elevation than to move the object upward. The principle is usually implemented by proper layout and arrangement of the workplace, and so it is often associated with the workplace arrangement principles of motion economy.
9. ***The method should achieve a natural rhythm of the motions involved***. Rhythm refers to motions that have a regular recurrence and flow from one to the next. Basically, the worker learns the rhythm and performs the motions without thinking, much like the natural and instinctive motion pattern that occurs in walking.
10. ***The lowest classification of hand and arm motions should be used***. The five classifications of hand and arm motions are presented in Table 5. With each lower classification, the worker can perform the hand and arm motion more quickly and with less effort. Therefore, the work method should be composed of motions at

**TABLE 5** Five Classifications of Hand and Arm Motions

| Classification | Defined as: |
|---|---|
| 1 | Finger motions only |
| 2 | Finger and wrist motions |
| 3 | Finger, wrist, and forearm motions |
| 4 | Finger, wrist, forearm, and upper arm motions |
| 5 | Finger, wrist, forearm, upper arm, and shoulder motions |

the lowest classification level possible. This can often be accomplished by locating parts and tools as close together as possible in the workplace.

The two remaining human body principles of motion economy are recommendations for using body members other than the hands and arms.

11. ***Minimize eye focus and eye travel activities***. In work situations where hand-eye coordination is required, the eyes are used to direct the actions of the hands. ***Eye focus*** occurs when the eye must adjust to a change in viewing distance—for example, from 25 in. to 10 in. with little or no change in line of sight. ***Eye travel*** occurs when the eye must adjust to a line-of-sight change—for example, from one location in the workplace to another, but the distances from the eyes are the same. Since eye focus and eye travel each take time, it is desirable to minimize the need for the worker to make these adjustments as much as possible. This can be accomplished by minimizing the distances between objects (e.g., parts and tools) that are used in the workplace.
12. ***The method should be designed to utilize the worker's feet and legs when appropriate***. The legs are stronger than the arms, although the feet are not as dexterous as the hands. The work method can sometimes be designed to take advantage of the greater strength of the legs, for example, in lifting tasks.

### 2.2 Principles Related to Workplace Arrangement

The principles of motion economy in this section are directed at the design of the workplace—for example, the layout of the workstation and the arrangement of tools and parts in it. The first three principles deal with the immediate work area and contribute to a natural rhythm in the work cycle. The other principles cover the use of gravity and the general conditions of the workplace.

1. ***Tools and materials should be located in fixed positions within the work area***. As the saying goes, "a place for everything, and everything in its place." The worker eventually learns the fixed locations, allowing him or her to reach for the object without wasting time looking and searching.
2. ***Tools and materials should be located close to where they are used***. This helps to minimize the distances the worker must move (travel empty and travel loaded) in the workplace. In addition, any equipment controls should also be located in close proximity. This guideline usually refers to a normal and maximum working area, as shown in Figure 2 and clarified further in Table 6. It is generally desirable to keep the parts and tools used in the work method within the normal working area, as defined for each hand and both hands working together. If the method requires the worker to move beyond the maximum working area, then the worker must move more than just the arms and hands. This expends additional energy, takes more time, and ultimately contributes to greater worker fatigue.
3. ***Tools and materials should be placed in locations that are consistent with the sequence of work elements in the work cycle***. Items should be arranged in a logical pattern that matches the sequence of work elements. Those items that are

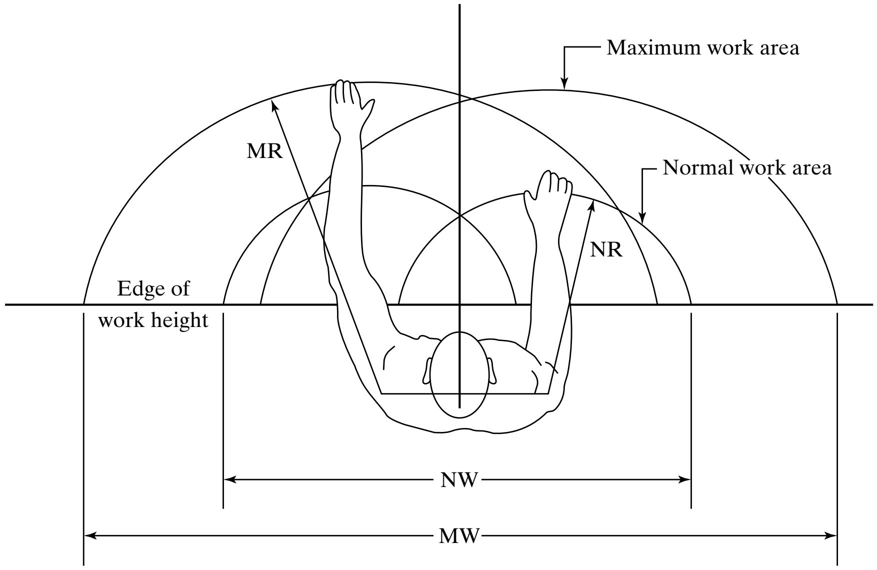


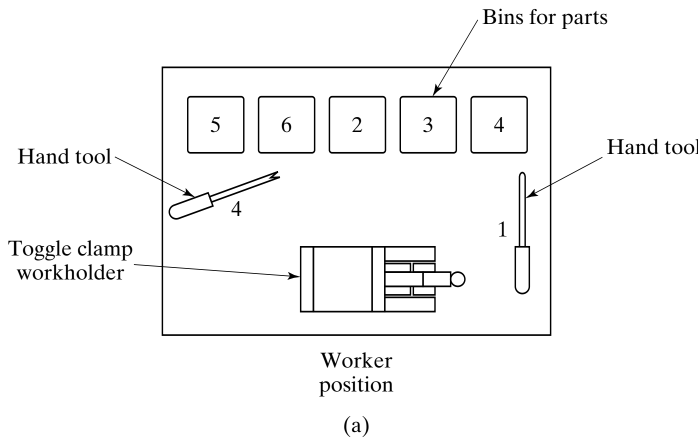

**Figure 2** Normal and maximum working areas in the workplace. See Table 6 for working area dimensions of average male and female workers. (Adapted from [4]. Original source was R. Farley, "Some Principles of Methods and Motion Study as Used in Development Work," *General Motors Engineering Journal* 2, no. 6: 20–25.)

**TABLE 6** Normal and Maximum Working Area Dimensions in Figure 2

| Symbol in Figure 2 | Dimension in Working Area for Worker Seated at Worktable | Male Worker [cm (in)] | Female Worker [cm (in)] |
|---|---|---|---|
| NR | Normal radius of arm reach | 39 (15.5) | 36 (14.0) |
| MR | Maximum radius of arm reach | 67 (26.5) | 60 (23.5) |
| NW | Normal width of arm reach | 109 (43.0) | 102 (40.0) |
| MW | Maximum width of arm reach | 163 (64.0) | 147 (58.0) |

used first in the cycle should be on one side of the work area, the items used next should be next to the first, and so on, until the last items are obtained on the other side of the work area. The alternative to this sequential arrangement is to locate items randomly about the work area. This increases the amount of searching required and detracts from the rhythm of the work cycle.

Figure 3 shows the top view of a workplace layout that illustrates these first three principles. Note that the layout in (b) locates bins in a more accessible pattern that is consistent with the sequence of work elements.



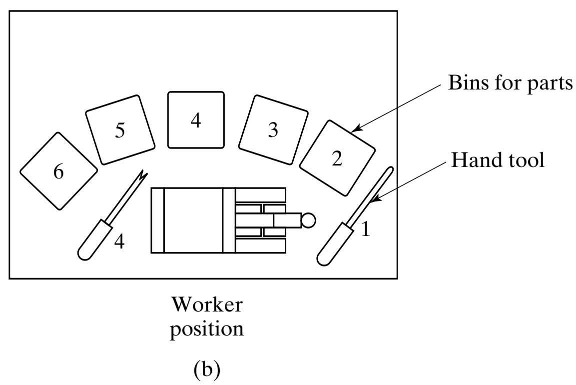




**Figure 3** Two workplace layouts. The layout in (b) illustrates principles of good workplace arrangement while the one in (a) does not. Numbers indicate sequence of work elements in relation to locations of hand tools and parts bins.

4. ***Gravity feed bins should be used to deliver small parts and fasteners***. A gravity feed bin is a container that uses gravity to move the items in it to a convenient access point for the worker. One possible design is shown in Figure 4(a). It generally allows for quicker acquisition of an item than a conventional rectangular tray shown in Figure 4(b).
5. ***Gravity drop chutes should be used for completed work units where appropriate***. The drop chutes should lead to a container adjacent to the worktable. The entrance to the gravity chute should be located near the normal work area, permitting the worker to dispose of the finished work unit quickly and conveniently. There are obvious limitations on the use of gravity chutes. They are most appropriate for lightweight work units that are not fragile.

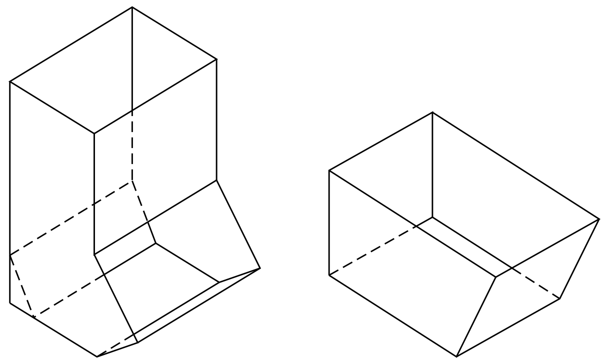



**Figure 4** Two types of bins used for small parts and fasteners in the workplace: (a) gravity feed bin and (b) conventional rectangular bin.

6. ***Adequate illumination must be provided for the workplace***. The issue of illumination is normally associated with ergonomics. However, illumination has long been known to be an important factor in work design, well before ergonomics and human factors became a field of study in the scientific and engineering community. Illumination is especially important in visual inspection tasks. Adequate illumination means not only enough light on the work area; it also refers to color, absence of glare, contrast among items in the field of view, and the location of the light source(s). Other aspects of workplace environment are also important, such as noise and temperature.
7. ***A proper chair should be provided for the worker***. This usually means an adjustable chair that can be fitted to the size of the worker. The adjustments usually include seat height and back height. Both the seat and back are padded. Many adjustable chairs also provide a means of increasing and decreasing the amount of back support. The chair height should be in proper relationship with the work height. An adjustable chair for the workplace is shown in Figure 5.

### 2.3 Principles Related to the Design of Tooling and Equipment

The principles of motion economy in this section are helpful for designing special tools and controls on equipment used in the workplace.

1. ***Workholding devices should be designed for the task; a worker's hand should not be used as a workholder***. A mechanical workholder with a fast-acting clamp permits the work unit to be loaded quickly and frees both hands to work on the task productively. Typically, the workholder must be custom-designed for the

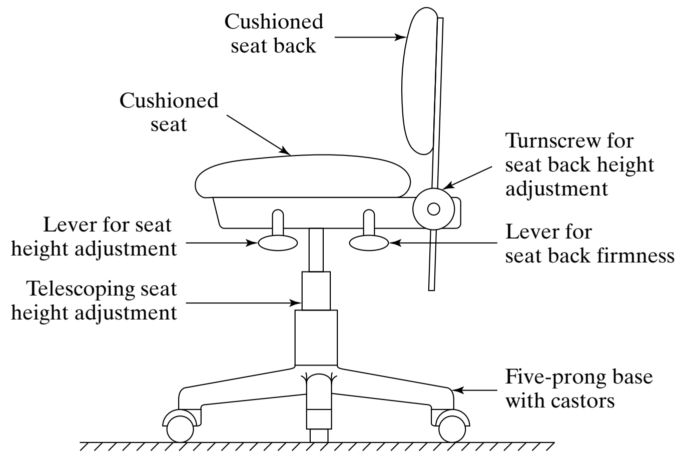



**Figure 5** An adjustable chair for the workplace. The adjustments provide comfortable seating related to the size of the worker and the work height.

work part processed in the task. A conventional vise is not recommended for a repetitive work cycle because of its slow-acting screw clamping mechanism.

2. ***The hands should be relieved of work elements that can be performed by the feet***. Foot pedal controls can be provided instead of hand controls to operate certain types of equipment. Sewing machines and church organs are two examples in which foot pedals are used as integral components in the operation of the equipment. As our examples suggest, training is often required for the operator to become proficient in the use of the foot pedals.
3. ***Multiple tooling functions should be combined into one tool whenever possible***. Many of the common hand tools and implements exhibit this principle. The head of a claw hammer is designed for both striking and pulling nails. Nearly all pencils are designed for both writing and erasing. Less time is usually required to reposition such a double-function tool than to put one tool down and pick another one up.
4. ***Multiple operations should be performed simultaneously rather than sequentially whenever possible***. A work cycle is usually conceptualized as a sequence of work elements or steps. The steps are performed one after the other by the worker and/or machine. In some cases, the work method can be designed so that the steps are accomplished at the same time rather than sequentially. Special tooling and processes can often be designed to simultaneously accomplish the multiple operations. Examples include the following:

   - Pneumatically powered, multiple-spindle lug nut runner used to attach wheels to the car in automobile final assembly plants
   - Multiple-spindle drill presses to drill holes in printed circuit boards
   - Wave soldering in electronics assembly to solder all connections on a printed circuit board simultaneously.

5. ***Where feasible, an operation should be performed on multiple parts simultaneously***. This usually applies to cases involving the use of a powered tool such as a machine tool. A good example is the drilling of holes in a printed circuit board (PCB). The PCBs are stacked three or four thick, and a numerically controlled drill press drills each hole through the entire stack in one feed motion.
6. ***Equipment controls should be designed for operator convenience and error avoidance***. Equipment controls include dials, cranks, levers, switches, push buttons, and other devices that regulate the operation of the equipment. All of the controls needed by the operator should be located within easy reach, so as to minimize the body motions required to access and actuate them. In addition, the controls should be designed to be mistake-proof, so that the operator does not cause an equipment action that is different from the one intended.
7. ***Hand tools and portable powered tools should be designed for operator comfort and convenience***. For example, the tools should have handles or grips that are slightly compressible so that they can be held and used comfortably for the duration of the shift. The location of the handle or grip relative to the working end of the tool should be designed for maximum operator safety, convenience, and utility of the tool. If possible, the tool should accommodate both right-handed and left-handed workers.
8. ***Manual operations should be mechanized or automated wherever economically and technologically feasible***.[4] Mechanized or automated equipment and tooling that are designed for the specific operation will almost always outperform a human worker in terms of speed, repeatability, and accuracy. This results in higher production rates and better quality products. The economic feasibility depends on the quantities to be produced. In general, higher quantities are more likely to justify the investment in mechanization and automation. The technological feasibility relates to the time and difficulty involved in mechanizing or automating a given operation. The time issue is the amount of time required to design and build an automated machine. It may take more than a year before the equipment can be installed, but the company must begin production right away on a product with a short life cycle. The difficulty issue refers to problems such as (1) physical access to the work location, (2) adjustments required in the task, (3) requirements of manual dexterity, and (4) demands on hand-eye coordination.

## REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] Akiyama, M., and H. Kamata. "Method Engineering and Workplace Design." Pp. 4.3–4.20 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[4]***Mechanization*** refers to the use of machinery, usually powered machinery, in place of human (or animal) labor. A human is usually required to operate and/or oversee the machinery. ***Automation*** refers to the use of powered equipment that operates under its own control and without human assistance or oversight.

U3

# Introduction to Work Measurement



Time is important in work systems because of its economic significance. Most workers are paid for the time they are on the job. The labor content to produce a product or deliver a service is often a major determinant of the cost of the product or service. For any organization to operate efficiently and effectively, it is critical to know how much time should be required to accomplish a given amount of work. For the reader's amusement, a list of quotations and sayings about time is compiled in Table 1.

The terms *time study* and *work measurement* are often used interchangeably. Both are concerned with how much time it should take to complete a unit of work. ***Work measurement*** refers to a set of four techniques that are concerned with the evaluation of a task in terms of the time that should be allowed for an average human worker to perform that task: (1) direct time study, (2) predetermined motion time systems, (3) standard data systems, and (4) work sampling, an alternative work measurement technique in which statistical measures are determined about how workers allocate their time among

From Chapter 12 of *Work Systems and the Methods, Measurement, and Management of Work,* First Edition. Mikell P. Groover. 

**TABLE 1** Quotations and Sayings About Time

| |
|---|
| Time flies when you're having fun. |
| Take care of the minutes, and the hours will take care of themselves. (Lord Chesterfield) |
| Time heals all things—except leaky faucets. |
| Those who kill time eventually mourn the corpse. |
| Time changes with age: in youth, time marches on; in middle age, time flies; and in old age, time runs out. |
| All things come to him who waits, except the precious time lost while waiting. |
| Some people count time, others make time count. |

*Source:* Compiled from E. Esar, *20,000 Quips and Quotes* (New York: Barnes and Noble 1995) and other sources.

multiple activities. The task that is measured is usually a repetitive work cycle. The objective of these work measurement techniques is to determine a ***standard time*** for the task.

Because of its emphasis on time, work measurement is often called time study. However, in modern usage and in our coverage in this text, time study has a slightly broader meaning. Whereas work measurement is generally focused on human work effort, ***time study*** refers to all of the ways in which time is investigated and analyzed in work situations, whether the work is accomplished by human workers or automated systems. For example, in an automated work system in which a human worker periodically attends to the system but is not present every cycle, it is still important to know the automatic cycle time. Time study also includes the learning curve phenomenon, which is not usually included in the definition of work measurement.

Here, we discuss the various techniques used to measure work and other job activities of human workers. Time is an important variable used to evaluate performance in the variety of work systems. In manual work, worker-machine systems, assembly lines, logistics work, service operations, and projects, time is the common factor that determines performance. In this chapter, we provide an overview of work measurement, the basic definitions and principles that apply to all of the techniques, and the issues that are faced in their application.

## 1 TIME STANDARDS AND HOW THEY ARE DETERMINED

Most workers are paid on the basis of time. The common work shift is 8 hours per day, and the worker is paid an hourly rate for those 8 hours. Several questions may be asked by a worker's employer: How much work did the worker accomplish in those 8 hours? Has the worker done a fair day's work? Time standards provide a way to answer these questions.[1] The most meaningful and useful measure of work is the amount of time it takes to accomplish it. Time is objective. It is quantifiable. People understand time.

[1]The terms *time standard* and *standard time* are used interchangeably and synonymously in this text. Other terms for time standards include *work standards* and *labor standards*.

The ***standard time*** for a given task is the amount of time that should be allowed for an average worker to process one work unit using the standard method and working at a normal pace. The standard time includes some additional time, called the ***allowance***, to provide for the worker's personal needs, fatigue, and unavoidable delays during the shift. The standard time is sometimes referred to as the ***allowed time***, because it indicates how much time is allowed the worker to process each work unit so that by the end of the shift a fair day's workload has been accomplished, despite the various interruptions that may occur.

How does an organization know whether it needs time standards for its operations? The following characteristics are typical of industrial situations in which time standards would be beneficial:

- *Low productivity.* If the current level of productivity is low, then there are significant opportunities for improvement.
- *Repeat orders.* Once the time standard is set during the first order, the same standard can be used for successive orders.
- *Long production runs.* A long production run means that the time invested to set the standard is apportioned over more parts, thus reducing the average cost of work measurement.
- *Repetitive work cycles.* Work measurement can be justified more readily when the work cycle is highly repetitive.
- *Short cycle times.* Short work cycles require less time to set standards.

When accurately determined, time standards serve the following important functions and applications in an organization:

- They define a "fair day's work." If the quantity of work units completed by a worker, when multiplied by their respective standard times, adds up to the number of hours in the shift, then the worker has put in a fair day's work.
- They provide a means of converting the workload to be produced into the staffing and equipment resources needed to accomplish the workload. They help to determine manpower requirements and capacity limitations.
- They provide an objective way to compare alternative methods for accomplishing the same task.
- They provide a basis for wage incentives and for evaluating worker performance.
- They provide time data for production planning and scheduling, cost estimating, material requirements planning, and other management functions that depend on accurate task time data.

### 1.1 Methods to Determine Time Standards

There are several methods by which "standard times" can be established for a task. They vary in terms of the accuracy and reliability of the values that are derived from them

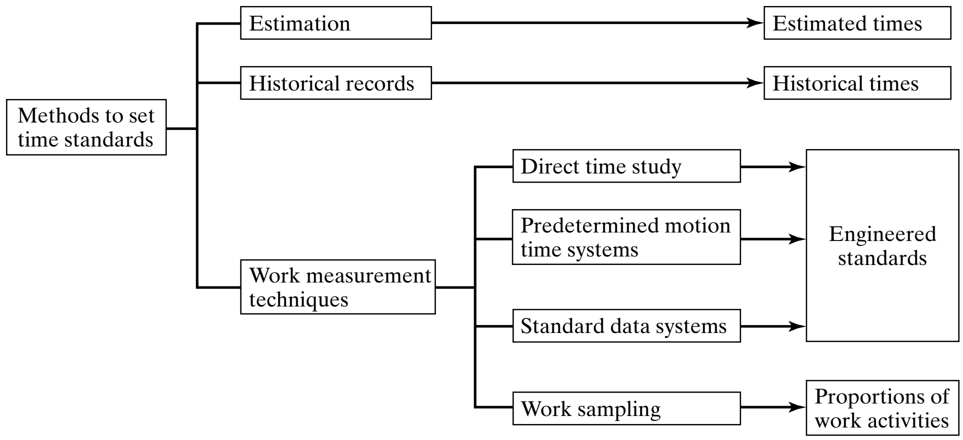



**Figure 1** Classification of methods to determine time standards.

and in the amount of time required to apply them. The following methods can be used to determine time standards, as shown in Figure 1:

- *Estimation.* In this method, the department foreman or other person familiar with the jobs performed in the department is asked to judge how much time should be allowed for the given task. Because this method depends on the estimator's judgment, it is the least accurate of the techniques for determining time standards.[2] On the other hand, it is better than making no estimate of the time and simply assigning a worker to the task with no accountability.
- *Historical records of previous production runs.* In this method, the actual times and production quantities from records of previous identical or similar job orders are used to determine the time standards. Most companies require workers to submit time records (e.g., "time cards") indicating the time spent on orders or jobs and the quantities of parts turned in against that time. From these records a calculation is made to determine the average time per part, and this can be used as a "time standard" for the next order. Historical records are an improvement over estimates because they represent actual times for amounts of work completed. Their limitation is that they do not include any indication of the efficiency with which the work was accomplished.
- *Work measurement techniques.* These are the four techniques identified in the chapter introduction: direct time study, predetermined motion time systems, standard data systems, and work sampling. These techniques are described in Section 1.2.

The work measurement techniques are more time consuming to implement, but they are more accurate than estimation or historical records. It is the author's opinion that task time values determined by estimation or historical records should not even be called "time standards." They should be called "estimated times" and "historical times," respectively, and in fact they are sometimes identified by these names.

[2]The issue of accuracy in work measurement is discussed in Section 4.1.

Among the four work measurement techniques, work sampling should be differentiated from the other three. The primary purpose of work sampling is to determine proportions of time spent in various categories of work activity using randomized observations of the subjects of interest. On the other hand, the principal purpose of direct time study, predetermined motion time systems, and standard data systems is to establish time standards. Standards set by these three techniques are sometimes referred to as ***engineered standards***, because (1) they are based on measured time values that have been adjusted for worker performance, and (2) some effort has been made to determine the best method to accomplish the task. Although work sampling can be used to determine time standards, the technique is not as reliable because of statistical errors, and no attempt is made to improve the method by which the work is accomplished.

### 1.2 Work Measurement Techniques

The four work measurement techniques are described briefly in the following sections. In the pyramidal structure of work, a task consists of work elements, and how each work element consists of basic motion elements. The various work measurement techniques measure work at different levels of this task hierarchy. Figure 2 shows the correspondence between the task hierarchy and work measurement techniques.

**Direct Time Study.** Direct time study (DTS) involves direct observation of a task using a stopwatch or other chronometric device to record the time taken to accomplish the task. The task is usually divided into work elements and each work element is timed separately. While observing the worker, the time study analyst evaluates the worker's performance (pace), and a record of this pace is attached to each work element time. This evaluation of the worker's pace is called ***performance rating***. The observed time is multiplied by the performance rating to obtain the ***normal time*** for the element or the task:[3]

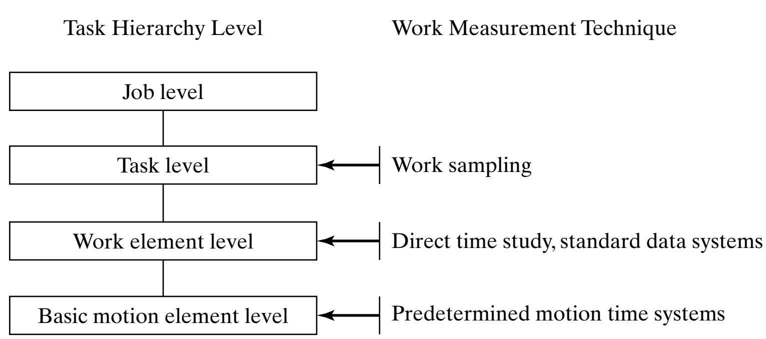



**Figure 2** How the work measurement techniques correspond to different levels in the general task hierarchy.

[3]Some time study analysts attach an overall performance rating to the entire task while others prefer to rate the worker's performance during each work element.

$$T_n = T_{obs}(PR) \tag{1}$$

where $T_n$ = normal time (the time required by a worker working at 100% pace for one cycle), min; $T_{obs}$ = observed time, min; and $PR$ = performance rating of the worker's pace observed by an analyst, usually recorded as a percentage during the observation but applied as a decimal fraction in the equation. If the task is repetitive, several work cycles are observed and the normalized time values are averaged to improve statistical accuracy.

To determine the standard time for the task, an allowance is added to the normal time to provide for personal time, fatigue, and delays. The calculation of the standard time is summarized in the following equation:

$$T_{std} = T_n(1 + A_{pfd}) \tag{2}$$

where $T_{std}$ = standard time, min.; $T_n$ = normal time, min.; and $A_{pfd}$ = allowance factor for personal time, fatigue, and delays (PFD). The PFD allowance is explained in Section 3.1.

**Predetermined Motion Time Systems.** A predetermined motion time system (PMTS) relies on a database of basic motion elements (therbligs) such as reach, grasp, and move that are common to nearly all manual industrial tasks. Associated with each motion element is a set of normal times, whose values depend on the conditions under which the motion element was performed. For example, the normal time for a reach depends on the distance reached. Longer distances take more time. The normal time for a move depends on the distance moved and the weight of the object being moved. And so on.

To use a predetermined motion time system to set a standard time for a given task, the analyst lists all of the basic motion elements that comprise the task, noting their respective conditions, and retrieves the normal time for each element from the database. The normal times for the motion elements are then summed to obtain the normal time for the task. Finally, the standard time is calculated by adding a PFD allowance. This final step is the same as that expressed by equation (2) for direct time study. There are two advantages related to predetermined motion time systems: (1) performance rating is not required and (2) they can be applied to determine the time standard for a task before production.

**Standard Data Systems.** A standard data system (SDS) is a compilation of normal time values for work elements used in tasks that are performed in a given facility. These normal times are used to establish time standards for tasks that are composed of work elements similar to those in the database. An advantage of a standard data system is that a time standard can be set before the job is in production.

The normal time values in a standard data system are usually compiled from previous direct time studies, but they may be based on predetermined motion time data, work sampling data, or even historical time records. Large amounts of data are generally required to compile the database, and the work elements are performed under different work variables that may affect their normal time values. Accordingly, the normal time for a given element is a function of the work variable(s), and this functionality must be included in the database in the form of tables, charts, or mathematical equations.

To use a standard data system, the analyst first identifies the work elements that make up the task together with the values of the work variables respectively for each element. He or she then accesses the database to find the normal time for each element. The work element values are summed to determine the normal time for the task. As in the other work measurement techniques, an allowance is added to the normal time to compute the standard time.

**Work Sampling.** Work sampling uses random sampling techniques to study work situations so that the proportions of time spent in different activities can be estimated with a defined degree of statistical accuracy. Examples of the activities in a work sampling study might include setting up for production, producing parts, machine idle, and so on. The activities must be defined specifically for the work situation that the study is intended to address. The capability to include multiple subjects (e.g., workers, machines) in the study is one of the advantages of work sampling. Direct time study and predetermined motion time systems are generally limited to one worker for each study. A large number of observations over an extended period of time are usually made in a work sampling study in order to achieve the desired level of statistical accuracy. The period of the study must be representative of the activities normally performed by the subjects, and the observations must be made at random times in order to minimize bias; for example, if the workers knew when the observations would be made, it might influence their behavior.

The objectives in a work sampling study may be to measure machine utilization in a plant or to determine an appropriate allowance factor for use in setting standards in direct time study. The objective is not always to determine time standards. But when used to establish standards, the statistical errors inherent in the sampling procedure cause the "time standards" to be less accurate than those obtained by the other work measurement techniques.

**Computerized Work Measurement.** The purpose of work measurement and time standards is to improve the productivity of the workers who perform the value-adding tasks of the organization. But work measurement itself can be very time-consuming for the analysts who do it. A number of hardware and software products have been introduced commercially to improve the productivity of the analysts who perform work measurement. These products reduce the amount of time required by the analyst to set a time standard. Thus fewer analysts can set more standards.

Computerized products have been developed for all four of the work measurement techniques: direct time study, predetermined motion time systems, standard data systems, and work sampling. They make use of the ubiquitous personal computer (PC) as well as portable devices such as personal digital assistants (PDAs). In general, these products reduce the time and effort to perform work measurement by means of the following conveniences:

- Facilitating the collection of data at the work site in direct time study and work sampling
- Automatically performing the routine computations that previously had to be performed by the analyst

- Organizing the time standards files and databases
- Retrieving data from databases in predetermined motion time systems and standard data systems
- Assisting in the preparation of the documentation required in work measurement (e.g., methods descriptions, reports).

## 2 PREREQUISITES FOR VALID TIME STANDARDS

The time to perform one work cycle of a given manual task depends on the worker (his or her physical size and strength, as well as mental abilities), the worker's pace, the method used (hand and body motions, tooling, equipment, and work environment), and the work unit. As a prerequisite for establishing a standard time for a task, all of these factors must be standardized. The standardized factors are the following:

- The task is performed by an ***average worker***
- The worker's pace represents ***standard performance***
- The worker uses the ***standard method***
- The task is performed on a ***standard work unit*** that is defined before and after processing.

Establishing these definitions for the task is a requirement for setting a valid time standard, as depicted in Figure 3. And the definitions must be fully documented for future reference. If any of the factors change in future replications of the task, then it can be argued that the previously determined time standard is no longer valid. The only exception is the worker. Different workers are all expected to work by the same definition of standard time, whether they represent the "average worker" or not.

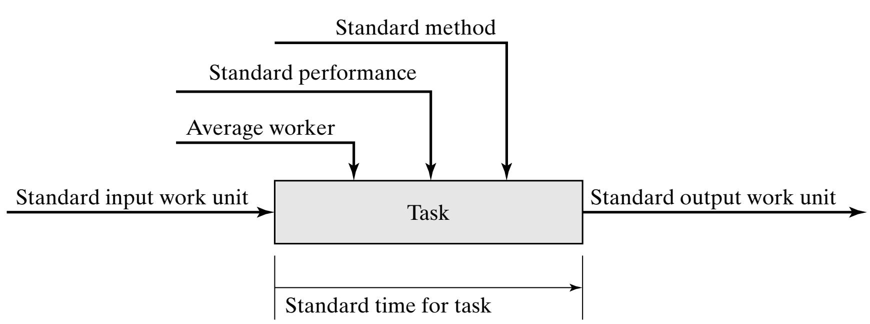



**Figure 3** Model indicating the factors that must be standardized before a standard time can be set for a task.

### 2.1 Average Worker and Standard Performance

For the purpose of defining standard time, an ***average worker*** is a person who is representative of those who perform tasks similar to the task being measured. If similar tasks are performed mostly by men, and gender makes a difference in the ability to perform the task, then the average worker should be defined as a male. If performed mostly by women, then average worker means a female worker. Sometimes men are better suited to certain types of work, and sometimes women are better suited. In addition, the average worker is assumed to have learned the task and is practiced and proficient at it. He or she is well into the learning curve and is capable of performing the task consistently throughout the shift, which includes occasional rest breaks as befits the work situation.

To accomplish the task in the standard time, the average worker works at standard performance. Performance refers to the rate or speed with which a worker does the task. It is the pace of working. ***Standard performance*** is a pace of working that can be maintained by the average worker throughout an entire work shift without harmful effects on the worker's health or physical well-being. The work shift is usually considered to be an 8-hour workday during which the worker is allowed periodic rest breaks and may experience other interruptions. Standard performance is a pace the worker can achieve day after day. It is the normal pace of an average worker trained to perform the given task.

Several benchmarks of "standard performance" have been developed over the years. Two of the most popular are the following:

- Walking at 3 miles per hour (or 4.82 km per hour) on level flat ground
- Dealing four hands of cards from a 52 card deck in exactly 30 seconds.

The term ***normal performance*** is often used in place of standard performance. Basically, the two terms mean the same thing. Normal performance is 100% pace while the worker is working, while standard performance is 100% performance but with the proviso that periodic breaks are taken and other delays are likely to occur during the shift. A worker could not maintain 100% pace for 8 hours straight (imagine walking at 3 miles per hour for 8 straight hours, 24 miles, without an occasional rest stop). However, with periodic breaks a worker is able to work at 100% pace when he or she is working (with occasional rest breaks amounting to total of an hour or so, walking 24 miles in a day can be readily accomplished by a person who has trained for it[4]).

When an individual work cycle is performed at 100% performance, the time taken is the ***normal time*** for the cycle. The normal time is less than the standard time because the normal time does not include an allowance for time losses during the shift. When this allowance is added to the normal time, we have the ***standard time***. This is captured concisely in our equation for standard time, equation (2):

$$T_{std} = T_n\,(1 + A_{pfd})$$

[4]There are many accounts during the American Civil War (1861–1865) of entire armies marching distances more than 24 miles per day over terrain that was hardly flat and level.

The allowance factor $A_{pfd}$ is intended to provide for the average time losses for personal needs, fatigue, and delays that occur during the shift. The allowance factor must be determined for the particular task or category of work to which it is applied. If the work is physically demanding, then a greater allowance factor must be provided for rest breaks. Allowances are discussed in Section 3.

Worker performance varies among individual workers. The variations occur due to age, sex, size and physical strength, physical conditioning, skill and aptitude for the task, experience and training, and motivation. For large numbers of workers, studies have shown that worker performance is normally distributed, and the ratio of the highest performing and lowest performing worker ranges between about 2.3 to 1.[5] The results (hypothetical) are presented in Figure 4. Worker performance in these studies is usually indicated by the hourly or daily production rate for a representative manual task.

It is common industrial practice to define standard performance as a pace that can be readily attained by the majority of workers. The companies (and unions representing the workers) want most workers to be able to achieve the standard performance fairly easily, so they define standard to be a performance that is less than the average in Figure 4. This is an additional condition on our preceding definition of standard performance. A typical multiplier is 1.30, although some companies prefer not to use a multiplier at all (in effect, the value of the multiplier for these companies is 1.0). A multiplier of 1.30 means that an average worker should be able to work at a pace that is 130% of the defined standard pace. This standard pace is sometimes referred to as ***day work pace***.

The production rate is equivalent to the reciprocal of the time to complete the task. Thus, for an output of 100 pieces per day (480 min) at average performance, the average task time would be 4.80 min (480 min/100 pc). Using a multiplier of 1.30 to define standard performance, this means that 100 pc/day is 130% of standard performance, and the corresponding standard time is therefore 4.80 (1.30) = 6.24 min, or a daily production rate of 77 pc. These values can be seen in Figure 5, a revision of our

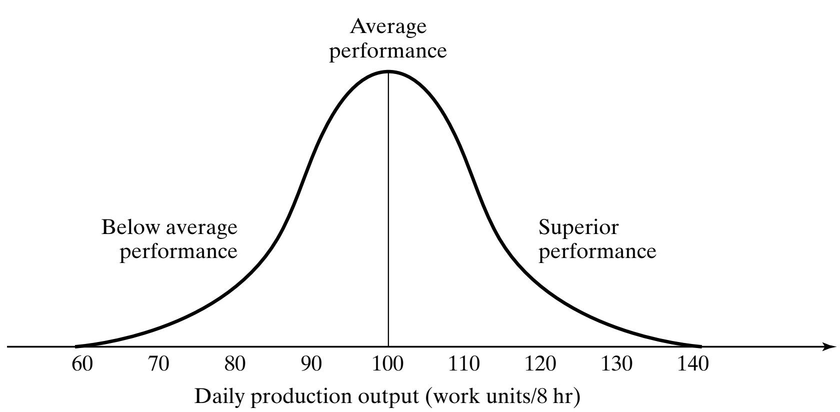



**Figure 4** Distribution of performance (daily production output) for large numbers of workers.

[5]Several studies are cited in [8].

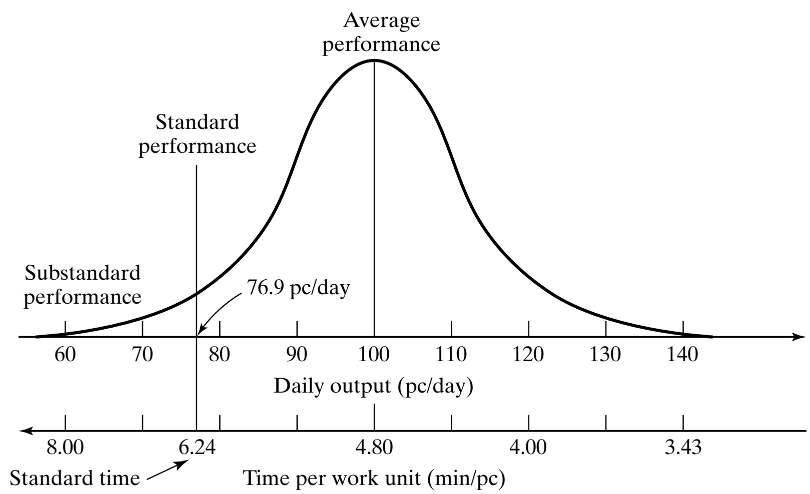



**Figure 5** Distribution of performance and task time for large numbers of workers, indicating how standard time is typically defined as a time that can be readily achieved by most workers.

previous performance distribution. Accordingly, we can define ***standard performance*** in common industrial practice to be the pace of working at which the standard time for the task is 130% (or some other multiple, depending on company policy) of the mean time that can be maintained throughout an entire work shift by an average worker without harmful effects on the worker's health or physical well-being.

## 2.2 Standard Method

The ***standard method*** is the procedure that has been determined to be the optimum method for processing a work unit. The standard method satisfies the "one best method principle." It is the procedure that is the safest, quickest, most productive, and least stressful to the worker. It is constrained only by the practical conditions and economic limitations of the particular work environment. If the task is loading coal into a furnace in a steel mill, it is impractical and uneconomical to perform such work in a clean, air-conditioned environment that is void of safety hazards.

The standard method should include the following details about how a task is performed:

- *Procedure (actions and motions) used by the worker.* This is best described by listing the work elements that comprise the work cycle. The predetermined time systems provide a natural means for doing this because they list the basic motions that comprise the procedure. For direct time study and standard data systems, the procedure must be included in the statement of the standard method.
- *Tools.* This includes hand tools and portable power tools, fixtures, and gauges used in the procedure.

- *Equipment.* This includes machinery and safety equipment used to perform the task.
- *Workplace layout.* How is the workplace arranged? What are the locations of the parts, tools, etc., that will allow for the most efficient method? Where do the work units enter and exit the workplace? An overhead layout sketch of the workplace is a useful means of documentation.
- *Irregular work elements.* This includes tasks that are not performed every work cycle. Depending on the task, these elements may include changing tools as they wear out, changing parts containers as they become empty or full, and any periodic maintenance of the equipment that the worker is required to perform.
- *Working conditions.* Is the work performed outside or inside? What is the surrounding temperature, noise level, and other conditions that might affect the work environment?
- *Setup.* What setup of the physical tools and equipment is required to perform the task? How much time is allowed for the setup?

All of these factors should be thoroughly documented in a written statement of the standard method. At some future time, when the task is performed again after a long delay, the document will provide an invaluable and timesaving description of the task for which the time standard applies. If there is some dispute about the standard method or the standard time, the written statement will be a useful resource in resolving the dispute. Good documentation, a burden to prepare, is a blessing to practitioners.

### 2.3 Standard Input and Output Work Units

Most tasks involve a process or action performed on a work unit. The state or condition of the work unit is altered in some way by the task. Value is added to the work unit by performing the task on it (at least we hope so). The time required to accomplish the task is likely to depend on the condition of the input work unit. Therefore this condition must be specified as completely as possible in the standard method statement. Similarly, the condition of the output work unit must be specified. Exactly what changes have been made in the work unit as a result of the task performed on it? What is the final state of the completed work unit?

The starting and completed work units are usually easy to define in production because they deal with objects being processed. The status or condition of the starting work unit is usually defined explicitly by engineering documents that can be referenced in the standard method statement. The completed work unit is specified by similar documents. If an actual work part does not conform to these specifications, either before or after processing, then this nonconformance may necessitate a deviation from the standard method. For example, if a starting sand casting has too much metal to be machined due to variations or errors in the casting process, an additional machining pass across the surface may be required, which will take more time. This condition can be determined by comparing the actual part dimensions with those specified in the engineering drawing. If inspection reveals that the incoming part from the foundry is oversized beyond the tolerance, so that an extra machining pass is required, then additional time must be allowed for the extra pass (and the foundry should be notified to take corrective action for future deliveries).

In service work, the input and output work units are usually more difficult to define. However, most service tasks consist of a standard approach or method, which is the best procedure to accomplish the task. And the standard method may also have a "standard time" associated with it. An example is the routine automotive repair work performed in a car dealer's service department. Changing motor oil, replacing the oil filter, rotating four tires, and state inspection are all pretty standard jobs. For a given car model, each of these tasks has a standard method and a standard time. You can sometimes find the standard time on your bill from the service department.

## 3 ALLOWANCES IN TIME STANDARDS

In all of the work measurement techniques, the normal time is adjusted by an allowance factor to obtain the standard time. Allowances are used because there will be periods during the regular work shift when the worker is not working. The purpose of the allowance factor is to compensate for this lost time by providing a small increment of "allowance time" in each cycle. This way, even with the time losses, the operator will still be able to complete a fair day's work during the hours of the shift.

### 3.1 Accounting for Lost Time in the Workplace

There are various reasons why people do not work continuously during a regular shift. Some of the interruptions are work-related while others are not. Table 2 lists some typical causes of interruptions that occur in industrial work.

There is no denying that these interruptions occur during the work shift, and there is no denying that allowance time in some form should be provided to deal with them. The question is how much should be allowed, and how should it be provided to the worker. Companies address the allowance issue using two basic approaches, often in combination: (1) scheduled break periods during the shift and (2) a PFD allowance added to the normal time.

**Scheduled Break Periods.** Scheduled breaks are planned periods set aside during the shift as break time from work. Lunch breaks (or supper break for evening and night shifts) are almost always handled this way. Many companies treat rest breaks the same

**TABLE 2** Typical Causes of Interruptions in the Workplace

| Causes of Work-Related Interruptions | Reasons for Nonwork Related Interruptions |
|---|---|
| Machine breakdowns or malfunctions | Personal needs (e.g., restroom interruption) |
| Waiting for materials, parts, or other items necessary to proceed with work | Talking to coworkers about matters unrelated to work |
| Receiving instructions from foreman | Lunch break |
| Talking to coworkers about job-related matters | Smoke break |
| Waiting at a tool crib for the attendant to deliver a tool | Beverage break |
| Rest breaks to overcome fatigue for physiologically demanding work | Personal telephone call |
| Cleaning up at the end of the shift | |

way. There is a specified rest break in the morning and one in the afternoon. The duration of these breaks is typically 5 to 15 minutes. All workers take their breaks during these specified times, and the workers are paid during these breaks.

In this approach, the shift is scheduled with breaks to divide the work day into periods of roughly equal length. For example, the day shift begins at 8:00 A.M. and ends at 4:30 P.M. There is a 15-minute rest break in midmorning, a lunch break from noon to 12:30 P.M., and another 15-minute rest break in midafternoon. Accordingly, the shift lasts 8.5 hours, which includes 8 hours of paid time (clock time).

**The PFD Allowance.** In this approach, an appropriate value of the ***personal time, fatigue, and delay*** (PFD) allowance factor ($A_{pfd}$) is determined for use in converting the normal time into the standard time. It results in extra time prorated in the time standard to compensate for the interruptions experienced by the worker during the shift, both work related and non-work related. Some companies include rest breaks in the allowance rather than using a scheduled break time for everyone as described above.

***Personal time*** includes restroom breaks, phone calls, water fountain visits, and similar interruptions of a personal nature. A typical allowance for this category of interruption is 5%. A larger value would be appropriate if the work environment is hot and uncomfortable, and a lower value would be used for very favorable working conditions.

The ***fatigue*** or ***rest*** allowance is intended to compensate the worker for time that must be taken to overcome fatigue due to work-related stresses and conditions. For example, if the work is physiologically very demanding and the worker expends significant metabolic energy in performing it, then relaxation time should be allowed periodically for the body to recover. Some of the factors that cause fatigue in human workers are listed in Table 3. The allowance for fatigue for a given task is often determined by a combination of (1) the rest formulas and (2) negotiation between labor and management. The rest formulas provide a more or less scientific basis to make the determination. In practice, rest allowances are often the result of negotiated agreements between workers, perhaps represented by their union, and company management. An allowance of 5% is typical for fatigue in light and medium work. For heavy work it may be much higher, perhaps 20% or more.

***Delays*** are unavoidable interruptions from work that occur at random times during the day. They usually refer to work-related events, such as, some of the interruptions listed in Table 2. It should be mentioned that work delays associated with a worker's personal needs (e.g., a sudden urge for a cigarette break) should not be counted in this

**TABLE 3** Factors Causing Fatigue in Workers

| Physical Factors | Mental and Cognitive Factors | Environmental and Work Factors |
|---|---|---|
| Standing | Concentration and attention | Poor lighting |
| Abnormal body position | Mental strain | Noise |
| Use of force | Monotony and tediousness | Fumes |
| Expenditure of muscular energy | Eyestrain | Heat |
| | | Atmospheric conditions |

*Source:* Adapted from a publication by the International Labour Office, *Introduction to Work Study* (1964) and other sources, as cited in [1].

category of interruption, since they are already included in the personal needs category. The delay category is generally reserved for delays that are ultimately the responsibility of management. For example, a machine breakdown may seem like a random occurrence, but management is responsible for the maintenance of the machine so that breakdowns are avoided. Other examples include waiting for parts or tools, waiting for an elevator, and waiting for the foreman's instructions. In general, these kinds of work-related delays must be controlled by company management. Certainly the worker is not responsible.

Determining the delay allowance is more challenging than determining allowances for personal needs or fatigue. There are traditions and/or formulas, often adjusted by negotiations, for determining these other allowances. But delays are usually random events that vary from day to day, and determining an allowance factor to account for them boils down to a problem of collecting data in the workplace so that an average lost-time value can be found. Two techniques are used to collect the data: (1) intensive observation over several days and (2) work sampling. Of the two methods, work sampling is usually preferred for several reasons: (1) it can be conducted over a longer period of time (e.g., weeks instead of days), so the plant activity observed is more likely to represent normal operations; (2) it can include a much larger scope of plant operations, because the observer is not limited to one location; (3) it does not require continuous observation, so it is more comfortable for workers and observer; and (4) statistical measures can be calculated to assess the accuracy of the results derived from the data.

The three allowances for personal needs, fatigue, and delays are then combined into a single allowance factor, which we symbolize $A_{pfd}$.

### 3.2 Other Types of Allowances

In addition to the PFD allowance factor, there are other reasons for adding allowances to the standard time of a task. They are not as common as the PFD allowance, but in some work situations they are quite appropriate. These other allowances are applied in addition to $A_{pfd}$, not as a substitute for it.

**Contingency Allowances.** The contingencies for which these allowances are provided are usually because of some kind of problem with the task or the production equipment used to perform it. Some examples of these problems are listed in Table 4. The

**TABLE 4** Examples of Problems for Which Contingency Allowances Might Be Provided

| Problem Area | Problems and Examples |
|---|---|
| Materials or parts | Starting materials or parts are out of specification, and extra time is needed to correct the nonconformance (e.g., oversized casting that requires an extra machining pass or slower feed rate). |
| Process | The manufacturing process is not in statistical control (too much variability), and additional time is required to inspect every piece rather than inspect on a sampling basis. |
| Equipment | Equipment is malfunctioning or breaking down more frequently than what is provided by the unavoidable delay factor, and additional time is needed to compensate the worker to make adjustments, lubricate the machine more frequently, or other extra task(s) not included in the standard time. |

problems are beyond what is covered by the regular PFD allowance, and so an additional allowance is provided. Contingency allowances should not be greater than 5% [1]. The problems that contingency allowances are intended to address should not occur in the first place, but if they occur, the problem itself should be solved. Contingency allowances may be needed on a temporary basis to keep production going, but the long-term solution is to fix the underlying problem.

**Policy Allowances.** These allowances are intended to cover special work situations that are usually associated with a wage incentive system. An example is the ***machine allowance*** that is added to the machine-paced portion of a work cycle in the operation of a worker-machine system. The machine allowance provides an opportunity for the worker to maintain a high rate of earnings even though he or she has control over only a portion of the cycle. It is company policy that the worker be given this opportunity. The application of the machine allowance to compute the standard time is as follows:

$$T_{std} = T_{nw}\,(1 + A_{pfd}) + T_m\,(1 + A_m) \tag{3}$$

where $T_{nw}$ = normal time of the worker during the worker-controlled portion of the cycle, min; $A_{pfd}$ = PFD allowance; $T_m$ = machine cycle time, min; and $A_m$ = machine allowance. If company policy does not recognize the need for a separate machine allowance when computing the time standard, then either $A_m = 0$ or it is set equal to the value of the PFD allowance. The following example illustrates the way the worker benefits from a machine allowance.

**Example 1 Use of Machine Allowance in a Wage Incentive Plan**

A wage incentive plan pays workers a daily wage at a rate of \$15.00/hr multiplied by the number of standard hours accomplished during the shift. One worker-machine task in the plant includes worker-paced elements totaling a normal time of 1.00 min and machine-paced elements with a time of 3.00 min. The PFD allowance is 15%. Determine the standard time for the task given that (a) $A_m = 0$ and (b) $A_m = 30\%$. (c) What does a worker earn for the day under each policy if he or she produces 115 parts that day?

**Solution:** **(a)** $T_{std} = 1.00\,(1 + 0.15) + 3.00\,(1 + 0) = 1.15 + 3.00 = 4.15$ min.

**(b)** $T_{std} = 1.00\,(1 + 0.15) + 3.00\,(1 + 0.30) = 1.15 + 3.90 = 5.05$ min.

**(c)** For the standard of 4.15 min, the number of standard hours is $H_{std} = 115(4.15)/60 = 7.95$ hr and the worker's earnings would be \$15.00 (7.95) = \$119.25.[6]

For $T_{std} = 5.05$ min, $H_{std} = 115(5.05)/60 = 9.68$ hr and the worker's earnings would be \$15.00(9.68) = \$145.19.

**Comment:** Which is fair? A PFD allowance of 15% equates to a lost time per day of 62.6 min in an 8-hour shift, which leaves an actual working time of 417.4 min. Of this time, machine time took 115 cycles x 3.00 min/cycle = 345 min, which leaves 417.4 – 345 = 72.4

[6]Most incentive plans pay a guaranteed base rate, so the worker's wage for the day would be 8(\$15.00) = \$120.00.

min for worker time. Given that 115 cycles were completed by the worker, his performance during the worker-paced portion of the cycle must have been 115(1.00)/72.4 = 1.59 = 159%.

Certainly this level of performance would seem to merit some bonus above the base rate of \$15/hr. The standard time in (a), $T_{std}$ = 4.15 min, does not even pay the worker \$15/hr (unless there is a guaranteed base rate, see footnote). The standard time in (b), $T_{std}$ = 5.05 min, does earn a respectable bonus of \$30.19. The worker's efficiency is 9.68/8.0 = 1.21 or 121%.

The other side of the argument is that the worker has free time for 345 min during the shift (because this is machine cycle time), so why should he or she be paid a premium bonus for the entire shift for only working 72.4 min of total shift time? This is a difficult issue, one on which common interest can often be found by both labor and management in keeping the machinery operating as high a proportion of time as possible (a win-win situation for both labor and management). Labor wins by earning higher bonuses. Management wins by using the equipment more efficiently. ■

Other types of policy allowances include ***training allowances*** for workers whose responsibilities include teaching other workers their jobs, and ***learning allowances*** for workers who are learning a new job or new employees who are just beginning work. In nearly all of the situations covered by policy allowances, a worker's earnings would be adversely affected if the allowances were not applied. Workers would be reluctant to train others or to learn new jobs unless some form of compensation were provided to cover the losses in earnings they would otherwise suffer for doing so. If a wage incentive plan is not used, there is less reason to have policy allowances since the worker's wage is not affected. The reason they are called policy allowances is because they are based on company policy, rather than any time interruptions actually experienced during the work shift.

## 4 ACCURACY, PRECISION, AND APPLICATION SPEED RATIO IN WORK MEASUREMENT

Work measurement is a measurement process. There are similarities and differences between the measurement of work and the measurement of physical variables in science and technology. Let us begin by defining measurement as it is applied in the physical sciences. ***Measurement*** is a procedure in which an unknown quantity is compared to a known standard, using an accepted and consistent system of units. The measurement provides a numerical value of the quantity of interest, within certain limits of accuracy and precision. All measurement systems are based on seven basic physical quantities: (1) length, (2) mass, (3) time, (4) electrical current, (5) temperature, (6) luminous intensity, and (7) matter. All other physical quantities (e.g., area, volume, velocity, force, electrical voltage) are derived from these seven basic quantities.

Time is one of the seven basic physical quantities, and time is the common quantity of interest in the measurement of work. The standard unit for time is the second.[7] Although time standards in work measurement often use alternative units (e.g., minutes, hours), they all can be related back to seconds.

[7]The ***second*** is defined in the System International as the "duration of 9,192,631,770 cycles of the radiation associated with a change in energy level of the cesium atom."

Three important attributes of a measurement system are accuracy, precision, and speed of response. These attributes are discussed in the following sections, and comparisons are made between the measurement of quantities in science and the measurement of work.

### 4.1 Accuracy and Precision

Measurement ***accuracy*** is the degree to which the measured value agrees with the true value of the quantity of interest. The measurement system is accurate if it is free of ***systematic errors***, which are positive or negative deviations from the true value that are consistent from measurement to measurement.

Measurement ***precision*** refers to the repeatability of the measurement system. High precision means that the random errors in the measurement procedure are small. It is generally assumed that random errors follow a normal statistical distribution with mean ($\mu$) equal to zero and standard deviation ($\sigma$) indicating the amount of dispersion in the measured data. The normal distribution has well-defined properties, one of which is that 99.73% of the population is included within $\pm 3\sigma$, and this is often used as a quantitative indicator of the measurement system's precision.

The distinction between accuracy and precision is depicted in the three normal distributions of Figure 6. In distribution (a), the random errors in the measurement are large (large variance), which indicates that the precision is low, but the mean value agrees with the true value of the variable of interest, which indicates perfect accuracy. In (b), the random errors are low (high precision), but the measured value differs from the true value (low accuracy). In (c), we have high accuracy and high precision.

Accuracy and precision are important in work measurement, but these terms that are so readily applied to physical measurements are fraught with difficulty when applied to the measurement of work, especially direct time study. Accuracy is concerned with closeness to the true value, but how can that definition be applied to the time that should be allowed a worker to perform a given task? What is the true value of a task time? Measurement is a procedure in which an unknown quantity is compared to a known standard. In work measurement, there is no way to make that comparison because the closest that we can come to a known standard is in the minds of the time study analysts.

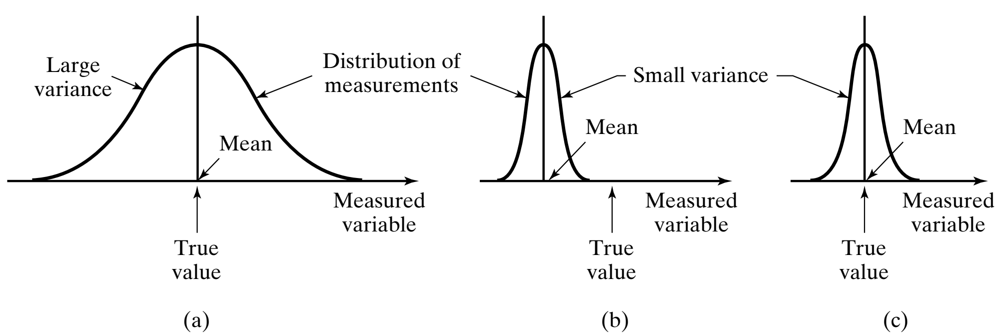



**Figure 6** Accuracy versus precision in measurement: (a) high accuracy but low precision, (b) low accuracy but high precision, and (c) high accuracy and high precision.

And that standard is not based on time; it is based on a definition of performance (the organization's definition of standard performance). The issue becomes more confounded in direct time study because of performance rating. The time study analyst records the observed time of a given work element and then applies a performance rating to that time in order to align the observed time with the analyst's concept of standard performance.[8] We must conclude that accuracy in work measurement is elusive because it is impossible to scientifically determine the true value of the quantity of interest.

Although there are difficulties in determining accuracy in work measurement, the precision of the time standard resulting from a work measurement study can be determined, and this is sometimes used as a pseudonym for accuracy. The precision of the time standard is determined at a certain reliability level. For example, we state that the standard time for the task is 4.00 min, and that we are 95% certain that the actual time is within $\pm$ 5% of that time. In other words, in 95 out of 100 time studies performed on the task, the resulting standard time values will lie between 3.80 min and 4.20 min. This is analogous to confidence intervals in statistics, and we make use of them in Chapter 13 to determine how many cycles must be observed in direct time study in order to achieve a known level of reliability and precision in the time standard. The term ***consistency*** also applies here, and it refers to the variations in standard time values among different time study analysts who study the same task. Good consistency means small variations in the resulting time standards by different analysts. The term ***precision*** refers to the expected variability within a single time study or the time standard resulting from that study.

The relative accuracies of the various methods of determining time standards are of interest. These are summarized in Figure 7. Estimation is likely to be the least accurate because it is not based on any actual data or measurement. It relies solely on the estimator's judgment, which may not be very good. By comparison, historical records provide hard data on the average time required for the previous production order, and that is generally an improvement over judgment. Nevertheless, the accuracy of the record is sometimes questionable. Were the parts counted correctly, or did workers inflate

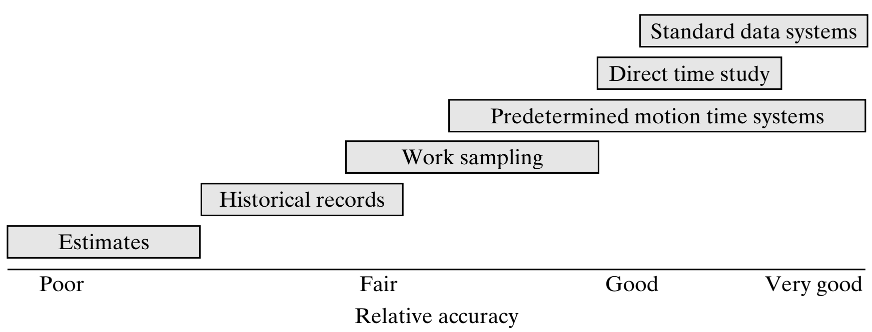



**Figure 7** Methods of determining time standards ranked by relative accuracy.

[8]Mitchell Fein, a leading industrial engineering consultant for many years, made the following observation about direct time study: "No other measurement system which tests data for statistical validity alters the observed data" [4].

numbers to make themselves look more productive? Did the time sheet provide an accurate accounting of time spent on the particular order, or was other work done and charged to this order? How long did the setup take for this order? Although historical records are likely to be more accurate than pure estimation, they suffer from the fact that they are not based on a measurement of the actual work; instead, they are based on measures of the work output and the time that was recorded for that output.

In terms of relative accuracy, the work measurement techniques are more accurate than estimation or historical records. Of the four work measurement techniques, work sampling is not as accurate for determining time standards as the other three techniques. This due to (1) the statistical errors that result from the sampling process and (2) the absence of any attempt to analyze and improve the method used in the task.

### 4.2 Application Speed Ratio

Accuracy and precision are not the only criteria in selecting a work measurement technique. The amount of time required to perform the work measurement procedure is also important. In other measurements, the ***speed of response*** refers to the lag between the time when the measuring device is applied and when the measured value is available to the user. We would like to obtain our measurement of the variable of interest as quickly as possible, whether we are measuring a physical variable in science or a task time in a work situation.

In work measurement, the related term is ***application speed ratio***, which is the ratio of the time required to determine a time standard relative to the standard time itself. For example, an application speed ratio of 100 means that it takes 100 min of analyst time to determine a 1.0-min time standard. In general there is an inverse relationship between application speed ratio and "accuracy." Methods for setting time standards that can be applied quickly tend to be less accurate than those that take more time to apply.

The time required to perform any of the work measurement techniques is significantly greater than the time required to estimate a task time or to compute the task time from historical records. If accuracy of the task time is not important, or if the work consists of a variety of "odd jobs," where it would be impractical to set a time standard for each job, then estimation or historical records are preferable as a way of developing the time values. Figure 8 indicates the relative application speeds of the various methods for setting time standards. The PMT systems have a large range of application speeds

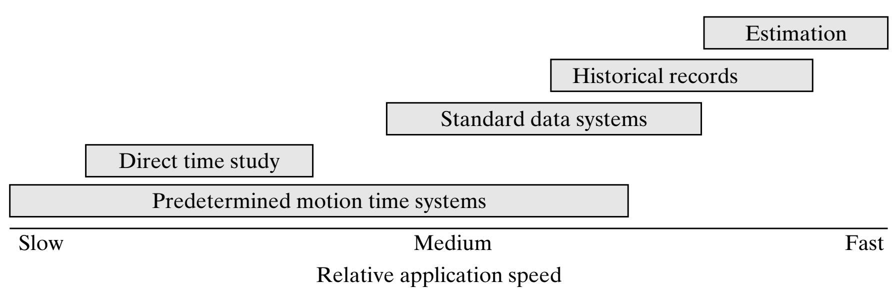



**Figure 8** Methods of determining time standards ranked by relative application speed.

because of the variety of systems available, some of which are very accurate but very time consuming, while others are less accurate and take less time. The application speeds for standard data systems and historical records are relatively fast, but these systems require a substantial investment of time to develop before they are operational. Work sampling is omitted from the chart because it requires an extended period of data collection before standards can be determined.

## REFERENCES

[1] Allerton, L. J. "Allowances." Pp. 5.101–119 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[2] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley, 1980.

[3] Bishop, G. "Purpose and Justification of Engineered Labor Standards." Pp. 5.23–40 in *Maynard's Industrial Engineering Handbook*, 5th ed., K. Zandin. New York: McGraw-Hill, 2001.

[4] Fein, M. "How 'Reliability,' 'Precision,' and 'Accuracy' Refer to Use of Work Measurement Data." Pp. 30–37 in *Work Measurement, Principles and Practice*, edited by R. L. Shell. Norcross, GA: Industrial Engineering and Management Press, 1986.

[5] Hancock, W. M., and Bayha, F. H. "The Learning Curve." Pp. 1585–98 in *Handbook of Industrial Engineering*, 2nd ed., edited by Gavrial Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 1992.

[7] Konz, S. "Time Standards." Pp. 1391–408 in *Handbook of Industrial Engineering*, 3rd ed., edited by Gavrial Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 2001.

[8] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.

[9] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 10th ed. New York: McGraw-Hill, 1999.

[10] Panico, J. A. "Work Standards: Establishment, Documentation, Usage, and Maintenance." Pp. 1549–84 in *Handbook of Industrial Engineering*, 2nd ed., edited by Gavrial Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 1992.

[11] Sellie, C. N. "Predetermined Motion-Time Systems and the Development and Use of Standard Data." Pp. 1639–98 in *Handbook of Industrial Engineering*, 2nd ed., edited by Gavrial Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 1992.

[12] Smith, G. S., "Developing Engineered Labor Standards." Pp. 5.73–100. *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill 2001.

[13] Zandin, K. B., *MOST Work Measurement Systems*. 3rd ed. New York: Marcel Dekker, 2003.

## REVIEW QUESTIONS

**1** What is the standard time for a task? Provide a definition.

**2** What are some of the characteristics typical of industrial situations in which time standards would be beneficial?

**3** What are the functions and applications of accurately established time standards in an organization?

**4** Identify the three basic methods to determine time standards.

**5** What are the four basic work measurement techniques?

**6** What is the difference between the normal time and the standard time for a task?

**7** How does computerized work measurement reduce the time and effort of the time study analyst?

**8** What are the prerequisites for valid time standards?

**9** Identify the two common benchmarks of standard performance that are often used.

**10** What are the details that should be included when defining the standard method?

**11** What is a PFD allowance in time standards?

**12** What are some of the reasons why workers experience lost time during a work shift?

**13** What is a contingency allowance?

**14** What is the difference between accuracy and precision in a measurement system?

**15** Why is accuracy an elusive quality in work measurement?

**16** What is precision in work measurement?

**17** What is consistency in work measurement?

**18** What is the application speed ratio in work measurement?

## PROBLEMS

### Determining Time Standards

**1** The average observed time for a repetitive work cycle in a direct time study was 3.27 min. The worker was performance rated by an analyst at 90%. The company uses a PFD allowance factor of 13%. What is the standard time for this task?

**2** A time study analyst observed a repetitive work cycle performed by a worker and then studied the task using a predetermined motion time system. During the observation, the analyst made a record of the work elements. He also made a note of the worker's performance and rated it as 85%. Back in the office, the analyst listed the basic motion elements for the task and retrieved from his computer the corresponding normal time values. The sum of the normal times for the basic motion elements was 1.38 min. The company uses a PFD allowance factor of 15%. Determine the standard time for the task.

**3** The ABC Company uses a standard data system to set time standards. One of the time study analysts listed the three work elements for a new task to be performed in the shop and then determined the normal time values to be 0.73 min, 2.56 min, and 1.01 min. The company uses a PFD allowance factor of 16%. Determine the standard time for the task.

### Allowances

**4** In the ABC machine shop, workers punch in at 8:00 A.M. and punch out at 4:30 P.M. The labor-management agreement allows 30 min for lunch, which is not counted as part of the 8-hour shift. In determining the allowance for computing time standards, two 15 min breaks are provided (personal time and fatigue), one in the morning and one in the afternoon; and 20 min have been negotiated as lost time due to supervisor interruptions and equipment malfunctions. What PFD allowance factor should be added to the normalized time to account for these losses in the computation of a standard time, so that if workers work at standard performance, they will produce exactly 8 standard hours?

**5** In the WS&FP plant, workers punch in at 8:00 A.M. and punch out at 5:00 P.M. The labor-management agreement allows 1 hour for lunch, which is not counted as part of the 8-hour shift. In determining the allowance for computing time standards, two 12 min breaks are included (personal time and fatigue), one in the morning and one in the afternoon; and 35 min are included as lost time due to interruptions and delays. What PFD allowance factor should be added to the normalized time to account for these losses in the computation of a standard time, so that if workers work at standard performance, they will earn exactly 8 standard hours?

**6** Determine the PFD allowance to be used for computing time standards in the following situation. Second shift workers punch in at 3:30 P.M. and punch out at 12:00 midnight. They are provided 30 min for supper at 6:00 P.M., which is not counted as part of the 8-hour shift. For purposes of determining the allowance, 30 min of break time (personal time and fatigue) are allowed each worker. In addition, the plant allows 35 min for lost time due to unavoidable delays. What should the PFD allowance factor be?

**7** The work shift at the ABC Company runs from 7:30 A.M. to 4:15 P.M. with a 45 min break for lunch from 11:30 to 12:15 P.M. that does not count as part of the work shift (workers are not paid for this time). The company provides two 12-min rest breaks during working hours (paid time), one in the morning and one in the afternoon. The company also allows 25 min per day for personal needs (paid time). In addition, a work sampling study has shown that on average, unavoidable delays in the plant result in 20 min lost time per worker per day (paid time). Determine the PFD allowance factor for the following two management policies on allowances: (a) the two 12-min breaks are both scheduled breaks that all workers take at the same time and (b) the two 12-min breaks are included in the allowance factor so that workers can take their breaks whenever they please.

# Direct Time Study



***Direct time study***, also known as ***stopwatch time study,*** involves the direct and continuous observation of a task using a stopwatch or other timekeeping device to record the time taken to accomplish a task.[1] While observing and recording the time, an appraisal of the worker's performance level is made. These data are then used to compute a standard time for the task, adding an allowance for personal time, fatigue, and delays.

Direct time study was the first work measurement technique to be used, dating back to around 1883. It is inextricably connected with the origins of industrial engineering. Computerized techniques have improved the accuracy and application speed of direct time study as well as the database management functions that support it.

The use of direct time study is most appropriate for tasks that involve a repetitive work cycle, at least a portion of which is manual. This kind of work is common in

[1]The Work Measurement and Methods Standards Committee (ANSI Standard Z94.11-1989) defines time study as follows: "A work measurement technique consisting of careful time measurement of the task with a time measuring instrument, adjusted for any observed variation from normal effort or pace and to allow adequate time for such items as foreign elements, unavoidable or machine delays, rest to overcome fatigue, and personal needs. Learning or progress effects may also be considered. If the task is of sufficient length, it is normally broken down into short, relatively homogeneous work elements, each of which is treated separately as well as in combination with the rest."

batch and mass manufacturing. Also, standards for routine office work can often be established using direct time study. Performing a direct time study is time-consuming and is therefore best justified when the job will have a relatively long production run, and/or there will be repeated orders in the future. A limitation of this work measurement technique is that it cannot be used to set a time standard prior to the start of production.

## 1 DIRECT TIME STUDY PROCEDURE

The procedure for determining the standard time for a task using direct time study can be summarized in the following five steps:

1. Define and document the standard method.
2. Divide the task into work elements.
3. Time the work elements to obtain the observed time for the task.
4. Evaluate the worker's pace relative to standard performance, a procedure called ***performance rating***. This is used to determine the normal time. Steps 3 and 4 are accomplished simultaneously.
5. Apply an allowance to the normal time to compute the standard time.

Steps 1 and 2 are preliminary steps before actual timing begins, during which the analyst becomes familiar with the task and attempts to improve the work procedure before defining the standard method. In steps 3 and 4, several work cycles are timed, each one performance rated independently. Finally, the values collected in steps 3 and 4 are averaged to determine the normalized time. An appropriate allowance factor for the kind of work involved is then added to compute the standard time for the task. Let us examine each of the steps in more detail.

### 1.1 Define and Document the Standard Method

Before defining and documenting the standard method, a methods engineering study should be undertaken to ensure that the standard method obeys the "one best method" principle—the best method that can be devised under the present economic and technological circumstances. All of the steps in the method should be defined. Any special tools, gauges, or equipment that can improve the task should be designed and included in the method. If there are irregular elements in the work cycle, the frequency with which these elements are to be performed should be stated explicitly. If the labor-management climate in the facility allows, the worker's advice and opinion should be sought in developing the standard method. Once the standard method has been defined, it should be difficult or impossible for the operator to make further improvements.

The standard method should be thoroughly documented. The company should have forms and/or checklists to make certain that all information about the method is included. An example of a methods description form is shown in Figure 1. The documentation should enumerate details about the procedure (hand and body motions), tools, equipment and the machine settings used for the equipment (e.g., feeds and speeds

| Date | **Standard Method Description for Direct Time Study** | Page of |
|---|---|---|

| Operation | Dept. | Part No. |
|---|---|---|
| Machine | Analyst | |

| Methods Improvements (check if implemented) | Sketch of Workplace: |
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| Work Element No. and Description with Machine Parameters for Machine Cycles | Freq. | Tools and Gauges |
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| Additional Notes |
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| |

**Figure 1** Form for documenting the standard method.

on machine tools), workplace layout, irregular work elements, working conditions, and setup. If there is a work unit associated with the task, then it should also be specified, both its starting and its completed conditions. A videotape (or other video medium) of the standard method can also be made as part of the documentation. This is especially helpful for complex tasks in which details of the method are difficult to explain in writing.

There are several reasons why thorough documentation of the standard method is important:

- *Batch production*. If the task is associated with batch production, then it is likely to be repeated at some time in the future. The time lapse may be significant between the previous batch and the next batch. A different worker may be assigned to perform the task on the next batch. The statement of the standard method provides the worker and the foreman with a complete description of the task and the procedure for doing it, as well as any tooling or equipment needed. This avoids "reinventing the wheel."
- *Methods improvement by the operator*. At some future time, the operator may discover a way to improve the method.[2] A question then arises: Should the methods improvement be incorporated into a new standard method (which would require a change in the standard time), or should the operator be allowed to benefit from the improvement without formally changing the standard method? The typical labor-management contract only allows for retiming the task if it can be demonstrated that a real change in the method has been made. If the operator has indeed made a methods improvement, this can be identified by comparing the method in use with the standard method description.
- *Disputes about the method*. If the operator complains that the standard for the task is too tight, or there is some other reason for a dispute about the standard method, documentation of the standard method can be used to settle the dispute.[3] Perhaps the operator is using a less efficient method than the standard method, or a certain hand tool that should be used in the operation has been neglected. These problems can be addressed by reference to the standard method documentation.
- *Data for standard data system*. Time standards developed by direct time study are sometimes used in standard data systems. Good documentation of the standard method, especially regarding the work elements and associated normal times, is essential in developing the database for a standard data system.

[2]This possibility seems at odds with the fact that the one best method has already been devised and has been defined as the standard method. However, the intelligence and ingenuity of the worker should not be underestimated. It must be acknowledged that the worker may be able to figure out some motion shortcut or hand tool or other gimmick that reduces the time for the work cycle, even though the work cycle has been subjected to a thorough methods study.

[3]A "tight" standard means that the standard time is too short, making it difficult or impossible to accomplish the task within the time standard. A "loose" standard is one in which the time is relatively long, making it easy to achieve a high worker efficiency.

### 1.2 Divide the Task into Work Elements

Any task can be divided into work elements. A ***work element*** is a series of motion activities that are logically grouped together because they have a unified purpose in the task. The description of the standard method can be organized into work elements suitable for use in direct time study. Indeed, the most natural way to describe the standard method is often as a list of work elements. Some practical guidelines for defining work elements in direct time study are presented in Table 1. An important reason for defining the

**TABLE 1** Guidelines for Defining the Work Elements in Direct Time Study

| Guideline | Explanation and Examples |
|---|---|
| Each work element should consist of a logical group of motion elements. | The work element should have a unified purpose, such as reaching for an object and moving it to a new location (e.g., reach, grasp, move, and place). There would be no purpose in separating the reach from the move motions since they both involve the same object. |
| Beginning point of one element should be end point of preceding element. | There should be no gap between one element and the next in the task sequence. Otherwise, the time of the gap is omitted from the recorded total time. |
| Each element should have a readily identifiable end point. | A readily identifiable end point can be easily detected during the study. It can often be anticipated to allow reading of the watch more conveniently.<br>An audible sound, such as the actuation of a pneumatic device, provides a readily identifiable end point. |
| Work elements should not be too long. | If a work element is very long (i.e., several minutes), it should probably be divided into multiple elements that are timed separately. Machine semiautomatic cycle time is an exception. Some machine cycles can take several minutes and should be identified as one element. |
| Work elements should not be too short. | A practical lower limit in direct time study is around 3 sec. Below this, reading accuracy may suffer.<br>If a video camera is used for timing purposes, shorter elements may be possible. |
| Irregular work elements should be identified and distinguished from regular elements. | Irregular elements are work elements that do not occur every cycle.<br>The frequency with which they should be performed must be noted.<br>The time(s) for the irregular element(s) are prorated across the regular work cycle when the standard time is computed. |
| Manual elements should be separated from machine elements. | Manual elements depend on the operator's performance (pace) and therefore vary over time.<br>Machine elements are generally constant values that depend on machine settings. Once the settings are established, the actuation time shows no perceptible variation. |
| Internal elements should be separated from external elements. | Internal elements are performed by the operator during the machine cycle. In most cases, they do not affect the overall work cycle time.<br>External elements are performed outside of the machine cycle. They contribute to the overall work cycle time. |

*Source:* Adapted from guidelines in [13] and other sources.

work elements is that the worker may exhibit different performance levels on different elements. Accordingly, these performance levels are rated and recorded separately by the time study analyst.

### 1.3 Time the Work Elements

Once the work elements have been defined, the analyst is ready to collect data. The time data are usually recorded on a time study form, similar to the one shown in Figure 2. Space is provided for a listing of the work elements, which can be referenced to the more complete standard method documentation. Each element should be timed over several cycles to obtain a reliable average, and the form is designed for recording multiple cycles of the task. The appropriate number of cycles can be determined using the statistical techniques described in Section 2. For the convenience of the analyst, the time study form is usually held in a special clipboard that also holds the stopwatch used in the study.

There are several pieces of equipment that can be used to record the times for the work cycles. We describe the products in Section 4. The traditional instrument in direct time study is the stopwatch, which is usually calibrated in decimal minutes. There are two principal methods for using a stopwatch in direct time study: (1) snapback timing method and (2) continuous timing method. In the ***snapback timing method***, the watch is started at the beginning of every work element by snapping it back to zero at the end of the previous element. The reader must therefore note and record the final time for that element just as the watch is being zeroed. (Most stopwatches, especially electronic watches, have features that facilitate this reading.) In the ***continuous timing method***, the watch is zeroed at the beginning of the first cycle and allowed to run continuously throughout the duration of the study. The analyst records the running time on the stopwatch at the end of each respective element. Some analysts prefer to adapt the continuous method by zeroing at the beginning of each work cycle, so that the starting time of any given work cycle is always zero. This facilitates cycle-to-cycle comparisons during the study.

There are two advantages to the snapback method: (1) the analyst can readily see how the element times vary from one cycle to the next, and (2) no subtraction is necessary, as in the continuous timing method, to obtain individual element times. The advantages of the continuous method include the following: (1) when the clock is continuously running, elements are not as easily omitted by mistake, (2) regular and irregular elements can be more readily distinguished, and (3) not as much manipulation of the stopwatch is required as in the snapback method.

### 1.4 Rate the Worker's Performance

While observing and recording the time data, the analyst must simultaneously observe the performance of the worker and rate this performance relative to the definition of standard performance used by the organization. Other terms for performance include pace, speed, effort, and tempo. Standard performance is given a rating of 100%. A performance rating greater than 100% means that the worker's performance is better than standard (which results in a shorter observed work cycle time), and less than 100% means poorer performance than standard (and a longer observed time).

| Date | Direct Time Study Observation Form | | Page of |
|---|---|---|---|
| Operation | | Dept. | Part No. |
| Machine | | | Tooling |
| Worker | | | Worker No. |
| Analyst | Start Time | Finish Time | Elapsed Time |

| Work Elements, Machine Settings, and Observations | | | | Cycle No. (regular elements) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Element Number and Description | Feed | Speed | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | Avg $T_n$ |
| 1 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 2 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 3 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 4 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 5 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 6 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 7 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| 8 | | | $T_{obs}$ | | | | | | | | | | | |
| | | | $PR$ | | | | | | | | | | | |
| | | | $T_n$ | | | | | | | | | | | |
| Normal time = Sum of $T_n$ (regular work elements) | | | | | | | | | | | | | | |

| Irregular Element and Description | Freq | $T_0$ | $T_f$ | $PR$ | $T_n$ | | **Calculation of Standard Time $T_{std}$** | |
|---|---|---|---|---|---|---|---|---|
| A | | | | | | | Sum of $T_n$ (regular work elements) | |
| B | | | | | | | Sum of freq x $T_n$ (irregular elements) | |
| C | | | | | | | Total $T_n$ per cycle | |
| D | | | | | | | PFD allowance $A_{pfd}$ | |
| E | | | | | | | Standard time $T_{std} = T_n(1 + A_{pfd})$ | |
| Additional Notes | | | | | | | | |

**Figure 2** Direct time study form.

The observed time is subsequently multiplied by the performance rating to obtain the ***normal time*** (other names include ***normalized time*** and ***base time***) for the element or cycle. The calculation is summarized in the following equation:

$$T_n = T_{obs}(PR) \tag{1}$$

where $T_n$ = normal time, min; $T_{obs}$ = observed time, min; and $PR$ = performance rating, usually expressed as a percentage but used in the equation as a decimal fraction. The symbols $T_n$ and $T_{obs}$ can be used to represent individual work elements or the entire work cycle, depending on how the data are taken and recorded.

Performance rating is the most difficult and controversial step in direct time study. The reason is that it requires the judgment of the analyst to assess the value of $PR$. The analyst's judgment of standard performance may differ from that of the worker who is being observed. It is in the worker's interest and advantage to be rated at a high performance level during the study, because that will mean that the normal time and ultimately the standard time for the task will be longer (resulting in a looser standard). Thus, it will be easier for the worker to achieve a higher efficiency level as the job continues. This is especially important to the worker if he or she is paid on a wage incentive plan. We consider performance rating and the issues surrounding it in Section 3.

### 1.5 Apply Allowance to Compute Standard Time

To obtain the standard time for the task, a PFD allowance is added to the normal time, as calculated in the following equation:

$$T_{std} = T_n(1 + A_{pfd}) \tag{2}$$

where $T_{std}$ = standard time, min; $T_n$ = normal time, min; and $A_{pfd}$ = allowance factor for personal time, fatigue, and delays. This is often expressed as a percentage but used as a decimal fraction in our equation. The function of the allowance factor is to inflate the value of the standard time relative to the normal time in order to account for the various reasons why the operator loses time during the work shift. The allowance factor represents an average for the type of work, equipment, and conditions under which the operator works. Some days the worker may lose more time, other days less time, than what is provided by the allowance factor. In the long run it is intended to average out to a realistic value.

**Example 1 Determining a Standard Time for Pure Manual Work**

A direct time study was taken on a manual work cycle using the snapback timing method. The regular work cycle consisted of three elements, identified as *a*, *b*, and *c* in the following table. Element *d* is an irregular element performed every 5 cycles. Observed times and performance ratings of the elements are also given. Determine (a) the normal time and (b) the standard time for the work cycle, using an allowance factor of 15%.

| Work Element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time | 0.56 min | 0.25 min | 0.50 min | 1.10 min |
| Performance rating | 100% | 80% | 110% | 100% |

**Solution** **(a)** The normal time for the cycle is obtained by multiplying the observed element times by their respective performance ratings and summing. In the case of element *d*, this normal time is prorated over 5 cycles.

$$T_n = 0.56(1.00) + 0.25(0.80) + 0.50(1.10) + 1.10(1.00)/5$$
$$= 0.56 + 0.20 + 0.55 + 0.22 = 1.53 \text{ min}$$

**(b)** The standard time is computed by adding the allowance.

$$T_{std} = 1.53(1 + 0.15) = 1.76 \text{ min.}$$ ■

If the task includes a machine cycle, then company policy may provide for a machine allowance factor to be included in the standard time computation. The standard time equation becomes

$$T_{std} = T_{nw}(1+A_{pfd})+T_m(1+A_m) \quad (3)$$

where $T_{nw}$ = normal time of the worker external elements, min; $T_m$ = machine cycle time, min; $A_m$ = machine allowance factor; and the other terms have the same meanings as before. If the company does not include a separate machine allowance, then $A_m = 0$ in the equation, or it uses the regular allowance $A_{pfd}$. Equation (3) assumes that there are no internal elements in the cycle. If the work cycle includes internal elements, then it must be determined whether the sum of the worker internal elements or the machine cycle time is larger in order to determine the normal time and the standard time.

**Example 2 Determining a Standard Time for a Task That Includes a Machine Cycle**

The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements *a*, *b*, *c*, and *d* are performed by the operator, and element *m* is a machine semi-automatic cycle. Element *b* is an internal element performed simultaneously with element *m*, and element *d* is an irregular element performed once every 15 cycles. Observed times and performance ratings are given in the table below. The PFD allowance factor is 15%, and the machine allowance is 20%. Determine (a) the normal time and (b) the standard time for the work cycle.

| Worker element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time, manual | 0.22 min | 0.65 min | 0.47 min | 0.75 min |
| Performance rating | 100% | 80% | 100% | 100% |
| Machine element | | *m* | | |
| Observed time, machine | (idle) | 1.56 min | (idle) | (idle) |

**Solution** **(a)** The normal time must take account of which element, *b* or *m*, has the larger value. Also, element *d* must be prorated across 15 cycles.

$$T_n = 0.22(1.00) + \text{Max}\{0.65(0.80), 1.56\} + 0.47(1.00) + 0.75(1.00)/15$$
$$= 0.22 + 1.56 + 0.47 + 0.05 = 2.30 \text{ min}$$

**(b)** The same comparison between elements *b* and *m* must be made in computing the standard time.

$$T_{std} = (0.22 + 0.47 + 0.05)(1 + 0.15)$$
$$+ \text{Max}\{0.52(1 + 0.15), 1.56(1 + 0.20)\}$$
$$= 0.85 + 1.87 = 2.72 \text{ min}$$ ■

## 2 NUMBER OF WORK CYCLES TO BE TIMED

One of the practical issues in taking a time study is determining how many work cycles should be timed. The reason this issue arises is that there is statistical variation in the times of respective elements from one work cycle to the next. Direct time study involves a sampling procedure, and the objective is to determine a value for the population work element time as accurately as is possible and practical. As we increase the sample size, we expect the accuracy of the estimate to improve. On the other hand, increasing the sample size also increases the cost of taking the time study. It seems reasonable to try to find a balance between these competing factors.

There is inherent variability in any human activity, and performing manual work is a human activity. Work element times vary from cycle to cycle because of the following reasons:

- Variations in hand and body motions
- Variations in the placement and location of parts and tools used in the cycle
- Variations in the quality of the starting work units (e.g., a plastic molded part with flash that must be trimmed)
- Mistakes by operator (e.g., operator accidentally drops the work part)
- Errors in timing the work elements by the analyst
- Variations in worker pace

All of these variations are manifested in the work element times recorded for the cycle. Performance rating is supposed to compensate for the last item, variations in worker pace. However, because performance rating requires judgment by the analyst, that also introduces error and variation.

For analysis purposes, we assume that the observed work element times are normally distributed about the true value of the work element time. For practical purposes, we identify the longest work element in the cycle, or the most critical element (the one in whose accuracy we are most interested), as the element to focus on. Let us call this element time $T_e$. Our objective is to be able to identify the true value of $T_e$ within a certain confidence interval. For example, we might state that we want to be 95% confident that the true value of $T_e$ lies within $\pm 10\%$ of the observed average value of the element time. Let us identify the average value simply as $\bar{x}$. This is illustrated in Figure 3, which shows the distribution of observed time values taken during the time study.

The area under the normal curve between $\pm 10\%$ of $\bar{x}$ represents the probability that the true value of element time $T_e$ lies within $\pm 10\%$ of $\bar{x}$. The general statement of the confidence interval can be expressed as follows:

$$\Pr\left(T_e \text{ lies within } \bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = (1-\alpha) \tag{4}$$

where $z_{\alpha/2}$ = standard normal variate, s = standard deviation of the population element time, $n$ = sample size, and (1 − a) = confidence level. The term $z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$ is made equal to our desired interval size (e.g., $\pm 10\%$ of $\bar{x}$), and we solve for the value of $n$, the number of work cycles to time.

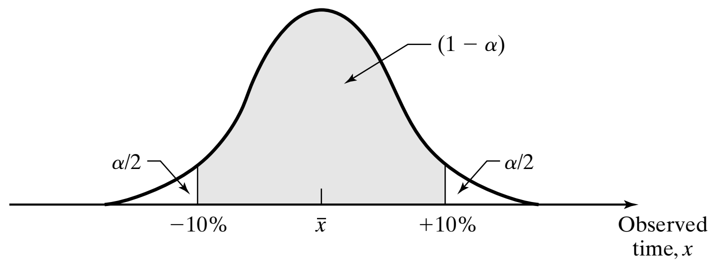


**Figure 3** Distribution of observed element times in a direct time study.

There are two difficulties with the preceding analysis: (1) we do not know the population standard deviation s and (2) the sample size $n$ that we are usually dealing with is relatively small. The standard deviation must be estimated from the sample itself, just as the mean $\bar{x}$ must be determined from the sample. The value of the sample standard deviation is given by

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} \tag{5}$$

where $s$ = sample standard deviation, $x$ = individual values of the observed times collected during the study, and the other terms have previously been defined. Because the sample size $n$ is usually small (e.g., less than 30), the sample values are distributed according to the student $t$ rather than the normal distribution.

Taking these adjustments into account, we can make the following confidence interval statement to replace equation (4):

$$\Pr\left(T_e \text{ lies within } \bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}\right) = (1-\alpha) \tag{6}$$

Our objective is to find the value of sample size $n$ that will satisfy our specification of $\alpha$ and interval size for the values of $\bar{x}$ and $s$ that have been determined from the data collected. Let us express the interval size in more general terms:

$$\text{Interval size} = \bar{x} \pm k\bar{x}$$

where $k$ = a proportion that specifies the interval size (e.g., $k$ = 10% or 0.10). We will set that equal to $\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$, from which the following can be established:

$$k\bar{x} = t_{\alpha/2}\frac{s}{\sqrt{n}} \tag{7}$$

Rearranging and solving for $n$, we have

$$n = \left(\frac{t_{\alpha/2}s}{k\bar{x}}\right)^2 \tag{8}$$

where values of $t_{\alpha/2}$ can be found in the Appendix: Statistical Tables.

**Example 3 Determining the Number of Work Cycles to Be Timed**

A time study analyst has collected 10 readings on a particular work element of interest and would like to consider how many more cycles to time. Based on the sample, the mean time for the element is $\bar{x} = 0.40$ min and the sample standard deviation $s = 0.07$ min. At a 95% confidence level, how many cycles should be timed to ensure the actual element time is within $\pm 10\%$ of the mean?

**Solution:** We have $10 - 1 = 9$ degrees of freedom in the $t$ distribution, so the value of $t_{\alpha/2}$ at the 95% confidence level ($\alpha/2 = 0.025$) is 2.262. Using equation (8), we obtain

$$n = \left(\frac{2.262(0.07)}{0.10(0.40)}\right)^2 = 15.7$$

This should be rounded up to 16 cycles total. Since 10 cycles have already been timed, the analyst needs data from 6 more work cycles. ■

The time study analyst may wish to turn the problem around; that is, to determine the size of the confidence interval at a certain desired confidence level, given that the observed times have already been recorded for a completed time study. This can be done using equation (8) but rearranging to solve for $k$.

Notwithstanding the preceding statistical approach, the problem of determining how many work cycles should be timed often boils down to the practical issue of how much time can we afford to spend doing the time study. Certain observations are germane here:

- The statistical accuracy of the time data increases when more observations are taken. The equations tell us that. If the task being studied is an important one in the factory operations, then more time should be spent on the time study to ensure the highest possible accuracy of the standard. More time should be spent studying high production operations than low production operations.
- Shorter work cycles allow more cycles to be timed during the study. For work cycles under 1 minute, 30 or more cycles are practical. For longer work cycles, fewer cycles can be included. For work cycles of 30 minutes, it may only be practical to observe six work cycles (which is 3 hours or more of time study observation).
- It is a judgment call by the time study department as to the appropriate number of work cycles to be timed. The judgment attempts to balance the desire for statistical accuracy against the cost of the time study.

## 3 PERFORMANCE RATING

***Performance rating***, also called ***performance leveling***, is the step in direct time study in which the analyst observes the worker's performance and records a value representing that performance relative to the analyst's concept of standard performance. A number of different performance rating methods have been proposed over the years, but the simplest and most common method is based on speed or pace.[4] It is sometimes referred to as speed rating. The other methods are rarely used today.

As its name indicates, ***speed rating*** involves rating the operator's pace of working relative to the concept of standard performance. A rating greater than 100% means that the worker is working faster than standard pace, and less than 100% means the worker is slower than standard. Pace is the only factor that is rated. However, pace depends on the type of work involved. To rate the operator's pace, the analyst must use judgment that is based on previous training and experience in rating similar work. An analyst who is experienced in machine shop time studies would find this experience of limited value if suddenly faced with the job of conducting time studies in a garment factory. The concept of standard performance is different for different kinds of work. Standard performance for a highly repetitive manual assembly task in which the cycle time is 1 minute should be defined differently than standard performance in a design prototype modeling shop in which the worker must apply technical skill and artistic talent to the job. And within a given category of work, some tasks are more difficult than others. Accordingly, the analyst must observe two aspects of the work scene during performance rating: (1) the degree of difficulty of the task and what 100% pace would be for that task, and (2) the subject worker's pace relative to the concept of 100% pace.

Let us examine these two aspects of performance using one of our accepted definitions of standard performance, walking at 3 miles/hr (4.83 km/hr). This is a 20-minute mile. The more complete definition is walking at 3 miles/hr on level flat ground, using 27-inch steps.[5] Also, it is assumed that the walker carries no load. If any of these conditions are not satisfied (i.e., level ground, 27-inch steps, no load), then it represents a change in the degree of difficulty, and so the definition of 100% performance might need adjusting. For example, walking uphill on a 3% slope might have a standard pace of only 2.5 miles/hr. But if the conditions of the definition are satisfied, so that standard performance is 3 miles/hr, then an individual's walking speed can be measured to assess his or her performance rating relative to this standard. In fact, walking is a situation where we can very precisely determine performance level if we assume that performance level is directly proportional to the speed of the walker. For example, suppose the walker is able to complete 1 mile in 18 min (0.30 hr) instead of 20. This is a speed of 3.33 miles/hr (1 mile ÷ 0.3 hr). Compared to 3.0 miles/hr, the performance level is 111%. We do not necessarily know that the walker's performance was a constant 111% (3.33 miles/hr) throughout the entire mile, but we know that on average it was 111%.

Unfortunately, very few work situations lend themselves to such a precise measurement of performance. Thus, performance rating comes down to a matter of the analyst judging the worker's pace relative to his or her concept of standard performance for the type of work involved. Direct time study is not the only instance where judgment is used to rate people. There are many areas of human activity in which judgments are used by experts to decide the relative performance of subjects, and the expert's judgments are readily accepted. Table 2 presents a list of activities in which judges rate the

[4]For the interested reader, the other performance rating methods are described in [14].

[5]This definition was apparently proposed by Ralph Presgrave, one of the early workers in time and motion study.

**TABLE 2** Activities in Which Trained and Experienced Judges Rate the Contestants

| Human Activities That Are Judged | Other Activities That Are Judged |
|---|---|
| Figure skating | Dog shows |
| Diving events | Farm shows |
| Gymnastics | Gourmet food contests |
| Bodybuilding contests | Art contests |
| Beauty contests | High school science fair contests |
| Umpire calling balls and strikes behind home plate in a baseball game | Grading of (college) freshman English essays |

contestants, often reducing the judgment to a numerical score. Just as in these other areas, time study analysts must be trained and become practiced in performance rating. Training is accomplished through the use of training films (videotapes) that show scenes of workers performing various kinds of industrial tasks at a variety of performance levels. The workers' pace in these scenes have been performance rated by experts, so that a trainee can judge each scene and compare his or her ratings with those of the experts. On-the-job practice is obtained by accompanying experienced analysts during actual time studies.

Depending on the type of work and length of the work cycle, the analyst must decide whether to rate the performance of individual elements (called ***elemental rating***) or to rate the entire cycle (***overall rating***). If the work cycle is relatively short (e.g., less than a minute) and the work content is similar throughout the cycle, then convenience may favor rating the entire cycle. If the cycle is longer and involves different kinds of work elements, on which the operator exhibits different pace levels, then elemental rating is more appropriate.

Operator selection can influence the performance rating procedure. The pace observed by the analyst depends to some extent on the worker's skill, experience, exertion level, and attitude toward time study. If there is more than one worker performing the same task that is to be time studied, then it is in the analyst's interest to select a skilled worker who is familiar with the job (has had time to learn the task) and who accepts time study as a necessary management tool. Such workers are usually willing to cooperate with the time study analyst, even to the extent that they will attempt to work at a pace consistent with the company's definition of standard performance, thus relieving the analyst of being forced to attach a rating that is significantly different from 100%. It has been found that when the subject performs at a pace that is within about 15% of standard, the resulting standard time value is much more likely to be fair and accurate and accepted by the worker. If the subject's performance during observation deviates significantly from standard, not only does this make life more difficult for the analyst, it increases the likelihood that the standard will not be an accurate measure of the task.

Potential difficulties are encountered in performance rating because a potential conflict of interest exists between the subject worker and the time study analyst. It is always in the worker's interest for the performance rating to be high due to its effect on the computation of standard time. Notwithstanding the difficulties, we can identify the following characteristics of a well-implemented performance rating system:

- *Consistency among tasks.* The performance rating system should provide consistent ratings from one task to another. A worker who is able to perform at 125% efficiency on one task should be able to achieve the same efficiency level on any other task. The performance rating used to establish the normal time is a key factor in obtaining this consistency among tasks.
- *Consistency among analysts.* The performance rating for a task should not depend on which time study analyst does the rating. Consistency within a group of time study analysts can be measured using the deviations of individual *PR* values about the group average. Niebel and Freivalds indicate that a deviation of $\pm 5\%$ is considered adequate [14]. Proper initial training and periodic refresher training for the analysts in a company should provide this level of consistency.
- *Easily understood.* The rating system should be easy to explain by the analyst and simple to understand by the worker.
- *Related to standard performance.* The company should have a well-defined concept of standard performance. To the extent possible, this concept should be documented for reference by workers. Time study analysts should apply this definition when performance rating a task.
- *Machine-paced elements rated at 100%.* The operator has no control over machine-paced elements and therefore deserves no performance rating for these elements. Any adjustment for machine-cycle elements should be done using a machine allowance when computing the standard time.
- *Performance rating recorded during observation of task.* The performance should not be rated and recorded afterward.
- *Worker notification.* At the conclusion of the time study observation session, the analyst should inform the worker of the performance rating that was observed. This sets a proper tone for the relationship between the two people. It puts pressure on the analyst to be as fair as possible in the rating process, and it allows the worker to express agreement or disagreement with the rating.

## 4 TIME STUDY EQUIPMENT

In direct time study, the analyst directly observes the task as a worker performs it. The equipment used by the analyst ranges from simple to sophisticated. In our coverage we divide the range into three categories: (1) traditional time study using a stopwatch, (2) video camera to record the observation on tape, and (3) computerized time study techniques.

### 4.1 Stopwatch Time Study

The traditional equipment used in direct time study consists of a stopwatch and a time study form on which to record the times during observation. There are alternatives to the stopwatch, such as a wristwatch or wall clock, but the proper professional instrument

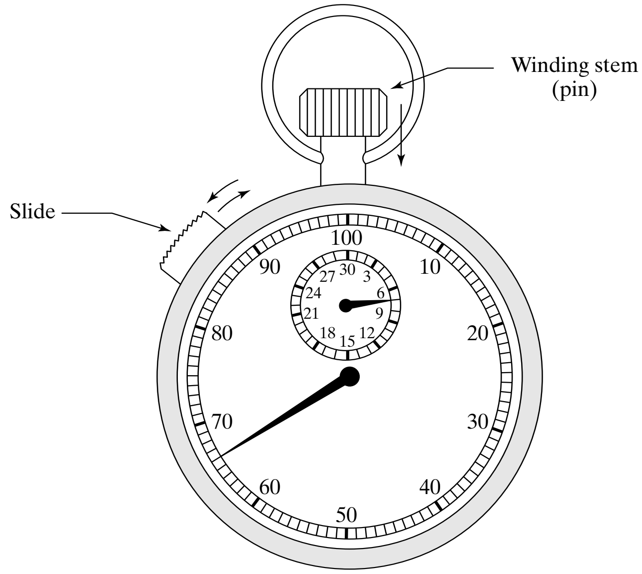


**Figure 4** Mechanical stopwatch calibrated in decimal minutes (0.01 min) on the large dial. The small dial reads up to 30 minutes. Depressing the pin (also used as the winding stem) zeros the watch. The slide is used to start and stop the timing of individual work elements.

is a stopwatch. The time study form is usually held on a clipboard, which is designed to hold the stopwatch as well. Following the observation, the collected time data are analyzed and a time standard is calculated for the task.

Stopwatches can be classified as mechanical or electronic. ***Mechanical stopwatches*** have the familiar round face with one or more rotating hands, as shown in Figure 4. They are designed to be held in the time study clipboard or the palm of one's hand so the fingers can actuate the pin and slide at the top. Mechanical stopwatches are not a recent development. They have been used for time study since the late 1800s. Modern stopwatches for time study are typically graduated in one of three time measurement scales: (1) decimal minutes, shown in Figure 4, (2) decimal hours, and (3) TMUs (Time Measurement Units); 1 TMU = 0.00001 hr or 0.036 sec). The reason for having different scales is that different organizations have adopted different units to express their time standards. Decimal minute watches seem to be the most common, decimal hours less common, and TMUs are sometimes used because they are usually the time units in certain predetermined motion time systems.

***Electronic stopwatches*** (Figure 5) provide a digital display of the time; otherwise their operating features are similar to those of mechanical watches. They have largely replaced mechanical stopwatches in time study for several compelling reasons. Because they display a digital readout, they are easier to read than the graduated mechanical dial, especially if the mechanical hand is moving. Because they are easier to read, reading errors occur less frequently. They are lighter in weight and generally less

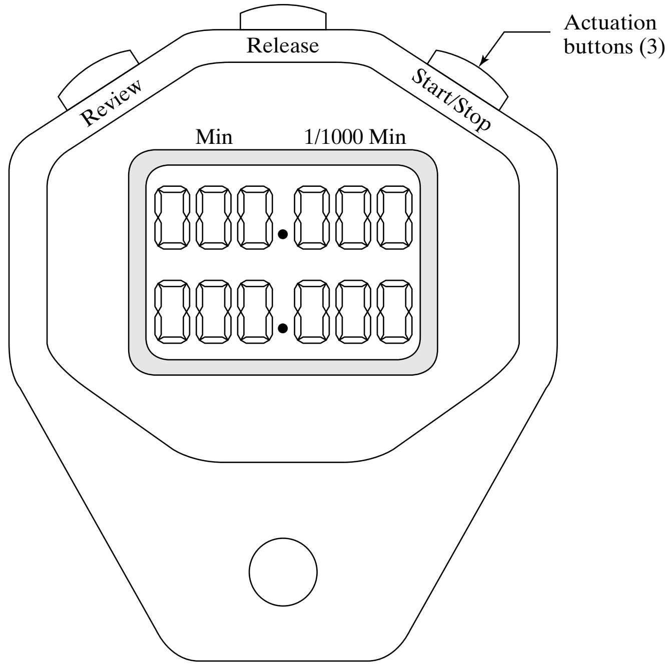


**Figure 5** Electronic stopwatch with LED read-out, that can be used for continuous or snapback timing methods.

susceptible to damage when dropped or otherwise abused. Electronic watches are also more accurate and precise, which allows them to be used to read shorter work elements more conveniently. Some electronic watches can be switched back and forth between different time scales, unlike their mechanical counterparts. They can be used for either the continuous timing mode or the snapback mode. Finally, electronic stopwatches are less expensive than mechanical stopwatches.

The direct time study observation form (Figure 2) provides space for the analyst to record the observed times and the performance rating of the operator. Either the continuous timing method or the snapback method can be used. Observed times for multiple work cycles can be entered on the form.

### 4.2 Video Cameras

Video cameras have become a familiar piece of electronic equipment not only to consumers but to time study analysts as well. They are useful in direct time study because they provide a complete visual and audio record of the method used by the worker. The method is captured in much greater detail than the analyst can observe with the naked eye or is willing to include in any written documentation of the method. The tape also provides an accurate record of the times taken by each work element (accurate to 1/30 sec based on a frame-to-frame frequency of 30 Hz in the United States, or 1/25 sec based on 25 Hz in Europe). The tape can be subsequently played and replayed (in slow motion if desired) to analyze the method, perhaps uncovering possible improvements in the task. Work element times can be analyzed for cycle-to-cycle variations. Operator performance can be rated in a much more relaxed and objective setting than what is possible in the shop floor environment. And finally, a standard time can be determined for the task from the tape.

Later, if a dispute arises about the standard time, the method, or the performance rating, the tape can be viewed to help settle the dispute, perhaps avoiding a formal grievance. If the operator wants to watch his own work performance, perhaps to decide whether he agrees with the rating, this can be accommodated. The tape can be used for training purposes for new time study analysts. The videotape provides an objective and accurate record of the worker performing the task using the method for which the time standard was determined.

Video cameras can be used for a variety of work situations in motion and time study: short repetitive work cycles, long cycles with variable elements, worker-machine systems, crews of workers, and several workers working independently but captured in the field of vision of the camera.[6]

## REFERENCES


[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] Aft, L. S. "Measurement of Work." Pp. 5.3–5.22 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[3] ANSI Standard Z94.0-1989, *Industrial Engineering Terminology*. Norcross, GA: Industrial Engineering and Management Press, Institute of Industrial Engineers, 1989.

[4] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley, 1980.

[5] Bishop, G. "Purpose and Justification of Engineered Labor Standards." Pp. 5.23–5.40 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[6] Jay, T. A. *Time Study*. Poole, England: Blandford Press, 1981.

[7] Kanawaty, G., ed. *Introduction to Work Study*. 4th ed. Geneva, Switzerland: International Labour Office, 1992.

[8] Karger, D. W., and F. H. Bayha. *Engineered Work Measurement.* New York: Industrial Press Inc., 1987.

[9] Konz, S., and S. Johnson. *Work Design: Industrial Ergonomics*. 5th ed. Scottsdale, AZ: Holcomb Hathaway Publishers, 2000.

[10] Konz, S. "Time Standards." Pp. 1391–408 in *Handbook of Industrial Engineering*, 3rd ed., edited by G. Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 2001.

[11] Matias, A. C. "Work Measurement: Principles and Techniques." Pp. 1409–462 in *Handbook of Industrial Engineering*, 3rd ed., edited by G. Salvendy. New York: Wiley; Institute of Industrial Engineers, Norcross, GA, 2001.

[12] Meyers, F. E., and J. R. Stewart. *Motion and Time Study for Lean Manufacturing*. 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[13] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.

[14] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 11th ed. New York: McGraw-Hill, 2003.


[6]Most of the items in this list were suggested by [13].

[15] Polk, E. J. *Methods Analysis and Work Measurement*. New York: McGraw-Hill, 1984.
[16] Sellie, C. N. "Stopwatch Time Study." Pp. 17.21–17.46 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.
[17] Smith, G. L., Jr. *Work Measurement: A Systems Approach.* Columbus, OH: Grid Publishing, 1978.

## REVIEW QUESTIONS

**1** Define direct time study.
**2** Identify the five steps in the direct time study procedure.
**3** Why is it so important to define and document the standard method as precisely and thoroughly as possible?
**4** What is the snapback timing method when using a stopwatch during direct time study?
**5** What is the continuous timing method when using a stopwatch during direct time study?
**6** Why is performance rating a necessary step in direct time study?
**7** Why is an allowance added to the normal time to compute the standard time?
**8** What are some of the causes of variability in the observed work element times that occur from cycle to cycle?
**9** Why is the student *t* distribution rather than the normal distribution used in the calculation of the number of work cycles to be timed?
**10** What is the difference between elemental performance rating and overall performance rating?
**11** What are the characteristics of a well-implemented performance rating system?
**12** What are the advantages of electronic stopwatches compared to mechanical stopwatches?

## PROBLEMS

*Note*: Some of the problems in this set require the use of parameters and equations.

### Determining Standard Times for Pure Manual Tasks

**1** The observed average time in a direct time study was 2.40 min for a repetitive work cycle. The worker's performance was rated at 110% on all cycles. The personal time, fatigue, and delay (PFD) allowance for this work is 12%. Determine (a) the normal time and (b) the standard time for the cycle.

**2** The observed element times and performance ratings collected in a direct time study are indicated in the table below. The snapback timing method was used. The PFD allowance in the plant is 14%. All elements are regular elements in the work cycle. Determine (a) the normal time and (b) the standard time for the cycle.

| Work element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time (min) | 0.22 | 0.41 | 0.30 | 0.37 |
| Performance rating(%) | 90 | 120 | 100 | 90 |

**3** The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 min. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers, which took 1.60 min, rated at 120%. The PFD allowance for this class of work is 15%. Determine (a) the normal time, (b) the standard time for the cycle, and (c) the worker's efficiency if the worker produces 123 work units during an 8-hour shift.

**4** The snapback timing method was used to obtain the average times and performance ratings for work elements in a manual repetitive task. (See table below). All elements are worker-controlled and were performance rated at 80%. Element *e* is an irregular element performed every five cycles. A 15% allowance for personal time, fatigue, and delays is applied to the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 120% on all manual elements for 7 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

| Work element | *a* | *b* | *c* | *d* | *e* |
|---|---|---|---|---|---|
| Observed time (min) | 0.32 | 0.85 | 0.48 | 0.55 | 1.50 |

**5** The continuous timing method is used to direct time study a manual task cycle consisting of four elements: *a*, *b*, *c*, and *d*. Two parts are produced each cycle. Element *d* is an irregular element performed once every six cycles. All elements were performance rated at 90%. The PFD allowance is 11%. Determine (a) the normalized time for the cycle, (b) the standard time per part, (c) the worker's efficiency if the worker completes 844 parts in an 8-hour shift during which she works 7 hours and 10 min.

| Element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time (min) | 0.35 | 0.60 | 0.86 | 1.46 |

**6** The readings in the table below were taken by the snapback timing method of direct time study to produce a certain subassembly. The task was performance rated at 85%. In addition to the above regular elements, an irregular element must be included in the standard: each rack holds 20 mechanism plates and has universal wheels for easy movement. After completing 20 subassemblies, the operator must move the rack (which now holds the subassemblies) to the aisle and then move a new empty rack into position at the workstation. This irregular element was timed at 2.90 min and the operator was performance rated at 80%. The PFD allowance is 15%. Determine (a) the normalized time for the cycle, (b) the standard time, and (c) the number of parts produced by the operator, if he or she works at standard performance for a total of 6 hr and 57 min during the shift.

| Element and Description | Observed Time (min) |
|---|---|
| 1. Pick up mechanism plate from rack and place in fixture. | 0.42 |
| 2. Assemble motor and fasteners to front side of plate. | 0.28 |
| 3. Move to other side of plate. | 0.11 |
| 4. Assemble two brackets to plate. | 0.56 |
| 5. Assemble hub mechanism to brackets. | 0.33 |
| 6. Remove plate from fixture and place in rack. | 0.40 |

**7** The time and performance rating values in the table below were obtained using the snapback timing method on the work elements in a certain manual repetitive task. All elements are worker-controlled and were performance rated at 85%. Element *e* is an irregular element performed every five cycles. A 15% PFD allowance is applied to the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 125% on all manual elements for 7 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

| Work element | *a* | *b* | *c* | *d* | *e* |
|---|---|---|---|---|---|
| Observed time (min) | 0.61 | 0.42 | 0.76 | 0.55 | 1.10 |

## Determining Standard Times for Worker-Machine Tasks

**8** The snapback timing method was used to obtain average times for work elements in one work cycle. The times are given in the table below. Element *d* is a machine-controlled element and the time is constant. Elements *a*, *b*, *c*, *e*, and *f* are operator-controlled and were performance rated at 80%; however, elements *e* and *f* are performed during the machine-controlled element *d*. The machine allowance is zero (no extra time is added to the machine cycle), and the PFD allowance is 14%. Determine (a) the normal time for the cycle and (b) the standard time for the cycle.

| Element | *a* | *b* | *c* | *d* | *e* | *f* |
|---|---|---|---|---|---|---|
| Observed time (min) | 0.24 | 0.30 | 0.17 | 0.76 | 0.26 | 0.14 |

**9** The continuous timing method in direct time study was used to obtain the element times for a worker-machine task as indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, *e*, and *f* are operator-controlled and external to the machine cycle, and they were performance rated at 80%. If the machine allowance is 25%, and the worker PFD allowance is 15%, determine (a) the normal time, (b) the standard time for the cycle, and (c) the worker's efficiency if the worker completes 360 work units working 7.2 hr on an 8-hour shift.

| Element | *a* | *b* | *c* | *d* | *e* | *f* |
|---|---|---|---|---|---|---|
| Observed time (min) | 0.18 | 0.30 | 0.88 | 1.12 | 1.55 | 1.80 |

**10** A worker-machine cycle is direct time studied using the continuous timing method. One part is produced each cycle. The cycle consists of five elements: *a*, *b*, *c*, *d*, and *e*. Elements *a*, *c*, *d*, and *e* are manual elements, external to machine element *b*. Every 16 cycles the worker must replace the parts container, which was observed to take 2.0 min during the time study. All worker elements were performance rated at 80%. The PFD allowance is 16%, and the machine allowance is 20%. Determine (a) the normalized time for the cycle, (b) the standard time per part, and (c) the worker's efficiency if the worker completes 220 parts in an 8-hour shift during which he or she works 7 hr and 12 min.

| Element | Description | Cumulative Observed Time (min) |
|---|---|---|
| *a* | Worker loads machine and starts automatic cycle. | 0.25 |
| *b* | Machine is engaged in automatic cycle. | 1.50 |
| *c* | Worker unloads machine. | 1.75 |
| *d* | Worker files part to size. | 2.30 |
| *e* | Worker deposits part in container. | 2.40 |

**11** In the preceding problem, a recommendation has been submitted for elements *d* and *e* to be performed as internal elements (accomplished simultaneously) with machine element *b*. The worker would file the part from the previous cycle and deposit it in the container while the current part is being processed in the machine automatic cycle. Performance rating and allowances are the same as in the previous problem. Determine (a) the normal time for the cycle, (b) the standard time per part, and (c) the number of parts that will be produced if the worker's efficiency is 115% and he works a total of 7 hr and 12 min during an 8-hour shift.

**12** The snapback method was used to time study a worker-machine cycle consisting of elements *a*, *b*, and *c*. Elements *a* and *b* are worker-controlled and were performance rated at 100% during the time study. Element *c* is machine-controlled. Elements *b* and *c* are performed simultaneously. The PFD allowance is 12% and the machine allowance is 10%. One work piece is produced each cycle. Determine (a) the normal time, (b) the standard time for the cycle. (c) the number of pieces completed if the worker works 7 hr and 10 min during an 8-hour shift, and his performance level is 135%.

| Element | *a* | *b* | *c* |
|---|---|---|---|
| Observed time (min) | 1.25 | 0.90 | 0.80 |

**13** The snapback timing method in direct time study was used to obtain the times for a worker-machine task. The recorded times are listed in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, and *d* are operator-controlled and were performance rated at 90%. Elements *a* and *b* are external to machine-controlled element *c*. Element *d* is internal to the machine element. The machine allowance is zero, and the PFD allowance is 13%. Determine (a) the normal time and (b) the standard time for the cycle. The worker's actual time spent working during an 8-hour shift was 7.08 hours, and he produced 420 units of output during this time. Determine (c) the worker's performance during the operator-controlled portions of the cycle and (d) the worker's efficiency during this shift.

| Element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time (min) | 0.34 | 0.25 | 0.68 | 0.45 |

**14** The table below lists the average work element times obtained in a direct time study using the snapback timing method. Elements *a* and *b* are operator-controlled. Element *c* is a machine-controlled element and its time is constant. Element *d* is a worker-controlled irregular element performed every five cycles. Elements *a*, *b* and *d* were performance rated at 80%. The worker is idle during element *c*, and the machine is idle during elements *a*, *b*, and *d*. One product unit is produced each cycle. To compute the standard, no machine allowance is applied to element *c*, and a 15% PFD allowance is applied to elements *a*, *b*, and *d*. (a) Determine the standard time for this cycle. (b) If the worker produces 220 units on an 8-hour shift during which 7.5 hours were actually worked, what was the worker's efficiency? (c) For the 220 units in (b), what was the worker's performance during the operator-paced portion of the cycle?

| Element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time (min) | 0.60 | 0.45 | 1.50 | 0.75 |

**15** The continuous timing method in direct time study was used to obtain the times for a worker-machine task as indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, *e*, and *f* are operator-controlled and were performance rated at 95%; they are all external elements performed in sequence with machine element *c*. The machine allowance is 30%, and the PFD allowance is 15%. Determine (a) the normal time and (b) standard time for the cycle. (c) If the operator works at 100% of standard performance in production and one part is produced each cycle, how many parts are produced if the total time worked during an 8-hour day is 7.25 hours? (d) For the number of parts computed in (c), what is the worker's efficiency for this shift?

| Element | *a* | *b* | *c* | *d* | *e* | *f* |
|---|---|---|---|---|---|---|
| Observed time (min) | 0.22 | 0.40 | 1.08 | 1.29 | 1.75 | 2.10 |

**16** The snapback timing method was used to obtain average time and performance rating values for the work elements in a certain repetitive task. The values are given in the table below. Elements *a*, *b*, and *c* are worker-controlled. Element *d* is a machine-controlled element and its time is the same each cycle (N.A. means performance rating is not applicable). Element *c* is performed while the machine is performing its cycle (element *d*). Element *e* is a worker-controlled irregular element performed every six cycles. The machine is idle during elements *a*, *b*, and *e*. Four product units are produced each cycle. The machine allowance is zero, and a 15% PFD allowance is applied to the manual portion of the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 140% on all manual elements for 7 actual hours worked on an 8-hour shift, determine (c) how many units will be produced and (d) what the worker's efficiency will be.

| Work Element | *a* | *b* | *c* | *d* | *e* |
|---|---|---|---|---|---|
| Observed time (min) | 0.65 | 0.50 | 0.50 | 0.55 | 1.14 |
| Performance rating (%) | 90 | 100 | 120 | N.A. | 80 |

**17** The continuous timing method was used to obtain the times for a worker-machine task. Only one cycle was timed. The observed time data are recorded in the table below. Elements *a*, *b*, *c*, and *e* are worker-controlled elements. Element *d* is machine controlled. Elements *a*, *b*, and *e* are external to the machine-controlled element, while element *c* is internal. There are no irregular elements. All worker-controlled elements were performance rated at 80%. The PFD allowance is 15% and the machine allowance is 20%. Determine (a) the normal time and (b) standard time for the cycle. (c) If worker efficiency is 100%, how many units will be produced in one 9-hour shift? (d) If the actual time worked during the shift is 7.56 hours, and the worker performance is 120%, how many units would be produced?

| | | | | |
|---|---|---|---|---|
| Worker element (min) | *a* (0.65) | *b* (1.80) | *c* (4.25) | *e* (5.45) |
| Machine element (min) | | | *d* (4.00) | |

**18** The continuous stopwatch timing method was used to obtain the observed times for a worker-machine task. Only one cycle was timed. The data are recorded in the table below. The times listed indicate the stopwatch reading at the end of the element. Elements *a*, *b*, and *d* are worker-controlled elements. Element *c* is machine controlled. Elements *a* and *d* are external to the machine-controlled element, while element *b* is internal. Every four cycles, there is an irregular worker element that takes 1.32 min rated at 100% performance. For determining the standard time, the PFD allowance is 15% and the machine allowance is 30%. Determine (a) the normal time and (b) standard time for the cycle. (c) If worker efficiency is 100%, how many units will be produced in one 8-hour shift? (d) If the actual time worked during the shift is 6.86 hours, and the worker performance is 125%, how many units will be produced?

| Worker Element | Description of Worker Element | Time (min) | Performance Rating (%) | Machine Element | Description of Machine Element | Time (min) |
|---|---|---|---|---|---|---|
| *a* | Acquire workpart from tray, cut to size, and load into machine | 1.24 | 100 | (idle) | | |
| *b* | Enter machine settings for next cycle | 4.24 | 120 | *c* | Automatic cycle controlled by machine settings entered in previous cycle | 4.54 |
| *d* | Unload machine and place part on conveyor | 5.09 | 80 | (idle) | | |

**19** The snapback timing method in direct time study was used to obtain the times for a worker-machine task. The recorded times are listed in the table below. Element *d* is a machine-controlled element and the time is constant. Elements *a*, *b*, *c*, *e*, and *f* are operator-controlled and were performance rated at 90%. Element *f* is an irregular element, performed every five cycles. The operator-controlled elements are all external to machine-controlled element *d*. The machine allowance is zero, and the PFD allowance is 13%. Determine (a) the normalized time for the cycle and (b) the standard time for the cycle. The worker's actual time spent working during an 8-hour shift was 7.08 hours, and he produced 400 units of output during this time. Determine (c) the worker's performance during the operator-controlled portions of the cycle and (d) the worker's efficiency during this shift.

| Element | *a* | *b* | *c* | *d* | *e* | *f* |
|---|---|---|---|---|---|---|
| Observed time (min) | 0.14 | 0.25 | 0.18 | 0.45 | 0.20 | 0.62 |

**20** For a worker-machine task, the continuous-timing method was used to obtain the times indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, and *e* are operator-controlled and were performance rated at 100%; however, element *d* is performed during the machine-controlled element *c*. The machine allowance is 16%, and the PFD allowance is 16%. Determine (a) the normal time and (b) standard time for the cycle. (c) If the operator works at 140% of standard performance in production and two parts are produced each cycle, how many parts are produced if the total time worked during an 8-hour day is 7.4 hours? (d) For the number of parts computed in (c), what is the worker's efficiency for this shift?

| Element | *a* | *b* | *c* | *d* | *e* |
|---|---|---|---|---|---|
| Observed time (min) | 0.30 | 0.65 | 1.65 | 1.90 | 2.50 |

**21** For a certain repetitive task, the snapback timing method was used to obtain the average work element times and performance ratings listed in the table below. Elements *a*, *b*, and *c* are worker-controlled. Element *d* is a machine-controlled element and its time is the same each cycle (N.A. means performance rating is not applicable). Element *c* is performed while the machine is performing its cycle (element *d*). Element *e* is a worker-controlled irregular element performed every six cycles. The machine is idle during elements *a*, *b*, and *e*. One part is produced each cycle. The machine allowance is 15%, and a 15% PFD allowance is applied to the manual portion of the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 130% on all manual elements for 7.3 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

| Work Element | *a* | *b* | *c* | *d* | *e* |
|---|---|---|---|---|---|
| Observed time (min) | 0.47 | 0.58 | 0.70 | 0.75 | 2.10 |
| Performance rating (%) | 90 | 80 | 110 | N.A. | 85 |

**22** The work element times for a repetitive work cycle are listed in the table below, as determined in a direct time study using the snapback timing method. Elements *a* and *b* are operator-controlled. Element *c* is a machine-controlled element and its time is constant. Element *d* is a worker-controlled irregular element performed every 10 cycles. Elements *a* and *b* were performance rated at 90%, and element *d* was performance rated at 75%. The worker is idle during element *c*, and the machine is idle during elements *a*, *b*, and *d*. One product unit is produced each cycle. No special allowance is added to the machine cycle time (element *c*), but a 15% allowance factor is applied to the total cycle time. (a) Determine the standard time for this cycle. If the worker produced 190 units on an 8-hour shift during which 7 hours are actually worked, (b) what was the worker's efficiency, and (c) what was his performance during the operator-paced portion of the cycle?

| Work element | *a* | *b* | *c* | *d* |
|---|---|---|---|---|
| Observed time (min) | 0.75 | 0.30 | 1.62 | 1.05 |

## Number of Cycles

**23** Seven cycles have been observed during a direct time study. The mean for the largest element time is 0.85 min, and the corresponding sample standard deviation *s* is 0.15 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10% of the true mean, how many more observations should be taken?

**24** Nine cycles have been observed during a time study. The mean for the largest element time is 0.80 min, and the corresponding sample standard deviation *s* is 0.15 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10 min of the true mean, how many more observations should be taken?

**25** Six cycles have been observed in a direct time study. The mean for the largest element time is 0.82 min, and the corresponding sample standard deviation *s* is 0.11 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10 min of the true mean, how many more observations should be taken?

**26** Ten cycles have been observed during a direct time study. The mean time for the longest element was 0.65 min, and the standard deviation calculated on the same data was

0.10 min. If the analyst wants to be 95% confident that the mean of the sample was within ±8% of the true mean, how many more observations should be taken?

**27** Six cycles have been observed during direct time study. The mean time for the longest element was 0.82 min, and the standard deviation calculated on the same data was 0.13 min. If the analyst wants to be 90% confident that the mean of the sample was within ±0.06 min of the true mean, how many more observations should be taken?

**28** Six cycles have been observed during a direct time study. The mean for the largest element time is 1.00 min, and the corresponding sample standard deviation *s* is 0.10 min. (a) Based on these data, what is the 90% confidence interval on the 1.0 min element time? (b) If the analyst wants to be 90% confident that the mean of the sample was within ±10% of the true mean, how many more observations should be taken?

**29** A total of 9 cycles has been observed during a direct time study. The mean for the largest element time is 1.30 min, and the corresponding sample standard deviation *s* is 0.20 min. (a) Based on these data, what is the 95% confidence interval on the 1.30 min element time? (b) If the analyst wants to be 98% confident that the mean of the sample was within ±5% of the true mean, how many more observations should be taken?

## Performance Rating

**30** One of the traditional definitions of standard performance is a person walking at 3.0 miles per hour. Given this, what is the performance rating of a long-distance runner who breaks the four-minute mile?

**31** In 1982, the winner of the Boston Marathon was Alberto Salazar, whose time was 2 hr, 23 min and 3.2 sec. The marathon race covers 26 miles and 385 yards. Given that one of the traditional definitions of standard performance is a person walking at 3.0 miles per hour, what was Salazar's performance rating in the race?

# Predetermined Motion Time Systems



The controversial step in direct time study is performance rating. It is based on a judgment by the time study analyst, and there is always room for disagreement between the worker doing the task and the analyst evaluating the worker's pace. An alternative to direct time study that does not include the performance-rating step, at least not directly, is the family of work measurement techniques known as predetermined motion time systems.

A ***predetermined motion time system*** (PMTS), also called a ***predetermined time system***, is a database of basic motion elements and their associated normal time values, together with a set of procedures for applying the data to analyze manual tasks and establish standard times for the tasks.[1] The PMTS database is most readily conceptualized as a set of tables listing time values that correspond to the basic motion elements, the lowest level in our hierarchy of manual work activity. They include motions such as reach, grasp, move, and release. The time required to perform these basic motions usually depends on certain work variables. For example, the time to reach for an

[1]The Work Measurement and Methods Standards Committee (ANSI Standard Z94.11-1989) defines a predetermined motion time system as follows: "An organized body of information, procedures, techniques, and motion times employed in the study and evaluation of manual work elements. The system is expressed in terms of the motions used, their general and specific nature, the conditions under which they occur, and their previously determined performance times."

object increases with the distance of the reach, among other factors. The time to move an object depends on its weight and the distance it is moved. The normal time values in the database have been determined based on extensive research of manual work activity, usually consisting of frame-by-frame analysis of motion pictures of the activity.

In this chapter we provide a general discussion of predetermined motion time systems, which were first identified around 1925 (Historical Note 1). We then examine two important examples of these systems: methods-time measurement (MTM) and the Maynard Operation Sequence Technique (MOST).

### HISTORICAL NOTE 1 PREDETERMINED MOTION TIME SYSTEMS

The notion that manual work consists of basic motion elements is attributed to Frank B. Gilbreth, who was a pioneer in the subject of motion study. His studies resulted in the first methodical classification of motion elements, which he called ***therbligs*** (Gilbreth spelled backwards—well, almost). The 17 therbligs consist mostly of arm and hand motions.

Asa B. Segur is credited with developing the first commercial predetermined motion time system, called Motion-Time Analysis (MTA). Segur was aware of Gilbreth's classification scheme for basic motion elements, and MTA's 17 basic motion elements agree closely with Gilbreth's 17 therbligs. Segur began the development of MTA in 1922 by analyzing motion picture films taken during World War I of operators performing factory tasks. The films were originally intended for use as training materials for blind and otherwise handicapped workers, but Segur constructed his system of motions and times from them. Supposedly for commercial reasons, he limited public access to MTA, and this is one of the reasons why his system is no longer used. However, his work influenced those who followed him in developing other predetermined motion time systems.

One of these analysts was Joseph H. Quick, who developed the Work-Factor system between 1934 and 1938. The system was based on the analysis of large numbers of motion picture films, snapshots, stroboscopic lighting techniques, and stopwatch studies taken of many different kinds of industrial operations. Among Quick's contributions in PMTS research were his studies involving cognitive work, such as visual inspection and similar technical labor.

Harold B. Maynard stands as one of the more important figures in the development of predetermined motion time systems. He is largely responsible for the Methods-Time Measurement (MTM) system, and the Pittsburgh-based consulting firm of H. B. Maynard and Company bears his name. In 1946, Maynard began development of MTM with colleagues G. J. Stegemerten and J. L. Schwab under the sponsorship of the Westinghouse Electric Company. Their study began by analyzing movies of drilling operations and relating the work patterns to therbligs. The therblig classification was found to be wanting in some respects, and many of the basic motion elements were revised, renamed, or removed. MTM became the most successful and widely used first level PMTS after its release in 1948. The MTM database of motion elements was used in the development of many higher-level systems, including those in the MTM family (see Table 5 later in this chapter).

Other individuals who should be mentioned for their efforts in developing PMT systems include G. B. Bailey and R. Presgrave, who developed Basic Motion Timestudy in 1950; R. M. Crossan and H. W. Nance, who developed Master Standard Data in the late 1950s; and G. C. Heyde, who developed the Modular Arrangement of Predetermined Time Standards (MODAPT) in the mid 1960s.

Kjell B. Zandin is largely responsible for the development of the Maynard Operation Sequence Technique (MOST). Starting in the late 1960s, he led a group at the Swedish Division of H. B. Maynard & Company in studying applications of MTM. It was observed that there were similarities in the motion patterns and MTM sequences of many manual operations. This ultimately resulted in the definition of three principal motion groups, which were called sequence models in the MOST system. MOST was first introduced in Sweden in 1972 and in the United States in 1974. Today, it is one of the most widely used PMT systems.

Subsequent further development of predetermined motion time systems has been in the computerization of several of these systems as commercial products. Examples include MOST for Windows and MTM-LINK.

## 1 OVERVIEW OF PREDETERMINED MOTION TIME SYSTEMS

This section provides a general introduction to PMT systems. First, we discuss the usual procedure in applying a predetermined motion time system to a task. One of the significant advantages of a PMTS is that it can be used to determine a standard time for a task before that task is in production. It can also be used to set a standard for an existing task. We then indicate the differences among PMT systems and how they can be classified.

### 1.1 The PMTS Procedure

All PMT systems are applied using the same general procedure. The differences are in the details. To apply a predetermined motion time system to a task, the analyst must accomplish the following steps:

1. Synthesize the method that would be used by a worker to perform the task (or analyze the method that is being used by a worker in an existing task). The method is described in terms of the basic motions comprising the task, based on a defined workplace layout and set of tools (if tools are used).[2]
2. Retrieve the normal time value for each motion element, based on the work variables and conditions under which the element will be (is) performed. Sum the normal times for all motion elements to determine the normal time for the task.

[2]Different predetermined motion time systems have different basic motion definitions, and the analyst must describe the method using the definitions and conventions of the particular system.

3. Evaluate the method to make improvements by eliminating motions, reducing distances, introducing special tools, using simultaneous right and left hand motions, and so on. The objective of this evaluation is to reduce the normal time. The evaluation process is facilitated by the detailed listing of basic motion elements and corresponding time values.
4. Apply allowances to the normal time to determine the standard time for the task.

If the emphasis in the application is methods improvement, then step 4 may be omitted. If the emphasis is setting a time standard, then step 4 is required.

The basic time values in PMT systems do not include any allowances, so the user organization must add an allowance for personal time, fatigue, and delays according to its policies on allowances. This is step 4 in our PMTS procedure outlined above. The following equation is the usual way in which allowances are applied to determine the time standard for a task:

$$T_{std} = T_n(1 + A_{pfd}) \tag{1}$$

where $T_{std}$ = standard time, min; $T_n$ = normal time, which is the sum of the basic motion time values that make up the task (step 2 with possible adjustments resulting from step 3 in the PMTS procedure), min; and $A_{pfd}$ = PFD allowance factor.

A performance rating step is not necessary in a predetermined motion time system, because the basic motion times are already rated at standard (100%) performance. However, different PMT systems use different definitions of standard performance. MTM and MOST use the definition of "daywork" standard performance. If the organization using either of these systems defines standard performance differently, then a universal adjustment must be made to all time values in the database to bring these data into agreement with the organization's definition.

Similarly, for predetermined motion time systems other than MTM or MOST, an adjustment in all data values may be necessary to achieve agreement with the organization's definition of standard performance. For example, Motion Time Analysis (MTA) time values were determined based on a faster working pace, and therefore the basic motion times are less than MTM times by about 20%. Therefore, if a company were to use the MTA system but wanted to set standards using the "daywork" as the standard performance level, then all of the MTA times must be multiplied by 125%.

### 1.2 PMTS Levels and Generations

There is a variety of predetermined motion time systems. Although we might think that the basic motions of the human body should be the same in all PMT systems, some PMT systems combine basic motions into larger motion elements. Thus, we have various levels of PMT systems. First-level systems use the basic motion elements (e.g., reach, grasp), while higher-level systems combine several basic motion elements into ***motion aggregates***, combined motion sequences that are commonly used in work situations. For example, the basic motions ***reach*** and ***grasp*** might be combined into one element called

***get*** for use in a higher-level PMT system. This motion aggregate would be common in assembly work, for example.

First-level PMT systems tend to be very detailed, with body motions differentiated very precisely in their databases. For example, ***reach*** and ***grasp*** are typically distinguished as two separate motion elements, even though they are commonly carried out as one smooth arm and hand movement. The times for the basic motions tend to be very short, sometimes much less than a second. Higher-level systems use condensed databases, with fewer body motions contained in the tables and longer time values for each motion sequence. The tables are simplified, with fewer work variables that must be identified about the task by the analyst.

Chronologically, first-level PMT systems were the first to be developed, and then second- and higher-level systems were subsequently constructed based on the first-level systems. Because of this chronological development of the systems, the level of the system usually corresponds to the ***generation*** of the system. First-level PMT systems are called first generation systems, and the subsequent systems are second and third generations. For example, MTM-1 is first generation. MTM-2 is second generation and is based on MTM-1. MTM-3 is a third generation MTM system.

Methods descriptions in first-level systems consist of long lists of elements, and the time to determine a standard for the task is long (high application speed ratio). For example, using MTM-1, a first-level system, the time to set a standard is a factor of about 250 compared to the duration of the task time [7]. For a task of 1 minute, it takes over 4 hours of analyst time to establish the standard time. Higher-level PMT systems require less application time due to their use of elements that combine basic motions. For example, MTM-2 is a higher-level system that is based on MTM-1, and the time to set a standard is a factor of 100 compared to the task time. The cost of this greater convenience is that the accuracy of higher-level systems tends to be less than for a first-level system. The general characteristics of PMT system levels are summarized in Table 1.

More than 50 predetermined motion time systems have been developed over the years. Most of these systems are no longer used. Table 2 provides a brief description of the major PMT systems that were developed, with an emphasis on those that are commercially important today. The most widely used systems today are based on MTM. Section 2 discusses MTM, both the first-level system (MTM-1) and some of its higher-level versions. Section 3 discusses MOST, a widely used higher-level PMTS based on MTM.

**TABLE 1** Characteristics of PMT System Levels

| Characteristic | First-Level PMTS | Higher-Level PMTS |
|---|---|---|
| Accuracy | Most accurate | Less accurate |
| Application time | Much time to set standard | Less time to set standard |
| Suited to specific types of tasks | Highly repetitive | Repetitive or batch |
| Cycle times | Short cycle (e.g., 1 min) | Longer cycle times feasible |
| Motion elements | Basic motions | Aggregates of basic motions |
| Methods description | Very detailed | Less detailed, easier to apply |
| Flexibility of application | Highest flexibility | Less flexibility |

**TABLE 2** Summary of Predetermined Motion Time Systems

***Motion-Time Analysis (MTA)***: A first-level system developed by A. B. Segur in the mid- and late 1920s. Its primary importance today stems from the fact that it was the first PMTS to be developed and applied. Today it is seldom used if at all. There are 17 basic motion elements in MTA, involving one or more body movements. The elements are classified as positive (considered useful to the task), negative (motions that should be reduced or eliminated), or either (might be positive or negative).

***Work-Factor (WF)***: A first-level PMTS developed by J. H. Quick, W. J. Shea, and R. E. Koehler in the mid-1930s, with applications starting in 1938. It is still used today and at least one simplified version has been developed to reduce application time. Coverage includes both factory and office work.

***Methods-Time Measurement (MTM)***: Developed by H. B. Maynard, G. J. Stegemerten, and J. L. Schwab in the mid-1940s. The original MTM was a generic, first-level system published as a book in 1948. Today, that original PMTS is known as MTM-1, and several higher-level and specific PMT systems have been developed based on MTM-1. We describe the MTM family in Section 2.

***Basic Motion Timestudy (BMT)***: Developed by R. Presgrave, G. B. Bailey, and J. A. Lowden in the early 1950s. Its initial applications were in 1952, and it was published in book form in 1958. It is considered a first-level system, but some of its motion elements are combinations of therbligs. At least one second-level PMTS has been developed based on BMT.

***Master Standard Data (MSD)***: One of the first second-level systems. It is based on MTM-1 and was developed in Canada by R. M. Crossan and H. W. Nance in the late 1950s (published as a book in 1962) to reduce the analyst's time needed to set a time standard for manual production operations with fewer than 100,000 cycles per year. Its coverage emphasizes reach-grasp-move-release motions that are common in such operations.

***Maynard Operation Sequence Technique (MOST)***: Developed by K. B. Zandin and others at the Swedish Division of H. B. Maynard Company as a higher-level PTMS based on MTM-1 and MTM-2. Its coverage emphasizes production operations and material handling. We describe MOST in Section 3.

***Modular Arrangement of Predetermined Time Standards (MODAPTS)***: A second-level PTMS developed by C. Heyde and others in Australia during the mid-1960s. It is based on MSD, MTM-1, and MTM-2. The basic motion element in MODAPTS is a finger movement, and other body motions are expressed in terms of this element. Coverage includes production operations and material handling, with a special version covering office work.

*Source:* Compiled from [1], [7], and other sources.

## 2 METHODS-TIME MEASUREMENT

Methods-Time Measurement (MTM) is a PMTS that can be used to analyze the method for performing a given task and to set a time standard for the task. MTM is defined by its developers as follows [4]:

> Methods-Time Measurement is a procedure which analyzes any manual operation or method into the basic motions required to perform it and assigns to each motion a predetermined time standard which is determined by the nature of the motion and the conditions under which it is made.

The hyphenation of "Methods-Time" was intended by its developers to emphasize the important connection between the basic motions used to perform a task and the time values associated with these motions. MTM is an analysis tool by which the method for a task is divided into its component elements to determine if any methods improvements can be made (step 3 in the PMTS procedure). Only after this analysis should a standard time be established.

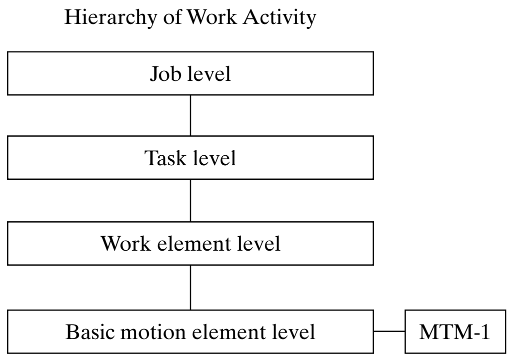


**Figure 1** The position of MTM motion elements in our work hierarchy.

MTM is actually a family of predetermined motion time systems. The original MTM is now called MTM-1. It is the first-level PMTS upon which all of the other MTM systems are based. We describe MTM-1 in Section 2.1. Other versions of MTM are briefly discussed in Section 2.2. Training in the members of the MTM PMTS family is available through the MTM Association, located in Des Plaines, Illinois.[3]

### 2.1 MTM-1

In our hierarchy of work activity, MTM-1 operates at the basic motion element level, as illustrated in Figure 1. Most of the MTM-1 basic motions involve hand and arm movements, although elements are also provided for eye, leg, foot, and body actions. Many of the basic motions correspond to the original therbligs as developed by Frank Gilbreth. However, the definitions of these basic motions have been adjusted to conform to the needs of MTM, and new elements not included among the original therbligs have been added.

Time units in MTM are called TMUs (time measurement units). MTM was developed by studying motion pictures of work activity, and the time units for MTM were originally defined as the time per frame of motion picture film. This proved awkward, and the TMU is now defined as follows:

$$1 \text{ TMU} = 0.00001 \text{ hr} = 0.0006 \text{ min} = 0.036 \text{ sec}$$

Conversely, there are 100,000 TMUs in 1 hr, 1667 TMUs in 1 min, and 27.8 TMUs in 1 sec.

Table 3 defines the MTM-1 motion elements, and Table 4 presents a tabulation of their time values. As with other PMT systems, the analyst must describe the task in terms of these basic motion elements, taking note of the work variables that influence the element time and accounting for simultaneous right-hand and left-hand motions,

[3]MTM Association, 1111 East Touhy Avenue, Des Plaines, IL 60018; their Web site can be accessed at www.mtm.org.

**TABLE 3** Summary of MTM-1 Motion Elements

Motion Element (Symbol) and Description

**Reach (R)**: A basic motion element involving movement of the hand or fingers. Its purpose is to move the hand or finger to a new destination. The corresponding therblig is "transport empty." If the hand is holding something during the motion, it is still classified as a reach if its primary purpose is to reposition the hand or fingers and not to move the object—for example, reaching for an object while holding a cigarette. Reach depends on two work variables: (1) the distance moved and (2) the case under which the reach is performed. The case refers to such factors as whether the reach is toward an object in a known location or the object is jumbled amongst other objects and must be searched out. The five cases are identified in Table 4 (a), which provides time values as a function of distance and case. If the hand is in motion before and/or after the reach, then the time is reduced slightly, as indicated in the table under "hand in motion" for cases A and B. The complete symbol for a given reach element in MTM-1 includes the distance and the case (e.g., R10C).

**Grasp (G)**: A motion element used by the fingers and hand to gain control of one or more objects. It is commonly employed as a prerequisite motion for performing the next basic motion, which is likely to be a move—for example, grasping an object prior to moving it. There are five categories of grasp: (1) pickup, (2) regrasp, (3) transfer, (4) select, and (5) contact. Within the pickup and select categories, there are several cases that must be distinguished. Table 4 (b) defines the various categories and cases and lists the times for them. The complete symbol for a grasp includes category and case (e.g., G1C3).

**Move (M)**: A hand or finger motion whose primary purpose is to relocate an object. The corresponding therblig is "transport loaded." The move element includes pushing or sliding an object across a surface, so long as the hand controls the object. Move depends on three work variables: (1) the weight of the object being moved, (2) the distance of the move, and (3) the case under which the move is performed. There are three cases, as described in Table 4 (c). The weight variable is divided into classes, and the user selects the next higher weight given in the table for the object weight in the given application. The complete symbol for move includes the distance, case, and weight—for example, M6B10 symbolizes a move of 6 in. under case B for an object weight of 10 lb. The following formula must be used to determine the TMU value from Table 4 (c):

$$\text{TMU} = \text{Constant} + \text{Factor} \times (\text{TMU value from table})$$

where Constant is the constant value from the table for the object weight in the application, TMU; Factor is the factor value from the table for the object weight; and (TMU value from table) is the TMU value from the table for the corresponding distance of the move and the move case. For example, given the symbol M6B10, the formula would be used as follows:

$$\text{TMU} = 3.9 + 1.11(8.9) = 13.8 \text{ TMU (about } \tfrac{1}{2} \text{ sec)}$$

**Position (P)**: A relatively short hand motion (no greater than 1 in.) employed to align, orient, or engage one object relative to another. It is usually preceded by a move. ***Align*** refers to the relative positioning of longitudinal axes of the objects; ***orient*** refers to the relative positioning of rotational axes of the objects; and ***engage*** refers to the insertion of one object into the other following either an alignment or an orientation. Accordingly, the position element may occur several times together depending on the number of separate positioning moves required. The work variables that affect the time for a position motion are the following: (1) pressure required to achieve the fit, (2) symmetry of the objects, and (3) ease of handling. Pressure to achieve fit is divided into three categories: loose (class 1), close (class 2), and exact (class 3). Symmetry is also divided into three categories: symmetrical (S)—for example, a round peg in a round hole; nonsymmetrical (NS)—for example, a key inserted into a lock; and semisymmetrical (SS), which is any case between S and NS. Ease of handling is simply divided into easy (E) and difficult (D). The categories are listed in Table 4 (d) along with the time to perform the element. The complete symbol for a position includes the three work variables (e.g., P3NSD).

**Release (RL)**: A hand and finger motion element whose purpose is to surrender control of an object. There are only two cases of release: (1) the fingers open to effect the release and (2) contact release with no finger motion. The complete symbol for release identifies the case (e.g., RL1 or RL2). Times are given in Table 4 (e).

**Disengage (D)**: A hand and finger motion that causes the separation of two objects from one another, where the objects were previously held together by some force. Thus, disengagement breaks the force, resulting in a recoil action by the hand. The work variables that affect the time of a disengage motion are (1) class of fit and (2) ease of handling. Classes

(*continued*)

**TABLE 3** (*continued*)

| Motion Element (Symbol) and Description |
|---|
| of fit are loose (slight effort required, class 1), close (normal effort, class 2), and tight (considerable effort, class 3). Ease of handling is either easy (E) or difficult (D). The disengage symbol indicates the two variables (e.g., D3E). Times are given in Table 4 (f). |
| **Turn (T)**: A basic motion element that involves rotation of the hand and wrist about the long axis of the forearm. The hand can either be holding an object (e.g., turning a dial on a machine) or empty. Turn depends on two work variables: (1) the degrees of turn and (2) resistance to the turn. Table 4 (g) lists the time values as a function of these factors. The maximum turn in the table is 180°, which is imposed by the anatomical structure of the elbow and forearm. Resistance to turn is indicated by force of resistance, divided into three categories in the table. The complete symbol for turn includes both variables; for example, T90L symbolizes a turn of 90° and a large resisting force. |
| **Apply pressure (AP)**: An MTM-1 element that involves the application of force, but the force results in little or no movement. The force is usually applied by the hand and/or fingers, but the element applies to a force applied by any body member. There are two categories of apply pressure: (1) apply pressure alone (APA) and (2) apply pressure preceded by a regrasp (APB). Time values for these two cases are given in Table 4 (h). |
| **Eye travel (ET)**, **eye focus (EF)**, and **reading**: Basic eye usage motions that are important enablers for performing manual work. However, the effect of the eyes during a motion element is usually accounted for in the tabulated TMU values. For example, a reach motion that requires greater hand-eye coordination takes longer; for example, compare Case C and Case A in Table 4 (a). On the other hand, there are situations when eye travel and/or eye focus are prerequisites for subsequent motions. In these cases, they must be counted as separate basic elements. Times for the two basic eye usage motions are given in Table 4 (i). In addition, MTM-1 uses a normal time for reading text as 5.05 TMUs per word. Thus, reading a 200 word paragraph would be allowed 1010 TMUs (about 36 sec). |
| **Body, leg, and foot motions**: Additional basic motions that involve moving the body, one or both legs, and/or one or both feet. The motions include walking, bending, stooping, standing from a seated position, and sitting from a standing position. These activities are distinct from those performed at a workbench, most likely with the worker in a sitting position. These elements and others are identified in Table 4 (j), together with their symbols and normal times. |
| **Simultaneous motions**: Basic motion elements that are performed at the same time. In general, it is desirable for more than one body member to be moving simultaneously. For example, it is desirable for both the right hand and left hand to be active during the task, not only sequentially but also simultaneously. When basic motion elements can be combined simultaneously, the time required for the combination is the greater of the two parallel motions. This is called the "limiting principle," which was first proposed by the original developers of MTM (Maynard, Stegemerten, and Schwab, Historical Note 1). Certain combinations of motions are more difficult to perform simultaneously than others. The degrees of difficulty can be classified as follows: (1) easy to perform simultaneously, (2) can be performed simultaneously with practice, and (3) difficult to perform simultaneously. Table 4 (k) indicates the degree of difficulty for various combinations of basic motion elements. In degrees of difficulty (1) and (2), the limiting principle applies, assuming sufficient practice is allowed for the worker in (2). In degree of difficulty (3), both times should be added. |

internal elements when the task involves operation of a machine, and other special circumstances. The analyst then retrieves the time for each basic motion element and sums the times to obtain the normal time for the task. Since the MTM time values do not include any allowances, the analyst must add the proper allowance according to company policy for the type of work being measured. Having stated this, we are obligated to mention that MTM-1 should not be used in industrial work measurement applications until the analyst has received proper training in the technique.

### 2.2 Other MTM Systems

Additional members of the MTM family have been introduced to satisfy various user needs. Several functional systems have been developed for specific work situations such as clerical activity, machine shop work, and electronic testing. Also, second- and third-level

**TABLE 4 (a)** Normal Time Values for MTM-1 Motion Element: **Reach** (R)

| Distance | | Time in TMU | | | | Hand in Motion | | Case and Description |
|---|---|---|---|---|---|---|---|---|
| cm | inches | A | B | C or D | E | A | B | |
| < 2.0 | < 0.75 | 2.0 | 2.0 | 2.0 | 2.0 | 1.6 | 1.6 | **A** Reach to object in fixed location, or to object in other hand or on which other hand rests. |
| 2.5 | 1 | 2.5 | 2.5 | 3.6 | 2.4 | 2.3 | 2.3 | |
| 5.1 | 2 | 4.0 | 4.0 | 5.9 | 3.8 | 3.5 | 2.7 | |
| 7.6 | 3 | 5.3 | 5.3 | 7.3 | 5.3 | 4.5 | 3.6 | **B** Reach to single object in location that may vary slightly from cycle to cycle. |
| 10.1 | 4 | 6.1 | 6.4 | 8.4 | 6.8 | 4.9 | 4.3 | |
| 12.5 | 5 | 6.5 | 7.8 | 9.4 | 7.4 | 5.3 | 5.0 | |
| 15.2 | 6 | 7.0 | 8.6 | 10.1 | 8.0 | 5.7 | 5.7 | |
| 17.8 | 7 | 7.4 | 9.3 | 10.8 | 8.7 | 6.1 | 6.5 | **C** Reach to object jumbled with other objects in a group so that search and select occur. |
| 20.3 | 8 | 7.9 | 10.1 | 11.5 | 9.3 | 6.5 | 7.2 | |
| 22.9 | 9 | 8.3 | 10.8 | 12.2 | 9.9 | 6.9 | 7.9 | |
| 25.4 | 10 | 8.7 | 11.5 | 12.9 | 10.5 | 7.3 | 8.6 | |
| 30.5 | 12 | 9.6 | 12.9 | 14.2 | 11.8 | 8.1 | 10.1 | **D** Reach to a very small object or where accurate grasp is required. |
| 35.6 | 14 | 10.5 | 14.4 | 15.6 | 13.0 | 8.9 | 11.5 | |
| 40.6 | 16 | 11.4 | 15.8 | 17.0 | 14.2 | 9.7 | 12.9 | |
| 45.7 | 18 | 12.3 | 17.2 | 18.4 | 15.5 | 10.5 | 14.4 | |
| 50.8 | 20 | 13.1 | 18.6 | 19.8 | 16.7 | 11.3 | 15.8 | **E** Reach to indefinite location to get hand in position for body balance or next motion or out the way. |
| 55.9 | 22 | 14.0 | 20.1 | 21.2 | 18.0 | 12.1 | 17.3 | |
| 61.0 | 24 | 14.9 | 21.5 | 22.5 | 19.2 | 12.9 | 18.8 | |
| 66.0 | 26 | 15.8 | 22.9 | 23.9 | 20.4 | 13.7 | 20.2 | |
| 71.1 | 28 | 16.7 | 24.4 | 25.3 | 21.7 | 14.5 | 21.7 | |
| 76.2 | 30 | 17.5 | 25.8 | 26.7 | 22.9 | 15.3 | 23.2 | |
| Additional | | 0.4 | 0.7 | 0.7 | 0.6 | TMU per 2.54 cm > 76 cm (per 1.0 in > 30 in.) | | |

**TABLE 4 (b)** Normal Time Values for MTM-1 Motion Element: **Grasp** (G)

| Type of Grasp | Case | Time, TMU | Description and Object Dimensions | |
|---|---|---|---|---|
| Pickup | 1A | 2.0 | Any size object, by itself | |
| | 1B | 3.5 | Object very small or lying close against a flat surface | |
| | 1C1 | 7.3 | Interference with grasp on bottom and one side of cylindrical object | Diameter > 1.3 cm (0.5 in.) |
| | 1C2 | 8.7 | | Diameter 0.6 to 1.3 cm (0.25 to 0.5 in.) |
| | 1C3 | 10.8 | | Diameter < 0.6 cm (0.25 in.) |
| Regrasp | 2 | 5.6 | Change grasp without relinquishing control | |
| Transfer | 3 | 5.6 | Control transferred from one hand to other | |
| Select | 4A | 7.3 | Object jumbled with other objects so that search and select occur | Size larger than 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.) |
| | 4B | 9.1 | | 0.6 × .6 × .3 cm (.25 × .25 × .12 in) to 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.) |
| | 4C | 12.9 | | Size smaller than .6 × .6 × .3 cm (.25 × .25 × .12 in.) |
| Contact | 5 | 0 | Contact, sliding, or hook grasp | |

**TABLE 4 (c)** Normal Time Values for MTM-1 Motion Element: **Move** (M)

| Distance | | Time in TMU | | | Hand in motion | Weight up to | Formula Parameters | | Case and Description |
|---|---|---|---|---|---|---|---|---|---|
| cm | inches | A | B | C | B | kg (lb) | Constant | Factor | |
| < 2.0 | < 0.75 | 2.0 | 2.0 | 2.0 | 1.7 | | | | **A** Move object to other hand or against stop. |
| 2.5 | 1 | 2.5 | 2.9 | 3.4 | 2.3 | 1.1 (2.5) | 0 | 1.00 | |
| 5.1 | 2 | 3.6 | 4.6 | 5.2 | 2.9 | | | | |
| 7.6 | 3 | 4.9 | 5.7 | 6.7 | 3.6 | 3.4 (7.5) | 2.2 | 1.06 | |
| 10.1 | 4 | 6.1 | 6.9 | 8.0 | 4.3 | | | | **B** Move object to approximate or indefinite location. |
| 12.5 | 5 | 7.3 | 8.0 | 9.2 | 5.0 | 5.7 (12.5) | 3.9 | 1.11 | |
| 15.2 | 6 | 8.1 | 8.9 | 10.3 | 5.7 | | | | |
| 17.8 | 7 | 8.9 | 9.7 | 11.1 | 6.5 | 7.9 (17.5) | 5.6 | 1.17 | |
| 20.3 | 8 | 9.7 | 10.6 | 11.8 | 7.2 | | | | |
| 22.9 | 9 | 10.5 | 11.5 | 12.7 | 7.9 | 10.2 (22.5) | 7.4 | 1.22 | **C** Move object to exact location. |
| 25.4 | 10 | 11.3 | 12.2 | 13.5 | 8.6 | | | | |
| 30.5 | 12 | 12.9 | 13.4 | 15.2 | 10.0 | 12.5 (27.5) | 9.1 | 1.28 | |
| 35.6 | 14 | 14.4 | 14.6 | 16.9 | 11.4 | | | | |
| 40.6 | 16 | 16.0 | 15.8 | 18.7 | 12.8 | 14.7 (32.5) | 10.8 | 1.33 | |
| 45.7 | 18 | 17.6 | 17.0 | 20.4 | 14.2 | | | | |
| 50.8 | 20 | 19.2 | 18.2 | 22.1 | 15.6 | 17.0 (37.5) | 12.5 | 1.39 | |
| 55.9 | 22 | 20.8 | 19.4 | 23.8 | 17.0 | | | | |
| 61.0 | 24 | 22.4 | 20.6 | 25.5 | 18.4 | 19.3 (42.5) | 14.3 | 1.44 | |
| 66.0 | 26 | 24.0 | 21.8 | 27.3 | 19.8 | | | | |
| 71.1 | 28 | 25.5 | 23.1 | 29.0 | 21.2 | 21.5 (47.5) | 16.0 | 1.50 | |
| 76.2 | 30 | 27.1 | 24.3 | 30.7 | 22.7 | | | | |
| Additional | | 0.8 | 0.6 | 0.85 | TMU per 2.54 cm > 76 cm (per 1.0 in. > 30 in.) | | | | |

**TABLE 4 (d)** Normal Time Values for MTM-1 Motion Element: **Position** (P)

| Class | Description of Fit | Symmetry | Time in TMU: Easy to Handle | Time in TMU: Difficult to Handle |
|---|---|---|---|---|
| 1 | Loose (no pressure required) | S | 5.6 | 11.2 |
| | | SS | 9.1 | 14.7 |
| | | NS | 10.4 | 16.0 |
| 2 | Close (light pressure required) | S | 16.2 | 21.8 |
| | | SS | 19.7 | 25.3 |
| | | NS | 21.0 | 26.6 |
| 3 | Exact (heavy pressure required) | S | 43.0 | 48.6 |
| | | SS | 46.5 | 52.1 |
| | | NS | 47.8 | 53.4 |

*Key*: S = symmetrical, SS = semi-symmetrical, NS = nonsymmetrical.

**TABLE 4 (e)** Normal Time Values for MTM-1 Motion Element: **Release** (RL)

| Case | Time in TMU | Description |
|---|---|---|
| 1 | 2.0 | Normal release performed by opening fingers as an independent motion |
| 2 | 0 | Contact release with no finger motion |

**TABLE 4 (f)** Normal Time Values for MTM-1 Motion Element: **Disengage** (D)

| | | | Time in TMU | |
|---|---|---|---|---|
| Class | Description of Fit | Height of Recoil | Easy to Handle | Difficult to Handle |
| 1 | Loose (very slight effort, blends with subsequent move) | Up to 2.5 cm (1 in) | 4.0 | 5.7 |
| 2 | Close (normal effort, slight recoil) (1 to 5 in) | 2.5 to 12.7 cm | 7.5 | 11.8 |
| 3 | Tight (considerable effort, hand recoils markedly) | 12.7 to 30 cm (5 to 12 in) | 22.9 | 34.7 |

**TABLE 4 (g)** Normal Time Values for MTM-1 Motion Element: **Turn** (T)

| | Time in TMU for Degrees Turned | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Weight, kg (lb) | 30° | 45° | 60° | 75° | 90° | 105° | 120° | 135° | 150° | 165° | 180° |
| Small, up to 0.9 (2) | 2.8 | 3.5 | 4.1 | 4.8 | 5.4 | 6.1 | 6.8 | 7.4 | 8.1 | 8.7 | 9.4 |
| Medium, 1 to 4.5 (2 to 10) | 4.4 | 5.5 | 6.5 | 7.5 | 8.5 | 9.6 | 10.6 | 11.6 | 12.7 | 13.7 | 14.8 |
| Large, 4.5 to 16 (10 to 35) | 8.4 | 10.5 | 12.3 | 14.4 | 16.2 | 18.3 | 20.4 | 22.2 | 24.3 | 26.1 | 28.2 |

**TABLE 4 (h)** Normal Time Values for MTM-1 Motion Element: **Apply Pressure** (AP)

| Symbol | Time in TMU | Description |
|---|---|---|
| APA | 10.6 | Apply pressure alone |
| APB | 16.2 | Apply pressure preceded by regrasp |

**TABLE 4 (i)** Normal Time Values for MTM-1 Motion Element: **Eye Travel** (ET) and **Eye Focus** (EF)

| Eye motion | Symbol | Time in TMU | Key to Symbols |
|---|---|---|---|
| Eye travel | ET | $\frac{15.2L}{D}$ | $L$ = distance between points from and to which eye travels, $D$ = perpendicular distance from the eye to the line of travel. Maximum time allowed = 20 TMU |
| Eye focus | EF | 7.3 | |
| Reading | (none) | $5.05N$ | $N$ = number of words read (330 words/min) |

**TABLE 4 (j)** Normal Time Values for MTM-1 Motion Element: **Body, leg**, and **foot motions** (various symbols given in table)

| Motion | Symbol | Time in TMU | Description and Conditions |
|---|---|---|---|
| Sit | SIT | 34.7 | From standing position |
| Stand | STD | 43.4 | From seated position |
| Turn body | TBC1 | 18.6 | Turn body 45° to 90°, Case 1 – Lagging foot not aligned with leading foot |
| Turn body | TBC2 | 37.2 | Turn body 45° to 90°, Case 2 – Lagging foot aligned with leading foot |
| Bend | B | 29.0 | Bend body forward so hands can reach knees |
| Stoop | S | 29.0 | Stoop body forward so hands can reach floor |
| Arise | AB | 31.9 | Arise from bent position |
| Arise | AS | 31.9 | Arise from stooped position |
| Kneel | KOK | 29.0 | Kneel on one knee |
| Kneel | KBK | 69.4 | Kneel on both knees |
| Arise | AKOK | 31.9 | Arise from kneeling position on one knee |
| Arise | AKBK | 76.7 | Arise from kneeling position on both knees |
| Walk | W*X*FT | 5.3 per ft | Walking in ft of distance, $X$ = distance in ft |
| Walk | W*N*P | 15.0/pace | Walking in number of paces, $N$ = number of paces |
| Walk | W*N*PO | 17.0/pace | Walking in number of paces with weight or obstruction, $N$ = number of paces |
| Leg motion | LM6 | 7.1 | Move leg up to 6 in. any direction |
| Leg motion | LM*X* | 7.1 + 1.2($X$-6) | Move leg more than 6 in. any direction, where $X$ = distance of movement |
| Foot motion | FM | 8.5 | Foot moves up to 4 in. hinged at ankle |
| Foot motion | FMP | 19.1 | Foot moves up to 4 in. hinged at ankle, apply heavy pressure with leg muscles |

**TABLE 4 (k)** MTM-1 Simultaneous Hand and Arm Motion Elements

| Motion of One Hand | | Reach | | | Grasp | | | Move | | | Position | | | Disengage | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Motion of other hand | Case or Class | A, E | B | C, D | 1A, 2, 5 | 1B, 1C | 4 | A, Bm[a] | B | C | 1S | 1SS, 2S | 1NS, 2SS, 2NS | 1E, 1D | 2 |
| Reach | A, E | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 1 |
| | B | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 1 | 1 |
| | C, D | 1 | 1 | 1 | 1 | 2 | 3 | 2 | 2 | 3 | 3 | 3 | 3 | 2 | 3 |
| Grasp | 1A, 2, 5 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 3 | 1 | 3 |
| | 1B, 1C | 1 | 1 | 2 | 1 | 3 | 2 | 1 | 1 | 2 | 3 | 3 | 3 | 3 | 3 |
| | 4 | 1 | 2 | 3 | 1 | 2 | 3 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 3 |
| Move | A | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 1 | 1 |
| | B | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | 1 | 2 | 2 | 3 | 1 | 1 |
| | C | 2 | 2 | 3 | 1 | 2 | 3 | 1 | 1 | 2 | 3 | 3 | 3 | 2 | 3 |
| Position | 1S | 1 | 2 | 3 | 1 | 3 | 3 | 1 | 2 | 3 | 2 | 3 | 3 | 3 | 3 |
| | 1SS, 2S | 1 | 2 | 3 | 1 | 3 | 3 | 1 | 2 | 3 | 3 | 3 | 3 | 3 | 3 |
| | 1NS, 2SS, 2NS | 2 | 3 | 3 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| Disengage | 1E, 1D | 1 | 1 | 2 | 1 | 3 | 3 | 1 | 1 | 2 | 3 | 3 | 3 | 1 | 1 |
| | 2 | 1 | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 3 | 3 | 3 | 3 | 1 | 1 |

[a]Bm is Case B with hand in motion.

(*continued*)

**TABLE 4 (k)** (*continued*)

| |
|---|
| *Key:* The cell numbers indicate the degree of difficulty when motions are performed simultaneously.<br>1 = Easy to perform simultaneously. Use the longest motion element time.<br>2 = Can be performed simultaneously with practice. Use the longest motion element time.<br>3 = Difficult to perform simultaneously. Add the times of the two simultaneous motion elements. |
| *Assumptions:* All Reach, Grasp, and Move motions are performed within the area of normal vision. In the Position and Disengage motion elements, objects are assumed easy to handle. In general, the degree of difficulty increases if these assumptions are violated. |
| *Motions not included in the table:*<br>Turn: normally degree of difficulty = 1 except when Turn is controlled or with Disengage.<br>Position Class 3: degree of difficulty = 3.<br>Disengage Class 3: normally degree of difficulty = 3.<br>Release: degree of difficulty = 1. |

**TABLE 5** Members of the MTM Family of Predetermined Motion Time Systems

**MTM-1**: The first level MTM, in which basic motion elements are used to describe, analyze, and determine the normal time for a manual task. The MTM-1 motion elements and associated times are listed in Table 3 and 4. MTM-1 is best suited to high production operations with relatively short cycle times. The analyst time required to apply MTM-1 is about 250 times the task cycle time.

**MTM-2**: A second-level MTM system, in which the basic motion elements are combined into motion aggregates in order to reduce the analyst's time to apply the technique. MTM-2 consists of 11 motions and motion aggregates, called *motion categories*. The two most important MTM-2 categories are GET, which combines Reach, Grasp, and Release; and PUT, which combines Move and Position. MTM-2 is suited to operations that are not highly repetitive and the cycle times are greater than 1 min. The analyst time required to apply MTM-2 is about 100 times the task cycle time.

**MTM-3**: A third-level MTM system designed to further reduce the analyst's time to set a time standard but at some sacrifice in accuracy. MTM-3 has only four motion categories: (1) Handle, (2) Transport, (3) Step and foot motions, and (4) Bend and arise. Tasks that include additional elements, such as eye movements, should not be analyzed using MTM-3. The analyst time required to apply MTM-3 is about 35 times the task cycle time.

**MTM-UAS**: A third-level MTM system that is suited to applications in batch production. It includes seven basic motion categories: (1) Get and place, (2) Place, (3) Handle tool, (4) Operate, (5) Motion cycles, (6) Body motions, and (7) Visual control. Additional second-level standard data in MTM-UAS cover activities such as fastening, marking, packing, and assembling. The analyst time required to apply MTM-UAS is about 30 times the task cycle time.

**MTM-MEK**:A third-level system intended for work measurement applications in small lot production with long cycle times and other tasks performed infrequently. Time standards are usually not established in these work situations by conventional time study because the cost of setting the standard is too high. MTM-MEK can be used for these cases and the analyst time is about 5 to 15 times the cycle time.

**MTM-HC**: A functional PMTS designed for work activities found in the health-care industry. It is described as a standard database by the MTM Association.

**MTM-C**: A functional MTM system used for work measurement applications of clerical work activity, such as typing or keypunching, filing, reading, and writing. It was developed by an association of banking and other service industries. There are two levels of MTM-C: MTM-C1 and MTM-C2. The difference is that MTM-C1 emphasizes precision and motion detail in its applications, while MTM-C2 emphasizes speed of application. The analyst time to apply MTM-C1 is about 125 times the task cycle time, and about 75 times for MTM-C2.

**MTM-V**: A functional standard data system developed using MTM-1 for work measurement in machine tool operation. Its work elements include handling of work parts, operating a machine tool and other equipment (e.g., cranes,

(*continued*)

**TABLE 5** (*continued*)

fixtures, chucks), and setting up a job for production. The analyst time required to apply MTM-V is about 10 times the task cycle time.

**MTM-TE**: A functional standard data system designed for work measurement in electronic testing applications. Its work elements cover both manual and mental activities.

**MTM-M**: A functional MTM system for measuring assembly work that is performed under a stereoscopic microscope.

*Source:* Estimates of analyst time to apply the various MTM systems are based on [7].

systems are available to reduce the time required to develop time standards. Finally, several computerized systems have been developed based on MTM. Table 5 lists many of these MTM systems with a brief description of each. Readers interested in learning more about the other MTM systems can consult the coverage provided in [6] or the MTM Web site at www.mtm.org.

## 3 MAYNARD OPERATION SEQUENCE TECHNIQUE

The Maynard Operation Sequence Technique (MOST) is a high-level predetermined motion time system that is based on MTM. It uses the same time units as in MTM: the TMU (time measurement unit), which is 0.00001 hr. MOST is a product of H. B. Maynard and Company, an educational and consulting firm located in Pittsburgh, Pennsylvania.[4] The basic version of MOST, which is now referred to as Basic MOST, was developed at Maynard's Swedish Division around 1967 under the direction of Kjell Zandin (Historical Note 1). Section 3.1, describes the Basic MOST package. Several additional versions of MOST have been developed since Basic MOST was introduced, and these are summarized in Section 3.2. Section 3.3 describes a computerized version of MOST.

### 3.1 Basic MOST

The focus of Basic MOST is on work activity involving the movement of objects. In fact, the majority of industrial manual work does involve moving objects (e.g., parts, tools) from one location to another in the workplace. The underlying premise of Basic MOST is that the movement of objects consists of a pattern of body motions and actions that is nearly the same for all moves. Only the details of the pattern differ, depending on the object being moved and the circumstances of the work activity. Consistent with this premise, Basic MOST uses motion aggregates (collections of basic motion elements) that are concerned with moving things. The motion aggregates are called ***activity sequence models*** in Basic MOST. In our hierarchy of work activity, an activity sequence model is either at the same level as a work element, or slightly below. Depending on how a work element is defined for a given task, it may include one or more activity sequence models. Rarely would a work element be defined as less than a complete activity sequence model. The relationships are indicated in Figure 2.

[4]MOST is a registered trademark of H. B. Maynard and Company, Inc.

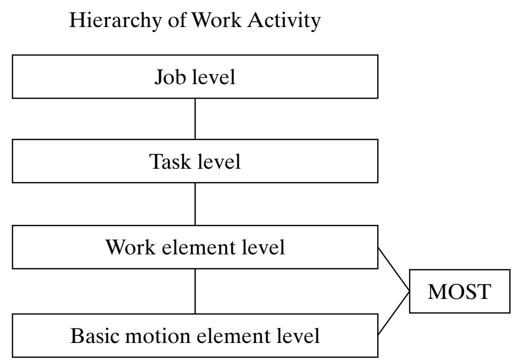


**Figure 2** The position of the Basic MOST activity sequence model in our work hierarchy.

There are three activity sequence models in Basic MOST, each of which consists of a standard sequence of actions:

- *General move.* This sequence model is used when an object is moved freely through space from one location to the next (e.g., picking something up from the floor and placing it on a table).
- *Controlled move.* This sequence model is used when an object is moved while it remains in contact with a surface (e.g., sliding the object along the surface) or the object is attached to some other object during its movement (e.g., moving a lever on a machine).
- *Tool use.* This sequence model applies to the use of a hand tool (e.g., a hammer or screwdriver).

The actions in an activity sequence model, called ***sequence model parameters*** in Basic MOST, are similar to basic motion elements in MTM, but there is not a one-to-one correspondence. Some parameters map directly into MTM basic motions, while others are unique to Basic MOST. Let us examine the three sequence models and indicate the standard sequence of model parameters for each.[5]

**General Move.** The General Move sequence is applicable when an object is moved through the air from one location to another. There are four parameters (actions) in the General Move, symbolized by letters of the alphabet:

A—Action distance, usually horizontal. This parameter is used to describe movements of the fingers, hands, or feet (e.g., walking). The movement can be performed either loaded or unloaded.

B—Body motion, usually vertical. This parameter defines vertical body motions and actions (e.g., sitting, standing up).

[5]Once again, we must caution against the application of MOST without proper training. Guidelines can be obtained from H. B. Maynard & Company in Pittsburgh, Pennsylvania; their Web site can be accessed at www.hbmaynard.com.

G—Gain control. This parameter is used for any manual actions involving the fingers, hands, or feet to gain physical control of one or more objects. It is closely related to the grasp motion element in MTM (e.g., grasp the object).

P—Placement. The placement parameter is used to describe the action involved to lay aside, position, orient, or align an object after it has been moved to the new location (e.g., position the object).

These parameters occur in the following standard sequence in the General Move:

| A | B | G | A | B | P | A |
|---|---|---|---|---|---|---|
| Action distance | Body motion | Grasp | Action distance | Body motion | Placement | Action distance |

where the first three parameters (A B G) represent basic motions to ***get*** an object, the next three parameters (A B P) represent motions to ***put*** or move the object to a new location, and the final parameter (A) applies to any motions at the end of the sequence, such as ***return*** to original position.

To complete the activity sequence model, each parameter is assigned a numerical value in the form of a subscript or index number that represents the time to accomplish that action. The value of the index number depends on the type of action, its motion content, and the conditions under which it is performed. Table 6 lists the parameters and possible circumstances for the action, together with the corresponding values of the index numbers. When the index values have been entered for all parameters, the time for the sequence model is determined by summing the index values and multiplying by 10 to obtain the total TMUs. The procedure is illustrated in the following example.

**Example 1 General Move**

Develop the activity sequence model and determine the normal time for the following work activity: A worker walks 5 steps, picks up a small part from the floor, returns to his original position, and places the part on his worktable.

**Solution:** Referring to Table 6, the indexed activity sequence model for this work activity would be the following:

$$A_{10}\,B_6\,G_1\,A_{10}\,B_0\,P_1\,A_0$$

where $A_{10}$ = walk 5 steps, $B_6$ = bend and arise, $G_1$=gain control of small part, $A_{10}$ = walk back to original position, $B_0$ = no body motion, $P_1$ = lay aside part on table, and $A_0$ = no motion. The sum of the index values is 28. Multiplying by 10, we have 280 TMUs (about 10 sec). ■

The General Move is the most common way in which materials are moved and is therefore the most frequently used activity sequence model in Basic MOST. Approximately half of all manual work occurs as General Moves. For assembly work and material handling, the proportion is higher. For machine shop work, the proportion is lower.

**TABLE 6** MOST Parameters and Index Values for the General Move Activity Sequence Model

| General Move activity sequence model = **A B G A B P A** | | | | |
|---|---|---|---|---|
| Index | **A** = Action distance | **B** = Body motion | **G** = Gain control | **P** = Placement |
| **0** | Close ≤ 5 cm (2 in.) | | | Hold, Toss |
| **1** | Within reach (but > 2 in.) | | Grasp light object using one or two hands | Lay aside Loose fit |
| **3** | 1 or 2 steps | Bend and arise with 50% occurrence | Grasp object that is heavy, or obstructed, or hidden, or interlocked | Adjustments, light pressure, double placement |
| **6** | 3 or 4 steps | Bend and arise with 100% occurrence | | Position with care, or precision, of blind, or obstructed, or heavy pressure |
| **10** | 5, 6, or 7 steps | Sit or stand | | |
| **16** | 8, 9, or 10 steps | Through door, or Climb on or off, or Stand and bend, or Bend and sit | | |

**Controlled Move.** The Controlled Move sequence model is used when an object is moved through a path that is somehow constrained. The object cannot be freely moved through the air. This situation arises when an object is slid across a surface or when the object is attached to something else and it can be moved only through a controlled path. The second case is commonly encountered in the operation of machinery, and the move involves a lever, crank, switch button, or similar entity that is attached to the machinery. The parameters in the Controlled Move include the A, B, and G parameters from the General Move. In addition, three new parameters are introduced:

M—Move, controlled. This parameter is used to describe any manual body motions required to move an object over a controlled path.

X—Process time. Since the Controlled Move can include the operation of machinery, there is often a process time associated with the machinery (e.g., time to turn the workpiece in a lathe). No manual motions are included in this parameter.

I—Align. This parameter is used when manual motions are performed at the end of the controlled move to align objects.

These parameters occur in the following standard sequence in the Controlled Move:

| A | B | G | M | X | I | A |
|---|---|---|---|---|---|---|
| Action distance | Body motion | Grasp | Move, controlled | Process time | Align | Action distance |

where the first three parameters (A B G) are the basic motions to ***get*** an object, the next three parameters (M X I) represent moving or actuating the object followed by a process time and alignment, and the final parameter (A) provides for a possible ***return*** at the end of the sequence.

**TABLE 7** MOST Parameters and Index Values for the Controlled Move Activity Sequence Model

| Controlled Move activity sequence model = **A B G M X I A** | | | | |
|---|---|---|---|---|
| Index | **M** = Move, controlled | **X** = Process time[a] | | **I** = Alignment |
| | | Seconds | Minutes | |
| **1** | Push, pull, pivot: button, switch, knob (≤ 12 in.) | 0.5 | 0.01 | Align to one point |
| **3** | Push and pull, turn, open, seat, shift, press: resistance encountered, or high control required, or 2 stages of control (≤ 12 in.); 1 crank of lever. | 1.5 | 0.02 | Align to 2 points, Close align (≤ 4 in.) |
| **6** | Open and shut, operate, push or pull: with 1 or 2 steps (> 12 in.); 3 cranks of lever. | 2.5 | 0.04 | Align to 2 points, Close align (> 4 in.) |
| **10** | Manipulate, maneuver, push, or pull with 3, 4, or 5 steps; 6 cranks of lever. | 4.5 | 0.07 | Precision align |
| **16** | Push or pull with 6, 7, 8, or 9 steps included; 11 cranks of lever. | 7.0 | 0.11 | High precision align |

[a]For process times longer than those listed in the table, the actual process time in seconds can be multiplied by 2.78 and rounded to the next higher value to obtain the index for the X parameter.

Index values are applied to each parameter as in the General Move. The same A, B, and G index values from Table 6 apply to the Controlled Move. Table 7 lists the index values for the M, X, and I parameters used exclusively for the Controlled Move.

**Example 2 Controlled Move**

Develop the activity sequence model and determine the normal time for the following work activity: A worker takes 2 steps, grasps the waist-level feed lever on the lathe, pulls up the lever approximately 15 cm to engage the feed. The process time to turn the part is 25 sec. There is no alignment and no action by the worker at the end of the process time.

**Solution:** Referring to Tables 6 and 7, and multiplying the 25 sec process time by 2.78 to obtain a value of 69.5 for the X parameter (rounded up to 70), we obtain the following indexed activity sequence model for this work activity:

$$A_3\ B_0\ G_1\ M_1\ X_{70}\ I_0\ A_0$$

where $A_3$ = take 2 steps, $B_0$ = no body motion, $G_1$ = gains control of lever, $M_1$ = pull the lever up 15 cm, $X_{70}$ = process time of 25 sec, $I_0$ = no alignment, and $A_0$ = no motion. The sum of the index values is 75. Multiplying by 10, we have 750 TMUs (about 27 sec). The lathe process time accounts for all but 2 sec of the activity sequence. ■

**Tool Use.** The Tool Use sequence model applies to the variety of work situations that involve the use of a hand tool. Hand tools covered by this model include screwdrivers, ratchets, hammers, scissors, knives, various kinds of wrenches, and common measuring instruments. When a worker's hand is used to perform a motion such as turning a screw, Tool Use is employed. The sequence model also covers writing, marking, reading, thinking, and inspecting. The parameters in the Tool Use sequence include A, B, G, and

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# Lean Production



In this chapter, we examine several additional approaches related to methods engineering. These approaches are aimed at increasing productivity, reducing cost, improving quality, and solving problems in the operations of an organization. These objectives are basically the same as those in methods engineering. In many ways, the topics discussed here are adaptations and modifications of the previous methods engineering techniques and principles. However, these alternative approaches are of more recent vintage than the traditional methods engineering programs, and they represent a more contemporary view of work and the social environment in which it is performed. One of the distinguishing characteristics of this current view is that the active participation of the worker should be enlisted in the improvement activities. The new approaches can be presented in two ways: (1) lean production, a highly efficient production system developed at Toyota Motors in Japan, and (2) Six Sigma, a quality improvement procedure that seeks perfection.

From Chapter 20 of *Work Systems and the Methods, Measurement, and Management of Work,* First Edition. Mikell P. Groover. 

***Lean production*** means doing more work with fewer resources. It is an adaptation of mass production in which work is accomplished in less time, smaller space, with fewer workers and less equipment, and yet achieving higher quality levels in the final product. Lean production also means giving customers what they want and satisfying or surpassing their expectations. Lean production is based on the way manufacturing operations were organized at the Toyota Motors Company in Japan during the 1980s. Toyota had pioneered a system of production that was quite different from the mass production techniques used by automobile companies in the United States and Europe. The Toyota Production System, as it was called before the term "lean" was applied to it, began to evolve in the 1950s to cope with the realities of Japan's postwar economy (e.g., a much smaller domestic auto market than in the United States or Europe and a scarcity of Japanese investment capital for plant and equipment). To deal with these challenges, Toyota developed a production system that could produce a variety of car models with fewer quality problems, lower inventory levels, smaller manufacturing lot sizes for the parts used in the cars, and reduced lead times to produce the cars. Development of the Toyota Production System was led by Taiichi Ohno (Historical Note 1). In this chapter, we examine the methods used at Toyota that have come to be called lean production.

### HISTORICAL NOTE 1 LEAN PRODUCTION

The person credited with developing the Toyota Production System is Taiichi Ohno (1912–1990), an engineer and executive at Toyota Motors. In the post–World War II period, the Japanese automotive industry had to basically start over. Ohno visited a U.S. auto plant to learn American production methods. However, the car market in Japan was much smaller than its U.S. counterpart, so a Japanese plant could not afford the large production runs and huge work-in-process inventories that were common in the United States. Ohno knew that Toyota's plants needed to be more flexible. Also, space was (and is) very precious in Japan. These conditions, as well as Ohno's aversion to waste in any form (***muda***, as the Japanese call it), motivated him to develop some of the basic ideas and procedures that have come to be known as lean production. He and his colleagues then proceeded to perfect these ideas and procedures over the next several decades, including just-in-time production and the ***kanban*** system of production control, production leveling, setup time reduction, quality circles, and dedicated adherence to quality control.

Ohno himself did not coin the term "lean production" to describe the collection of methods and systems used at Toyota to improve production efficiency. In fact, he titled his book ***The Toyota Production System: Beyond Large-Scale Production*** [13]. The term "lean production" was coined by researchers at Massachusetts Institute of Technology to describe the activities and programs that seemed to explain Toyota's success in the efficiency with which they produced cars and the quality of the cars they produced. In an MIT research project that came to be known as the International Motor Vehicle Program (IMVP), a survey was conducted of 87 automobile assembly plants throughout the world. The research was popularized by the book ***The Machine that Changed the World*** [23]. In the subtitle of the book was the term "lean production."

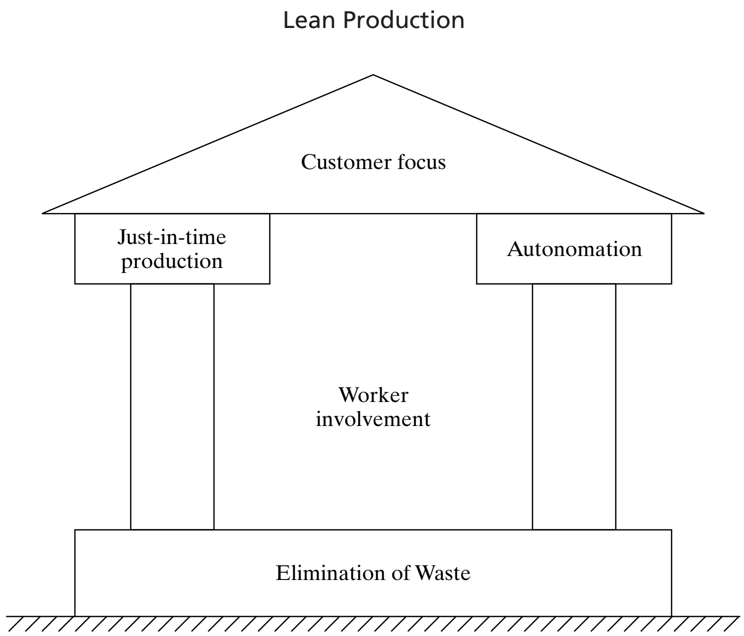


**Figure 1** The structure of a lean production system.

**TABLE 1** The Elements of Just-in-Time Production, Worker Involvement, and Autonomation in the Lean Production Structure

| Just-in-Time Production | Worker Involvement | Autonomation |
|---|---|---|
| Pull system of production control using kanbans | Continuous improvement (kaizen) | Stop the process when something goes wrong (e.g., production of defects) |
| Setup time reduction for smaller batch sizes | Quality circles | Prevention of overproduction |
| Production leveling | Visual management | Error prevention and mistake proofing |
| On-time deliveries | The 5S system | Total productive maintenance for reliable equipment |
| Zero defects | Standardized work procedures | |
| Flexible workers | Participation in total productive maintenance by workers | |

The ingredients of a lean production system are shown in the structure in Figure 1.[1] At the base of the structure is the foundation of the Toyota system: elimination of waste in production operations. Taiichi Ohno was driven to reduce waste at Toyota. Standing on the foundation are two pillars: (1) just-in-time production and (2) autonomation (automation with a human touch).[2] The two pillars support a roof that symbolizes a focus on the customer. The goal of lean production is customer satisfaction. Between the two

[1]Various forms of the "lean structure" have appeared in the literature. We believe the one shown here is representative.
[2]Taiichi Ohno describes just-in-time and autonomation as the two pillars needed to support the Toyota production system [13].

pillars and residing inside the structure is an emphasis on worker involvement: workers who are motivated, flexible, and continually striving to make improvements. Table 1 identifies the elements of just-in-time production, worker involvement, and autonomation in the lean production structure.

## 1 ELIMINATION OF WASTE

The underlying basis of the Toyota Production System is elimination of waste—called ***muda*** in Japanese. The very word has the sound of something messy (perhaps because it begins with the English word "mud"). In manufacturing, waste abounds. Ohno identified seven forms of waste in manufacturing that he wanted to eliminate by means of the various procedures that made up the Toyota system. Ohno's seven forms of waste are as follows:

1. Production of defective parts
2. Production of more than the number of items needed (overproduction)
3. Excessive inventories
4. Unnecessary processing steps
5. Unnecessary movement of people
6. Unnecessary transport and handling of materials
7. Workers waiting.

### 1.1 Production of Defective Parts

Eliminating production of defective parts (waste form 1) requires a quality control system that achieves perfect first-time quality. In the area of quality control, the Toyota production system was in sharp contrast with the traditional QC systems used in mass production. In mass production, quality control is typically defined in terms of an acceptable quality level (AQL), which means that a certain minimum level of fraction defects is sufficient, even satisfactory. In lean production, by contrast, perfect quality is required. The just-in-time delivery discipline used in lean production necessitates a zero defects level in parts quality, because if the part delivered to the downstream workstation is defective, production is forced to stop. There is little or no inventory in a lean system to act as a buffer. In mass production, inventory buffers are used just in case these quality problems occur. The defective work units are simply taken off the line and replaced with acceptable units. However, the problem is that such a policy tends to perpetuate the cause of the poor quality. Therefore, defective parts continue to be produced. In lean production, a single defect draws attention to the quality problem, forcing corrective action and a permanent solution. Workers inspect their own production, minimizing the delivery of defects to the downstream production station.

### 1.2 Overproduction and Excessive Inventories

Overproduction (waste form 2) and excessive inventories (waste form 3) are correlated. Producing more parts than necessary means that there are leftover parts that must be stored. Of all of the forms of muda, Ohno believed that the "greatest waste of all is excess inventory."[3] Overproduction and excess inventories generate increased costs in the following areas:

- Warehousing (building, lighting and heating, maintenance)
- Storage equipment (pallets, rack systems, forklifts)
- Additional workers to maintain and manage the extra inventory
- Additional workers to make the parts that were overproduced
- Other production costs (raw materials, machinery, power, maintenance) to make the parts that were overproduced and stored
- Interest payments to finance all of the above.

The kanban system (Section 2.1) for just-in-time production provides a control mechanism at each workstation to produce only the minimum quantity of parts needed to feed the next process in the sequence. In so doing, it limits the amount of inventory that is allowed to accumulate between operations.

### 1.3 Other Forms of Waste

Waste forms 4 through 7 all represent inefficient use of resources. Most of these inefficiencies can be corrected through the use of good methods engineering and layout planning. Good equipment maintenance (total productive maintenance, Section 3.3) helps to reduce machine downtime that causes waiting time for workers.

**Unnecessary Processing Steps.** Unnecessary processing steps mean that energy is being expended by the worker and/or machine to accomplish work that adds no value to the product. An example of this waste form is a product that is designed with features that serve no useful function to the customer, and yet time and cost are consumed to create those features. Another reason for unnecessary processing steps is that the processing method for the given task has not been well designed. Perhaps no work design has occurred at all. Consequently, the method used for the task includes wasted hand and body motions, unnecessary work elements, inappropriate hand tools, inefficient production equipment, poor ergonomics, and safety hazards.

**Unnecessary Movement of Workers and Materials.** The movement of people and materials is a necessary activity in manufacturing. Body motions and walking are necessary and natural elements of the work cycle for most workers, and materials must be transported from operation to operation during their processing. It is when the movement of workers or materials is done unnecessarily and without adding value to the

[3] Ohno [13].

product that waste occurs. The following reasons explain why people and materials are sometimes moved unnecessarily:

- *Inefficient workplace layout.* Tools and parts are randomly organized in the work space, so that workers must search for what they need and use inefficient motion patterns to complete their tasks.
- *Inefficient plant layout.* Workstations are not arranged along the line of flow of the processing sequence.
- *Improper material-handling method.* For example, manual-handling methods are used instead of mechanized or automated equipment.
- *Improper spacing of production machines.* Greater distances mean longer transit times between machines.
- *Larger equipment than necessary for the task.* Larger machines need larger access space and greater distances between machines.
- *Conventional batch production.* In batch production, changeovers are required between batches that result in downtime during which nothing is produced.

**Workers Waiting.** The seventh form of muda is workers waiting. When workers are forced to wait, it means that no work (either value-adding or non–value-adding) is being performed. There is a variety of reasons why workers are sometimes forced to wait:

- Materials have not been delivered to the workstation.
- The assembly line has stopped.
- A machine has to be repaired.
- A machine is being serviced by the setup crew.
- A machine is performing its automatic processing cycle on a work part.

## 2 JUST-IN-TIME PRODUCTION

Just-in-time (JIT) production systems were developed to minimize inventories, especially work-in-process. Excessive WIP is seen in the Toyota Production System as waste that should be minimized or eliminated. The ideal ***just-in-time production system*** produces and delivers exactly the required number of each component to the downstream operation in the manufacturing sequence at the exact moment when that component is needed. Each component is delivered "just in time." This delivery discipline minimizes WIP and manufacturing lead time as well as the space and money invested in WIP. In the Toyota Production System, the just-in-time discipline was applied not only to its own production operations but to supplier delivery operations as well.

While the development of JIT production systems is attributed to Toyota, many U.S. firms have also adopted the just-in-time philosophy. Other terms are sometimes applied to the American practice of JIT to suggest differences with the Japanese practice. For example, ***continuous flow manufacturing*** is a widely used term in the United States that denotes a just-in-time style of production operations in which work parts are processed and transported directly to the next workstation one unit at a time. Each process is completed just before the next process in the sequence begins. In effect, this is JIT with a batch size of one work unit. Prior to JIT, the traditional U.S. practice might be described as a "just-in-case" philosophy—that is, to hold large in-process inventories to cope with production problems such as late deliveries of components, machine breakdowns, defective components, and wildcat strikes.

The just-in-time production discipline has shown itself to be very effective in high-volume repetitive operations, such as those found in the automotive industry [12]. The potential for WIP accumulation in this type of manufacturing is significant, due to the large quantities of products made and the large numbers of components per product.

The principal objective of JIT is to reduce inventories. However, inventory reduction cannot simply be mandated to happen. Three requisites must be in place for a just-in-time production system to operate successfully: (1) a pull system of production control, (2) setup time reduction for smaller batch sizes, and (3) stable and reliable production operations.

### 2.1 Pull System of Production Control

JIT is based on a ***pull system*** of production control, in which the order to make and deliver parts at each workstation in the production sequence comes from the downstream station that uses those parts. When the supply of parts at a given workstation is about to be exhausted, that station orders the upstream station to replenish the supply. Only upon receipt of this order is the upstream station authorized to produce the needed parts. When this procedure is repeated at each workstation throughout the plant, it has the effect of pulling parts through the production system. By comparison, in a ***push system*** of production control, parts at each workstation are produced irrespective of the immediate need for those parts at its respective downstream station. In effect, this production discipline pushes parts through the plant. The risk in a push system is that more parts get produced in the factory than it can handle, resulting in large queues of work in front of machines. The machines are unable to keep up with arriving work, and the factory becomes overloaded with work-in-process inventory. The difference between push and pull systems of production control is illustrated in Figure 2.

The Toyota Production System implemented its pull system by means of kanbans. The word *kanban* (pronounced kahn-bahn) means "card" in Japanese. The ***Kanban system*** of production control is based on the use of cards that authorize (1) parts production and (2) parts delivery in the plant. Thus, there are two types of kanbans: (1) production kanbans, and (2) transport kanbans. A ***production kanban*** (P-kanban) authorizes the upstream station to produce a batch of parts. As parts are produced, they are placed in containers, so the batch quantity is just sufficient to fill the container. Production of more than this quantity of parts is not allowed in the kanban system. A ***transport kanban*** (T-kanban) authorizes transport of the container of parts to the downstream station.

The operation of a kanban system is illustrated in Figure 3. The workstations shown in the figure (station $i$ and station $i + 1$) are only two in a sequence of multiple stations upstream and downstream. The flow of work is from station $i$ (the upstream station)

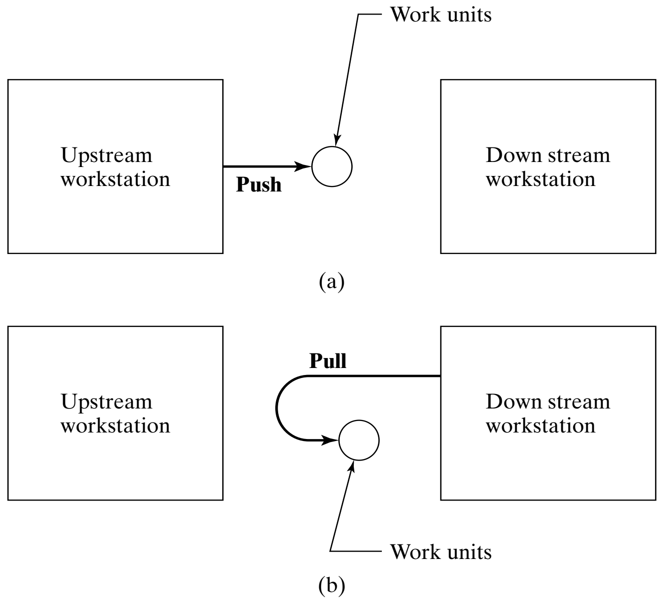


**Figure 2** The difference between push and pull systems of production control: (a) push system, in which upstream stations push work to downstream stations and (b) pull system, in which downstream stations pull work from upstream stations.

to station $i + 1$ (the downstream station). The sequence of steps in the kanban pull system is as follows (our numbering sequence is coordinated with Figure 3):

1. Station $i + 1$ removes the next P-kanban from the dispatching rack. This P-kanban authorizes it to process a container of part b. A material handling worker removes the T-kanban from the incoming container of part b and takes it back to station $i$.
2. At station $i$, the material-handling worker finds the container of part b, removes the P-kanban and replaces it with the T-kanban. He then puts the P-kanban in the dispatching rack at station $i$.
3. The container of part b that was at station $i$ is moved to station $i + 1$, as authorized by the T-kanban. The P-kanban for part b at station $i$ authorizes station $i$ to process a new container of part b, but it must wait its turn in the rack for the other P-kanbans ahead of it. Scheduling of work at each station is determined by the order in which the production kanbans are placed in the dispatching rack. Meanwhile, processing of the b parts at station $i + 1$ has been completed and that station removes the next P-kanban from the dispatching rack and begins processing that container of parts (it happens to be part d as indicated in the figure).

As mentioned, stations $i$ and $i + 1$ are only two adjacent stations in a longer sequence. All other pairs of upstream and downstream stations operate according to the

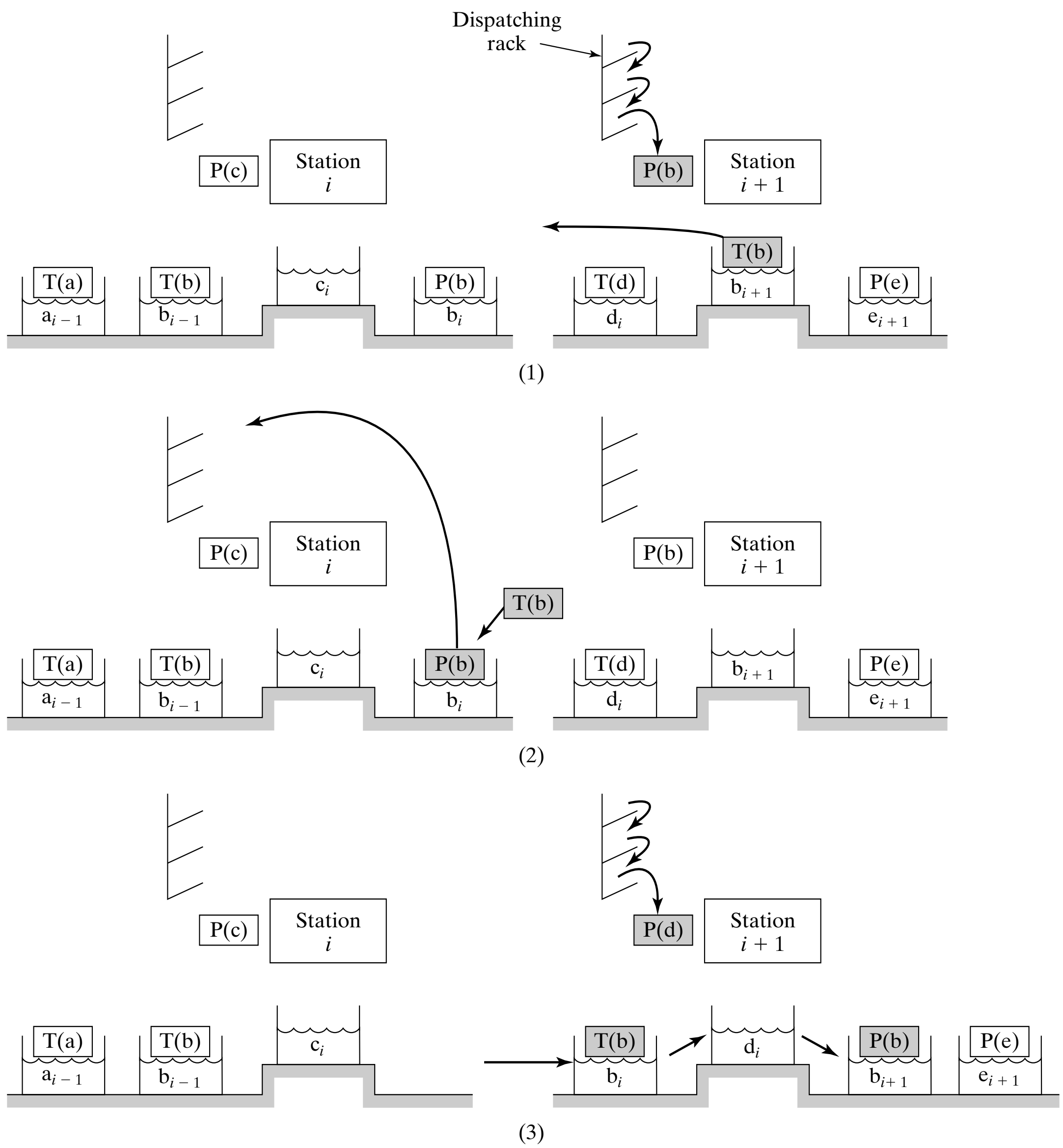


**Figure 3** Operation of a kanban system between workstations (see description of steps in the text).

same kanban pull system. This production control system avoids unnecessary paperwork. The kanban cards are used over and over again instead of generating new production and transport orders every cycle. Although considerable labor is involved in material handling (moving cards and containers between stations), this is claimed to promote teamwork and cooperation among workers.

Today's kanban installations in the automotive industry rely on modern communications technologies rather than cards [6]. These electronic kanban systems connect production workers to the material handlers who deliver the parts. For example, one system installed at Ford Motor Company uses battery-powered wireless buttons located at each operator workstation. When the supply of parts gets down to a certain level, the operator presses the button, which signals the material handlers to deliver another batch of parts. Each button emits a low power signal that is received by antennas attached to the plant ceiling and transmitted to a computer system that provides instructions to the material handlers about what to deliver, where, and when.

### 2.2 Setup Time Reduction for Smaller Batch Sizes

To minimize work-in-process inventories in manufacturing, batch sizes and setup times must be minimized. The relationship between batch size and setup time is given by the economic order quantity formula. In our mathematical model for total inventory cost, from which the EOQ formula is derived, average inventory level is equal to one-half the batch size. To reduce average inventory level, batch size must be reduced. And to reduce batch size, setup cost must be reduced. This means reducing setup times. Reduced setup times permit smaller batches and lower work-in-process levels.

Setup time reductions result from a number of basic approaches that are best described as methods improvements. The approaches are largely credited to the pioneering work of Shigeo Shingo, an industrial engineering consultant at Toyota Motors during the 1960s and 1970s.[4]

The starting point in setup time reduction is to recognize that the work elements in setting up a machine can be separated into two categories:

- *Internal elements.* These work elements can only be done while the production machine is stopped.
- *External elements.* These work elements do not require that the machine be stopped.

Examples of the two categories are listed in Table 2. By their nature, external work elements can be accomplished while the previous job is still running. For the setup time to be reduced in a given changeover, the setup tooling (e.g., die, fixture, mold) must be designed and the changeover procedure must be planned to permit as much of the setup as possible to consist of external work elements.

Although it is desirable to reduce the times required to accomplish both internal and external work elements, the internal elements must be given a higher priority, since they determine the length of time that a machine will not be producing during a changeover. The following approaches apply mostly to the internal elements in a setup:

- Use time and motion study and methods improvements to reduce the sum of the internal work elements to the minimum possible time.
- Use two workers working in parallel to accomplish the setup, rather than one worker working alone. This approach is not applicable to all changeover situations,

[4]They have been documented in [4], [12], [18], [19], and [22]. The approaches can be applied to virtually all batch production situations, but their applications in the automotive industry have emphasized pressworking and machining operations, owing to their widespread use in this industry.

**TABLE 2** Examples of Internal and External Work Elements During a Production Setup or Changeover

| Internal Work Elements | External Work Elements |
|---|---|
| Removing tooling (e.g., dies, molds, fixtures) used in previous production job from machine | Retrieving tooling for next job from tool storage |
| Positioning and attaching tooling for next job in machine | Assembling tooling components next to machine (if tooling consists of separate pieces that must be assembled) |
| Making final adjustments and alignments of tooling | Reading engineering drawings regarding new setup |
| Trying out the setup and making trial parts | Reprogramming machine for next job (e.g., downloading part program for new part) |

**TABLE 3** Examples of Setup Reductions in Japanese and U.S. Industries

| Industry | Equipment Type | Setup Time Before Reduction | Setup Time After Reduction | Reduction (%) |
|---|---|---|---|---|
| Japanese automotive | 1000 ton press | 4 hr | 3 min | 98.7 |
| Japanese diesel | Transfer line | 9.3 hr | 9 min | 98.4 |
| U.S. power tool | Punch press | 2 hr | 3 min | 97.5 |
| Japanese automotive | Machine tool | 6 hr | 10 min | 97.2 |
| U.S. electric appliance | 45 ton press | 50 min | 2 min | 96.0 |

*Source:* [21].

but where it is applicable, it theoretically reduces the downtime to one-half the time required for one worker.

- Eliminate or minimize adjustments in the setup. Adjustments are time-consuming.
- Use quick-acting fasteners instead of bolts and nuts where possible.
- When bolts and nuts must be used with washers, use U-shaped instead of O-shaped washers. A U-shaped washer can be inserted between the parts to be clamped without completely disassembling the nut from the bolt. To add an O-shaped washer, the nut must be removed from the bolt.
- Design modular fixtures consisting of a base unit plus insert tooling that can be quickly changed for each new part style. The base unit remains attached to the production machine, so that only the insert tooling must be changed.

Although methods for reducing setup time were pioneered by the Japanese, U.S. firms have also adopted these methods. Results of the efforts are sometimes dramatic. Table 3 presents some examples of setup reductions in Japanese and U.S. industries reported by Suzaki [21].

The economic impact of setup time reduction in a production operation can be assessed using the economic order quantity equations.

**Example 1 Effect of Setup Reduction on EOQ and Inventory Cost**

Let us determine the effect on economic batch size and total inventory costs of reducing setup time given the following information: annual demand = 15,000 pc/yr, unit cost = \$20, holding cost rate = 18%/yr, setup time = 5 hr, and cost of downtime during setup = \$150/hr. Suppose it were possible to reduce setup time from 5 hr to 5 min (this kind of reduction is not so far-fetched, given the data in Table 3). Determine (a) the economic order quantity and (b) total inventory cost for this new situation.

**Solution:** First, recall the results of the earlier example. Setup cost $C_{su}$ = (5 hr)(\$150/hr) = \$750 and holding cost $C_h$ = 0.18(\$20) = \$3.60. Using these values, the economic order quantity was computed as

$$EOQ = \sqrt{\frac{2(15{,}000)(750)}{3.60}} = 2500 \text{ units}$$

The corresponding total inventory costs are computed as

$$TIC = 0.5(2500)(3.60) + 750(15{,}000/2500) = \$9000$$

**(a)** Reducing the setup time to 5 min reduces the setup cost to $C_{su}$ = (5/60 hr)(\$150) = \$12.50. Holding cost remains the same, and the new economic order quantity is

$$EOQ = \sqrt{\frac{2(15{,}000)(12.50)}{3.60}} = 323 \text{ units}$$

This is a significant reduction from the 2500-piece batch size when setup time was 5 hr.

**(b)** Total inventory costs are also reduced, as follows:

$$TIC = 0.5(323)(3.60) + 12.50(15{,}000/323) = \$1161$$

This is an 87% cost reduction from the previous value. ■

## 2.3 Stable and Reliable Production Operations

Other requirements for a successful JIT production system include (1) production leveling, (2) on-time delivery, (3) defect-free components and materials, (4) reliable production equipment, (5) a workforce that is capable, committed, and cooperative, and (6) a dependable supplier base.

**Production Leveling.** If production is to flow as smoothly as possible, there must be minimum perturbations from the fixed schedule. Perturbations in downstream operations tend to be magnified in upstream operations. A 10% change in final assembly is often amplified into a 50% change in parts production operations, due to overtime, unscheduled setups, variations from normal work procedures, and other exceptions. By maintaining a constant master production schedule over time, smooth workflow is achieved and disturbances in production are minimized.

The trouble is that demand for the final product is not constant. Accordingly, the production system must adjust to the ups and downs of the marketplace using ***production***

***leveling***, which means distributing the changes in product mix and quantity as evenly as possible over time. Production leveling can be accomplished using the following approaches [5]:

- Authorizing overtime during busy periods
- Using finished product inventories to absorb daily ups and downs in demand
- Adjusting the cycle times of the production operations
- Producing in small batch sizes that are enabled by setup time reduction techniques. In the ideal, the batch size is reduced to one. Instead of producing parts A and B according to a schedule that looks like this:

  AAAAAAAAAABBBBBBBBBB,

  the parts are instead scheduled like this:

  ABABABABABABABABABAB.

The benefits of production leveling include greater responsiveness to changes in product demand, shorter lead times, smaller in-process and finished goods inventories, and regularity in the workload of production workers.

**On-Time Deliveries, Zero Defects, and Reliable Production Equipment.** Just-in-time production requires near perfection in on-time delivery, parts quality, and equipment reliability. Owing to the small lot sizes used in JIT, parts must be delivered before stock-outs occur at downstream stations. Otherwise, these stations are starved for work and production is forced to stop.

JIT requires high quality in every aspect of production. If parts are produced with quality defects, they cannot be used in subsequent processing or assembly stations, thus interrupting work at those stations and possibly stopping production. Such a severe penalty motivates a discipline of very high quality levels (zero defects) in parts fabrication. Workers are trained to inspect their own output to make sure it is right before it goes to the next operation. In effect, this means controlling quality during production rather than relying on inspectors to discover the defects later.

JIT also requires highly reliable production equipment. Low work-in-process leaves little room for equipment stoppages. Machine breakdowns cannot be tolerated in a JIT production system. The equipment must be "designed for reliability," and the plant that operates the equipment must employ total productive maintenance.

**Workforce and Supplier Base.** Workers in a just-in-time production system must be cooperative, committed, and cross-trained. Small batch sizes mean that workers must be willing and able to perform a variety of tasks and to produce a variety of part styles at their workstations. As indicated above, they must be inspectors as well as production workers in order to ensure the quality of their own output. They must be able to deal with minor technical problems that may be experienced with the production equipment, so that major breakdowns are avoided.

The suppliers of raw materials and components to the company must be held to the same standards of on-time delivery, zero defects, and other JIT requirements as the company itself. New policies such as the following are required for JIT for dealing with vendors:

- Reducing the total number of suppliers, thus allowing the remaining suppliers to do more business
- Entering into long-term agreements and partnerships with suppliers, so that suppliers do not have to worry about competitively bidding for every order
- Establishing quality and delivery standards and selecting suppliers on the basis of their capacity to meet these standards
- Placing employees into supplier plants to help those suppliers develop their own JIT systems
- Selecting parts suppliers that are located near the company's final assembly plant to reduce transportation and delivery problems.

## 3 AUTONOMATION

Appearing at first glance to be a misspelling of "automation" ***autonomation*** has been defined by Taiichi Ohno as "automation with a human touch" [13].[5] The notion is that the machines operate autonomously as long as they are functioning properly. When they do not function properly—for example, when they produce a defective part—they are designed to stop immediately. Another aspect of autonomation is that the machines and processes are designed to prevent errors, to be mistake-proof. Finally, machines in the Toyota Production System must be reliable, which requires an effective maintenance program. Three aspects of autonomation are discussed in this section: (1) stopping the process automatically when something is wrong, (2) error prevention, and (3) total productive maintenance.

### 3.1 Stopping the Process

Much of autonomation is embodied in the Japanese word ***jidoka***, which refers to machines that are designed to stop automatically when something goes wrong, such as a defective part being processed. Production machines in Toyota plants are equipped with automatic stop devices that activate when a defective work unit is produced.[6] Therefore, when a machine stops, it draws attention to the problem, requiring corrective action to be taken to avoid recurrences. Adjustments must be made to fix the machine, thereby eliminating or reducing subsequent defects and improving overall quality of the final product.

In addition to its quality control function, autonomation also refers to machines that are controlled to stop production when the required quantity (the batch size) has been completed, thus preventing overproduction (one of the seven forms of muda).

[5]An alternative definition of autonomation is "automation with a human mind" [12].
[6]The origins of ***jidoka*** in the Toyota Production System can be traced to Ohno's work experience early in his career in the textile industry, where the weaving machines were equipped with automatic stopping mechanisms that shut down the looms when abnormal operating conditions occurred. He implemented the idea at Toyota.

Although autonomation is often applied to automated production machines, it can also be used with manual operations as well. In either case, it consists of the following control devices: (1) a sensor to detect abnormal operation that would result in a quality defect, (2) a device to count the number of parts that have been produced, and (3) a means to stop the machine or production line when abnormal operation is detected or the required batch quantity has been completed.

The alternative to autonomation is when a production machine is not equipped with these control mechanisms and continues to operate abnormally, possibly completing an entire batch of defective parts before the quality problem is even noticed, or producing more parts than the quantity required at the downstream workstation. To avoid such a calamity in a plant that does not have automatic stop mechanisms on its machines, each machine must have a worker in continuous attendance to monitor its operation. Machines equipped with autonomation do not require a worker to be present all the time when it is functioning correctly. Only when it stops must the worker attend to it. This allows one worker to oversee the operation of multiple machines, thereby improving worker productivity significantly.

Because workers are called upon to service multiple machines, and the machines are frequently of different types, the workers must be willing and able to develop a greater variety of skills than those who are responsible for only a single machine type. The net effect of more versatile workers is that the plant becomes more flexible in its ability to shift workers around among machines and jobs to respond to changes in workload mix.

At Toyota, the jidoka concept is extended to its final assembly lines. Workers are empowered to stop the assembly line when a quality problem is discovered, using pull cords located at regular intervals along the line. Downtime on final assembly lines in the automotive industry is expensive. Managers desperately want to avoid it. They accomplish this by making sure that the quality problems that cause it are eliminated. Pressure is applied on the parts fabrication departments and suppliers to keep parts and subassemblies that are defective out of the final assembly area.

### 3.2 Error Prevention

This aspect of autonomation is derived from two Japanese words: ***poka***, which means error, and ***yoke***, which means prevention. Together, ***poka-yoke*** refers to the prevention of errors through the use of low cost devices that detect and/or prevent them. The poka-yoke concept was developed by Shigeo Shingo, who also did pioneering work in setup time reduction (Section 2.2). The use of poka-yoke devices relieves the worker of constantly monitoring the process for errors that might cause defective parts or other undesirable consequences (e.g., unsafe working conditions, omitting a processing step).

Mistakes in manufacturing are common, and they often result in the production of defects. Examples include the following:

- Omission of processing steps, such as neglecting to spray lubricant on a mold cavity to prevent sticking of the molded part
- Incorrectly locating a work part in a fixture
- Using the wrong tool (for example, using the wrong cutting tool material for a given work material)

- Not aligning jigs and fixtures properly on the machine tool table (which could result in the entire batch of parts being processed incorrectly)
- Neglecting to add a component part in an assembly

When a poka-yoke identifies that an error or other exception has occurred, it responds in either or both of the following ways:

- *Stops the process.* The mechanized or automated cycle is stopped when a problem is detected in the operation of a production machine. For example, a limit switch installed in a workholder detects that the workpiece is incorrectly located and is interlocked with the milling machine to prevent the process from starting.
- *Provides an alert.* This response provides an audible or visible warning signal that an error has occurred. This signal alerts the operator and perhaps other workers and supervisors about the problem. The use of andon boards (Section 4.2) is a means of implementing this type of response.

### 3.3 Total Productive Maintenance

Production equipment in the Toyota Production System must be highly reliable. The just-in-time delivery system cannot tolerate machine breakdowns, since there is little buffer stock between workstations to keep upstream and downstream stations producing when the middle station stops. Lean production requires an equipment maintenance program that minimizes machine breakdowns. ***Total productive maintenance*** (TPM) is a coordinated group of activities whose objective is to minimize production losses due to equipment failures, malfunctions, and low utilization through the participation of workers at all levels of the organization. Worker teams are formed to solve maintenance problems (such as the kaizen projects, discussed in Section 4.1). Workers who operate equipment are assigned the routine tasks of inspecting, cleaning, and lubricating their machines. This leaves the regular maintenance workers with time to perform the more demanding technical duties, such as emergency maintenance, preventive maintenance, and predictive maintenance (see definitions in Table 4). In TPM, the goal is zero breakdowns.

**TABLE 4** Some Maintenance Definitions

| Term | Definition |
|---|---|
| Emergency maintenance | Repairing equipment that has broken down and returning it to operating condition. Action must be taken immediately to correct the malfunction. Also known as reactive maintenance. |
| Preventive maintenance | Performing routine repairs on equipment (e.g., replacement of key components) to prevent and avoid breakdowns. |
| Predictive maintenance | Anticipating equipment malfunctions before they occur based on computerized machine monitoring, machine operator being attentive to the way the machine is running, historical data, and other predictive techniques. |
| Total productive maintenance | Integration of preventive and predictive maintenance to avoid emergency maintenance. |

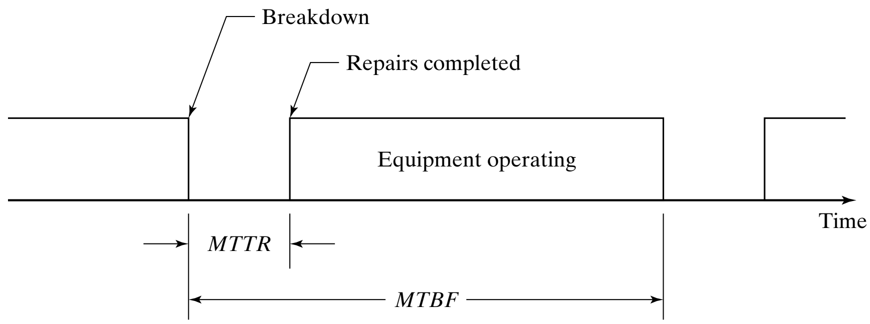


**Figure 4** Time scale showing MTBF and MTTR for a piece of machine that breaks down occasionally.

The traditional measure of machine reliability is ***availability***, which is the proportion of the total desired operating time that the machine is actually available and operating.[7] To be more precise, availability is defined in terms of two other reliability measures: ***mean time between failures*** ($MTBF$) and ***mean time to repair*** ($MTTR$). $MTBF$ indicates the average length of time the piece of equipment runs between breakdowns. $MTTR$ indicates the average time required to service the equipment and put it back into operation after a breakdown occurs. $MTBF$ and $MTTR$ are illustrated in Figure 4. Using these terms, ***availability*** is defined as follows:

$$A = \frac{MTBF - MTTR}{MTBF} \tag{1}$$

where $A$ = availability, $MTBF$ = mean time between failures, hr; and $MTTR$ = mean time to repair, hr. Availability is typically expressed as a percentage. When a piece of equipment is brand new (and being debugged), and later when it begins to age, its availability tends to be lower. This results in a typical U-shaped curve for availability as a function of time over the life of the equipment, as shown in Figure 5.

There are other reasons besides breakdowns that might cause a piece of production equipment to operate at less than its full capability. Such reasons include low equipment utilization, production of defective parts, and operating at less than the machine's designed speed.

***Utilization*** refers to the amount of output of a production machine during a given time period (e.g., week) relative to its capacity during that same period. This can be expressed as an equation,

$$U = \frac{Q}{PC} \tag{2}$$

where $U$ = utilization of the facility; $Q$ = actual quantity produced by the facility during a given time period—for instance, pc/wk; and $PC$ = production capacity for the same period, pc/wk. Utilization can also be measured as the number of hours of productive

[7]Availability is similar to line efficiency or uptime proportion.

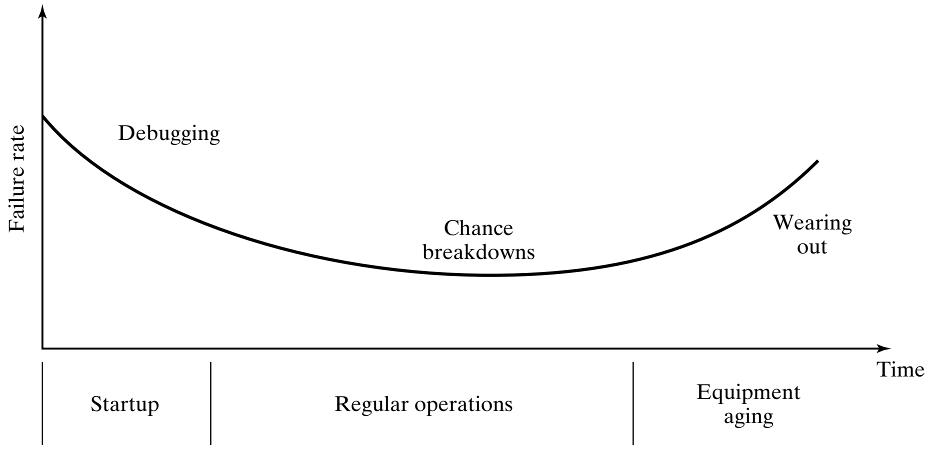


**Figure 5** Typical U-shaped availability curve for a piece of equipment during its life.

operation relative to the total number of hours the machine is available. Reasons for poor machine utilization include poor scheduling of work, machine starved for work by upstream operation, downtime due to setups and changeovers between production batches, worker absenteeism, and low demand for the type of process performed by the machine. Utilization can be assessed for a single machine, an entire plant, or any other productive resource (e.g., labor). It is often expressed as a percentage (e.g., the plant is operating at 83% of capacity).

Production of defective parts may be due to incorrect machine settings, inaccurate adjustments in the setup, or improper tooling. All of these reasons are related to equipment maintenance problems. Additional reasons for producing defects that may not be related to equipment problems include defective starting materials and human error. The ***fraction defect rate*** is defined as the proportion of defective parts that are produced in a given process. The yield of an individual process is defined as

$$Y = 1 - q \tag{3}$$

where $Y$ = process yield (ratio of conforming parts produced to total parts processed).

Finally, running the equipment at less than its designed speed also reduces its ***operating capability***, which is the ratio of the actual operating speed divided by the designed speed of the machine, symbolized as $r_{os}$.

All of these factors can be combined as follows to obtain a measure of the overall equipment effectiveness:

$$OEE = AUYr_{os} \tag{4}$$

where $OEE$ = overall equipment effectiveness, $A$ = availability, $U$ = utilization, $Y$ = process yield, and $r_{os}$ = operating capability. The objective of total productive maintenance is to make $OEE$ as close as possible to unity (100%).

## 4 WORKER INVOLVEMENT

Between the two pillars of lean production in Figure 1 are workers who are motivated and flexible and who participate in continuous improvement. Our discussion of worker involvement in the lean production system consists of three topics: (1) continuous improvement, (2) visual management and 5S, and (3) standardized work procedures.

### 4.1 Continuous Improvement

In the context of lean production, the Japanese word ***kaizen*** means continuous improvement of production operations. Kaizen is usually implemented by means of worker teams, sometimes called ***quality circles***,[8] which are organized to address specific problems that have been identified in the workplace. The teams address not only quality problems but also problems relating to productivity, cost, safety, maintenance, and other areas of interest to the organization. The term ***kaizen circle*** is also used, suggesting the broader range of issues that are usually involved in team activities.

Kaizen is a process that attempts to involve all workers as well as their supervisors and managers. Workers often are members of more than one kaizen circle. Although a principal purpose of organizing workers into teams is to solve problems in production, there are other less obvious but also important objectives. Kaizen circles encourage a an individual sense of responsibility, allow workers to gain acceptance and recognition among colleagues, and improve workers' technical skills [12].

Kaizen is applied on a problem-by-problem basis by worker teams. The team is constituted to deal with a specific problem. As mentioned above, the problem may relate to any of various areas of concern to the organization (e.g., quality, productivity, maintenance). Team members are selected according to their knowledge and expertise in the problem area and may be drawn from various departments. They serve part-time on a project team in addition to fulfilling their regular operational duties. On completion of the project, the team is disbanded. The usual expectation is that the team will meet two to four times per month, and each meeting will last about an hour.

The steps in each project depend on the type of problem being addressed. Details of the approaches recommended by different authors also vary [10, 15, 16]. Basically, the approach is the same as our systematic approach in methods engineering.

### 4.2 Visual Management and 5S

The principle behind ***visual management*** is that the status of the work situation should be evident just by looking at it.[9] If something is wrong, this condition should be obvious to the observer, so that corrective action can be taken immediately. Also called the ***visual workplace***, the principle applies to the entire plant environment. Objects that obstruct the view inside the plant are not allowed, making the entire interior space visible. The buildup of work-in-process is limited to a specified height (e.g., five feet). Thus, the visual workplace provides visibility throughout the plant and encourages continuous improvement and good housekeeping.

[8] The terms ***quality control circle*** and ***QC circle*** have also been used for these teams [9, 12].
[9] This is what statistician L. Tippett discovered in England during the 1920s while he was studying downtime of loom operations. The discovery resulted in the development of work sampling.

The use of kanbans in just-in-time production systems can be considered an example of visual management. The kanbans provide a visual mechanism for authorizing production and transportation of parts within the plant.

An important means of implementing visual management involves the use of an ***andon board***, a light panel positioned above a workstation or production line that is used to indicate its operating status. Its operation is commonly associated with the pull cords along a production line that permit a worker to stop the line. If a problem occurs, such as a line stoppage, the andon board identifies where the problem is and the nature of the problem. Different colored lights are often used to indicate the status of the operation. For example, a green light indicates normal operation, yellow means a worker has a problem and is calling for help, and red shows that the line has stopped. Other color codes may be used to indicate the end of a production run, shortage of materials, the need for a machine setup, and so on.

The visual workplace principle can also be applied in worker training. It includes the use of photographs, drawings, and diagrams to document work instructions, as opposed to lengthy text that is void of illustrations. "A picture is worth a thousand words" can be a powerful training tool for workers. In many cases, an actual example of the work part is used to convey the desired message; for example, examples of good parts and defective parts are shown to teach inspectors in quality control.

A means of involving workers in the visual workplace is a system called ***5S***, a set of procedures that is used to organize work areas in the plant. The name represents the first letters of five Japanese words as they would be spelled in English: ***seiri*** (sort), ***seiton*** (set in order), ***seiso*** (shine), ***seiketsu*** (standardize), and ***shitsuke*** (sustain).

The steps in 5S, briefly described in Table 5, provide an additional means of implementing visual management to provide a clean, orderly, and visible work environment that promotes high morale among workers and encourages continuous improvement. The steps are usually carried out by worker teams, and the 5S system must be a continuing process to sustain the accomplishments that have been made.

**TABLE 5** The Five Steps in the 5S System

1. ***Sort.*** Sort things in the workplace; in particular, identify the items that are not used and dispose of them. This eliminates the clutter that usually accumulates in a workplace after many years.
2. ***Set in order.*** Organize the items remaining in the work area after sorting according to frequency of use, and provide easy access to those items that are most often needed.
3. ***Shine.*** Clean the work area and inspect it to make sure that everything is in its proper place. The alternative S-words for this step are "sweep" and "scrub," which may also be required in the work area.
4. ***Standardize.*** Document the standard locations for items in the workplace. For example, use a "shadow board" for hand tools, in which the outline of the tool is painted on the board to indicate where it belongs. Looking at the shadow board, workers can immediately tell whether a tool is present and where to return it.
5. ***Sustain.*** Establish a plan for sustaining the gains made in the previous four steps, and assign individual responsibilities to team members for maintaining a clean and orderly work environment. Make workers responsible for taking care of the equipment they operate, which includes cleaning and performing minor maintenance tasks.

*Source:* [5] and [14].

### 4.3 Standardized Work Procedures

Standardized work procedures are established in the Toyota Production System, using approaches that are similar to the traditional methods engineering techniques. And time study is used to determine the length of time that should be taken to complete a given work cycle. However, there are differences in the Toyota approach to the development of standard methods and standard times, and these differences are emphasized in the discussion that follows. The following objectives are part of the standardized work procedures at Toyota:

- Maintain high productivity—that is, accomplish the required amount of production using the fewest possible number of workers.
- Balance the workload among all processes.
- Minimize the amount of work-in-process in the production sequence.

In the Toyota Production System, a standardized work procedure for a given task has three components [13]: (1) cycle time, (2) work sequence, and (3) standard work-in-process quantity. These components are documented using forms that emphasize Toyota's unique manufacturing procedures. The forms are sometimes quite different from those used in traditional methods engineering and work measurement.

**Cycle Time and Takt Time.** The ***cycle time*** is the actual time it takes to complete a task. This time is established using stopwatch time study. Closely related to the cycle time is the ***takt time***, the reciprocal of the demand rate for a given product or part, adjusted for the available shift time in the factory ("takt" is a German word meaning cadence or pace). For a given product or part,

$$T_{takt} = \frac{EOT}{Q_{dd}} \tag{5}$$

where $T_{takt}$ = takt time, $EOT$ = effective daily operating time, and $Q_{dd}$ = daily quantity of units demanded. The effective daily operating time is the shift hours worked each day, without subtracting any allowances for delays, breakdowns, or other sources of lost time. The daily quantity of units demanded is the monthly demand for the item divided by the number of working days in the month, without increasing the quantity to allow for defective units that might be produced. The reason why the effective daily operating time is not adjusted for lost time and the daily quantity is not adjusted for defects is to draw attention to these deficiencies so that corrective action will be taken to minimize or eliminate them.

**Example 2 Takt Time**

The monthly demand for a certain part is 10,000 units. There are 22 working days in the month. The plant operates two shifts, each with an effective operating time of 440 min. Determine the takt time for this part.

**Solution:** With two shifts, the effective daily operating time is 2(440 min) = 880 min. The daily quantity of units demanded is 10,000/22 = 454.5 pc/day.

$$T_{takt} = 880/454.5 = 1.94 \text{ min}$$

The takt time provides a specification based on demand for the part or product. In the Toyota Production System, the work must be designed so that the operation cycle time is synchronized with the takt time. This is accomplished through planning of the work sequence and standardizing the work-in-process quantity.

**Work Sequence.** The ***work sequence*** defines the order of work elements or operations performed by a given worker in accomplishing an assigned task. For a worker performing a repetitive work cycle at a single machine, it indicates the actions that need to be carried out, such as pick up the work part, load it into the machine, engage the feed, and unload the completed part at the end of the cycle. For a multifunction worker responsible for several machines, each with its own semiautomatic cycle, the work sequence describes what must be done at each machine and the order in which the machines must be attended. Pictures and drawings are often included (the visual workplace, Section 4.2) to show the proper use of hand tools and other aspects of the work routine such as safety practices and correct ergonomic posture.

For the multimachine situation, the work sequence is documented by means of the standard operations routine sheet, as shown in Figure 6. This form lists the machines that must be visited by the worker during each work cycle. A horizontal time scale indicates how long each operation should take, a solid line for the worker and dashed lines for the machines, and squiggly, nearly vertical lines are used to depict the worker walking between machines. The time lines show how well the machine cycle times are matched to the work routine performed by the worker each cycle.

Most production parts require more than one process (see sequential operations). It is not unusual for a dozen or more processing steps to be required to complete a part. The cycle time for a given process may be different from the takt time, which is based on the demand for the item produced in the task, not the requirements of the task itself. In the Toyota system, attempts are made to organize the work in such a way that the cycle time for each processing step is equal to the takt time for the part. By matching the cycle times with the takt time, all of the processes in the sequence are balanced, and the amount of work-in-process is minimized.

How can the work to produce a given part be organized so that the cycle time for each processing step is equal to the takt time? In the Toyota Production System, the

| Sheet # 236 | Date | | Cycle time 1.8 min | Daily Quantity | Worker Name |
|---|---|---|---|---|---|
| Op # | Operation | Time | | | |
| | Machine # | Manual | Machine | | |
| 1 | M-23 | 0.25 | 1.31 | | |
| 2 | M-16 | 0.25 | 1.44 | | |
| 3 | M-68 | 0.25 | 0.87 | | |
| 4 | D-47 | 0.20 | 0.36 | | |
| 5 | D-33 | 0.20 | 0.54 | | |
| 6 | D-25 | 0.20 | 0.62 | | |
| 7 | D-42 | 0.20 | 0.67 | | |

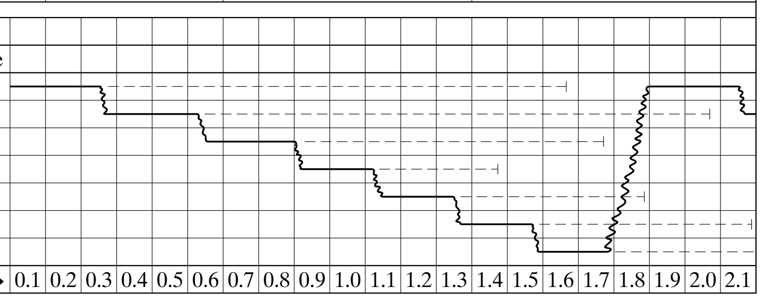


**Figure 6** The standard operations routine sheet.

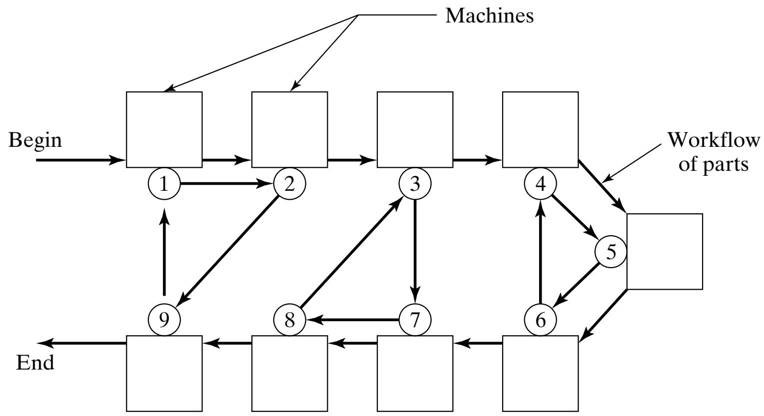


**Figure 7** Cell layout with three workers performing nine operations. Operations are indicated by numbers in circles. The first worker performs operations 1, 2, and 9; the second worker performs operations 3, 7, and 8; and the third worker performs operations 4, 5, and 6.

tasks are integrated into work cells. At Toyota, a ***work cell*** is a group of workers and processing stations that are physically arranged in sequential order so that parts can be produced in small batches, in many cases one at a time. The cells are typically U-shaped rather than straight to promote comradeship among the workers and to achieve continuous flow of work in the cell. An example of a work cell layout and the operations performed in it by three workers is shown in Figure 7.[10] A total of nine operations are performed by the three workers, as depicted by the arrows in the figure. The corresponding standard operations routine sheets are illustrated in Figure 8. The total workload is balanced among the workers and the operations are allocated so the corresponding cycle times are equal to the takt time for the part produced in the cell.

**Standard Work-in-Process Quantity.** ***Work-in-process*** (*WIP*) is the quantity of work parts that have been started in the production sequence but are not yet completed. *WIP* consists of both the parts that are currently being processed (e.g., parts that are fixtured in the machines) as well as those that are between processing steps (e.g., parts waiting to be loaded into machines). WIP does not include parts that have been completed, nor does it include the starting raw materials.

In the Toyota Production System, the standard work-in-process quantity is the minimum number of parts necessary to avoid situations where workers are waiting. For example, if the first worker in Figure 7 had only one work part for machines 1 and 2, he would have to wait for machine 1 to finish its cycle before moving the part to machine 2. But if there were two work parts, one in front of each machine, then he could load the first machine and move immediately to load the second machine without waiting for the first machine to finish.

[10]Based on an example in [12].

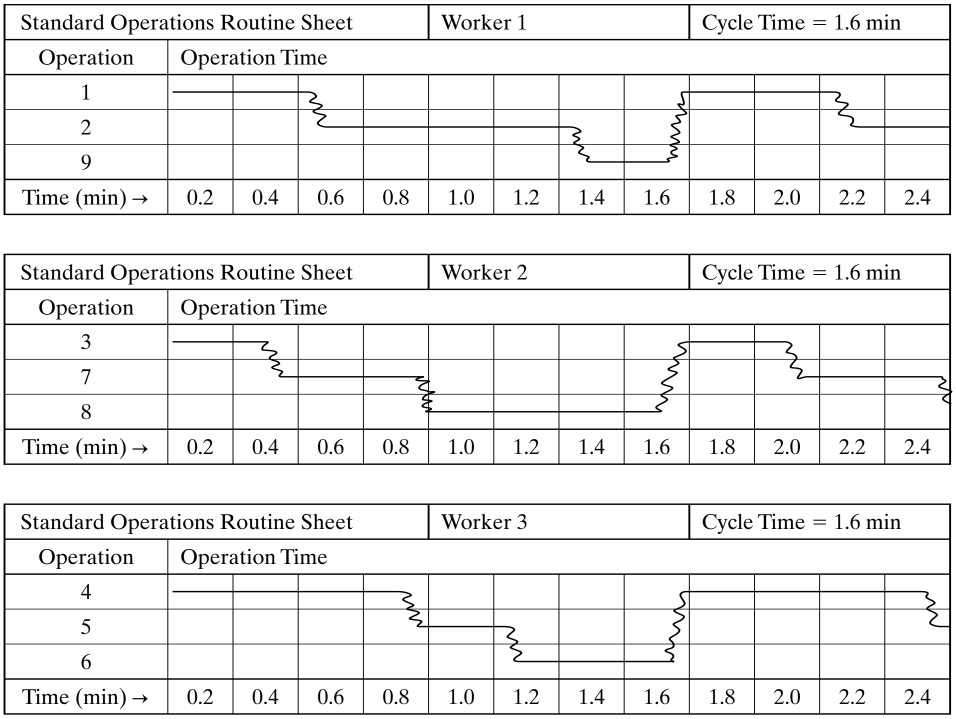


**Figure 8** The standard operations routine sheets for the three workers in Figure 7.

## REFERENCES

[1] Bodek, N. "Quick and Easy Kaizen." *IIE Solutions.* (July 2002): 43–45.
[2] Box, G. E. P., and N. R. Draper. *Evolutionary Operation*. New York: Wiley, 1969.
[3] Brecker, J. "Leaning Toward Six Sigma." Paper presented at a meeting of East Coast Region 3 Conference, Society of Manufacturing Engineers, Bethlehem, Pa., October 10–11, 2003.
[4] Claunch, J. W. *Setup Time Reduction.* Inc., Chicago: Irwin, 1996.
[5] Dennis, P. *Lean Production Simplified*. New York: Productivity Press, 2002.
[6] "Ford's Electronic Kanban Makes Replenishment Easy." *Lean Manufacturing Advisor.* 5, no. 5 (October 2003): 6–7.
[7] Groover, M. P. *Automation, Production Systems, and Computer Integrated Manufacturing*. 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.
[8] Hirai, Y. "Continuous Improvement (*Kaizen*)." Pp. 4.21–4.32 in *Maynard's Industrial Engineering Handbook.* 5th ed. edited by K. Zandin. New York: McGraw-Hill, 2001.
[9] Imai, M. *Kaizen, The Key to Japan's Competitive Success*. New York: McGraw-Hill 1986.
[10] Juran, J. M., and F. M. Gryna. *Quality Planning and Analysis*. 3rd ed. New York: McGraw-Hill, 1993.
[11] Monden, Y., and H. Aigbedo, "Just-in-time and Kanban Scheduling." Pp. 9.63–9.86 in *Maynard's Industrial Engineering Handbook,* 5th ed. edited by K. Zandin. New York: McGraw-Hill, 2001.
[12] Monden, Y. *Toyota Production System*. Norcross, GA: Industrial Engineering and Management Press, Institute of Industrial Engineers, 1983.

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# Introduction to Ergonomics and Human Factors



***Ergonomics*** is an applied scientific discipline that is concerned with how humans interact with the tools and equipment they use while performing tasks and other activities.[1] A common focus of ergonomics is the interface between humans and equipment—for example, the controls, displays, and other devices that are used by workers to operate the equipment. Our primary interest is on the activities that humans perform while working. These work activities impose a continuum of physical and cognitive demands on the human worker, as represented by the scale in Figure 1. In fact, most human activities require a combination of physical and cognitive exertions. Ergonomics is also concerned with the physical and social environment in which the tasks and activities are performed and how humans and machines interact with the environment. The word is derived from two Greek words, *ergon*, meaning work, and *nomos*, meaning rules or laws.[2] Combined, the words mean (roughly) the study of work. A professional whose practice is primarily focused on ergonomics is called an ***ergonomist.***

A term frequently used in place of ergonomics is ***human factors.*** Until recently, this term was most commonly used in the United States, while "ergonomics" was the

[1]The Work Measurement and Methods Standards Subcommittee (ANSI Standard Z94.12-1989) defines ergonomics as follows: "The application of a body of knowledge (life sciences, physical science, engineering, etc.) dealing with the interactions between man and the total working environment, such as atmosphere, heat, light and sound, as well as all tools and equipment of the workplace."
[2]The word ***ergonomics*** entered the English language in 1949. It was coined by British scientist K. F. H. Murrell, although it had been used first by Polish scientist W. Jastrzebowski in an 1857 newspaper article.

From Chapter 22 of *Work Systems and the Methods, Measurement, and Management of Work,* First Edition. Mikell P. Groover. 

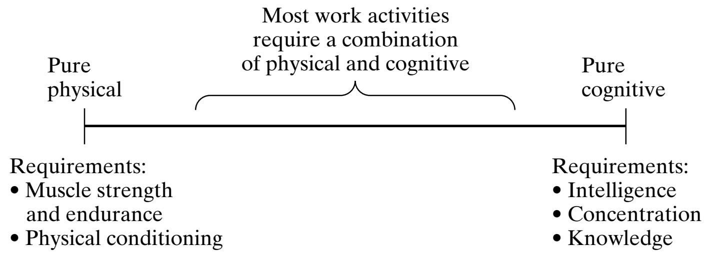


**Figure 1** Continuum of physical and cognitive demands that work activities impose on humans. Based on a figure in [13].

preferred term in Europe.[3] Today it is generally accepted that ergonomics and human factors are synonymous (Historical Note 1). If there was a difference in these terms, the difference was that ergonomics emphasized work physiology and anthropometry,[4] while human factors emphasized experimental psychology and systems engineering [7]. Also, the interest and applications in European ergonomics was more on industrial work systems, while the interest and applications in human factors in the United States was more on military systems.

### HISTORICAL NOTE 1 HISTORY OF ERGONOMICS

Ergonomics as an identified field of study can be traced back to around 1945, although some of the motivating influences precede this date. One of the most important was no doubt the scientific management movement, or ***Taylorism***,[5] with its emphasis on task planning, motion and time study, and worker efficiency. Critics have argued that Taylorism created industrial jobs that were simplistic and repetitive, and that it dehumanized labor. On the other hand, it made possible an industrial system in the United States that was the most productive in the world. And it focused attention on work as an area of study and research. Included in the scientific management movement were the pioneering studies of Frank and Lillian Gilbreth in the area of motion study, which has been referred to as "one of the forerunners" of human factors [15].

In the early 1900s, there was considerable effort in industry devoted to the selection of workers for a given job. This was called "fitting the man to the job" (FMJ). The idea was to choose from the pool of job applicants those individuals who were best suited to the requirements of the position on the basis of their physical and/or mental

[3]It is of interest to note that the Human Factors Society in the United States changed its name in 1992 to include the word ergonomics. The society is now called the Human Factors and Ergonomics Society.
[4]These terms are defined in Section 3.
[5]Named after Frederick W. Taylor, the father of scientific management.

aptitudes. Psychological testing was often used in the selection process when intelligence and personality were among the job requirements.

Studies carried out in the late 1920s at the Hawthorne Works of Western Electric Company near Chicago drew attention to the importance of social factors in the workplace. The Hawthorne experiments as they came to be known represented the beginning of interest in “human relations” research.

Finally, another important influence leading up to ergonomics was the growth in the use of machinery and mechanization between 1900 and 1945. The automobile industry was the significant growth industry at the beginning of this period, and the production of cars was enabled by mechanized machine tools and conveyorized assembly lines. In the 1930s, World War II forced nations to develop modern production technologies to meet the demand for munitions. Although the United States did not officially enter the war until the end of 1941, it was supplying weapons to its future allies during the 1930s.

Recognition of an emerging discipline related to humans working with machines crystallized around the end of World War II. It was called *ergonomics* in Europe and *human factors* in the United States. The war had witnessed a dramatic increase in the technological complexity and sophistication of weapons systems, such as aircraft, artillery, tanks, radar, and sonar. Significant problems were sometimes encountered in the operation of this equipment by humans, with loss of life too often the result. These problems only increased with the new jet aircraft that were being developed after the war. There was a need to deal with a new problem area known as *human–machine systems*. The interest in the United States was on aircraft and other weapons systems, and human factors research was frequently associated with “knobs and dials” between 1945 and 1960. The Human Factors Society was established in the United States in 1957.

Meanwhile, the field of ergonomics was starting in Britain. K. F. H. Murrell, an early contributor in the field, is credited with giving the discipline its name in 1949. The emphasis of ergonomics in Europe was on industrial work systems, including equipment and workspace design used in such systems. In 1950, British researchers met to discuss a new professional society to represent the discipline, and the name Ergonomics Research Society was adopted in that year (the name was later shortened to Ergonomics Society).

By the 1960s, the space age had arrived and the computer industry had begun its growth spurt. In the United States, human factors started to be applied outside of the military. The National Aeronautics and Space Administration (NASA) had been established in 1958 for the exploration of space, and human factors were quickly identified as a concern in manned space flight. Industrial applications of human factors also grew during the decades following 1960, to design both work systems and consumer products. A human factors group was established in the early 1960s at Eastman Kodak Company, the first company to form such a department.

Between 1980 and the present, interest in ergonomics and human factors has continued to grow, motivated on the one hand by advances in computer and automation technologies and on the other hand by several disasters that have highlighted the critical importance of the human in the operation of advanced human–machine systems.

In a computer system, the computer is the machine with which the human interacts. Human factors research has focused on the human-computer interface and the effective exchange and processing of information between these components. Several textbooks have been written on this subject, including two that are listed among our references [4, 6].

Since 1979, the following significant human-related disasters have contributed to public awareness about the importance of ergonomics and human factors:

- Three Mile Island nuclear power plant accident (near Harrisburg, Pennsylvania, March 1979)
- Bhopal pesticide plant, Union Carbide Company (Bhopal, India, December 1984)
- *Challenger* space shuttle (above Cape Kennedy, Florida, January 1986)
- Chernobyl nuclear power plant explosion (Kiev, Ukraine, April 1986)
- *Exxon Valdez* oil spill (Prince William Sound, Alaska, March 1989)
- *Columbia* space shuttle (over Texas, February 2003).

Many of these disasters were the result of human errors, sometimes caused by inadequate attention to ergonomics and human factors in system design and system management, and other times caused by flaws in decisions by humans.

This chapter provides an overview of the field of ergonomics and human factors, including its history and main topical areas. It then focuses on the human–machine system, an important model in the application of ergonomics.

## 1 OVERVIEW OF ERGONOMICS

In this section, we discuss the objectives and applications of ergonomics, and how the field has evolved into an important area in work design that attempts to adapt the task to the worker's capabilities.

### 1.1 Objectives and Applications of Ergonomics

The main objective of ergonomics is to improve the performance of systems consisting of people and equipment. Such systems are often referred to as human–machine systems (Section 2), although the word *machine* includes a variety of objects that are not normally thought of as machines—for example, aircraft, appliances, automobiles, chairs, computers, hand tools, and sports equipment. Better performance of human–machine systems in an ergonomics context means the following:

- Greater ease of interaction between the user and the equipment
- Avoidance of errors and mistakes by the user
- Greater comfort and satisfaction in the use of the equipment

- Reduced stress and fatigue while using the equipment
- Greater efficiency and productivity
- Safer operation of the equipment
- Avoidance of accidents and injuries.

These objectives are similar to those in methods engineering. Methods engineering and ergonomics are closely related and their general objectives are the same: (1) to improve the performance of existing systems and (2) to design new systems for optimum performance. Whereas methods engineering emphasizes issues such as efficiency, cost reduction, and elimination of waste, ergonomics emphasizes the human aspects such as comfort and safety and the interactions that occur between the human, the equipment, and the environment.

Ergonomics has a wide variety of applications, nearly all of which can be classified into two main areas:

- *Work system design.* This area is concerned with the interaction between worker and the equipment used in the workplace. Objectives include safety, accident avoidance, and related performance attributes. Work system design includes consideration of factors related to the work environment such as lighting and noise levels.
- *Product design.* This area deals with the design of products that are safer, more comfortable, and more user-friendly and mistake-proof. In addition to providing greater customer satisfaction by means of these kinds of features, an issue in product design is product liability lawsuits and their avoidance through consideration of ergonomics.

In our treatment of ergonomics here, the focus is on work systems, which in fact overlap with product design in the sense that work systems usually include equipment, and equipment is designed and fabricated by companies that are selling it as a product. A production machine is a product that must be designed with ergonomics in mind if it is to operate productively and safely.

### 1.2 Fitting the Job to the Person

A common employment practice prior to ergonomics was based on a philosophy called "fitting the person to the job" (FPJ), which recommended that workers be selected on the basis of their mental aptitudes and physical characteristics for a particular job opening.[6] Psychometric testing (e.g., tests for intelligence and personality characteristics) had begun to be used following World War I in the hiring process to improve productivity by trying to identify workers who had the right psychometric capabilities for the particular job. A worker's physical attributes were also used in the selection process for jobs requiring characteristics such as size and strength. The FPJ approach is still considered among the eligibility factors for certain positions in many hiring situations today; for example, firefighters must meet physical endurance requirements and

[6]The original term used was "fitting the man to the job" (FMJ). To observe modern gender-neutral language conventions, we have changed the name to "fitting the person to the job" (FPJ).

candidates seeking to become military pilots must meet height requirements. Educational qualifications are also widely used in selecting employees for certain positions such as engineering and public school teaching, and professional certification is required for many occupations, including medical doctors, dentists, and lawyers. All of these examples represent situations in which mental, physical, and/or educational requisites are used to ensure that the person is qualified for a given job.

The ergonomics approach is diametrically opposite of FPJ. The philosophy in ergonomics is "fitting the job to the person" (FJP)—that is, designing the job so that nearly any member of the workforce can perform it. There are several factors that explain why the new philosophy has evolved and now occupies a position that operates in parallel with and sometimes supersedes the FPJ approach: (1) changes in worker skill requirements, (2) demographic changes, and (3) social and political changes.

In the early decades of the 1900s, the general skill requirements for many jobs were much lower than today and there were many applicants, including many young people, for those low-skilled positions. Companies could be much more selective when hiring their workers. Today, the skill requirements of employment are much higher. Whereas the work in earlier times was largely physical and clerical, today it involves the operation of machinery and computers. The equipment is often sophisticated, and the work processes are complex. The average age of the available workforce is older than in the early 1900s. Companies are often faced with a smaller pool of applicants to choose from, and they cannot afford to be as choosy. They must hire from the pool those who are qualified but not necessarily optimal in the FPJ sense. Training must be provided, and the work itself must be designed to be readily understood so that the necessary amount of training can be minimized and workers can be productive as soon as possible after employment begins.

One significant manifestation of the demographic changes in employment has been the increase in the number of working women in the latter half of the twentieth century. More women have been entering the workforce in positions heretofore dominated by male workers. Women have been hired as assembly line workers and machine operators, and they have attended college in growing numbers to occupy positions as engineers and managers. Table 1 provides some statistical data to show the demographic changes that have taken place between 1930 and 2000.

At the same time that these changes in skill requirements and demographics were occurring, social and political pressures were effecting changes in cultural attitudes and government regulations. Equal opportunities legislation was enacted to prevent discrimination on the basis of race, gender, and age. Laws were enacted at the state and

**TABLE 1** Changes in U.S. Population and Workforce Between 1930 and 2000

| | 1930 | 2000 |
|---|---|---|
| Total population in the United States | 123 million | 281 million |
| Life expectancy | 60 years | 77 years |
| Median age | 27 years | 35 years |
| Number of people aged 65 and over | 7 million | 35 million |
| Proportion of women in the labor force | 22% | 61% |

*Source:* U.S. Census Bureau. (www.census.gov/pubinfo/www/1930_factsheet.html, March 28, 2002.)

federal levels to mandate that physically handicapped persons be provided with access ramps and other conveniences at public facilities and in transportation. Hiring of the handicapped was encouraged. In some cases, organizations went beyond the legislation by pursuing proactive policies on the hiring of minorities and the handicapped.

## 2 HUMAN–MACHINE SYSTEMS

Worker–machine systems, from an operational perspective, can focus on issues such as the kind of work done and the time factors involved. In this section, we consider the same basic work systems but from an ergonomics viewpoint. The basic model in ergonomics is the ***human–machine system***, defined as a combination of humans and equipment interacting to achieve some desired result. The number of humans can range from one to many, and the type and amount of equipment can range from a single hand tool to a complex and sophisticated collection of machines and computers. The desired result achieved by the system refers to its purpose and/or function—for example, to produce a product or provide a service. A human–machine system can have multiple objectives—for example, to produce a product, to contribute to the profit of the organization, and to provide a safe and comfortable workplace.

Human–machine systems can be classified into three basic categories [15]:

1. *Manual systems.* This system involves a person using some hand tool or other nonpowered implement to perform an activity. An example is a farmer using a pitchfork to load hay into a wagon.
2. *Mechanical systems.* This system refers to one or more humans using powered equipment to accomplish some job. The typical case is that the equipment provides the mechanical power for the job, and the function of the human(s) is to control the equipment. An example is a farmer driving a tractor to harvest a crop.
3. *Automated systems.* This system involves the performance of a job with a minimum of human attention. The truth is that automated systems do require occasional human attention, for purposes of maintenance and repair, reprogramming, or loading and unloading of parts from a production machine. In addition, humans are required to design and install automated systems. Thus, even automated systems are human–machine systems.

A key feature of a human–machine system is that interactions occur between the humans and machines, as depicted in Figure 2. The human controls the machine operation, and the machine displays its actions. The display may consist of the operator simply observing the operation, or it may involve some sort of display device such as a television monitor. Whatever the case, the human senses the operation, processes the sensed information, and performs actions on the machine controls that regulate its operation. The interactions consist of a cycle of activities accomplished by the human and the machine. The interactions occur in an environment—both physical and social—that may influence system performance. For example, poor lighting may impede a worker's ability to perform an inspection task; an unfriendly supervisor may reduce a worker's motivation to work.

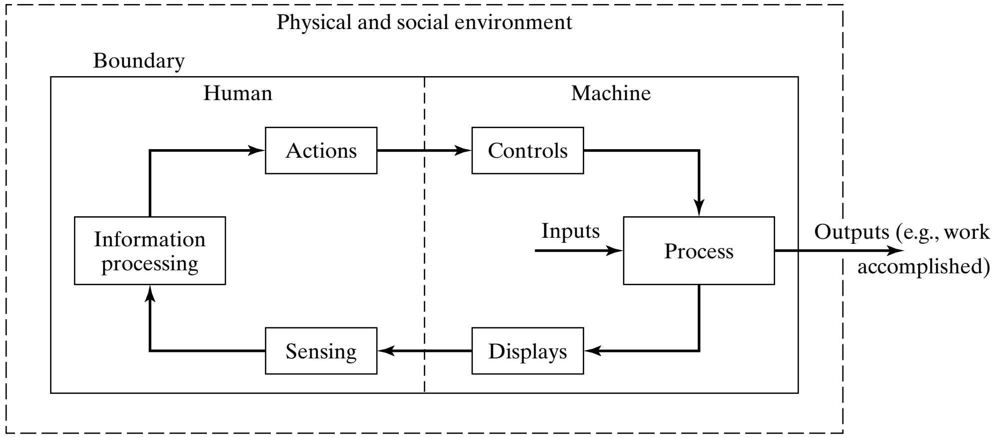


**Figure 2** Block diagram model of the interactions in a human–machine system.

From an ergonomics design viewpoint, a human–machine system has boundaries, as shown in Figure 2, that define what components are included within the scope of the system for purposes of analysis and design. Most systems are components of a larger system. There is a systems hierarchy. A worker–machine production cell is one component in the larger production department, which is only one of many departments in the company. The ergonomist must decide where to draw the boundaries of the human– machine system of interest. The boundaries separate the human–machine system from its environment.

The two main components of a human–machine system are the human and the equipment. In turn, the human and the machine have components that must be considered in the context of the interactions occurring between them. The third component is the environment, which interacts with the human–machine system. The ergonomic objective is to optimize the performance of the system by making the interactions between the human and the equipment as seamless as possible while the system operates in its environment.

**Human Components.** As suggested by the model of the human–machine system in Figure 2, the human components of interest are those that perform three functions: (1) sensing the operation, (2) information processing, and (3) actions. The corresponding human components are the following:

1. *Human senses.* The five basic human senses are vision, hearing, touch, taste, and smell. They are the sensors by which humans are able to be aware of where they are and what is happening around them. Vision and hearing are the most important in ergonomics, although the others are sometimes useful—for example, sensing that a machine is running too hot or that it is emitting noxious fumes. People use a combination of their senses in many of their activities. For example, when we eat a meal at a fine restaurant, we enjoy not only its taste but also the visual presentation of the food and its aroma.

2. *Human brain.* The human brain can be considered as a data- and information-processing unit, analogous in some ways to the operation of a computer. It receives data through the human senses as well as feedback from the joints and muscles of the body (data entry). It stores the data in memory (computer storage). It operates on the data (as in the operation of a computer's central processing unit) by thinking, planning, calculating, solving problems, and making decisions about what actions to take. These thought processes include both reflex and cognitive activities (low-level and high-level software). The decisions are communicated to and executed by the human effectors (computer peripherals).
3. *Human effectors.* An effector is a body part such as a muscle or group of muscles that actuates in response to some stimulus. The principal human effectors in the human–machine system are the fingers, hands, feet, and voice. These are supported by the musculoskeletal system of the body, and the stimulus is provided by the information processing occurring in the human brain.

These human components operate in a coordinated fashion and interact with the machine components in the human–machine system.

**Machine Components.** As indicated previously, the machine in the human–machine system can range from a simple hand tool to a complex and sophisticated system of equipment. The typical model in ergonomics is one in which the interactions between the human and the machine are directly coupled. The following are three common examples:

- A worker using a shovel to dig a hole in the ground. The digger maintains continuous control over the shovel, using arm and body strength to manipulate it and accomplish the hole-digging activity.
- A person driving a car. The driver continuously steers the car using the steering wheel and controls its speed using the throttle pedal, and the car provides feedback such as direction, speed data, and engine rpm. The connections between the car and driver are tangible as the car moves down the road.
- A student writing a term paper on a PC. The student types in the text using the alphanumeric keyboard, and the computer responds by displaying the text on the monitor and identifying spelling and grammar errors.

In other cases, the coupling is less direct and the interactions are less tangible, as in the following two examples:

- A worker monitoring the operation of an automated chemical process on a computer terminal. The automated process proceeds without human intervention. The worker's function is to make sure that each process variable remains within a defined tolerance of its target value and to take some action (directly interact with the process) only if the tolerance is exceeded or some other exception occurs.
- A researcher using the Internet to search for articles on a topic of interest. The researcher enters keywords about the topic, and a search engine looks on the

Internet to find Web sites that provide links with the keywords. The search engine is guided by its own software, independent of any further input from the researcher.

As Figure 2 indicates, there are three principal components of the machine in the human–machine system:

1. *The process.* The process refers to the function or operation performed by the human–machine system, such as digging a hole, driving a car, writing a term paper, performing a chemical process automatically, or searching the Internet.
2. *Displays.* For processes that are simple and the worker participates directly, the display function is accomplished by the worker observing the process as it is being performed. No additional display of information is needed. For example, the worker digging a hole sees the depth and shape of the hole as each shovelful is removed and proceeds with digging until the proper hole size has been achieved. For processes that are more mechanized and automated, there is a greater need for artificial displays to allow the human to see and understand what is going on. Driving a car requires a combination of natural and artificial displays. The driver receives input about the process directly by observing how the car is tracking the road ahead and indirectly by observing the speedometer and other dashboard gauges. Highly automated processes like the chemical process in our example require the worker to obtain the necessary information entirely from the display monitor because the process itself cannot be directly observed.
3. *Controls.* The human interacts with the equipment using machine controls that can be operated by the human effectors, usually the fingers and hands. When the equipment is simple (e.g., a hand tool), the controls often consist of one of the equipment components. In our example of a worker digging a hole, the handle of the shovel is the only control that is manipulated by the worker. As the machine becomes more complex, the controls tend to be more remote from the actual operation performed by the human–machine system. Examples of these remote controls include the steering wheel of an automobile, the alphanumeric keyboard of a personal computer, the foot pedals of a church organ, and the levers on an engine lathe for feed and speed.

**Environmental Components.** A human–machine system operates in an environment that includes two components:

- *Physical environment.* The physical environment includes the immediate area of the human–machine system, separated from the system by a defined boundary in Figure 2. Components of the physical environment usually include the location and surrounding lighting, noise, temperature, and humidity. These environmental factors can affect the performance of the human–machine system and are of interest to the ergonomist. For example, the workspace of a fighter aircraft is its cockpit, which imposes severe limitations on the freedom of movement of the pilot. The many controls of the aircraft must be located within easy reach of the pilot.

- *Social environment.* Work serves a social function. It is a chance for workers to meet and interact with other people. The primary work interactions are job-related, but social interactions occur also. The social environment at work is determined by coworkers and colleagues, immediate supervisors, organizational culture, and the work organization. The work organization consists of factors such as whether workers work alone or in teams, the required pace of the work, and whether the pace is set by the worker or by a machine.

## 3 TOPIC AREAS IN ERGONOMICS

As an academic subject, ergonomics has a breadth and depth that are much greater than we can do justice to in the space we have allocated to the subject in this text. For the interested reader, there are a number of good textbooks listed among our references as well as a handbook devoted to the subject. In our coverage, we emphasize the following topics.

**Physical Ergonomics.** Many work activities require physical exertion. The term *manual labor* is often used to refer to these activities. ***Physical ergonomics*** is mostly about manual labor. It is concerned with (1) how the human body functions during physical exertion and (2) how the physical dimensions of the body affect the capabilities of the worker. These two areas are related to the sciences of human physiology and anthropometry.

***Physiology*** is the branch of biology that deals with the vital processes carried out by living organisms and how their constituent tissues and cells function. Our interest in physiology is directed at how the human muscles function during work activities. There are strength and endurance limitations of the muscles that must be considered in the design of manual work. The energy expenditure that is required in the activity must be in balance with the capacity of the body's cardiovascular and respiratory systems. Rest periods must be allowed for work periods involving strenuous physical output. There are guidelines and formulas that can be used to determine the appropriate rest periods for work of a given physical energy expenditure.

***Anthropometry*** is the branch of anthropology that is concerned with the physical dimensions and other data related to the human body, such as height, reach, and weight. The data for a given dimension is normally distributed for a defined population (e.g., heights of females living in Northern Europe), but there are differences in these dimensions between defined populations (e.g., males versus females living in Northern Europe). There are several ergonomic design principles that relate to anthropometric data. One widely applied principle is "design for extreme individuals," which recommends that the designed entity should accommodate virtually all of the user population. For example, doorways should be designed for very tall people, even though the vast majority of users who pass through them are not very tall.

**Cognitive Ergonomics.** Most work activities also include cognitive elements as well as physical elements. Even manual labor usually requires some cognitive effort. Human activities with a high cognitive component include thinking, reading, speaking, learning, problem solving, and decision making. ***Cognitive ergonomics*** is concerned with the capabilities and limitations of the human brain and sensory system while performing activities that include a significant amount of information processing. It is the ergonomics area that considers how the human brain receives information from the environment, processes that information, and determines appropriate responses and actions. The human cognitive processes include the following components:

- *Sensory system.* This system consists of the five senses—vision, hearing, touch, smell, and taste—that are activated by external stimuli, such as sights and sounds. The vast majority of human information input is by sight and hearing.
- *Perception.* This is the stage of cognition that follows the sensing of external stimuli. It occurs when the human mind becomes aware of the sensation and interprets it based on previous experience and knowledge.
- *Memory.* Human memory consists of two main components: working memory and long-term memory. Working memory is temporary memory that holds a limited amount of information while being processed. It transfers information to and retrieves information from long-term memory, which is the warehouse of one's knowledge and experiences.
- *Response selection and execution.* This is the cognitive process of figuring out what actions are needed in the light of information perceived through the sensory system and processed through working memory and long-term memory.

The functions and operations of these components are not described completely here, nor are the number of ergonomic design guidelines that are derived from an understanding of them.

**The Physical Work Environment.** As noted in Figure 2, work is performed in an environment that includes both physical and social aspects. The physical work environment can be classified into the following components: (1) the visual environment, (2) the auditory environment, and (3) the climate.

The ***visual environment*** is what the worker sees while working. Most of the information input perceived by the worker is through visual stimuli. The important factor enabling the proper functioning of a worker's sense of vision is lighting. The term that describes how well things can be seen is ***visibility***. The visibility of tools, materials, and other items in the workplace depends on lighting levels, brightness contrast, color, and glare. The job of the ergonomist is to determine the optimum illumination levels for the task and to implement other design details in the workplace (e.g., types of lamps, colors, glare reduction) in order to achieve the best possible visual environment.

The ***auditory environment*** is what the worker hears while working. Much of the information input received by the worker is through the sense of hearing. Auditory information input is the second most important after visual in most work situations. Included in the auditory environment is ***noise***, which can be defined as sound that is undesired and possibly harmful to a worker's sense of hearing. The possible effects of noise range from minor distraction to major deafness. The most important factors about noise in terms of its effects on humans are (1) intensity, perceived as loudness, and (2) duration of exposure. The Occupational Safety and Health Administration has established permissible noise levels that define the duration of exposures allowed for various intensity levels. Employers must make provisions to control noise in the workplace through various administrative and engineering approaches.

The ***climate*** is what the worker feels while working. Climate in the workplace environment is defined by four primary variables: (1) air temperature, (2) humidity, (3) air movement, and (4) radiation from surrounding objects (e.g., machines that operate at elevated temperatures and radiate heat). To a limited extent, the human body can compensate for variations in the normal body temperature of 35°C (98.6°F) through such mechanisms as perspiration when body temperature is above normal and shivering when body temperature is below normal. However, the climatic conditions that cause heat stress and cold stress often require interventions beyond the body's natural responses. The interventions include proper clothing for the conditions, frequent rest breaks, and heating and air conditioning of the facility.

**Occupational Safety and Health.** Every year in the United States, approximately 5000 lives are lost due to occupational injuries, and about 50,000 lives are lost due to work-related diseases and illnesses. Occupational safety and health constitute an important national issue that affects virtually every person who works. Several government agencies have been established to address the issue. The following federal agencies were established with the enactment of the Occupational Safety and Health Act in 1971:

- Occupational Safety and Health Administration (OSHA), whose responsibilities include enforcing the provisions of OSHAct, implementing safety and health programs, and setting occupational safety and health standards.
- National Institute for Occupational Safety and Health (NIOSH), whose responsibility is to engage in research, training, and education in the area of occupational safety and health.

The topics of occupational safety and occupational health are certainly related, but distinctions can be made between them. ***Occupational safety*** is concerned with the avoidance of industrial accidents, which are one-time events that can cause injury or fatality. The occupational safety objective in ergonomics is to understand how and why accidents, injuries, and fatalities occur and to take the necessary steps to prevent them. ***Occupational health*** is concerned with avoiding diseases and disorders that are caused by workers being exposed to hazardous materials or conditions in the workplace. These diseases and disorders do not generally occur immediately upon exposure. Instead they tend to develop after prolonged periods of exposure to the materials or conditions. The effects of the exposure are cumulative and may take years before symptoms reveal the onset of the malady.

# *Occupational Safety and Health*



One of the important goals of ergonomics and human factors is to provide a safe workplace. Safety and a healthful working environment are not only desirable social objectives; they are also a significant cost issue for industry and government. Consider the following:[1]

- Each day, an average of 16 workers die from injuries that occurred while working. This amounts to approximately 5000 lives lost per year due to occupational injuries.
- These deaths and injuries are estimated to cost society more than $100 billion per year, not to mention the personal upheavals and tragedies inflicted upon the families involved.
- Each day, an average of 9000 workers in the United States suffer injuries on the job. This amounts to nearly 5 million workers per year.
- The number of workdays lost due to occupational injuries is between 50 million and 100 million per year.
- Each day, an average of 137 workers die from work-related diseases. This adds up to approximately 50,000 lives lost per year due to occupational illnesses.

[1]Compiled from Web sites for Bureau of Labor Statistics (www.bls.gov), Occupational Safety and Health Administration (www.osha.gov), and National Institute for Occupational Safety and Health (www.cdc.gov/niosh), based on 2002 data.

Although occupational safety and occupational health are related topics, they should be distinguished. ***Occupational safety*** is concerned with the avoidance of industrial accidents and in particular accidents that cause injury or fatality. These accidents are one-time events. The goal in ergonomics is to understand how and why they occur and to undertake steps to prevent them. ***Occupational health*** is concerned with avoiding diseases and disorders that are induced by exposures to materials or conditions in the workplace. These sicknesses and ailments do not generally occur immediately but instead develop after prolonged periods of exposure to the materials or conditions. The effects of the exposure are cumulative.

Our chapter begins with a discussion of industrial accidents and injuries: what causes them, and what steps can be taken to reduce their frequency and severity. The second section deals with disorders and diseases that result from exposures in the workplace. Next we review the important laws that have been enacted in the area of industrial safety and health. The federal agency that is most directly concerned with industrial safety and health is the Occupational Safety and Health Administration (OSHA). Finally, we define some of the measurements that are used to compile statistics about occupational safety and health.

## 1 INDUSTRIAL ACCIDENTS AND INJURIES

An ***industrial accident*** is an unexpected and unintentional event that disrupts work procedures and has the potential to cause damage to property and injury or death to workers. To be classified as an accident, the event does not have to actually cause damage, injury, or death; it only has to have the potential for these consequences. The objective of an occupational safety program is to reduce the incidence of accidents by reducing the hazards that precipitate them. A ***hazard*** is a condition or set of conditions that has the potential for causing an accident or other harmful outcome. The ***danger*** posed by a hazard is the relative exposure or liability to injury, death, and/or damage from that hazard. For example, a railroad crossing is a hazard, but it poses little danger so long as the crossing signal works and drivers pay attention to the signal.

Industrial accidents can cause injuries that are fatal or nonfatal. The Bureau of Labor Statistics maintains statistical records on occupational injuries, illnesses, and fatalities. The rates vary according to industry. A representative listing of fatal and nonfatal injury rates in the United States is compiled in Table 1. In Section 2, we provide a list of industrial illnesses. It should be noted that the statistics in these tables include only private industries (only private industries are covered by OSHA), and so it excludes government occupations, some of which are inherently dangerous (e.g., police, firefighters, military).

The causes of fatal and nonfatal injuries can be identified and classified, and the Bureau of Labor Statistics compiles statistics on the causes. Table 2 lists the major causes. Ranking at the top of the list are transportation incidents (e.g., highway accidents causing death or injury) that occurred during employment. So much industrial commerce involves transportation. Also note that homicides and suicides are high on the list. These are fatal injuries that occurred during work.

**TABLE 1** Private Industry Categories with Frequencies of Fatal and Nonfatal Injuries

| Industries in Order of Frequency of Fatal Injuries | Fatal Injuries per 100,000 Workers | Industries in Order of Frequency of Nonfatal Injuries | Nonfatal Injuries per 100 Workers |
|---|---|---|---|
| Mining | 23.5 | Construction | 6.9 |
| Agriculture | 22.7 | Manufacturing | 6.4 |
| Construction | 12.2 | Agriculture | 6.0 |
| Transportation | 11.3 | Transportation | 5.8 |
| Manufacturing | 3.1 | Wholesale, retail trade | 5.1 |
| Wholesale, retail trade | 2.4 | Services | 4.3 |
| Services | 1.7 | Mining | 3.8 |
| Finance, insurance, real estate | 1.0 | Finance, insurance, real estate | 1.5 |

*Source:* Bureau of Labor Statistics (www.bls.gov), based on 2002 data.

**TABLE 2** Causes and Exposures of Fatal and Nonfatal Injuries

| |
|---|
| Transportation incidents (e.g., collisions) |
| Overexertion (e.g., back problems from lifting) |
| Assaults and violent acts (includes homicides and suicides) |
| Contact with objects and equipment (e.g., struck by objects, compression) |
| Falls |
| Exposure to harmful substances and environments (electric shock, temperature extremes, hazardous substances) |
| Fires and explosions |

Why do industrial accidents occur? Three factors are commonly identified in the ergonomics literature as the primary reasons:

- *Human errors.* These factors relate to the worker or workers who are responsible for the operation and make mistakes that are sometimes the direct cause of an industrial accident.
- *Job factors.* This category refers to the kinds of tasks, methods, materials, equipment, and so on that are used in the operation. Some jobs are more dangerous than others.
- *Environmental conditions.* The environment includes lighting, noise, temperature, and other conditions that surround the operation. For example, the dim lighting in a plant made it difficult for a worker to see the obstacle in his path, and he tripped and broke his ankle.

These factors often act in combination, so that the cause of an accident cannot be isolated to a single reason.

In addition to these primary factors, there are organizational variables and management policies that influence the likelihood that an accident will occur in a given company. For example, if management shows little regard for safety in the plant, the company's workforce may develop a culture that disregards hazards, thus increasing the probability of accidents. A model that depicts all of these factors is shown in Figure 1.

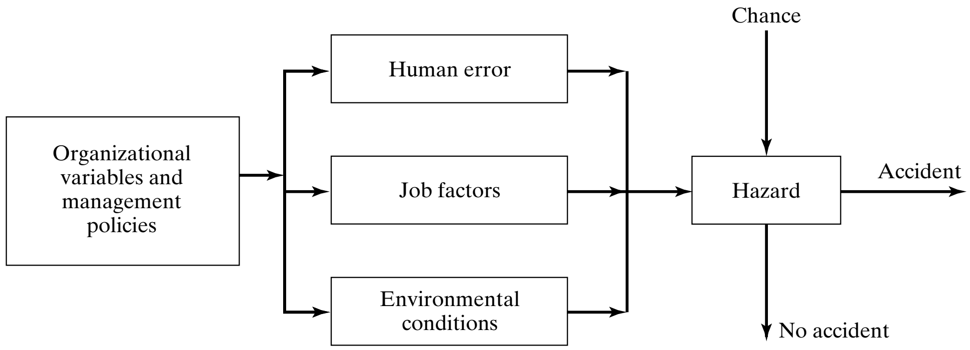


**Figure 1** A model depicting the possible factors that contribute to industrial accidents.

### 1.1 Human Error and Accidents

Human error is often identified as the cause of an industrial accident. In fact, human error probably plays a role in the various other factors identified in Figure 1 that ultimately contribute to an industrial accident. In the preceding example on environmental conditions, the plant engineer who neglected to bring the lighting in the plant up to an acceptable level caused the conditions that resulted in the worker breaking his ankle. Based on this kind of reasoning, it could be argued that all accidents are caused by human error, either directly or indirectly.

In the context of industrial accidents, ***human error*** can be defined as an improper and/or inadvertent human act or decision that has the effect or the potential to reduce effectiveness or safety in the workplace. Human errors can be classified as (1) errors of omission and (2) errors of commission. ***Errors of omission*** occur when a worker fails to take some action that is called for. For example, a home improvement contractor forgets to turn off the circuit breaker to the section of the house that he is rewiring and experiences an electrical shock (defined in Section 1.2). ***Errors of commission*** occur when a worker takes an action that is incorrect. For example, the same home improvement contractor remembers to turn off the circuit breaker, but he switches off the wrong circuit, causing him to suffer an electrical shock.

Among the factors that correlate with accident rates are (1) age, (2) time on the job, (3) fatigue, and (4) stress [9]. Age is a significant factor. Young persons have higher accident rates according to research studies. One can speculate that as people become more mature, they become more cautious and conservative, and their propensity for risk-taking decreases. The net effect is a reduction in accident rates. Contrasting with this trend, however, is that accident rates increase for the elderly, at least in work that requires a high level of physical and cognitive capacity.

Time on the job also correlates with accident rate. A high proportion of industrial accidents occur within the first three years of employment on a new job, with the rate peaking at two to three months [9]. This is explained by the fact that the new worker has completed whatever training program was provided but is still lacking in the experience needed to adequately recognize hazards and select optimal responses.

Finally, fatigue and stress are shorter-term factors that adversely affect accident rates. Fatigue reduces awareness, and stress reduces available attention resources for the task at hand. Both effects increase the likelihood of accidents. The reasons for stress need not be work-related. A worker's personal problems (e.g., financial, marital) may intrude on his or her capacity for safe job performance.

### 1.2 Job Factors

Accident rates are higher in certain industries because the jobs in those industries are inherently more dangerous. Construction work exposes employees to greater dangers than real estate work. Job factors include the characteristics of the task or job, such as the methods, equipment, and materials used.

**Methods.** The methods used in a job may be manual, mechanized, or automated, or there may be combinations of these categories for certain tasks. If completely manual, the job may require a high physical workload that leads to physical fatigue, a result that increases the likelihood of human error and accidents. Material handling represents a significant proportion of manual labor tasks in industry (e.g., moving cartons in warehouses, loading and unloading production machines, handling materials at a construction site), and it accounts for a large fraction of occupational accidents and injuries. Common injuries in material handling include (1) musculoskeletal strains and sprains resulting from lifting loads that are too heavy, (2) fractures and bruises caused by loads being dropped and similar mishaps, and (3) lacerations and bruises resulting from efforts to dislodge items in storage and cutting of packaging materials used for items in storage [2].

Mechanized and automated methods can often be used to mitigate the accident and injury rates due to manual material-handling tasks. These methods include the use of powered forklift trucks, conveyors, and cranes. However, material handling by machine carries its own risks. Forklift trucks are driven by humans, and humans commit errors. Aside from the bad driver issues, there are problems with loads not being centered on the forks, overloading, loads stacked too high, and so on. Conveyor systems provide an efficient method to move large amounts of materials in factories, warehouses, and distribution centers. Common causes of conveyor injuries include pinch points between moving and stationary components of the conveyor, loads jamming, loads falling from the conveyor, and workers falling as they try to cross the conveyor. Cranes are used for lifting and moving large loads in a facility, and hazards often arise simply because the loads are heavy. As the vice president of a crane company once said at a material-handling meeting, "We stand behind our product, but we don't stand beneath it."

**Equipment.** The equipment used by a worker can be the source of hazards in a job. The machine component often plays its role in the worker-machine system through its capacity to apply greater forces and power than a human can apply. The capacity for high forces and power gives rise to equipment hazards. These hazards can be classified into three categories: (1) electrical hazards, (2) mechanical hazards, and (3) temperature hazards.

**TABLE 3** Common Effects on Human Body of Various Current Levels for a 1-Second Duration

| Current Level (amps) | Effect on Human Body |
|---|---|
| 0.001 | Faint tingling sensation. |
| 0.005 | Slight shock felt that is disturbing but not painful. Most individuals can let go. |
| 0.006–0.025 | Painful shock with loss of muscular control. At the lower end of this range is the "let-go" current level, above which the individual is unable to voluntarily let go of the contact. |
| 0.050–0.150 | Extreme pain, respiratory arrest, muscular convulsions, and possible death. |
| 1 | Heart fibrillation (lack of rhythmic pumping action), severe muscular convulsions, and death likely. |
| 10 | Severe burns, cardiac arrest, and probable death. |

*Source: Controlling Electrical Hazards*, OSHA Publication 3075, U.S. Department of Labor, Washington, DC, 1986.

**TABLE 4** Techniques for Reducing Electrical Hazards

| |
|---|
| Insulate electrical circuits using materials that are suitable for the voltage and conditions (e.g., temperature, humidity, chemicals). |
| Isolate indoor electrical installations of more than 600 volts so that unqualified personnel cannot gain access (e.g., lock-controlled rooms or screens, elevation above the floor). |
| Provide proper grounding of equipment to provide a low-resistance path to the earth so that higher than normal voltages will avoid using the human body to complete the circuit to ground. |
| Use fuses and circuit breakers to limit excessive current flow. |
| Use appropriate personal protective equipment and clothing. |
| De-energize electrical equipment before performing maintenance. |

*Source:* Compiled from [2].

Electrical hazards arise because of ***electrical shock***, which is the discharge of electricity through the body that causes a sudden stimulation of the nerves and possible convulsive contraction of the muscles.[2] An electrical shock occurs when the body is part of an electrical circuit, and current passes through the circuit. The severity of the shock usually depends on the level of current, its duration, and the path of the current between the contact points of the body with the circuit. Table 3 describes the common effects on the human body of various current levels for a 1-second duration, where the path of the current is from the hand to the foot. It should be noted that these reactions vary for different individuals. Burns are the most common injury from electrical shock. Electrical burns are caused by (1) current passing through the body tissue, (2) electrical arcs and flashes, (3) contact with hot surfaces of overheated electrical components, and (4) burning clothing caused by electrical contact or arcing. Electrical shock sometimes induces secondary injuries due to involuntary muscle contractions. These muscle contractions can cause falls and impact injuries due to body reactions. Table 4 presents a list of actions that can be taken to reduce or prevent electrical shock hazards.

[2]Much of this section on electrical hazards is based on *Controlling Electrical Hazards*, OSHA Publication 3075, U.S. Department of Labor, Washington, DC, 1986.

Mechanical hazards arise due to the mechanical motions and forces applied by equipment components. Equipment producing these motions and forces include rotating machinery (e.g., lathes, motors, turbines), shearing equipment (e.g., power shears, brake presses), presses (e.g., press forges, stamping presses, injection molding machines), and impact equipment (e.g., stamping presses, forge hammers). The power generated by these kinds of equipment creates the conditions under which accidents can occur. If we add the potential for human error to the mix, there is an increased risk of fatalities and injuries such as scrapes, cuts, lacerations, broken bones, severed limbs, crushed body parts, and eye loss. Table 5 presents various means by which mechanical hazards are reduced.

The category of temperature hazards includes equipment that is too hot or too cold, but it does not include fire hazards, which are generally associated with the materials used in the job rather than equipment. However, it should be noted that fires are sometimes initiated by electrical problems in equipment, resulting in so-called "electrical fires." Equipment that normally operates at elevated temperatures includes processes such as arc welding, oxyfuel welding, brazing, soldering, heat treatment, flame cutting, metal casting, plastic molding, and glassworking. Burns are the obvious types of injuries that occur with equipment that operates hot. Cryogenic equipment operates at temperatures that are well below freezing, and human physical contact with the cold surfaces or materials (e.g., liquid nitrogen) used in the processes causes skin damage.

**Materials.** Many industries use materials in their normal operations that are classified as hazardous—that present health or safety hazards to workers, the facility in which they are used, or the environment. Materials that cause such hazards to humans can be classified into three categories: (1) corrosive materials, (2) toxic or irritant materials, and (3) flammable materials [5].

***Corrosive materials*** are usually acidic or caustic substances that can burn or damage human tissue. Exposure can occur due to skin contact or inhalation.

***Toxic or irritant materials*** are poisons that disrupt the normal body processes. They include liquids, gases, and solids. ***Toxicology*** is the science that studies poisons and their effects and problems. There are more than 50,000 known toxic substances. Their effects depend on the concentration of the toxic chemical, and in some instances the effects do not manifest themselves for years. Effects may include (1) cancerous tumors and other tumors, (2) embryo damage, (3) irritation of the skin, eyes, or respiratory tract, (4) reduction in mental alertness, (5) altered behavior, (6) general decline in health, (7) reduction in sexual function, and (8) other short-term or long-term illnesses [6].

**TABLE 5** Techniques and Devices for Reducing Mechanical Hazards

| |
|---|
| Design a machine so that its dangerous working parts are not accessible to operators. |
| Surround the entire machine with enclosures that isolate it from workers. |
| Use machine guards to enclose certain operations if enclosures for the entire machine are not feasible. |
| Install interlocks for doors that open to the operation, so that only when the doors are closed can the operating cycle proceed. |
| Place emergency stop buttons within easy reach of the operator. |
| Use two-hand start buttons that require both hands to begin the machine cycle. This ensures that both hands are out of the operation area of the machine. |

*Source:* Compiled from [2].

There are three ways in which toxic substances can enter the body: (1) inhalation, (2) absorption, and (3) ingestion. Inhalation is the most rapid entry mode, and since the respiratory and circulatory systems are so closely associated in the lungs, inhaled toxic materials are readily transported throughout the body. Absorption through the skin is less common than inhalation because the epidermis resists absorption, especially through the palms of the hands, where the skin is thicker. Finally, ingestion of toxic materials is also less common because humans tend to exercise reasonable judgments about the things they eat and drink.

***Flammable materials*** are those that present hazards of fire or explosion. Three ingredients are needed to ignite a flammable material: (1) an oxidizer, (2) heat, and (3) a chain chemical reaction. Preventing ignition of flammable materials means denying them of these conditions. There are several ways to remove these conditions: (1) separating the flammable material from the oxidizer, (2) quenching, and/or (3) inhibiting the chain chemical reaction. Table 6 describes the four classes of fires, based on the fuel involved. Different firefighting approaches are required for the each class.

### 1.3 Environmental Conditions

Environmental conditions are those factors in the immediate surroundings of an operation. They include (1) physical factors and (2) social or psychological factors. Table 7 organizes these environmental factors according to this distinction. If environmental factors are found to adversely affect workers, they will play a role in increasing the risk of industrial accidents.

**TABLE 6** Four Classes of Fires

| Class | Description |
|---|---|
| Class A | Burning of solid materials such as wood, coal, and paper. |
| Class B | Burning of gases or liquids that require vaporization for combustion. |
| Class C | Class A or B fires involving electrical equipment. |
| Class D | Burning of easily oxidized metals such as magnesium, titanium, or aluminum. |

**TABLE 7** Environmental Factors that May Affect the Likelihood of Industrial Accidents

| Physical Environmental Factors | Social/Psychological Environmental Factors |
|---|---|
| Lighting | Management policies and practices |
| Noise | Social and cultural norms |
| Temperature and humidity | Morale |
| Vibration | Training and instruction |
| Airborne pollutants | Incentives |
| Fire hazards | Safety programs in place |
| Radiation hazards | |
| Conditions that increase the likelihood of falls | |

*Source:* Based on a table in [9].

**TABLE 8** Some Common Cumulative Trauma and Other Occupational Disorders

| Disorder | Description |
|---|---|
| Carpal tunnel syndrome | Inflammation of the tendons in the carpal tunnel (in the wrist) resulting in compression of the median nerve, causing pain, swelling, and numbness. |
| Bursitis | Inflammation of the bursa, which are pouches containing fluid that reduce friction in the joints. |
| Tendonitis | Inflammation of certain tendons resulting from excessive use. |
| Trigger finger | Condition in which the person can curl the finger but is unable to straighten it out, except passively. |
| Neuritis | Inflammation of the nerve in a body member such as the hand. |
| Lower-back disorders | Condition of the lower back muscles and skeleton causing pain and accounting for a high proportion of all nonfatal occupational injuries. It is estimated that the jobs of as many as 30% of workers in the United States require tasks that risk lower-back disorders. |

*Source:* Compiled from [6] and [8].

## 2 OCCUPATIONAL DISORDERS AND DISEASES

The injuries and fatalities discussed in the previous section are one-time events or ***single-incident traumas***. Sometimes, the injury or health problem is one that results from a cumulative condition. These types of disorders, which may take years before their effects are evident, are called ***cumulative trauma disorders*** (CTDs). They are also known as repetitive-motion disorders because they are caused by repeated use of certain tendons and nerves, such as those in the fingers, wrist, forearm, upper arm, and shoulder. Occupations in which these body members are used repeatedly and in which CTDs are common include manual assembly with short cycle times, garment sewing, packaging, wiring, and carpentry [6]. In these kinds of manual operations, there are several causal factors leading to cumulative trauma disorders: (1) highly repetitive activity, (2) unnatural joint postures, and (3) application of forces through hinge joints. A list of cumulative trauma disorders is compiled in Table 8.

Occupational diseases usually result from exposure to a substance over an extended period of time. In some cases, the onset of the disease is delayed for many years. Some of these maladies are discussed in the context of toxic substances in Section 1.2. The more common occupational diseases are identified and briefly described in Table 9.

## 3 OCCUPATIONAL SAFETY AND HEALTH LAWS AND AGENCIES

This section focuses on the important legislation that has been enacted to promote safer and more healthful conditions in the workplace. In most cases, the laws have created governmental agencies to promulgate and enforce the policies and standards established by the legislation. Before these laws and agencies came into being, it is generally acknowledged that working conditions were neither safe nor healthful in many industries throughout the world. The two most important pieces of legislation enacted into law in the United States are (1) the various state workers' compensation laws and (2) the Occupational Safety and Health Act (OSHAct), a federal law.